

药材热湿耦合模型与干燥时间计算

完整附录

本附录给出固定半径与径向收缩模型的完整推导、离散与求解方法、支撑材料目录、计算结果及源程序。公式采用独立编号，源程序按文件完整列出；四问的全量径向与逐秒结果保存在所列 Excel 工作簿中。

物性关系使用题目给定经验式，计算环境与复现结论对应具体代码版本。检验与探索程序按实际使用范围列示，其历史试验不均属于最终模型。

附录目录

附录 A 连续模型、坐标变换与数值方法的完整推导	3
A.1 干基含水率、湿基含水率与参考质量	3
A.2 从组分质量平衡得到干基扩散方程	4
A.3 圆柱薄壳收支与固定域方程	5
A.4 四问物性与驱动条件	5
A.5 初始条件、表面边界与量纲	7
A.6 干物质密度、有效密度与收缩的一致性	7
A.7 一般移动坐标与网格速度项	8
A.8 节点控制体、轴心及表面权重	9
A.9 导热调和平均与 Kirchhoff 水分面量	11
A.10 干基质量与有效热收支的离散恒等式	12
A.11 解析 Jacobian：面量、容量、边界与累计量	13
A.12 隐式 BDF 与连续数值输出	15
A.13 全域最大含水率、临界根与严格报告时刻	16
A.14 固定物理位置查询与缩域输出	17
A.15 圆柱 Robin-Bessel 解析基准	18
A.16 解析比较的点值、单元平均及误差范围	19
A.17 方法比较、收缩尺度与参数敏感性	20
A.18 经验边界阻力及潜热情景的推导范围	21
A.19 公式与现行程序的对应关系及验证限度	22
A.20 推导依据	23
附录 B 支撑材料与复现说明	24
B.1 材料范围与代码覆盖	24
B.2 输入、单位与资料来源	24
B.3 计算环境与依赖	25

B.4 独立生产程序的复现步骤	26
B.5 已实际完成的数值复现与验证	27
B.6 检验源码的限制与未通过项目	29
B.7 正文六图的代码与数据衔接	30
B.8 当前 106 份正式支撑文件清单	31
附录 C 结果表与数值精度核查	37
C.1 临界事件与严格达标时刻	37
C.2 题目要求的径向采样结果	37
C.3 全量导出及一致性检查	40
附录 D 完整无注释源程序	41

附录 A 连续模型、坐标变换与数值方法的完整推导

本附录给出正文采用的有效热湿模型及其数值实现的推导，补充公式的成立条件、变量含义和计算检查范围。题给物性、环境观测及模型闭合分别说明；数学推导不改变正文已冻结的模型和结果。计算统一使用 m、s、K；含水率以 kg 水/kg 干物质计。题定长度为 $L_z=0.25$ m，初始半径为 $R_0=0.02$ m，初温为 $T_0=301.15$ K，初始干基含水率为 $C_0=2.55$ 。为避免与长度混淆，累计平均失水量统一记为 ℓ 。文献序号沿用正文参考文献；扩散与圆柱基准的理论出处为 Crank[4]，有限体积的一般构造见 Eymard 等[5]，隐式多步求解及具体实现见文献[6]、[7]。以下推导同时列明这些一般方法在当前题设中的使用条件。

A.1 干基含水率、湿基含水率与参考质量

设某一材料微团内干物质质量为 m_d 、水质量为 m_w 。干基含水率 C 与湿基水质量分数 w 定义为

$$C = \frac{m_w}{m_d}, \quad w = \frac{m_w}{m_d + m_w} = \frac{C}{1 + C}, \quad C = \frac{w}{1 - w} \quad (\text{A.1})$$

这里的 C 是质量比，可以大于 1； w 是总湿质量中的水质量分数，应在 0 与 1 之间。由式 (A.1)， $C_0=2.55$ 对应 $w_0=2.55/3.55 \approx 0.7183098592$ ，即初始水分约占湿质量的 71.83%。不能将 $C_0=2.55$ 直接写成 2.55% 或 255% 的湿基质量分数。

在当前物理体积内，分别记干物质质量密度为 ρ_d 、水质量密度为 ρ_w 。微团两组分占用同一统计体积时，干基定义等价于

$$\rho_w = \rho_d C, \quad m_d = \int_{\Omega(t)} \rho_d \, dV, \quad m_w = \int_{\Omega(t)} \rho_d C \, dV \quad (\text{A.2})$$

$\Omega(t)$ 为当前药材区域， ρ_d 与 ρ_w 的单位均为 kg/m^3 。水质量的计算必须有相应的干物质权重，不能把含水率节点的算术平均直接当作体积总水质量。

附件 1 的烘房水分指标仅标注 kg/kg ，记其观测数值为 $Y_\infty(t)$ 。基线采用 $C_{\text{eq}}(t)=Y_\infty(t)$ 的数值映射，构造表面有效平衡含水率。这是一项边界闭合假设。若进一步把 Y_∞ 解释为空气湿度比，则其分母是干空气质量，与药材 C 的干物质参考质量不同；两者同写 kg/kg 不能证明数值可直接相减。材料真实平衡含水率一般还依赖温度与吸附、解吸等温线，当前附件没有给出这种标定关系。

A.2 从组分质量平衡得到干基扩散方程

设干物质骨架速度为 v_s ，水相对骨架的质量通量为 j_w ，其单位为 $\text{kg}/(\text{m}^2\text{s})$ 。无干物质生成、损失及相对于骨架迁移时，干物质连续性方程为

$$\frac{\partial \rho_d}{\partial t} + \nabla \cdot (\rho_d v_s) = 0 \quad (\text{A.3})$$

水的实验室坐标总通量由随骨架运输的 $\rho_d C v_s$ 与相对通量 j_w 组成。无内部水源项时，水质量守恒为

$$\frac{\partial (\rho_d C)}{\partial t} + \nabla \cdot (\rho_d C v_s + j_w) = 0 \quad (\text{A.4})$$

将式 (A.4) 展开，得到

$$\rho_d \left(\frac{\partial C}{\partial t} + v_s \cdot \nabla C \right) + C \left[\frac{\partial \rho_d}{\partial t} + \nabla \cdot (\rho_d v_s) \right] = - \nabla \cdot j_w \quad (\text{A.5})$$

方括号为干物质连续性方程的左端，因而为零。定义随骨架运动的物质导数 $D_s/Dt = \partial_t + v_s \cdot \nabla$ ，即可得到干基含水率的输运方程

$$\rho_d \frac{D_s C}{Dt} = - \nabla \cdot j_w \quad (\text{A.6})$$

式 (A.6) 显示，骨架压缩导致的干密度变化已经在质量比的求导过程中相消。若没有相对水通量，则 $D_s C/Dt = 0$ ；纯几何收缩不使同一材料微团的干基质量比自行增加。

采用相对骨架的有效 Fick 闭合。扩散通量与浓度梯度关系的一般理论可参见 Crank[4]；这里对于骨架参考系和有效扩散率的选取仍是本模型的具体闭合：

$$j_w = - \rho_d D(T, C) \nabla C, \quad \rho_d \frac{D_s C}{Dt} = \nabla \cdot [\rho_d D(T, C) \nabla C] \quad (\text{A.7})$$

D 的单位为 m^2/s ，负号表示水沿含水率下降方向迁移。将散度展开后，若 ρ_d 空间不均匀，一般有

$$\frac{D_s C}{Dt} = \nabla \cdot (D \nabla C) + D \nabla (\ln \rho_d) \cdot \nabla C \quad (\text{A.8})$$

因此只有在干骨架密度空间均匀时，才能从散度内约去 ρ_d 。本文以初始干物质均匀分布、固定骨架或同比收缩保证这一条件；若采用非均匀收缩，须回到式 (A.7)，不能沿用已经约去密度的简式。

A.3 圆柱薄壳收支与固定域方程

忽略轴向、周向差异及两端面的传递，所有场量仅随径向位置 r 和时间 t 变化。固定半径时 $v_s=0$ 。半径 r 至 $r+dr$ 、长度 L_z 的薄壳体积为 $2\pi r L_z dr$ ，两侧传递面积分别为 $2\pi r L_z$ 与 $2\pi(r+dr)L_z$ 。将水的净流入量除以薄壳体积并令 $dr \rightarrow 0$ ，有

$$\frac{\partial C}{\partial t} = \frac{1}{r} \frac{\partial}{\partial r} [r D(T, C) \frac{\partial C}{\partial r}], \quad 0 < r < R_0 \quad (\text{A.9})$$

这里 $1/r$ 来自圆柱薄壳面积随半径的变化。 D 随状态变化时，应保留在径向导数内；把右端写成 $D(C_{rr} + C_r/r)$ 会遗漏系数梯度贡献。

热侧采用 Fourier 通量 $q = -k(C) \nabla T$ ，并以题给物性构造有效显热容量 $B(C) = \rho_{\text{eff}}(C) c_p(C)$ 。选择 $B(C) D_s T / Dt$ 作为本模型的局部温升需求，薄壳导热收支给出

$$\begin{aligned} B(C) \frac{D_s T}{Dt} &= \nabla \cdot [k(C) \nabla T], \\ B(C) \frac{\partial T}{\partial t} &= \frac{1}{r} \frac{\partial}{\partial r} [r k(C) \frac{\partial T}{\partial r}] \quad (v_s=0) \end{aligned} \quad (\text{A.10})$$

k 的单位为 $\text{W}/(\text{m K})$ ， c_p 为 $\text{J}/(\text{kg K})$ ， ρ_{eff} 为 kg/m^3 ，故 B 为 $\text{J}/(\text{m}^3 \text{K})$ 。式 (A.10) 是本文采用的有效显热闭合；由 Fourier 定律只能确定导热通量，不能单独推出完整多组分能量方程恰好等于该容量形式。

特别地， ρ_{eff} 不自动等于质量守恒中的 ρ_d 。比热、密度依赖含水率时，有

$$\frac{D_s}{Dt} [B(C) T] = B(C) \frac{D_s T}{Dt} + T B'(C) \frac{D_s C}{Dt} \quad (\text{A.11})$$

因此不能把式 (A.10) 的左端改写为 $D_s(BT)/Dt$ 而省略第二项。基线未逐项建立水分迁移携焓、相变热及机械功，后面的有效热通量恒等式只检查所采用的方程，不证明真实完整过程的总能量平衡已经闭合。

A.4 四问物性与驱动条件

问题一采用题给附录 2 的常热物性及含水率相关扩散率：

$$\rho_{\text{eff}} = 820, \quad c_p = 2600, \quad k = 0.36, \quad D_1(C) = 7 \times 10^{-9} \exp(-0.89/C) \quad (\text{A.12})$$

密度、比热、导热率和扩散率依次使用 kg/m^3 、 $\text{J}/(\text{kg K})$ 、 $\text{W}/(\text{m K})$ 和 m^2/s 。问题一的热场可独立求解，水分扩散仍是非线性的； D_1 不含温度因子。

问题二与问题三从同一均匀初态开始，在整个过程采用题给附录 3：

$$\begin{aligned}\rho_{\text{eff}}(C) &= 650 + 128C, \\ c_p(C) &= 1450 + 2736 \frac{C}{1+C}, \\ k(C) &= 0.21 + 0.38 \frac{C}{1+C}, \\ D_{23}(T, C) &= 2.4 \times 10^{-3} \exp\left(-\frac{0.45}{C} - \frac{3850}{T}\right)\end{aligned}\quad (\text{A.13})$$

问题四从初始时刻整组使用题给附录 4，并同时引入半径变化：

$$\begin{aligned}\rho_{\text{eff}}(C) &= 760 + 90C, \\ c_p(C) &= 1850 + 2150 \frac{C}{1+C}, \\ k(C) &= 0.12 + 0.20 \frac{C}{1+C}, \\ D_4(T, C) &= 4.2 \times 10^{-4} \exp\left(-\frac{0.30}{C} - \frac{3850}{T}\right)\end{aligned}\quad (\text{A.14})$$

式 (A.13)、式 (A.14) 的单位与式 (A.12) 相同。 T 必须以 K 代入，3850 带有使 $3850/T$ 无量纲的温度尺度； a/C 中参数 a 与干基含水率采用同一数值口径。这些系数来自题给经验关系，不是由本次附件重新拟合，也不是从第一原理推导的常数。问题二、三不拼接问题一的末态，问题四也不在某一阶段才切换附录 4。

相同 T, C 下，两组全过程扩散率之比为

$$\frac{D_4(T, C)}{D_{23}(T, C)} = 0.175 \exp\left(\frac{0.15}{C}\right), \quad C_{\times} = \frac{0.15}{\ln(1/0.175)} \approx 0.08606 \quad (\text{A.15})$$

其中 $C_{\times}$ 是两扩散率相等的含水率。 $C > C_{\times}$ 时 $D_4 < D_{23}$ ， $0 < C < C_{\times}$ 时顺序反转。因此不能写“附录 4 扩散系数在全程整体更小”；收缩和整组物性变化的净作用需由保持条件明确的数值对照判断。

环境表有 241 个 0—4 h 记录，半径表有 145 个 0—72 h 记录。对相邻时间节点 $t_j \leq t \leq t_{j+1}$ ，以 f_j 表示同一观测变量，分段线性插值为

$$I_f(t) = f_j + \frac{f_{j+1} - f_j}{t_{j+1} - t_j} (t - t_j) \quad (\text{A.16})$$

温度观测先换成 K；半径换成 m。正文基线环境为

$$(T_{\infty}, C_{\text{eq}})(t) = \begin{cases} (I_T(t), I_Y(t)), & 0 \leq t \leq 14400 \text{ s}, \\ (323.15, 0.05), & t > 14400 \text{ s}. \end{cases} \quad (\text{A.17})$$

式 (A.17) 后一段是平台延拓假设, 不是持续数天的观测。收缩半径在附件 2 的观测区间内使用 $R(t)=I_R(t)$ 。正式问题四的达标时刻位于 72 h 内, 未用到半径观测窗外延拓; 对其他长时情景则必须另行说明。

A.5 初始条件、表面边界与量纲

三条正式轨迹均从均匀初态开始, 轴心由对称性给零径向梯度:

$$T(r, 0)=T_0, \quad C(r, 0)=C_0, \quad T_r(0, t)=C_r(0, t)=0 \quad (\text{A.18})$$

记 $T_s=T(R(t), t)$ 、 $C_s=C(R(t), t)$ 为真实表面状态。向外为正时, 表面热、水通量分别为

$$q_{\text{out}}=-k T_r=h(T_s-T_\infty), \quad j_{w, \text{out}}=-\rho_d D C_r=\rho_d \beta(C_s-C_{\text{eq}}) \quad (\text{A.19})$$

$h=25 \text{ W}/(\text{m}^2\text{K})$, $\beta=8 \times 10^{-7} \text{ m/s}$ 。热边界两侧单位均为 W/m^2 ; 水质量边界两侧单位均为 $\text{kg}/(\text{m}^2\text{s})$ 。将水边界约去同一表面干密度得到源码使用的形式

$$-D C_r=\beta(C_s-C_{\text{eq}}) \quad (\text{A.20})$$

式 (A.20) 的两侧是以干密度归一化的传递量, 其长度—时间单位为 m/s , 尚不是质量通量。材料较空气冷时 $T_s < T_\infty$, 向外热通量为负, 热量进入材料; $C_s > C_{\text{eq}}$ 时向外水通量为正, 材料失水。该符号规则贯穿边界与控制体离散。

初值与瞬时表面 Robin 条件未必在 $t=0$ 完全相容。这表示启动阶段形成快速边界层, 不要求人为改动题给初态。数值求解采用较小的早期步长, 并另行检查接受状态和早期场误差。

A.6 干物质密度、有效密度与收缩的一致性

问题一至三令骨架固定; 问题四采用定长、径向同比收缩。每个初始材料半径 r_0 随时间变为 $r=r_0 R(t)/R_0$, 相应骨架速度为

$$v_s(r, t)=\left.\frac{\partial r}{\partial t}\right|_{r_0}=\frac{r \dot{R}(t)}{R(t)}, \quad \nabla \cdot \mathbf{v}_s=\frac{1}{r} \frac{\partial(r v_s)}{\partial r}=2 \frac{\dot{R}(t)}{R(t)} \quad (\text{A.21})$$

$\dot{R} < 0$ 表示收缩; 轴向速度取零。式 (A.3) 因而给出

$$\begin{aligned}
\frac{D_s \rho_d}{D t} &= -2 \frac{\dot{R}}{R} \rho_d, \\
\rho_d(t) &= \rho_{d0} \left(\frac{R_0}{R(t)} \right)^2, \\
M_d &= \pi L_z R(t)^2 \rho_d(t) = \pi L_z R_0^2 \rho_{d0}
\end{aligned} \tag{A.22}$$

这里 M_d 为恒定总干质量, ρ_{d0} 为初始均匀干物质密度。同比收缩使每个材料体积均按 R^2/R_0^2 缩小, 因此初始均匀干密度保持空间均匀而随时间变化。半径的表面观测不能单独证明内部运动同比, 式 (A.21) 是用于闭合内部运动的明确假设。

若同时强行把附录 4 的 760+90C 解释为真实湿密度, 则非负含水率下应有

$$\rho_d = \frac{760+90C}{1+C} = 90 + \frac{670}{1+C} \leq 760 \tag{A.23}$$

以初始关系 $\rho_{d0} = (760+90C_0)/(1+C_0) \approx 278.7323943662$ 归一化干质量, 再与式 (A.22) 联立, 得到

$$R(t) \geq R_0 \sqrt{\frac{\rho_{d0}}{760}} \approx 1.2112029565 \text{ cm} \tag{A.24}$$

附件 2 在 72 h 给出 1.198 cm, 小于上述下界。因此真实湿密度解释、非负含水率、固定长度和给定收缩轨迹不能在所给全时段同时成立。本文将 $\rho_{\text{eff}} c_p$ 用作有效热容量, ρ_d 独立守恒; 初始归一化不意味着随时重新设置 $\rho_d = \rho_{\text{eff}}(C)/(1+C)$ 。基线温湿方程约去均匀干密度后不依赖其绝对标度; 只有换算绝对水量或添加指定潜热负荷时才需要该标度。

A.7 一般移动坐标与网格速度项

为明确网格运动的来源, 先不假定网格速度等于骨架速度。取 $r = R(t)x$, $0 \leq x \leq 1$; 定义 $\hat{C}(x, t) = C(R(t)x, t)$, $\hat{T}(x, t) = T(R(t)x, t)$ 。固定 x 的网格速度为 $v_g = x \dot{R}$ 。链式法则给出

$$\left. \frac{\partial C}{\partial t} \right|_r = \left. \frac{\partial \hat{C}}{\partial t} \right|_x - \frac{x \dot{R}}{R} \frac{\partial \hat{C}}{\partial x}, \quad \frac{\partial C}{\partial r} = \frac{1}{R} \frac{\partial \hat{C}}{\partial x} \tag{A.25}$$

把物理方程中的骨架运输项同时变换, 得到

$$\begin{aligned}
\frac{D_s C}{D t} &= \frac{\partial \hat{C}}{\partial t} + \frac{v_s - v_g}{R} \frac{\partial \hat{C}}{\partial x}, \\
\frac{D_s T}{D t} &= \frac{\partial \hat{T}}{\partial t} + \frac{v_s - v_g}{R} \frac{\partial \hat{T}}{\partial x}
\end{aligned} \tag{A.26}$$

在干密度空间均匀的条件下，完整变换后的水分方程为

$$\frac{\partial \hat{C}}{\partial t} + \frac{v_s - x \dot{R}}{R} \frac{\partial \hat{C}}{\partial x} = \frac{1}{R^2 x} \frac{\partial}{\partial x} (x D \frac{\partial \hat{C}}{\partial x}) \quad (\text{A.27})$$

式 (A.27) 保留了网格相对骨架运动的项。如果假设骨架在空间固定而只有计算边界改变，则 $v_s=0$ ，左侧必须保留 $-x \dot{R} \hat{C}_x / R$ ；不能直接套用材料随动形式。这样的假设是否与边界处物质流失、干质量守恒相容，还需重新说明，不能只换坐标而省略物理解释。

本文问题四采用式 (A.21)，故 $v_s=v_g=x \dot{R}$ 。坐标变化项与骨架平流项恰好相消，材料坐标控制方程为

$$\begin{aligned} \frac{\partial \hat{C}}{\partial t} &= \frac{1}{R(t)^2 x} \frac{\partial}{\partial x} (x D(\hat{T}, \hat{C}) \hat{C}_x), \\ B(\hat{C}) \frac{\partial \hat{T}}{\partial t} &= \frac{1}{R(t)^2 x} \frac{\partial}{\partial x} (x k(\hat{C}) \hat{T}_x) \end{aligned} \quad (\text{A.28})$$

固定半径的前三问是 $R(t)=R_0$ 、 $v_s=v_g=0$ 的特例。材料坐标形式不再重复添加网格平流，也不增加 $-2 \dot{R} C / R$ 的体积浓缩项；后者只适用于相应体积密度的收支，不能施加在已由组分方程消去压缩项的干基比值上。

表面导数换元后，边界条件变为

$$\begin{aligned} \hat{T}_x(0, t) &= \hat{C}_x(0, t) = 0, \\ -\frac{k_s}{R} \hat{T}_x(1, t) &= h(T_s - T_\infty), \\ -\frac{D_s}{R} \hat{C}_x(1, t) &= \beta(C_s - C_{\text{eq}}) \end{aligned} \quad (\text{A.29})$$

k_s, D_s 为当前表面物性。 $1/R$ 来自空间导数换元，内部方程中的 $1/R^2$ 来自两次长度缩放。半径折线的 $\dot{R}$ 在观测节点处可不连续，但相消后的求解方程只需连续的 $R(t)$ ；各开区间上的推导可拼接为相容的分段解，不需对折线半径作数值微分。

A.8 节点控制体、轴心及表面权重

有限体积法以控制体积分收支和共用面通量建立离散方程，一般框架见 Eymard 等[5]；本题的具体圆柱权重和节点配置如下推导。取 N 个均匀区间， $\Delta x=1/N$ ，节点 $x_i=i \Delta x$ ， $i=0, \dots, N$ 。对偶控制体的内部边界为相邻节点中点，左、右外边界分别截在 0 和 1。记第 i 个控制体为 $[a_i, b_i]$ ，其圆柱权重为

$$\begin{aligned}
w_i &= \int_{a_i}^{b_i} x \, dx = \frac{b_i^2 - a_i^2}{2}, \\
w_0 &= \frac{\Delta x^2}{8}, \quad w_i = x_i \Delta x \quad (1 \leq i < N), \\
w_N &= \frac{\Delta x}{2} - \frac{\Delta x^2}{8}
\end{aligned} \tag{A.30}$$

中心控制体为 $[0, \Delta x/2]$ ，表面控制体为 $[1 - \Delta x/2, 1]$ 。由区间拼接可得 $\sum_i w_i = 1/2$ 。单位轴向长度的实际控制体积为 $2\pi R^2 w_i$ ，完整控制体体积为 $2\pi L_z R^2 w_i$ 。

未知量 T_i, C_i 位于节点，节点包括轴心与真实表面。用 $w_i C_i$ 近似积分 $\int_{a_i}^{b_i} x \hat{C} \, dx$ ，用 $w_i B(C_i) \dot{T}_i$ 近似有效热容量积分。这个集总近似不将节点值变成精确单元平均；点值基准与环体平均基准应分别比较。

定义便于装配的梯度面量

$$g^T = x k \hat{T}_x, \quad g^C = x D \hat{C}_x \tag{A.31}$$

g^T 单位为 W/m ， g^C 为 m^2/s ，它们与向外物理热流、水流的符号相反。对式 (A.28) 在控制体内乘 x 积分，并采用上述节点集总，得到半离散方程

$$\dot{T}_i = \frac{g_{i+1/2}^T - g_{i-1/2}^T}{R^2 w_i B_i}, \quad \dot{C}_i = \frac{g_{i+1/2}^C - g_{i-1/2}^C}{R^2 w_i}, \quad B_i = B(C_i) \tag{A.32}$$

这里 $i-1/2, i+1/2$ 表示第 i 个控制体的两侧面；中心外侧面和表面外侧面的坐标按 0、1 截断。内部面必须由相邻控制体共用同一个数值，左控制体取正贡献，右控制体取负贡献。

中心面直接令 $g_{-1/2}^T = g_{-1/2}^C = 0$ ，因此程序不计算 $1/x_0$ 。例如常系数热方程的第一行是

$$\dot{T}_0 = \frac{4k(T_1 - T_0)}{B R^2 \Delta x^2} \tag{A.33}$$

若光滑轴对称场在中心展开为 $T(x) = T(0) + \frac{1}{2} T_{xx}(0) x^2 + O(x^4)$ ，式 (A.33) 趋于 $2k T_{xx}(0)/(B R^2)$ ，与圆柱算子 $k(T_{xx} + T_x/x)/(B R^2)$ 的轴心极限一致。

由式 (A.29)，表面外侧面直接取

$$g_{N+1/2}^T = h R (T_\infty - T_N), \quad g_{N+1/2}^C = -\beta R (C_N - C_{eq}) \tag{A.34}$$

最后一个未知量本身位于实际表面，无须从末单元中心再外推一个表面值，也不额外附加半单元传质内阻。若改成单元中心网格，未知量位置与边界离散必须同步重建。

A.9 导热调和平均与 Kirchhoff 水分面量

对内部面 $i+1/2$ ，记 $\gamma_i = x_{i+1/2} / \Delta x$ 。两侧各半间距的导热阻相加，等效导热系数与热面量为

$$\begin{aligned}\frac{\Delta x}{k_H} &= \frac{\Delta x/2}{k_i} + \frac{\Delta x/2}{k_{i+1}}, \\ k_H &= \frac{2k_i k_{i+1}}{k_i + k_{i+1}}, \\ g_{i+1/2}^T &= \gamma_i k_H (T_{i+1} - T_i)\end{aligned}\tag{A.35}$$

这里的等效阻力解释针对面附近两段系数近似，圆柱面位置也按 $x_{i+1/2}$ 近似；不是对整个有限厚圆柱壳稳态导热的无误差积分。

三组扩散率可统一写成 $D = D_0 A(T) f_a(C)$ ，其中 $A(T) = \exp(-b/T)$ 、 $f_a(C) = \exp(-a/C)$ 。参数 (a, b, D_0) 依次为 $(0.89, 0, 7 \times 10^{-9})$ 、 $(0.45, 3850, 2.4 \times 10^{-3})$ 和 $(0.30, 3850, 4.2 \times 10^{-4})$ 。对 $C > 0$ 定义 Kirchhoff 浓度势

$$\Phi_a(C) = C \exp(-a/C) + a Ei(-a/C)\tag{A.36}$$

Ei 为指数积分函数。利用 $d Ei(z)/dz = e^z/z$ 和链式法则，分别有

$$\begin{aligned}\frac{d}{dC} [C e^{-a/C}] &= e^{-a/C} (1 + \frac{a}{C}), \\ \frac{d}{dC} [a Ei(-a/C)] &= -\frac{a}{C} e^{-a/C}, \\ \Phi_a'(C) &= e^{-a/C}\end{aligned}\tag{A.37}$$

因此 $\Phi_a(C_{i+1}) - \Phi_a(C_i) = \int_{C_i}^{C_{i+1}} e^{-a/c} dc$ 。在面上用当前两侧温度平均 $\bar{T}_i = (T_i + T_{i+1})/2$ ，水分面量为

$$g_{i+1/2}^C = \gamma_i D_0 \exp(-\frac{b}{\bar{T}_i}) [\Phi_a(C_{i+1}) - \Phi_a(C_i)]\tag{A.38}$$

式 (A.38) 对浓度因子的积分采用解析势差；温度面值、空间重构、几何面值及时间积分仍有数值误差。Q1 的 $b=0$ ；其余两组的温度因子在一个面内单独取值。“面温度冻结”指这一步空间近似，不表示 Newton 迭代中温度不更新。

不能对 $A(T)\Phi_a(C)$ 整体作差。因为其空间导数为

$$\frac{\partial}{\partial x}[A(T)\Phi_a(C)] = A(T)e^{-a/C}C_x + A(T)\frac{b}{T^2}\Phi_a(C)T_x \quad (\text{A.39})$$

第二项不在本题所用 Fick 闭合内。当两侧含水率相同、温度不同时，正确浓度梯度通量应为零，整体势差却会产生非零值，因而不符合原模型。

相邻浓度很接近时，两项势值相减存在消减误差。令 $\Delta C = C_{i+1} - C_i$ 、 $\bar{C}_i = (C_i + C_{i+1})/2$ ，则中点积分展开给出

$$\Delta\Phi = f_a(\bar{C}_i)\Delta C + \frac{f_a''(\bar{C}_i)}{24}(\Delta C)^3 + O((\Delta C)^5) \quad (\text{A.40})$$

现行源码在 $|\Delta C| < 10^{-7} \max(\bar{C}_i, 10^{-3})$ 时使用第一项，其局部差为 $O((\Delta C)^3)$ 。一般分支保留指数积分势差。系数求值采用 $C^+ = \max(C, 10^{-12})$ 保护 Newton 试探点，实际比较与势差分支以 C^+ 为准；它不裁剪求得的状态。正值保护是数值定义，不是干基含水率的物理下限。

A.10 干基质量与有效热收支的离散恒等式

全体水方程乘 $2w_i$ 求和，内部面两两相消，只有表面项保留。定义加权平均含水率 $\bar{C}_N = 2\sum_i w_i C_i$ 和累计平均失水量 $\ell(0)=0$ ，有

$$\begin{aligned} \frac{d\bar{C}_N}{dt} &= \frac{2g_{N+1/2}^C}{R^2} = -\frac{2\beta}{R}(C_N - C_{\text{eq}}), \\ \dot{\ell} &= \frac{2\beta}{R}(C_N - C_{\text{eq}}), \\ \bar{C}_N(t) + \ell(t) &= C_0 \end{aligned} \quad (\text{A.41})$$

连续材料域的对平均为 $\bar{C} = 2\int_0^1 x \hat{C} dx$ 。由 $\rho_d R^2$ 恒定，水质量为 $M_d \bar{C}$ ；故式 (A.41) 与移动体积下的干质量守恒相容。 ℓ 单位为 kg 水/kg 干物质， $M_d \ell$ 才是累计水质量。当前基线以失水为主；若边界条件改变而出现吸湿， $\dot{\ell}$ 可以为负，仍是带符号的净流出量。

热方程乘 $2\pi R^2 w_i B_i$ 求和，同样得到瞬时有有效热通量恒等式

$$2\pi R^2 \sum_i w_i B_i \dot{T}_i = 2\pi R h (T_\infty - T_N) \quad (\text{A.42})$$

式 (A.42) 两侧均为单位轴向长度的功率, 单位 W/m。若另定义单位长度的容量加权温升量 $H_{\text{eff}} = 2\pi R^2 \sum_i w_i B_i (T_i - T_{\text{ref}})$, 其中 T_{ref} 为固定参考温度, 则乘积法则给出

$$\begin{aligned} \frac{dH_{\text{eff}}}{dt} &= 2\pi R h (T_\infty - T_N) + 2\pi R^2 \sum_i w_i (T_i - T_{\text{ref}}) B_i' \dot{C}_i \\ &\quad + 4\pi R \dot{R} \sum_i w_i B_i (T_i - T_{\text{ref}}) \end{aligned} \quad (\text{A.43})$$

第二项来自容量变化, 第三项来自物理体积变化。二者是这一数学量的乘积求导项, 未经额外热力学本构不能认定为真实机械功或潜热。Q1 的 B, R 恒定时, 后两项为零, 可对式 (A.42) 积分核验相应常物性显热收支; Q23/Q4 不能省略它们后把残差解释为数值积分误差。

A.11 解析 Jacobian: 面量、容量、边界与累计量

将状态交错排列为 $y = (T_0, C_0, T_1, C_1, \dots, T_N, C_N, \ell)^T$, 维数为 $2N+3$, 半离散系统写为 $\dot{y} = F(t, y)$ 。Jacobian 为 $J = \partial F / \partial y$ 。固定求值时刻时, $R, T_\infty, C_{\text{eq}}$ 是外部驱动, 不对状态求导。

对内部热面, 记 $\Delta T = T_{i+1} - T_i$ 。调和系数偏导为

$$\frac{\partial k_H}{\partial k_i} = \frac{2k_{i+1}^2}{(k_i + k_{i+1})^2}, \quad \frac{\partial k_H}{\partial k_{i+1}} = \frac{2k_i^2}{(k_i + k_{i+1})^2} \quad (\text{A.44})$$

由此, 热面量对四个相邻状态的偏导是

$$\begin{aligned} \partial_{T_i} g^T &= -\gamma_i k_H, \quad \partial_{T_{i+1}} g^T = \gamma_i k_H, \\ \partial_{C_i} g^T &= \gamma_i \Delta T \frac{2k_{i+1}^2 k_i'}{(k_i + k_{i+1})^2}, \quad \partial_{C_{i+1}} g^T = \gamma_i \Delta T \frac{2k_i^2 k_{i+1}'}{(k_i + k_{i+1})^2} \end{aligned} \quad (\text{A.45})$$

其中 $k_i' = dk(C_i)/dC_i$ 。本题 k 不显式依赖温度, 故热通量的温度导数来自温差, 浓度导数来自导热系数。

对水面量定义 $G_i = \gamma_i D_0 e^{-b/\bar{T}_i}$ 、 $\Delta \Phi = \Phi_a(C_{i+1}) - \Phi_a(C_i)$ 。常规正值势差分支下

$$\begin{aligned} \partial_{C_i} g^C &= -G_i e^{-a/C_i}, \quad \partial_{C_{i+1}} g^C = G_i e^{-a/C_{i+1}}, \\ \partial_{T_i} g^C &= g^C \frac{b}{2\bar{T}_i^2}, \quad \partial_{T_{i+1}} g^C = g^C \frac{b}{2\bar{T}_i^2} \end{aligned} \quad (\text{A.46})$$

温度偏导不能因采用面温度近似而省去；两侧温度各以 $1/2$ 进入 $\bar{T}_i$ 。Q1 的 $b=0$ ，相应热对水耦合偏导才为零。

近等浓度分支必须对实际用到的 $f_a(\bar{C}_i)\Delta C$ 求导，得到

$$\begin{aligned}\partial_{C_i}\Delta\Phi &= \frac{1}{2}f'_a(\bar{C}_i)\Delta C - f_a(\bar{C}_i), \\ \partial_{C_{i+1}}\Delta\Phi &= \frac{1}{2}f'_a(\bar{C}_i)\Delta C + f_a(\bar{C}_i), \\ f'_a(C) &= \frac{a}{C^2}e^{-a/C}\end{aligned}\tag{A.47}$$

在系数保护区，势差与物性关于浓度的偏导还需乘对应节点的活跃标记 $1_{C_i>10^{-12}}$ 。表面驱动使用原状态 $C_N - C_{eq}$ ，其下述边界偏导不因此变为零。保护阈值处属于分段数值规则，不是处处光滑的物理本构；Jacobian 采用与右端相同的分支，不能用另一表达式的导数替代。

热方程除面量外还有依赖 C_i 的分母 B_i 。设 $H_i = g_{i+1/2}^T - g_{i-1/2}^T$ ，则

$$\partial_{C_i}\dot{T}_i = \frac{\partial_{C_i}H_i}{R^2w_iB_i} - \dot{T}_i\frac{B'_i}{B_i}, \quad B'_i = \rho'_{\text{eff}}(C_i)c_p(C_i) + \rho_{\text{eff}}(C_i)c'_p(C_i)\tag{A.48}$$

物性导数由题给表达式直接求得：

$$\begin{aligned}\text{Q1: } (\rho'_{\text{eff}}, c'_p, k') &= (0, 0, 0), \\ \text{Q23: } (\rho'_{\text{eff}}, c'_p, k') &= (128, \frac{2736}{(1+C)^2}, \frac{0.38}{(1+C)^2}), \\ \text{Q4: } (\rho'_{\text{eff}}, c'_p, k') &= (90, \frac{2150}{(1+C)^2}, \frac{0.20}{(1+C)^2})\end{aligned}\tag{A.49}$$

将每个面偏导向左、右控制体按正、负号加入，再乘各自的 $1/(R^2w_iB_i)$ 或 $1/(R^2w_i)$ ；热方程同时添加式 (A.48) 的容量项。由此得到邻近节点的 2×2 块带结构。

基线表面与累计量导数为

$$\partial_{T_N}g_s^T = -hR, \quad \partial_{C_N}g_s^C = -\beta R, \quad \partial_{C_N}\dot{\ell} = \frac{2\beta}{R}\tag{A.50}$$

累计量 ℓ 不反馈温湿场，因此 Jacobian 对 ℓ 的整列为零。令行向量 $\mathbf{m}^T$ 在每个 C_i 分量取 $2w_i$ 、在 ℓ 分量取 1、其余取零，则式 (A.41) 的微分结构要求

$$\mathbf{m}^T \mathbf{F} = 0, \quad \mathbf{m}^T \mathbf{J} = 0^T\tag{A.51}$$

前者检查共享通量的质量装配，后者检查其导数装配。方向差分检验采用 $[F(y+\varepsilon v)-F(y-\varepsilon v)]/(2\varepsilon)$ 与 Jv 对照；必须同时覆盖浓度、温度、边界、累计量及近等浓度分支，不能只检查对角项。

A.12 隐式 BDF 与连续数值输出

空间加密使扩散算子的快速尺度约随 $(R\Delta x)^{-2}$ 增大，状态相关扩散率又带来不同的局部时间尺度。系统因此采用隐式 BDF 推进。以当前状态和若干历史状态的插值多项式在当前时刻求导，可得一般后向差分残量

$$G(y_n)=\alpha_0 y_n+\sum_{j=1}^q \alpha_j y_{n-j}-\Delta t_n \beta_q F(t_n, y_n)=0 \quad (\text{A.52})$$

q 为阶数， α_j, β_q 是该历史时间布局对应的系数。等步长二阶仅作为原理示例：

$$\frac{3y_n-4y_{n-1}+y_{n-2}}{2\Delta t}=F(t_n, y_n) \quad (\text{A.53})$$

对式 (A.52) 作 Newton 线性化，每次修正求解

$$(\alpha_0 I - \Delta t_n \beta_q J) \delta y = -G \quad (\text{A.54})$$

实际采用 SciPy BDF 的 1—5 阶自适应实现及 NDF 精度修正；式 (A.53) 不表示全程固定二阶。隐式多步常微分方程求解的算法背景见 Byrne 与 Hindmarsh[6]，当前 BDF 的阶数、误差控制与 Jacobian 接口依 SciPy 官方说明[7]及本次使用版本源码核对。变步长、变阶时的历史差分更新、误差估计和失败重试由库实现，不能将固定等步长系数用于任意不等间距历史点。

时间局部误差按分量尺度归一化：

$$s_j = atol_j + rtol |y_j| \quad (\text{A.55})$$

正式参数为 $rtol=10^{-10}$ ，温度绝对容差 10^{-10} K，水分与累计失水绝对容差 10^{-12} 。前 4 h 最大步长 2 s，之后 120 s；程序在 14400 s 处切段，并用上一段接受末态作为下一段初态。这些参数控制时间积分局部误差，不能当作全时空场误差的严格上界。

BDF 每个接受步带有连续输出多项式。以一个接受步的右端 t_n 、多项式步长尺度 h_p 、阶数 q 和后向差分系数向量 $D_0, \dots, D_q$ 表示，其求值形式为

$$P_n(t) = D_0 + \sum_{j=1}^q D_j \prod_{k=0}^{j-1} \frac{t - (t_n - k h_p)}{(k+1) h_p} \quad (\text{A.56})$$

式 (A.56) 中的差分系数和尺度是求解器为该步输出对象保存的量, 不应另从稀疏采样反推。`diskDense.py` 保存原始双精度系数与接受状态, 并使用同一求值公式; 只改变存储位置。BDF 内部断点恰好落在查询点时, 保留与 SciPy 原 BDF 一致的右侧多项式选择; 跨显式分段边界则保持 `Run.state` 的段选择规则。

题定每 1 s 或 60 s 的数值直接从尚未关闭的连续数值轨迹查询。保存的 60 s 等审查用 NPZ 并不是逐秒输出的插值来源。连续输出多项式是离散 ODE 解的数值表示, 不是偏微分方程的解析解。

A.13 全域最大含水率、临界根与严格报告时刻

对连续模型定义

$$M(t) = \max_{0 \leq r \leq R(t)} C(r, t) = \max_{0 \leq x \leq 1} \hat{C}(x, t) \quad (\text{A.57})$$

$M(t)$ 表示时刻 t 最湿位置的干基含水率, 单位仍为 kg/kg。题目要求各处含水率均低于 $C_* = 0.15$, 等价于 $M(t) < C_*$; 平均含水率低于阈值不能替代这一条件, 也不应在求解前只指定中心为唯一检查点。

数值计算实际采用全部 $N+1$ 个节点的最大值

$$M_N(t) = \max_{0 \leq i \leq N} C_i(t), \quad g_N(t) = M_N(t) - C_* \quad (\text{A.58})$$

节点间作分段线性重构时, 每段值都是两个端点的凸组合, 因此重构场的最大值恰等于 M_N 。这使数值判据覆盖整个重构空间域, 但不等于证明真实连续场与重构场的最大值完全相同。二者差异仍需空间、时间误差检查; 若需要真实连续场的严格可行保证, 还需可用的统一误差界。

以 g_N 从正到负的第一次穿零定位临界时刻 $t_{c,N}$ 。在临界附近连续下降时, $M_N(t_{c,N}) = C_*$ 对应等号, 严格可行集合的下确界为该根, 严格集合本身未必具有最小时间点。最大值函数在控制节点切换时可以不可微, 但连续性仍使包围区间上的标量事件定位有意义; 不要求对最大值求解析导数。

正文式(14)对应指定报告网格上求严格可行的首点。本附录以离散数值模型明确写为

$$\begin{aligned} \mathcal{K}_N = \{k \in N_0 : k \Delta t_r \geq t_{c,N}, M_N(k \Delta t_r) < C_*\}, \\ t_{r,N} = \Delta t_r \min \mathcal{K}_N, \quad \Delta t_r = 0.36 \text{ s} = 10^{-4} \text{ h} \end{aligned} \quad (\text{A.59})$$

N_0 为非负整数集合, $\mathcal{K}_N$ 为满足时间与严格含水率条件的整数指标集合。所有集合内比较使用未舍入的数值解, $t_{c,N}, t_{r,N}, \Delta t_r$ 统一以秒计算。该定义在所搜索区间内有可行点时才成立; 未触发事件不能把预设计算上限当作烘干时长。

程序先从临界状态实际续算至

$$t_{\text{end}} = \Delta t_s \left[\left\lceil \frac{t_{c,N}}{\Delta t_s} \right\rceil + 1 \right], \Delta t_s = 1 \text{ s}, \quad (\text{A.60})$$

再生成报告候选首点

$$t_r^{(0)} = \Delta t_r \left\lceil \frac{t_{c,N}}{\Delta t_r} \right\rceil, \quad t_r^{(j)} = t_r^{(0)} + j \Delta t_r \quad (\text{A.61})$$

从 $j=0$ 起逐点查询 M_N , 第一个严格小于 0.15 的点作为 $t_{r,N}$; 若查询超出已实际积分的 t_{end} , 程序报错, 不外推状态。 t_{end} 是保守的事件后整数秒核对点, 不宣称是最早严格可行整数秒; $t_{r,N}$ 才是按 0.0001 h 精度报告的时长。若根恰在报告格点且该点仍为等号, 必须继续到下一格点。

现行正式轨迹给出问题三 $t_{c,N}/3600=57.47230195056044 \text{ h}$ 、 $t_{r,N}/3600=57.4724 \text{ h}$; 问题四相应为 51.09057478683054 h 和 51.0906 h。四位最近舍入得到的临界根显示值, 不能替代式 (A.59) 的严格可行时刻。表格中含水率显示 0.1500 也不直接否定未舍入值小于 0.15; 不通过手改末位改变判定。

A.14 固定物理位置查询与缩域输出

给定物理半径 r_j , 先在每个查询时刻计算 $x_j(t)=r_j/R(t)$ 。若 $0 \leq x_j \leq 1$ 且 $x_i \leq x_j \leq x_{i+1}$, 使用

$$\hat{C}_N(x_j, t) = (1 - \eta) C_i(t) + \eta C_{i+1}(t), \quad \eta = \frac{x_j - x_i}{\Delta x} \quad (\text{A.62})$$

温度查询采用同样的空间重构, 时间值先由式 (A.56) 所在轨迹查询。 $r_j > R(t)$ 时该位置已不在药材内, 输出留空, 不能填零或外推。真实表面另查 $x=1$, 不能用最近的 0.1 cm 固定列替代。代码对仅由浮点边界换算造成的约 10^{-12} m 容差内归属差异按表面值一致处理, 不把它扩展成物理域外预测。

完整逐秒或逐分钟工作簿、正文采样表与达标时刻均源自同一条轨迹。最后才转换 $K \rightarrow ^\circ\text{C}$ 、 $s \rightarrow h$, 并进行四位显示舍入。问题二和问题三复用 Q23 的同一个 Run, 避免场表和事件时间来源不同。

A.15 圆柱 Robin-Bessel 解析基准

常系数圆柱扩散的特征函数方法可参见 Crank[4]。本节只对问题一常热物性热方程展开当前时变边界下的推导，令 $\alpha = k/B$ 、 $x = r/R_0$ 、 $v(x, t) = T(x, t) - T_\infty(t)$ 。由于外界温度在空间均匀，有

$$\begin{aligned} v_t &= \frac{\alpha}{R_0^2} \frac{1}{x} (x v_x)_x - \dot{T}_\infty(t), \\ v_x(0, t) &= 0, \quad v_x(1, t) + Bi_h v(1, t) = 0, \\ Bi_h &= \frac{h R_0}{k} \end{aligned} \quad (\text{A.63})$$

Bi_h 为该常系数热问题的 Biot 数。空间特征方程为 $(x X')' + \mu^2 x X = 0$ ，轴心有界的解取第一类零阶 Bessel 函数 $X(x) = J_0(\mu x)$ ；第二类函数在轴心发散而舍去。利用 $J_0'(z) = -J_1(z)$ ，表面 Robin 条件给出特征根

$$\mu_n J_1(\mu_n) = Bi_h J_0(\mu_n), \quad \lambda_n = \frac{\alpha \mu_n^2}{R_0^2}, \quad n = 1, 2, \dots \quad (\text{A.64})$$

λ_n 为正的模态衰减率，单位 s^{-1} 。

本题 Bi_h 有限且大于零，设 $j_{v,n}$ 为 J_v 的第 n 个正零点，根的计算可在下列区间分别包围并求解：

$$0 < \mu_1 < j_{0,1}, \quad j_{1,n-1} < \mu_n < j_{0,n} \quad (n \geq 2) \quad (\text{A.65})$$

该区间选择与现行解析基准源码一致，避免用等距扫描漏根。绝热 $Bi_h = 0$ 时包含零模态，均匀初态保持不变，应单独处理；不能把这一极限直接代入只列正根的展开流程。对两个不同正根，将特征方程交叉相乘、相减并积分，边界项因同一 Robin 条件消失，得到带圆柱权重 x 的正交性：

$$\int_0^1 x J_0(\mu_m x) J_0(\mu_n x) dx = 0 \quad (m \neq n) \quad (\text{A.66})$$

均匀函数的投影系数需要以下两项积分：

$$\int_0^1 x J_0(\mu_n x) dx = \frac{J_1(\mu_n)}{\mu_n}, \quad \int_0^1 x J_0(\mu_n x)^2 dx = \frac{J_0(\mu_n)^2 + J_1(\mu_n)^2}{2} \quad (\text{A.67})$$

因此常数 1 的加权展开系数为

$$b_n = \frac{2 J_1(\mu_n)}{\mu_n [J_0(\mu_n)^2 + J_1(\mu_n)^2]} \quad (\text{A.68})$$

令 $v = \sum_n a_n(t) J_0(\mu_n x)$ ，将式 (A.63) 投影到每一特征函数，得

$$\dot{a}_n + \lambda_n a_n = -b_n \dot{T}_\infty(t), \quad a_n(0) = b_n [T_0 - T_\infty(0)] \quad (\text{A.69})$$

使用积分因子求解该一阶常微分方程，得到时变边界的圆柱热场基准

$$T(x, t) = T_\infty(t) + \sum_{n=1}^{\infty} b_n J_0(\mu_n x) [[T_0 - T_\infty(0)] e^{-\lambda_n t} - \int_0^t e^{-\lambda_n(t-\tau)} \dot{T}_\infty(\tau) d\tau] \quad (\text{A.70})$$

这里的解析基准包含附件给定时变温度的卷积，不是恒温阶跃公式。外界温度在一段 $[a, b]$ 内线性变化、斜率为 s 时，该段在 $t \geq b$ 的卷积贡献为

$$\int_a^b e^{-\lambda_n(t-\tau)} s d\tau = \frac{s}{\lambda_n} [e^{-\lambda_n(t-b)} - e^{-\lambda_n(t-a)}] \quad (\text{A.71})$$

若 t 落在当前段，则把上限截为 $b=t$ ；尚未开始的段不贡献。分段线性导数在节点处的跳变不影响该积分；若要推广到边界值本身跳变的时刻，必须另外处理对应阶跃，不能只积普通斜率。

对固定半径且常扩散率 D_* 的水分验证特例，同样令 $Bi_m = \beta R_0 / D_*$ 、 $\lambda_n = D_* \mu_n^2 / R_0^2$ ，将 T, T_∞, T_0 替换为 C, C_{eq}, C_0 即可构造相同类型的线性基准。这个特例只用于检验算法，不替代正式 $D_1(C)$ 非线性水分结果，也不能验证 Q23/Q4 的变物性收缩问题。

A.16 解析比较的点值、单元平均及误差范围

本文生产未知量按节点点值解释，主要比较 T_i 与 $T(x_i, t)$ 。若为了诊断另外比较解析环体平均，应明确使用

$$\begin{aligned} \langle J_0 \rangle_i &= \frac{1}{w_i} \int_{a_i}^{b_i} x J_0(\mu x) dx \\ &= \frac{b_i J_1(\mu b_i) - a_i J_1(\mu a_i)}{\mu w_i} \end{aligned} \quad (\text{A.72})$$

式 (A.72) 与节点点值 $J_0(\mu x_i)$ 不是同一个对象。尤其表面控制体为单侧区间，若梯度非零，平均值与端点值之差可为 $O(\Delta x)$ ；不能把这种表示差异直接称为物理误差。

对指定公共采样集合 $\mathcal{S}$ ，定义点值最大差和均方根差

$$\begin{aligned} E_\infty &= \max_{(t,r) \in \mathcal{S}} |U_N(t, r) - U_{\text{ref}}(t, r)|, \\ E_{\text{RMS}} &= \sqrt{\frac{1}{|\mathcal{S}|} \sum_{(t,r) \in \mathcal{S}} [U_N(t, r) - U_{\text{ref}}(t, r)]^2} \end{aligned} \quad (\text{A.73})$$

U 可表示温度或含水率, U_{ref} 是明确指定的解析解或更细数值轨迹。解析级数实际截为有限项, 需另比较 120、240、480 等截断项数; 截断差估计不自动成为全空间全时间的严格上界。正文 N3200 常热物性温度与解析基准在 1801 个整数秒、21 个半径点上的最大差约为 1.804×10^{-6} K, 只支持该比较范围内的实现一致性。

早期及本轮独立复核中, 冻结扩散率的 N400 密集样点最大水分误差为 $2.875083008730961 \times 10^{-4}$, 超过当时 10^{-4} 门槛; N800 对应约 $7.158659870798445 \times 10^{-5}$ 。程序外层退出 0 不改变该内部不通过结果。另一个历史脚本得到的点值/环体平均差随 N800→6400 逐次约减半, 而其静态解释称“不随细化减少”; 本附录按实际数值与式 (A.72) 解释, 不采纳该错误文字结论。

A.17 方法比较、收缩尺度与参数敏感性

在不同空间网格间比较时, 采用相同物性、环境、边界、初态及时间判据, 在共同物理位置上求差; 问题四排除较小当前域之外的位置并单列真实表面。若观察到幂律误差 $E_N \approx K N^{-p}$, 两倍加密下可计算观察阶

$$p_{\text{obs}} = \frac{\ln(E_N/E_{2N})}{\ln 2} \quad (\text{A.74})$$

该式仅在误差度量一致且进入渐近区时具有通常的阶数解释; 若只有两条未知误差数值解, 则它们的差本身并非真实误差。正式正文分别采用 Q1/Q23 的 N1600/N3200 和 Q4 的 N3200/N6400 比较, 并在最细网格保持空间设置不变、同时收紧容差和最大步长检查时间敏感性。本轮没有为本附录新增正式 PDE 重算; 旧全秒空间投影未全部保留的限制继续保留, 不将摘要差值冒充重新全量回放。

对另一守恒水分面格式, 可将相邻 D_i, D_{i+1} 的调和平均乘浓度差, 与式 (A.38) 对比。常 D 时两式退化为同一个线性面量; 变量 D 时差别属于空间离散。现行 `method_comparison.py --quick` 的 8 条 Q1 轨迹实际完成, 变量 D 的 N100 与 N200 两格式最大差约为 1.4891×10^{-5} 和 3.7288×10^{-6} , 观察阶约 1.9977。历史粗网格调和格式严重偏离时长的记录不能用来否定调和平均在所有问题中的适用性, 但以说明本题不能采用未加密检查的粗网格答案。

从式 (A.28) 看, 在给定参考扩散率 D_{ref} 下, 内部扩散自然累积尺度为

$$\tau(t) = \int_0^t \frac{D_{\text{ref}}}{R(s)^2} ds, \quad \tilde{t}(t) = R_0^2 \int_0^t \frac{1}{R(s)^2} ds \quad (\text{A.75})$$

τ 无量纲, $\tilde{t}$ 的单位为秒。该换算只揭示几何缩短路径的尺度, Robin 边界中的 R 、随状态变化的物性和温湿反馈同时存在, 所以不能由 R^2 比例直接得出精确全过程时长。使用仿真事件作积分上限的尺度核对也不构成独立求得的解析烘干时间。

分离收缩影响时, 应固定附录 4 的全部物性、初态、环境和数值设置, 只改变 $R(t)$ 。正文已有 N200 对照: 定半径临界时长约 129.8452 h, 给定收缩约 51.0910 h。问题三与问题四的正式时长差约 6.3818 h 同时含物性与几何变化, 不能全部归因于收缩。对照网格 N200 用于所列机理比较, 不替代 Q4 N6400 正式结果。

参数敏感性必须真正进入正在使用的面通量。以扩散前因子倍率 s_D 、导热率倍率 s_k 为例, 应分别将式 (A.38) 中的 D_0 和热面中的 k 同步改变。只修改未被 Kirchhoff 分支读取的节点 D 数组, 可能使所谓扩散参数扰动实际没有进入 RHS, 不能作为有效敏感性结果。已采用的修复后 N200 单参数扫描表明, s_D 从 0.7 增至 1.4 时, 问题三临界时长由约 79.4187 h 降至 43.0796 h, 问题四由 70.4059 h 降至 38.3160 h; k 在 $\pm 15\%$ 范围内变化时, 两问时长跨度均小于 0.011%。这些是指定工况的一维参数扫描, 不能扩大为完整全局敏感性排序, 未完成的 Sobol 和旧无效 Morris 注入不采纳。

A.18 经验边界阻力及潜热情景的推导范围

为展示气固映射未标定的影响, 可在 $C_s \geq C_{\text{ref}} = 0.05$ 、 $p \geq 1$ 的限定下定义如下经验关系。分式适用于 $C_s > C_{\text{ref}}$; 在 $C_s = C_{\text{ref}}$ 处按连续延拓定义 $K_p(C_{\text{ref}}) = 1/p$, 不直接计算 0/0:

$$\begin{aligned} K_p(C_s) &= \frac{1 - (C_{\text{ref}}/C_s)^{1/p}}{1 - C_{\text{ref}}/C_s} \quad (C_s > C_{\text{ref}}), \\ K_p(C_{\text{ref}}) &= \frac{1}{p}, \\ j_{w, \text{out}} &= \rho_d \beta K_p(C_s)(C_s - C_{\text{eq}}) \end{aligned} \quad (\text{A.76})$$

p 为无量纲经验参数。取 $z = C_{\text{ref}}/C_s \in (0, 1]$ 、 $q = 1/p \in (0, 1]$ 。由 $z^q \geq z$ 与凹函数 z^q 在 1 处的切线不等式 $z^q \leq 1 + q(z - 1)$, 得到

$$\begin{aligned} q(1 - z) &\leq 1 - z^q \leq 1 - z, \\ \frac{1}{p} &\leq K_p(C_s) \leq 1, \quad K_1(C_s) = 1, \\ \lim_{C_s \downarrow C_{\text{ref}}} K_p(C_s) &= \frac{1}{p} \end{aligned} \quad (\text{A.77})$$

这个族包含原边界并可控地减弱所设传质驱动，但不是由实测材料等温线推得的唯一边界。界限仅在上述 C_s, p 范围成立。正文历史 N800 的 $p=1, 2, 4$ 情景保留为条件数值比较，不据几个参数点证明所有 p 下时长单调，也不能作为真实烘干时长下界。当前旧经验闭合源码存在参数接口和结果数组方向问题，本轮 sandbox 没有把该批历史情景重新完整跑通； $p=1$ 代数恒等性检查不能替代其原入口复现。

若另指定比例 χ 的外流水在表面汽化，汽化潜热为 L_v ，则热边界梯度面量改为

$$g_s^T = h R (T_\infty - T_s) - R \chi L_v \rho_d \beta (C_s - C_{eq}) \quad (\text{A.78})$$

$\chi=0$ 恢复正式基线， L_v 单位为 J/kg；加入 K_p 情景时，蒸发质量项也须一致乘 $K_p(C_s)$ 。固定求值时刻下，不含 K_p 的潜热项对 C_s 的偏导为 $-R \chi L_v \rho_d \beta$ ，应加入表面热行，累计失水方程保持与实际水边界一致。这类情景约定了蒸发位置和比例，但没有识别真实气膜传递、凝结、吸附热或内部相变，不能声称补入一个潜热项就完成总热力学闭合。

A.19 公式与现行政程序的对应关系及验证限度

03_程序代码/dryingCore.py 对应物性、材料网格、面量、半离散 RHS、事件监测、物理位置查询及质量诊断；analyticJacobian.py 对应解析导数装配；diskDense.py 对应式 (A.56) 及断点选择；q1Model.py、q2Model.py、q4Model.py 分别给出正式 N3200、N3200、N6400 参数；q3Model.py 对应式 (A.59) 的上取候选及严格核验。exportOutputs.py 负责从同一未关闭轨迹导出、舍入和回读，runDelivery.py 与 runLogged.ps1 记录执行、输入及退出状态。现行支撑源码及追加的数据处理、绘图源码在后续源码附录完整列出，文件数与逐行清单以该附录为准；各文件仍应保持原有模块边界，不能合并成一个可运行文件。

2026-09-13 实际 Windows Sandbox 正式 03 重算采用独立运行库及禁网环境，原宿主工作区不可见；三条正式网格轨迹和四表回读完成，9,335,598 个工作簿格及 297 个正文格核验，21 个保存数组与冻结参照逐值一致，严格报告时长未变。该证据支持当前源码的数值复现，不提供内部温湿实验准确率。

05 检验目录的 31 个 Python 源码在实际 sandbox 中有原入口或函数调用覆盖，29 项 Python 任务中 24 项退出 0、5 项退出 1。其范围包含小网格与函数级检查，没有把所有默认长批次写成完整通过。Bessel 的 8 项自检、Jacobian 的 84 状态与 336 方向检查，以及三条 N40 内存/磁盘轨迹的 38,337 次查询一致性分别承担解析实现、局部线性化和存储等价性检查。Node 对照入口缺少旧 MATLAB 结果输入而失败；该

sandbox 没有 MATLAB 运行环境，历史 MATLAB 对照与本轮 Python 重算分别记录。

旧 M-K 顺序统计对常数序列产生伪趋势、经验闭合旧 B 参数接口失败、时间与空间数组反向访问、替代 Jacobian 注入没有被调用、N400 解析误差门槛未过等问题均保留。公式推导不将这些历史程序的存在或退出 0 视为科学结论。正文采用的有限网格差、参数扫描及条件情景，只在其明确输入、版本、网格、时间和采样范围内解释。

完整连续场误差界、真实材料等温线、内部温湿测量以及完整组分能量关系尚不由当前附件提供。以后若修改这些闭合、物性或收缩运动，须从组分守恒重新检查方程并重新求解；本附录的代数正确性与实现自洽性不能替代这些外部证据。

A.20 推导依据

本附录沿用正文文献序号。扩散理论支持通量和圆柱基准的一般构造；有限体积文献支持积分离散框架；常微分方程算法论文和 SciPy 文档支持时间求解方法。题给物性系数、气固映射、有效热容量与同比收缩假设均由本模型另行声明，不能用这些一般文献替代实验标定。

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附录 B 支撑材料与复现说明

B.1 材料范围与代码覆盖

本附录对应当前冻结的支撑材料目录，共 106 份正式文件。目录包括题面与原始数据、文献资料、独立生产程序、四问完整结果以及数值检验程序。Visual Studio 自动产生的本机缓存和升级日志不列入正式材料计数。附录 D 完整收录 47 份源码、9917 行：其中 44 份为支撑材料现有全部源码，另补充原始数据处理、六图数据导出和 R 绘图 3 份源码。语言构成为 Python 43 份，PowerShell、JavaScript、MATLAB 和 R 各 1 份。

独立生产程序位于 03_程序代码；05_数值检验与实验保留了模型依赖、交叉检验和历史诊断。这两部分有不同的运行目的和版本来源，不能互相混用模块。附录 D 各节列明用途和采用范围，并逐行给出完整源代码；实际验证范围与限制见 B.5、B.6。未采用的早期算法副本、论文排版脚本和资料包表格展示工具不作为新增建模代码重复收录。

44 份现有源码与已完成独立运行核查的去注释候选逐文件核对 SHA256。追加的两份 Python 脚本删除说明性 docstring 及注释后，非注释抽象语法树与原文件相同；R 脚本通过原生解析器确认去注释前后表达式相同。字符串中的颜色、格式符、文件名和运行提示保留。源码逐字节校验用于绑定版本，运行通过范围仍按本附录的实际证据分别说明。

B.2 输入、单位与资料来源

数值输入来自本届 A 题题面和官方附件。原始 PDF、Excel 与清洗后 CSV 分别保存；数据处理不删除观测、不填充缺失、不平滑、不拟合外推。附件 1、附件 2 均通过字段、有限值、重复时间和采样间隔检查，清洗操作仅为增加单位换算列。官方空白 result1.xlsx 至 result4.xlsx 用于规定导出结构；04_结果表格中的同名工作簿是已计算结果。

输入或变量	实际范围与单位	程序处理
附件 1 环境观测	241 条；0—14400 s；每 60 s 一条；温度为℃；空气水分指标为 kg/kg	增加小时列与开尔文列；观测间分段线性插值

输入或变量	实际范围与单位	程序处理
附件 2 半径观测	145 条；0—259200 s；每 1800 s 一条；半径为 cm	增加小时列与米制列；正值、非增和采样间隔检查
时间与空间坐标	求解内部使用 s、m	输出表及图按题意转为 h、cm 或保留 s
材料温度	求解内部为 K	温度输出由 $T-273.15$ 换算为 $^{\circ}\text{C}$
材料含水率 C	kg 水/kg 干物质，即干基 kg/kg	与空气水分指标的物理分母区别说明；二者映射需模型闭合
附录 2—4 物性	扩散系数 m^2/s ；导热系数 $\text{W}/(\text{m}\cdot\text{K})$ ；比热 $\text{J}/(\text{kg}\cdot\text{K})$	按各问给定函数计算，不从外部资料代入经验参数
对流与质量交换系数	热交换 $\text{W}/(\text{m}^2\cdot\text{K})$ ，质量交换 m/s	与模型的温度差及含水率通量定义一致

4 h 以后的环境温度 50°C 和空气水分指标 0.05 kg/kg 属于延拓假设。附件 1 在 4 h 的真实末点为 50.165°C 和 0.04986 kg/kg ，程序及图件保留此点，并将后续平台作为右侧延拓。空气指标映射为材料平衡含水率、干物质骨架径向同比收缩、轴向长度不变及有效热容量均是模型闭合的一部分。题面未提供内部温湿实测、吸附等温线、测量不确定度或轴向收缩观测，不能将数值自洽解释为物理预测准确率。

外部资料用于干燥背景、有限体积方法、隐式积分与接口依据，未向正式求解引入额外环境实测或直接移植其他物料的经验参数。支撑文献中的两篇中文综述实际 PDF 分别为 28 页和 8 页；Handbook of Industrial Drying 文件为 5 页节选；Finite Volume Methods 文件为 254 页作者更新稿；The MATLAB ODE Suite 扫描为 35 页，其期刊信息为 1997 年 18 卷 1 期 1—22 页。FAO 网页用于平衡含水率概念，SciPy 网页用于接口核查；文件范围与文献版本按 02 目录说明区分。

B.3 计算环境与依赖

正式重算在独立 Windows Sandbox 中实际执行。该环境关闭网络，原宿主 D 盘工作区不可见，运行库、源码、输入和模板显式提供，不继承宿主 Python 的 site-packages。Windows Sandbox 使用 CPU 完成计算；所记录耗时只对应本次设备和环境，不构成其他机器的性能保证。

环节	实际验证环境	范围
03 正式生产求解	Windows 11; 64 位 Python 3.14.7; NumPy 2.5.2; SciPy 1.18.1; openpyxl 3.1.5	Q1/Q23 为 N3200, Q4 为 N6400; 三条轨迹、四问完整导出与回读
03 辅助入口	Windows Sandbox; Windows PowerShell 5.1; 同一 Python 依赖	原 runLogged.ps1 的 quick 全流程, N40 三轨迹及四表
05 检验	Windows Sandbox; 同版 Python/NumPy/SciPy/openpyxl; Matplotlib 3.11.2	31 个 Python 源码有实际入口或函数调用覆盖, 结果和限制分别记录
原始数据预处理	Windows 11; Python 3.14.7; NumPy 2.5.2; openpyxl 3.1.5; Matplotlib 3.11.1	原版已实际处理两附件并回读 CSV; 不是本轮 VM 运行版本声明
最终六图	宿主 Windows 11; R 4.6.1 自带 graphics/grDevices; Cairo; Microsoft YaHei、Times New Roman	已实际生成六张 PNG/PDF/SVG 和合并 PDF; 本次附录仅核对代码与现存产物
JavaScript 对照入口	Windows Sandbox; Node.js 24.20.0	实际进程退出 2, 缺 MATLAB 结果文件; 未通过
MATLAB 对照源码	本机历史环境记录为 MATLAB R2026a; 本轮 VM 没有 MATLAB 入口	本轮未执行, 不能以 Python 对照替代

03 的 requirements.txt 固定三项数值依赖。diskDense.py 使用 SciPy 的 BDF 内部稠密输出接口, 复现时应按指定版本安装。程序固定计算线程数并顺序求解三条轨迹, 不要求 GPU。Q4 稠密输出缓存约 8.33 GB, 建议预留至少 20 GB 磁盘空间; 已有运行设备约 32 GB 内存, 最低内存需求未专门测定。Linux 和 macOS 未实际验证。

B.4 独立生产程序的复现步骤

在 03_程序代码目录打开终端, 先安装依赖, 再预检输入。每次运行使用新的结果目录; 若自行指定相同 run-id 且目录已存在, 程序会拒绝覆盖。

```
python -m venv .venv
.venv\Scripts\python.exe -m pip install -r requirements.txt
.venv\Scripts\python.exe -m pip check
.venv\Scripts\python.exe -X utf8 -B runDelivery.py --preflight-only
.venv\Scripts\python.exe -X utf8 -B runDelivery.py --profile quick
.venv\Scripts\python.exe -X utf8 -B runDelivery.py --profile final
./runLogged.ps1 -Python .venv\Scripts\python.exe -Profile final
```

quick 为 N40 小网格完整流程，final 为正文采用的 N3200/N3200/N6400 正式流程。辅助 PowerShell 命令与直接 Python 命令是两种启动方式，使用任一方式完成目标网格即可。判定完成必须同时检查真实进程退出码 0 与该次运行记录中的 PASS；preflight-only 的 PASS 仅表示输入预检通过。N40 全流程因问题二逐秒导出，仍需较长时间。

程序根目录按 runDelivery.py 位置确定。input_manifest.csv 逐项核查 12 份输入、模板与参照的大小和 SHA256；reference_data.py 保存冻结数值参照，reference 目录的三份 NPZ 用于轨迹比较。运行后在 results 下生成四份 Excel、对应原精度 CSV.GZ、七份正文 CSV、轨迹 NPZ 及运行日志。运行产物包含程序生成的 JSON 校验记录，与本附录列出的静态支撑文件清单区分。

Visual Studio 工程 A_CodeReview.sln 和 A_CodeReview.pyproj 可用于打开源码并选择本目录解释器。当前去注释源码已在 Visual Studio 中实际打开、命中 dryingCore.py 断点并单步核对初值和加载源码；本轮没有在 Visual Studio 完成正式全量求解，团队人工审查须由本人完成。

B.5 已实际完成的数值复现与验证

正式 Windows Sandbox 运行真实退出码为 0，程序内部 PASS。程序记录耗时 1810.989 s，外部监督记录 1812.629 s。二者分别计入程序主体和外部启动监督开销，应按同一口径比较。四份正式冻结 Excel 在本次去注释与复现过程中未被覆盖，其 SHA256 与冻结清单一致。

核查项目	实际检查规模	结果及适用范围
正式三轨迹	Q1/Q23 N3200；Q4 N6400	求解、四问导出、完整回读和冻结数值对照 PASS
四份工作簿	Q1 79,244 格；Q2 9,103,732 格；Q3 75,922 格；Q4 76,700 格	共 9,335,598 格全量回读并对照同次原精度 CSV 归档，297 正文数据格另由实际求解对象查询
七份正文 CSV	合计 297 个数据格	与同次求解对象逐格核对通过

核查项目	实际检查规模	结果及适用范围
三轨迹保存状态	共 21 个保存数组	与 final_v6a 冻结参照逐值相同；输入和源码运行前后 SHA 不变
严格阈值判定	Q3 报告 57.4724 h，Q4 报告 51.0906 h	未舍入全域最大值分别为 0.14999989718225315 与 0.14999995339076624，均严格小于 0.15
原 PowerShell 辅助入口	quick，N40 三轨迹、四表导出和回读	稳定重试真实退出 0、内部 PASS；监督耗时 892.825 s
Bessel 基准自检	原 8 项检查	满足原容差
解析稀疏 Jacobian	84 状态、336 方向、每方向 3 个差分步长；4 条短 BDF 轨迹	原检查全部通过，无警告
内存/磁盘稠密输出	N40 的 Q1/Q23/Q4 各两条轨迹；38,337 次稠密查询	接受步、状态与事件逐位相同，查询差为 0，关闭后缓存清除
Q1 数值检查	N20/N40 网格比较；N20 基准/紧容差双子进程	原比较链完成；用于相应小网格检查，不替代正式网格精度证明
独立 Q1 导出自检	N40；79,244 个工作簿格和 84 个正文 CSV 格	当前求解对象与导出回读一致，内部 PASS
二维关闭端面对照	nr=6，nz=4，端面交换关闭	二维与一维事件差约 -1.899×10^{-7} h；只对应本次小网格
事件独立复核	N20 Q23/Q4，事件根与二分核验	时刻差约 8.73×10^{-11} s、 5.82×10^{-11} s，严格报告判据通过
求解方法比较	quick 的 8 次 Q1 计算	N100/N200 误差约 1.489×10^{-5} 和 3.729×10^{-6} ，原门槛通过

PowerShell 首轮在内部数值检查完成后，外部观察器提前退出引发控制台状态错误，真实退出码为 1；该轮未计入通过结论。后续在独立隐藏控制台中保留监督进程至结束，原辅助脚本重新完整运行得到上述成功记录。核验同时保留失败与重试的来源，避免以内部日志中的单一 PASS 覆盖真实进程状态。

B.6 检验源码的限制与未通过项目

05 目录包含 33 份源码：31 份 Python、1 份 MATLAB 及 1 份 JavaScript。Windows Sandbox 中 29 项受控 Python 任务覆盖 31 份 Python 源码，24 项退出 0、5 项退出 1。源码覆盖、进程正常退出和数值判据通过是不同结论，不能据此概称 05 全部跑通。

这些历史检验使用原工作区路径。先验证未适配入口的路径问题后，仅在独立测试副本中修改 14 处根目录表达式，未改变物理模型、求解算法或误差门槛；正式支撑源码及附录 D 保留原路径。运行矩阵分别保存交付源码 SHA 和沙盒实测源码 SHA。重建这些历史测试时，须提供对应的源码层级、两份观测 CSV、四个模板和 Q23 参照 NPZ，另按入口要求提供必要配置；03 生产入口没有这些历史上级路径依赖。

首轮测试中 6 项因旧输出目录被带入而触发 FileExistsError，随后在只含源码和必要输入的新目录重试，5 项退出 0，另 1 项明确缺 model_route.json。旧输出与首轮目录错误不计为新的计算成果，也不将目录准备问题作为模型算法失败。

未通过任务	实际原因	处理与结论
run_modeling.py	缺少原要求的 paper_output/ plan/model_route.json	保留真实 FileNotFoundError， 未伪造配置或删除检查
isotherm_diagnose.py	向 ActivityModel 传入旧 B 参数	真实 TypeError；没有把 B 改 释为 p
isotherm_activity_closure.py	_extract 将时间×空间数组按相 反方向取值；真实形状 3×2	第三个时点 IndexError；p=1 恒 等式和 RHS 检查通过不除此 失败
isotherm_jacobian_probe.py	短时求解完成，替代 Jacobian 调用计数为 0	注入诊断未被调用，不能证明 已测试该 Jacobian
常数序列负例诊断	crossvalidate_series.py 的旧 sequential_mk 对 40 个相等值 产生伪趋势	诊断任务退出 1；主程序即使 退出 0，旧统计结论仍不成立

退出 0 的任务中也有明确限制。compare_analytic.py 在 N400 的密集采样冻结扩散水分最大误差为 $2.875083008730961 \times 10^{-4}$ ，超过原 10^{-4} 门槛，内部状态为 REVIEW_REQUIRED；N800 该误差为 $7.158659870798445 \times 10^{-5}$ 。analytic_metric_check.py 的点值与单元平均差随网格加密递减，原静态解释与计算趋势不符。publication_plots.py 完成 4 图生成，其状态仍为待视觉检查；这 4 张检验图不能替代正文 6 张 R 图的核验。

能量检查通过的明确范围是 Q1 N20：瞬时相对残差约 4.44×10^{-16} ，累计相对残差约 2.385×10^{-4} ，小于原 10^{-3} 门槛；Q23/Q4 短时能量诊断中的较大累计残差保留，不能据此声称四问能量守恒全部通过。潜热情景只实测 N20、latentFraction=0.25 的一条 Q23 事件；灵敏度脚本只核验 N200 控制对照两条事件；未完成完整 Morris 或 Sobol 批次，也不将小网格局部运行记作原大批次通过。

Node.js 原对照进程实际退出 2，原因是缺少 MATLAB 的 runSummary.json。Windows Sandbox 未提供 MATLAB 可执行环境，MATLAB 脚本本轮未运行。保留这两份完整源码用于追溯，其存在和历史对照材料不意味着本次独立沙盒验证通过。

B.7 正文六图的代码与数据衔接

附录 D.45 的 prepare_a_data.py 处理官方原始数据，D.46 的 export_six_figure_package_20260912.py 从冻结 NPZ/JSON 和观测 CSV 导出 14 张六图数据表，D.47 的 draw_figures.R 直接读取这些 CSV 绘图。14 张表共 4627 行；R 输入副本与原导出 CSV 逐字节一致，现稿六图的 DOCX 媒体均可按 SHA 直接对应 R 目录中的实际图件。图 1、3、4 采用 N3200 结果，图 5 采用 N6400 结果，图 6 的 6 个情景点统一为 N800、零潜热负荷。

图号	内容	数据与绘制约束
1	指定时刻径向温度和含水率	七个时刻；121 个径向展示样点；原值作图
2	环境观测与 4 h 后平台	保留 241 条观测和 4 h 末点；平台左端开放，观测与假设分开
3	中心、表面热湿响应	Q23 同一轨迹；温度 0—4 h，含水率至临界事件
4	全域阈值与临界放大	主图标示对数纵轴；中心与最大值重合处如实叠加；放大图只连接已有点，独立标注严格核验时刻
5	半径变化与收缩域内剖面	六条曲线各用其真实半径列；止于当时表面，域外留空
6	经验边界阻力情景	仅 p=1、2、4 的实际 6 点；不混入高分辨率生产结果，不添加误差棒

六图由 R 4.6.1 自带 `graphics/grDevices` 实际绘制，包含 PNG 300 dpi 和 PDF/SVG 矢量输出。原运行完成 11 组坐标范围检查及末点、开放平台、阈值和真实半径核验；本次编制附录没有重新积分或重新绘图。原始数据处理运行记录绑定了原脚本 SHA；六图导出和 R 绘图的既有记录未完整绑定原源码 SHA，因此本次对追加副本的结论限于语法、去注释等价及现存产物对应，不将其写作去注释版本已独立重跑。

追加脚本保留原项目相对层级和输入依赖，不能作为 03 独立生产入口的替代。复用已保存的 14 张 CSV 绘图时，在项目根目录执行以下命令；若迁移工程，应按源码中读取路径一并保留对应目录。

```
Rscript --vanilla paper_output/figures/review_20260913/draw_figures.R
```

原始数据处理脚本依赖项目 `input_manifest.json`；六图数据导出脚本依赖冻结结果和 `build_manifest.json` 中的来源索引。附录完整收录实际算法以便审阅，未把这些历史依赖删除或伪装成独立可运行条件。六图数值表来源、现存图件、程序验证范围和正文模型结论应结合阅读。

B.8 当前 106 份正式支撑文件清单

下列清单覆盖本次冻结支撑材料的全部 106 份正式文件，编号连续。各表文件路径以表前所列目录为相对根；“根目录”条目直接位于支撑材料目录内。追加的 3 份数据与绘图源码单列于附录 D.45—D.47，未据此改写原 106 文件目录。完整工作簿与轨迹参照保留在电子支撑中，其大型数值内容不在纸面逐格重印；七份正文结果表及关键高精度值见附录 C。

根目录（6 份）

编号	相对文件路径
1	00_可用于论文附录的支撑清单.txt
2	00_支撑材料总说明.txt
3	00_整理验收说明.txt
4	00_文件清单.txt
5	00_材料来源说明.txt
6	源码去注释与沙盒核查.txt

01_赛题与原始数据（10 份）

编号	相对文件路径
7	A 题.pdf
8	数据说明.txt
9	清洗后数据/A_environment_observed.csv
10	清洗后数据/A_radius_observed.csv
11	附件/附件 1.xlsx
12	附件/附件 2.xlsx
13	附件/附件 3/result1.xlsx
14	附件/附件 3/result2.xlsx
15	附件/附件 3/result3.xlsx
16	附件/附件 3/result4.xlsx

02_参考文献与网络资料（11 份）

编号	相对文件路径
17	Handbook of Industrial Drying.pdf
18	Handbook of numerical analysis.pdf
19	中药材干燥技术与装备研究现状.pdf
20	参考文献文件说明.txt
21	数值模拟仿真研究现状及其在中药干燥领域应用展望_王晓辉.pdf
22	补充网络资料/FAO_干燥原理_官方网页.html
23	补充网络资料/FAO_干燥原理_官方网页_离线文本.html
24	补充网络资料/SciPy_solve_ivp_官方 API 网页.html
25	补充网络资料/SciPy_solve_ivp_官方 API 网页_离线文本.html
26	补充网络资料/Shampine1997_BYU 原始扫描_35 页.pdf
27	补充网络资料/补充资料说明.txt

03_程序代码（30 份）

编号	相对文件路径
28	A_CodeReview.pyproj
29	A_CodeReview.sln
30	README.txt
31	analyticJacobian.py
32	diskDense.py
33	docs/源码变更说明.txt
34	docs/运行验收说明.txt
35	dryingCore.py
36	exportOutputs.py
37	input_manifest.csv
38	inputs/cleaned/A_environment_observed.csv
39	inputs/cleaned/A_radius_observed.csv
40	inputs/original/附件 1.xlsx
41	inputs/original/附件 2.xlsx
42	inputs/templates/result1.xlsx
43	inputs/templates/result2.xlsx
44	inputs/templates/result3.xlsx
45	inputs/templates/result4.xlsx
46	numerical_design.txt
47	q1Model.py
48	q2Model.py
49	q3Model.py
50	q4Model.py
51	reference/Q1/sampled_solution.npz
52	reference/Q23/sampled_solution.npz
53	reference/Q4/sampled_solution.npz
54	reference/reference_data.py
55	requirements.txt

编号	相对文件路径
56	runDelivery.py
57	runLogged.ps1

04_结果表格（12 份）

编号	相对文件路径
58	Reference Table Files/q1_paper_moisture.csv
59	Reference Table Files/q1_paper_temperature.csv
60	Reference Table Files/q2_paper_moisture.csv
61	Reference Table Files/q2_paper_temperature.csv
62	Reference Table Files/q3_paper_moisture.csv
63	Reference Table Files/q4_paper_moisture.csv
64	Reference Table Files/q4_paper_radius.csv
65	result1.xlsx
66	result2.xlsx
67	result3.xlsx
68	result4.xlsx
69	结果表说明.txt

05_数值检验与实验（37 份）

编号	相对文件路径
70	MATLAB 对照证据/compareMatlab.mjs
71	MATLAB 对照证据/runCrossCheck.m
72	Python 检验源码/paper_output/code/modeling/analytic_jacobian.py
73	Python 检验源码/paper_output/code/modeling/axisymmetric_check.py
74	Python 检验源码/paper_output/code/modeling/compare_analytic.py
75	Python 检验源码/paper_output/code/modeling/disk_dense.py
76	Python 检验源码/paper_output/code/modeling/drying_core.py
77	Python 检验源码/paper_output/code/modeling/export_outputs.py

编号	相对文件路径
78	Python 检验源码/paper_output/code/modeling/production_provenance.py
79	Python 检验源码/paper_output/code/modeling/publication_plots.py
80	Python 检验源码/paper_output/code/modeling/q1_model.py
81	Python 检验源码/paper_output/code/modeling/q2_model.py
82	Python 检验源码/paper_output/code/modeling/q3_model.py
83	Python 检验源码/paper_output/code/modeling/q4_model.py
84	Python 检验源码/paper_output/code/modeling/run_experiments.py
85	Python 检验源码/paper_output/code/modeling/run_modeling.py
86	Python 检验源码/paper_output/code/modeling/validate_bessel.py
87	Python 检验源码/paper_output/code/modeling/verify_convergence.py
88	Python 检验源码/paper_output/code/modeling/verify_dense_storage.py
89	Python 检验源码/paper_output/code/modeling/verify_time_accuracy.py
90	Python 检验源码/paper_output/code/verification/analytic_metric_check.py
91	Python 检验源码/paper_output/code/verification/crossvalidate_series.py
92	Python 检验源码/paper_output/code/verification/crossvalidate_solver.py
93	Python 检验源码/paper_output/code/verification/energy_balance_check.py
94	Python 检验源码/paper_output/code/verification/energy_balance_diagnose.py
95	Python 检验源码/paper_output/code/verification/energy_balance_diagnose2.py
96	Python 检验源码/paper_output/code/verification/isotherm_activity_closure.py
97	Python 检验源码/paper_output/code/verification/isotherm_diagnose.py
98	Python 检验源码/paper_output/code/verification/isotherm_jacobian_probe.py
99	Python 检验源码/paper_output/code/verification/latent_heat_scenarios.py
100	Python 检验源码/paper_output/code/verification/method_comparison.py
101	Python 检验源码/paper_output/code/verification/sensitivity_analysis.py
102	Python 检验源码/paper_output/code/verification/threshold_and_scaling_checks.py
103	数值检验说明.txt

编号	相对文件路径
104	源码文件与校验值.txt
105	独立沙盒运行矩阵.csv
106	独立沙盒逐源码核查.txt

附录 C 结果表与数值精度核查

C.1 临界事件与严格达标时刻

下表保留事件根和报告时刻的未舍入查询结果。报告时刻对应 0.0001 h 的时间网格，并逐点检查最大含水率严格小于 0.15 kg/kg；显示为 0.1500 的四位浓度单元格不作为等于阈值的判据。全域判定与离散最大值的关系见附录 A。

量	问题三	问题四
事件根 / h	57.47230195056044	51.09057478683054
报告烘干时长 / h	57.4724	51.0906
报告时刻最大 C / (kg/kg)	0.14999989718225315	0.14999995339076624
径向区间数 N	3200	6400

C.2 题目要求的径向采样结果

表 C.1 至表 C.7 按已冻结 CSV 原样显示，时间、位置和单位与正文结果表对应。完整结果工作簿另见附录 B 的文件清单。径向含水率均为干基含水率；径向列名中的 0、0.5、1、1.5、2 表示物理半径，单位为 cm。问题四的“药材表面”列查询当时真实表面 $r=R(t)$ ，位于当时材料范围之外的固定半径留空。

表 C.1 问题一温度 (°C)

时间/s	0	0.5	1	1.5	2
100.0000	28.0001	28.0003	28.0040	28.0327	28.1801
300.0000	28.0408	28.0635	28.1514	28.3681	28.8489
600.0000	28.4533	28.5360	28.8039	29.3159	30.1652
900.0000	29.3243	29.4583	29.8755	30.6161	31.7304
1200.0000	30.5427	30.7098	31.2223	32.1126	33.4276
1500.0000	31.9957	32.1867	32.7660	33.7463	35.1203
1800.0000	33.5753	33.7720	34.3642	35.3621	36.7856

表 C.2 问题一干基含水率 (kg/kg)

时间/s	0	0.5	1	1.5	2
100.0000	2.5500	2.5500	2.5500	2.5500	2.2470
300.0000	2.5500	2.5500	2.5500	2.5492	2.0508
600.0000	2.5500	2.5500	2.5500	2.5353	1.8770
900.0000	2.5500	2.5500	2.5497	2.5045	1.7547
1200.0000	2.5500	2.5500	2.5482	2.4646	1.6586
1500.0000	2.5500	2.5499	2.5445	2.4206	1.5787
1800.0000	2.5500	2.5497	2.5383	2.3755	1.5102

表 C.3 问题二温度 (°C)

时间/h	0	0.5	1	1.5	2
0.5000	32.1892	32.3821	32.9659	33.9608	35.4130
1.0000	40.3816	40.5536	41.0605	41.8782	42.9976
1.5000	45.8468	45.9348	46.1930	46.6051	47.1401
2.0000	48.4502	48.4881	48.5984	48.7736	49.0033
2.5000	49.4670	49.4792	49.5136	49.5653	49.6609
3.0000	49.8495	49.8553	49.8746	49.9101	49.9664

表 C.4 问题二干基含水率 (kg/kg)

时间/h	0	0.5	1	1.5	2
0.5000	2.5499	2.5489	2.5256	2.3257	1.6486
1.0000	2.5257	2.4948	2.3578	2.0230	1.4711
1.5000	2.3861	2.3256	2.1344	1.8020	1.3476
2.0000	2.1709	2.1084	1.9236	1.6259	1.2311
2.5000	1.9566	1.9006	1.7360	1.4720	1.1166
3.0000	1.7662	1.7165	1.5702	1.3333	1.0081

表 C.5 问题三干基含水率 (kg/kg)

时间/h	0	0.5	1	1.5	2
6.0000	1.0170	0.9881	0.9015	0.7550	0.5328
12.0000	0.4561	0.4432	0.4031	0.3297	0.1642
18.0000	0.2988	0.2915	0.2687	0.2253	0.0845
24.0000	0.2378	0.2327	0.2167	0.1855	0.0665
30.0000	0.2058	0.2018	0.1892	0.1641	0.0598
36.0000	0.1859	0.1825	0.1718	0.1504	0.0566
42.0000	0.1721	0.1691	0.1597	0.1407	0.0549
48.0000	0.1618	0.1592	0.1507	0.1334	0.0537
54.0000	0.1539	0.1514	0.1436	0.1277	0.0530
57.4724	0.1500	0.1477	0.1402	0.1249	0.0527

表 C.6 问题四干基含水率 (kg/kg)

时间/h	0	0.5	1	1.5	2	药材表面
6.0000	1.7196	1.5377	1.0226			0.4207
12.0000	0.7377	0.6546	0.4083			0.1670
18.0000	0.4087	0.3686	0.2397			0.0892
24.0000	0.2851	0.2611	0.1790			0.0673
30.0000	0.2263	0.2094	0.1493			0.0595
36.0000	0.1928	0.1797	0.1317			0.0560
42.0000	0.1712	0.1604	0.1200			0.0541
48.0000	0.1561	0.1468	0.1116			0.0530
51.0906	0.1500	0.1413	0.1081			0.0526

表 C.7 问题四半径变化

时间/h	表面半径/cm
6.0000	1.3740
12.0000	1.2480
18.0000	1.2140
24.0000	1.2040
30.0000	1.2010
36.0000	1.2000
42.0000	1.2000
48.0000	1.2000
51.0906	1.2000

C.3 全量导出及一致性检查

本轮独立 Windows Sandbox 正式计算使用 Q1、Q23 的 N=3200 和 Q4 的 N=6400，内部求解与导出记录耗时 1810.9893 s。四份工作簿合计核查 9,335,598 个数据单元格，另对正文 297 个结果单元格进行实际轨迹查询核对。工作簿全量回读结合逐行 CSV 对照，正文采样由求解轨迹直接查询；两项核查覆盖方式不同。三条保存轨迹的 21 个数组与冻结参照逐元素相同。

数值复现说明指定代码、输入与环境能够再现冻结结果，并不单独证明物理模型的实验预测精度。与离散方法相关的基准、网格和容差检验应在相应实际执行范围内解释，不能将历史检验脚本的成功退出直接等同于全部检验条件通过。

附录 D 完整无注释源程序

本附录按文件列出 47 份完整源码，共 9,917 行。源码正文不含解释性注释或 Python 文档字符串；文件路径、用途与采用范围在源码块前列明。长行仅由 Word 自动折行，行内字符串和缩进保持原文件内容。

D 中同时保留正式求解程序和实际使用过的检验、对照及绘图程序。历史失败或仅在有限设置下运行的程序不作为已通过模型验证证据，具体范围见附录 B。源码中的原有路径常量按原文件保留；复现入口及路径适配方式按附录 B 执行。

小节	程序文件	语言	行数
D.1	runDelivery.py	Python	236
D.2	q1Model.py	Python	7
D.3	q2Model.py	Python	7
D.4	q3Model.py	Python	27
D.5	q4Model.py	Python	7
D.6	dryingCore.py	Python	407
D.7	analyticJacobian.py	Python	315
D.8	diskDense.py	Python	114
D.9	exportOutputs.py	Python	720
D.10	reference_data.py	Python	216
D.11	runLogged.ps1	PowerShell	40
D.12	compareMatlab.mjs	JavaScript	199
D.13	runCrossCheck.m	MATLAB	491
D.14	analytic_jacobian.py	Python	311
D.15	axisymmetric_check.py	Python	283
D.16	compare_analytic.py	Python	234
D.17	disk_dense.py	Python	111
D.18	drying_core.py	Python	392
D.19	export_outputs.py	Python	720
D.20	production_provenance.py	Python	485
D.21	publication_plots.py	Python	395
D.22	q1_model.py	Python	7
D.23	q2_model.py	Python	7
D.24	q3_model.py	Python	26

小节	程序文件	语言	行数
D.25	q4_model.py	Python	7
D.26	run_experiments.py	Python	73
D.27	run_modeling.py	Python	439
D.28	validate_bessel.py	Python	319
D.29	verify_convergence.py	Python	126
D.30	verify_dense_storage.py	Python	68
D.31	verify_time_accuracy.py	Python	244
D.32	analytic_metric_check.py	Python	76
D.33	crossvalidate_series.py	Python	404
D.34	crossvalidate_solver.py	Python	174
D.35	energy_balance_check.py	Python	190
D.36	energy_balance_diagnose.py	Python	61
D.37	energy_balance_diagnose2.py	Python	74
D.38	isotherm_activity_closure.py	Python	404
D.39	isotherm_diagnose.py	Python	66
D.40	isotherm_jacobian_probe.py	Python	90
D.41	latent_heat_scenarios.py	Python	161
D.42	method_comparison.py	Python	160
D.43	sensitivity_analysis.py	Python	249
D.44	threshold_and_scaling_checks.py	Python	154
D.45	prepare_a_data.py	Python	246
D.46	export_six_figure_package_20260912.py	Python	178
D.47	draw_figures.R	R	197

D.1 runDelivery.py

路径: 03_程序代码/runDelivery.py

用途: 独立生产入口: 输入预检、三条轨迹调度、四题结果导出、完整回读及冻结数值参照对照

使用范围: 正式采用: 生产求解、导出或数值参照

语言: Python; 源文件共 236 行。

SHA256: 3e9936e94995fce5a8917c7e66721b6cece900b3c9d5a754f71a6c6002188511

```
from __future__ import annotations
import argparse
import csv
from contextlib import redirect_stderr, redirect_stdout
from datetime import datetime, timezone
import gc
import hashlib
import importlib
import importlib.metadata
import json
import os
from pathlib import Path
import re
import sys
import time
import traceback
import warnings

ROOT = Path(__file__).resolve().parent
os.environ['OPENBLAS_NUM_THREADS'] = '1'
os.environ['OMP_NUM_THREADS'] = '1'
sys.dont_write_bytecode = True
for key in ['TMPDIR', 'TEMP', 'TMP']:
    os.environ[key] = str(ROOT/'tmp/cache/runtime')
(ROOT/'tmp/cache/runtime').mkdir(parents=True, exist_ok=True)

def utcNow():
    return datetime.now(timezone.utc).isoformat()

def writeJson(path, value):
    path.parent.mkdir(parents=True, exist_ok=True)
    path.write_text(json.dumps(value, ensure_ascii=False, indent=2, allow_nan=False)+'\n',
encoding='utf-8')

def sha256(path):
    digest = hashlib.sha256()
    with path.open('rb') as stream:
        for block in iter(lambda: stream.read(1024*1024), b''):
            digest.update(block)
    return digest.hexdigest()

def fileRecord(path):
    return {'path': path.relative_to(ROOT).as_posix(), 'bytes': path.stat().st_size,
'sha256': sha256(path)}
```

```

class Tee:
    def __init__(self, screen, log):
        self.screen, self.log = screen, log
    def write(self, text):
        self.screen.write(text)
        self.log.write(text)
        self.flush()
        return len(text)
    def flush(self):
        self.screen.flush()
        self.log.flush()

def preflight():

    if sys.maxsize <= 2**32:
        raise RuntimeError('Use a 64-bit Python interpreter.')
    with (ROOT/'input_manifest.csv').open('r', encoding='utf-8-sig', newline='') as stream:
        manifest = list(csv.DictReader(stream))
    checked = []
    for expected in manifest:
        path = (ROOT/expected['path']).resolve()
        if not path.is_relative_to(ROOT) or not path.is_file():
            raise FileNotFoundError(expected['path'])
        actual = fileRecord(path)
        if actual['sha256'] != expected['sha256']:
            raise AssertionError('Input/reference checksum mismatch: '+expected['path'])
        checked.append(actual)
    import dryingCore
    import exportOutputs
    env, rad = dryingCore.loadInputs()
    if len(env) != 241 or len(rad) != 145:
        raise AssertionError('Observed input row counts differ from audited attachments')
    templates = {qid: exportOutputs.loadTemplate(qid)['sheet_names'] for qid in ['Q1',
'Q2', 'Q3', 'Q4']}
    return {'status': 'PASS', 'observedRows': {'environment':len(env), 'radius':len(rad)},
        'templates': templates, 'checkedFiles': checked}

def compareReference(run, question, summary, output):

    import numpy as np
    from reference.reference_data import REFERENCE_DATA
    reference = REFERENCE_DATA['questions'][question]
    tolerances = {'T_K':5e-6, 'C':5e-7, 'mean_C':5e-7, 'cumulative_loss':5e-7,
        'radius_m':1e-10, 'event_s':0.005}
    checks = []

    with np.load(ROOT/'reference'/question/'sampled_solution.npz', allow_pickle=False) as
frozen:
        ts = frozen['times_s']
        event = reference['diagnostics']['event_s']
        mask = ts < min(run.endS, reference['diagnostics']['end_s'])-1e-5
        if event is not None:
            mask &= np.abs(ts-event) > 1e-5
        selected = ts[mask]
        values = {key: [] for key in ['T_K', 'C', 'mean_C', 'cumulative_loss', 'radius_m']}
        for first in range(0, len(selected), 128):
            times = selected[first:first+128]
            T, C = run.fields(times, materialX=frozen['material_x'])
            raw = run.state(times)
            values['T_K'].append(T)

```

```

        values['C'].append(C)
        values['mean_C'].append(2*run.model.w@raw[1:-1:2])
        values['cumulative_loss'].append(raw[-1])
        values['radius_m'].append(run.model.radius(times))
    for key, arrays in values.items():
        actual = np.concatenate(arrays, axis=0)
        expected = frozen[key][mask]
        difference = float(np.max(np.abs(actual-expected)))
        if not np.isfinite(actual).all() or difference > tolerances[key]:
            raise AssertionError(f'Frozen reference mismatch {question}/{key}:
{difference}')
        checks.append({'array':key, 'valuesCompared':int(actual.size),
'maximumAbsoluteDifference':difference,
                        'absoluteTolerance':tolerances[key], 'status':'PASS'})
    with np.load(output/'sampled_solution.npz', allow_pickle=False) as actual_npz:
        exact = {key:bool(actual_npz[key].shape == frozen[key].shape and
np.array_equal(actual_npz[key], frozen[key], equal_nan=True)) for key
in frozen.files}
    sample = reference['reproduction_samples']
    T, C = run.fields(sample['times_s'], materialX=sample['material_x'])
    for key, actual in [(('T_K', T), ('C', C))]:
        difference = float(np.max(np.abs(actual-np.asarray(sample[key]))))
        if difference > tolerances[key]:
            raise AssertionError('Fixed reproduction sample mismatch: '+question+'/'+key)
        checks.append({'array':'reproduction_samples/'+key,
'maximumAbsoluteDifference':difference,
                        'absoluteTolerance':tolerances[key], 'status':'PASS'})
    if question != 'Q1':
        difference = abs(run.eventS-reference['diagnostics']['event_s'])
        if difference > tolerances['event_s']:
            raise AssertionError('Continuous event mismatch')
        completion = summary['completion']
        if completion['reported_drying_time_h'] != reference['completion']
['reported_drying_time_h']:
            raise AssertionError('Strict report time moved to a different 0.0001-hour grid
point')
        if not completion['max_C_at_reported_time'] < .15:
            raise AssertionError('Reported state does not satisfy the strict threshold')
        checks.append({'array':'event_s', 'maximumAbsoluteDifference':difference,
                        'absoluteTolerance':tolerances['event_s'], 'status':'PASS',
                        'strictReportGridMatches':True, 'strictUnroundedThreshold':True})
    return {'status':'PASS', 'source':'final_v6a', 'checks':checks,
            'supplementaryExactEqualityOfAllSavedArrays':exact,
            'scope':'Numerical reproducibility, not physical prediction accuracy'}

def runQuestion(question, intervals, output, profile):
    import dryingCore
    import exportOutputs
    import q3Model
    name = {'Q1':'q1Model', 'Q23':'q2Model', 'Q4':'q4Model'}[question]
    run = None
    try:
        print(f'{utcNow()} SOLVE {question} N={intervals}', flush=True)
        with warnings.catch_warnings(record=True) as caught:
            warnings.simplefilter('always')
            run = importlib.import_module(name).solve(intervals)
        if caught:
            raise RuntimeError('Solver warning: '+'; '.join(str(w.message) for w in
caught))
        directory = output/question
        summary = dryingCore.saveRun(run, directory)
        summary['solverWarningCount'] = 0

```

```

        if question != 'Q1':
            summary['completion'] = q3Model.completion(run)
            if not run.diagnostics()['strictly_dry_at_end']:
                raise AssertionError('Post-event state is not dry')
            writeJson(directory/'summary.json', summary)
            comparison = compareReference(run, question, summary, directory) if profile ==
'final' else {
                'status':'NOT_APPLICABLE', 'reason':'N40 quick checks are not the fine-grid
numerical result'}
            exports = []
            for qid in ([ 'Q2', 'Q3'] if question == 'Q23' else [question]):
                print(f'{utcNow()} EXPORT_AND_FULL_READBACK {qid}', flush=True)
                artifact = exportOutputs.exportQuestion(run, qid, output/'outputs')
                validation = exportOutputs.validateExports([artifact], runs={qid:run})
                if validation['status'] != 'PASS' or not
validation['fully_verified_with_live_Run']:
                    raise AssertionError('Workbook/archive/live Run verification failed')
                exports.append(validation)
                print(f'{utcNow()} VERIFIED {qid} cells={validation["exports"][0]
["checked_cells"]}', flush=True)
                checks = {'referenceComparison':comparison, 'exports':exports}
                writeJson(directory/'comparison.json', checks)
                return summary, checks
            finally:
                if run is not None:
                    run.close()
                gc.collect()
                print(f'{utcNow()} RELEASED {question}', flush=True)

def main():
    parser = argparse.ArgumentParser(description=__doc__)
    parser.add_argument('--profile', choices=['quick', 'final'], default='quick')
    parser.add_argument('--run-id', default=None)
    parser.add_argument('--preflight-only', action='store_true')
    args = parser.parse_args()
    run_id = args.run_id or
args.profile+'_'+datetime.now(timezone.utc).strftime('%Y%m%dT%H%M%SZ')
    if not re.fullmatch(r'[A-Za-z0-9][A-Za-z0-9_-]{0,79}', run_id):
        parser.error('run-id may contain only letters, digits, underscore, hyphen')
    output = ROOT/'results'/run_id
    output.mkdir(parents=True, exist_ok=False)
    started = time.perf_counter()
    sources = [fileRecord(path) for path in sorted(ROOT.glob('*.*py'))]
    report = {'status':'RUNNING', 'profile':args.profile, 'runId':run_id,
'startedAtUtc':utcNow(),
            'guiReproduced':'NOT_PERFORMED', 'humanReview':'PENDING_USER_REVIEW',
            'environment':{'python':sys.version, 'libraries':
{n:importlib.metadata.version(n)
                for n in ['numpy', 'scipy', 'openpyxl']}, 'cpuThreads':1},
            'sources':sources, 'randomSeed':None, 'deterministic':True,

'Historical107FileRegression':'NOT_RUN_NOT_REQUIRED_FOR_INDEPENDENT_REPRODUCTION'}
    with (output/'stdout.log').open('w', encoding='utf-8') as log:
        with redirect_stdout(Tee(sys.stdout, log)), redirect_stderr(Tee(sys.stderr, log)):
            try:
                report['preflight'] = preflight()
                print(f'{utcNow()} PREFLIGHT PASS profile={args.profile}', flush=True)
                if not args.preflight_only:
                    report['summaries'], report['comparisons'] = {}, {}
                    for q, n in [( 'Q1', 3200), ( 'Q23', 3200), ( 'Q4', 6400)]:
                        report['summaries'][q], report['comparisons'][q] = runQuestion(
                            q, 40 if args.profile == 'quick' else n, output, args.profile)

```

```

        if sources != [fileRecord(path) for path in sorted(ROOT.glob('*.py'))]:
            raise AssertionError('Source changed during run')
        report['status'] = 'PASS'
        print(f'{utcNow()} ALL_REQUESTED_CHECKS_COMPLETED', flush=True)
    except BaseException as error:
        report.update(status='FAIL', error=repr(error))
        traceback.print_exc()
        raise
    finally:
        report.update(finishedAtUtc=utcNow(), elapsedSeconds=time.perf_counter()-
started)
        writeJson(output/'runResult.json', report)
    return 0

if __name__ == '__main__':
    raise SystemExit(main())

```

D.2 q1Model.py

路径: 03_程序代码/q1Model.py

用途: 问题一的附录 2 物性与 1800 秒求解设置

使用范围: 正式采用: 生产求解、导出或数值参照

语言: Python; 源文件共 7 行。

SHA256: ba6c80a1711a4c4803bbc77ce3cf4b2a97c56a61997b5e5ffb38e733adc0a535

```
from dryingCore import Settings, solveCase

def solve(intervals=3200):
    return solveCase(Settings(question='Q1', intervals=intervals, faceScheme='kirchhoff',
                               rtol=1e-10, atolTemperature=1e-10, atolMoisture=1e-12,
                               earlyMaxStepS=2., maxStepS=120.))
```

D.3 q2Model.py

路径: 03_程序代码/q2Model.py

用途: 问题二与问题三共用的附录 3 物性及固定半径轨迹设置

使用范围: 正式采用: 生产求解、导出或数值参照

语言: Python; 源文件共 7 行。

SHA256: bb437d6ca5171d73b2530e7d84e84ba338f946b2a9dc4a10ffa48c5c2808170b

```
from dryingCore import Settings, solveCase

def solve(intervals=3200):
    return solveCase(Settings(question='Q23', intervals=intervals, faceScheme='kirchhoff',
                               rtol=1e-10, atolTemperature=1e-10, atolMoisture=1e-12,
                               earlyMaxStepS=2., maxStepS=120.))
```

D.4 q3Model.py

路径: 03_程序代码/q3Model.py

用途: 连续全域阈值事件、0.0001 小时格点上的严格达标时刻与原精度核验

使用范围: 正式采用: 生产求解、导出或数值参照

语言: Python; 源文件共 27 行。

SHA256: b46504b0dcd5411ee55c03dad9e36e9033cef0275d112d5642e939783ffbee21

```
import math
import numpy as np

def completion(run):
    if run.eventS is None:
        raise ValueError('No full-domain drying event within the solved horizon')

    count = math.ceil(run.eventS / 3600. * 10000.)

    while True:
        reportH = count/10000.
        t = reportH*3600.
        if t > run.endS:
            raise RuntimeError('Four-decimal reporting time is outside the verified
trajectory')
        values = run.state([t])[1:-1:2,0]
        if float(np.max(values)) < .15:
            break
        count += 1
    return {'critical_event_s':run.eventS,'critical_event_h':run.eventS/3600.,
        'reported_drying_time_h':reportH,'reported_time_s':t,
        'max_C_at_reported_time':float(np.max(values)),
        'slowest_material_coordinate':float(run.model.x[np.argmax(values)]),
        'conservative_post_verification_s':run.endS,
        'rounding_convention':'upward on 0.0001 h grid, followed by actual strict threshold
check',
        'interpretation':'A conditional numerical event, not a confidence bound on physical
drying time.'}
```

D.5 q4Model.py

路径: 03_程序代码/q4Model.py

用途: 问题四的附录 4 物性、观测半径收缩及正式网格设置

使用范围: 正式采用: 生产求解、导出或数值参照

语言: Python; 源文件共 7 行。

SHA256: ba62825c22d60a931b6568e425fabae88fae6f35ce1f485f45f8a9f74f5b4efc

```
from dryingCore import Settings, solveCase

def solve(intervals=6400):
    return solveCase(Settings(question='Q4', intervals=intervals, faceScheme='kirchhoff',
        shrink=True, rtol=1e-10, atolTemperature=1e-10, atolMoisture=1e-12,
        earlyMaxStepS=2., maxStepS=120.))
```

D.6 dryingCore.py

路径: 03_程序代码/dryingCore.py

用途: 材料坐标径向控制体、热湿物性、Kirchhoff 水通量、BDF 积分、状态查询、诊断及保存

使用范围: 正式采用: 生产求解、导出或数值参照

语言: Python; 源文件共 407 行。

SHA256: 8beb436df1acad7d541810b3f89b4aa87d207ad1f7a0b6d7880128bb717bf379

```
from __future__ import annotations

from dataclasses import asdict, dataclass
from pathlib import Path
import hashlib
import io
import json
import time

import numpy as np
from scipy.integrate import solve_ivp
from scipy.sparse import lil_matrix
from scipy.special import expi
import analyticJacobian
import diskDense

ROOT = Path(__file__).resolve().parent
LOADED_CODE_SHA256 = hashlib.sha256(Path(__file__).read_bytes()).hexdigest()
C0, T0, R0, LENGTH = 2.55, 301.15, 0.02, 0.25

def loadInputs(withRecords=False):
    arrays, records = [], []
    for name in ['A_environment_observed.csv', 'A_radius_observed.csv']:
        path = ROOT / 'inputs/cleaned' / name
        content = path.read_bytes()
        arrays.append(np.genfromtxt(io.StringIO(content.decode('utf-8-sig')),
                                    delimiter=',', names=True))
        records.append({'path':path.relative_to(ROOT).as_posix(), 'bytes':len(content),
                        'sha256':hashlib.sha256(content).hexdigest(), 'exists':True})
    return (*arrays, records) if withRecords else tuple(arrays)

@dataclass(frozen=True)

class Settings:
    question: str = 'Q23'
    intervals: int = 100
    rtol: float = 1e-7
    atolTemperature: float = 1e-7
    atolMoisture: float = 1e-9
    maxStepS: float = 600.0
    earlyMaxStepS: float = 30.0
    horizonH: float = 240.0
    shrink: bool = False
    boundaryExtension: str = 'nominal'
    tailTemperatureC: float = 50.0
    tailEquilibrium: float = 0.05
    h: float = 25.0
    beta: float = 8e-7
```

```

equilibriumScale: float = 1.0
surfaceLatentFraction: float = 0.0
latentJkg: float = 2.4e6
constantD: float | None = None
constantThermal: bool = False
method: str = 'BDF'
faceScheme: str = 'harmonic'
jacobianMode: str = 'analytic'
denseStorage: str = 'disk'

class RadialModel:
    def __init__(self, settings: Settings):
        self.settings = settings
        if settings.intervals < 2 or settings.jacobianMode not in ('analytic',
'finite_difference'):
            raise ValueError('At least two intervals and a supported Jacobian mode are
required')
        self.env, self.rad, self.inputRecords = loadInputs(withRecords=True)

        self.x = np.linspace(0., 1., settings.intervals + 1)
        self.dx = 1. / settings.intervals
        faces = np.r_[0., (self.x[1:] + self.x[:-1]) / 2., 1.]
        self.w = np.diff(faces ** 2) / 2.
        self.internalFaces = faces[1:-1]
        self.n = len(self.x)
        self.rhoD0 = (760 + 90 * C0) / (1 + C0) if settings.question == 'Q4' else (
            820. / (1 + C0) if settings.question == 'Q1' else (650 + 128 * C0) / (1 + C0))
        self evaluations = 0
        self.jacPattern = self.buildSparsity()

    def radius(self, t):
        if self.settings.shrink:
            return np.interp(t, self.rad['time_s'], self.rad['radius_m'])
        return np.asarray(t) * 0. + R0

    def environment(self, t):
        s = self.settings
        tair = np.interp(t, self.env['time_s'], self.env['temperature_K'])
        ceq = np.interp(t, self.env['time_s'], self.env['air_moisture_kg_per_kg'])
        after = np.asarray(t) > self.env['time_s'][-1]
        if s.boundaryExtension == 'nominal':
            tair = np.where(after, s.tailTemperatureC + 273.15, tair)
            ceq = np.where(after, s.tailEquilibrium, ceq)
        elif s.boundaryExtension == 'tail_mean':
            tail = self.env['time_s'] >= 10800
            tair = np.where(after, self.env['temperature_K'][tail].mean(), tair)
            ceq = np.where(after, self.env['air_moisture_kg_per_kg'][tail].mean(), ceq)
        elif s.boundaryExtension != 'last':
            raise ValueError('Unknown boundary extension')
        return tair, ceq * s.equilibriumScale

    def properties(self, T, C):
        s = self.settings

        positiveC = np.maximum(C, 1e-12)
        if np.any(T <= 0):
            raise FloatingPointError('Nonpositive absolute temperature')
        wet = positiveC / (1. + positiveC)
        if s.question == 'Q1':
            rho = np.full_like(C, 820.)

```

```

        cp = np.full_like(C, 2600.)
        k = np.full_like(C, .36)
        D = 7e-9 * np.exp(-.89 / positiveC)
    elif s.question in ('Q2', 'Q3', 'Q23'):
        rho, cp, k = 650 + 128 * positiveC, 1450 + 2736 * wet, .21 + .38 * wet
        D = 2.4e-3 * np.exp(-.45 / positiveC - 3850 / T)
    elif s.question == 'Q4':
        rho, cp, k = 760 + 90 * positiveC, 1850 + 2150 * wet, .12 + .20 * wet
        D = 4.2e-4 * np.exp(-.30 / positiveC - 3850 / T)
    else:
        raise ValueError(s.question)
    if s.constantD is not None:
        D = np.full_like(C, s.constantD)
    if s.constantThermal:
        rho, cp, k = np.full_like(C, 820.), np.full_like(C, 2600.), np.full_like(C,
.36)

    return rho, cp, k, D

@staticmethod
def harmonic(a):
    return 2 * a[:-1] * a[1:] / np.maximum(a[:-1] + a[1:], np.finfo(float).tiny)

def buildSparsity(self):
    p = lil_matrix((2 * self.n + 1, 2 * self.n + 1), dtype=int)
    for i in range(self.n):
        for j in range(max(0, i - 1), min(self.n, i + 2)):
            p[2*i:2*i+2, 2*j:2*j+2] = 1
    p[-1, 2 * (self.n - 1) + 1] = 1
    return p.tocsr()

def waterInternalFlux(self, T, C, D):
    if self.settings.faceScheme == 'harmonic' or self.settings.constantD is not None:
        return self.internalFaces * self.harmonic(D) * np.diff(C) / self.dx
    if self.settings.faceScheme != 'kirchhoff':
        raise ValueError('Unknown nonlinear face scheme')
    a, D0 = {'Q1':(.89,7e-9), 'Q23':(.45,2.4e-3), 'Q2':(.45,2.4e-3),
            'Q3':(.45,2.4e-3), 'Q4':(.30,4.2e-4)}[self.settings.question]
    cc = np.maximum(C, 1e-12)

    potential = cc * np.exp(-a/cc) + a * expi(-a/cc)
    difference = np.diff(potential)
    small = np.abs(np.diff(cc)) < 1e-7 * np.maximum((cc[:-1]+cc[1:])/2, 1e-3)
    difference[small] = (np.exp(-a/((cc[:-1][small]+cc[1:][small])/2)) * np.diff(cc)
[small])

    thermalFactor = 1. if self.settings.question == 'Q1' else
np.exp(-3850/((T[:-1]+T[1:])/2))

    return self.internalFaces * D0 * thermalFactor * difference / self.dx

def rhs(self, t, state):
    self.evaluations += 1
    T, C = state[:-1:2], state[1:-1:2]
    rho, cp, k, D = self.properties(T, C)
    radius = float(self.radius(t))
    tair, ceq = self.environment(t)
    heatG = np.zeros(self.n + 1)
    waterG = np.zeros(self.n + 1)
    heatG[1:-1] = self.internalFaces * self.harmonic(k) * np.diff(T) / self.dx
    waterG[1:-1] = self.waterInternalFlux(T, C, D)

```

```

waterG[-1] = -self.settings.beta * radius * (C[-1] - ceq)
heatG[-1] = -self.settings.h * radius * (T[-1] - tair)
if self.settings.surfaceLatentFraction:

    rhoD = self.rhoD0 * (R0 / radius) ** 2
    jEvap = rhoD * self.settings.beta * (C[-1] - ceq)
    heatG[-1] -= (radius * self.settings.surfaceLatentFraction *
                  self.settings.latentJKg * jEvap)
derivative = np.empty_like(state)

derivative[:-1:2] = np.diff(heatG) / (radius ** 2 * self.w * rho * cp)
derivative[1:-1:2] = np.diff(waterG) / (radius ** 2 * self.w)
derivative[-1] = 2 * self.settings.beta / radius * (C[-1] - ceq)
return derivative

def initial(self):
    state = np.empty(2 * self.n + 1)
    state[:-1:2], state[1:-1:2], state[-1] = T0, C0, 0.
    return state

class Run:
    def __init__(self, model, pieces, elapsedS, eventS, cache=None):
        self.model, self.pieces = model, pieces
        self.cache = cache
        self.elapsedS, self.eventS = elapsedS, eventS
        self.endS = float(pieces[-1].t[-1])

    def close(self):

        self.pieces.clear()
        if self.cache is not None:
            self.cache.close()

    def state(self, times):
        tt = np.atleast_1d(np.asarray(times, dtype=float))
        if np.min(tt) < -1e-10 or np.max(tt) > self.endS + 1e-7:
            raise ValueError('Requested time outside solved interval')
        out = np.empty((2 * self.model.n + 1, len(tt)))
        remaining = np.ones(len(tt), dtype=bool)
        for result in self.pieces:
            select = remaining & (tt >= result.t[0]-1e-7) & (tt <= result.t[-1]+1e-7)
            if np.any(select):
                out[:, select] = result.sol(tt[select])
                remaining[select] = False
        if np.any(remaining):
            raise RuntimeError('Missing dense solution segment')
        return out

    def fields(self, times, radiiM=None, materialX=None):
        tt = np.atleast_1d(np.asarray(times, dtype=float))
        state = self.state(tt)
        Ts, Cs = state[:-1:2].T, state[1:-1:2].T
        if materialX is not None:
            points = np.asarray(materialX, dtype=float)
            if np.any(~np.isfinite(points)) or np.any((points < 0.) | (points > 1.)):
                raise ValueError('Material coordinates must be finite and within [0,1]')
            return np.array([np.interp(points, self.model.x, row) for row in Ts]),
np.array([
    np.interp(points, self.model.x, row) for row in Cs])
if radiiM is None:

```

```

        return Ts, Cs
    radial = np.asarray(radiiM)
    Tout, Cout = [], []
    for i, t in enumerate(tt):
        xx = radial / self.model.radius(t)
        Tout.append(np.interp(xx, self.model.x, Ts[i], left=np.nan, right=np.nan))
        Cout.append(np.interp(xx, self.model.x, Cs[i], left=np.nan, right=np.nan))
    return np.asarray(Tout), np.asarray(Cout)

def diagnostics(self):

    minimumC = minimumT = minimumD = minimumProperty = np.inf
    maximumC = maximumT = maximumD = maximumRadialIncrease = -np.inf
    maximumMassResidual = 0.
    for piece in self.pieces:
        for first in range(0, len(piece.t), 256):
            raw = piece.y[:, first:first+256]
            T, C = raw[:-1:2], raw[1:-1:2]
            residual = 2*self.model.w*C + raw[-1] - C0
            rho, cp, k, D = self.model.properties(T.ravel(), C.ravel())
            minimumC, maximumC = min(minimumC, C.min()), max(maximumC, C.max())
            minimumT, maximumT = min(minimumT, T.min()), max(maximumT, T.max())
            minimumD = min(minimumD, D.min())
            maximumD = max(maximumD, D.max())
            minimumProperty = min(minimumProperty, rho.min(), cp.min(), k.min(), D.min())
            maximumRadialIncrease =
max(maximumRadialIncrease, np.max(np.diff(C, axis=0)))
            maximumMassResidual = max(maximumMassResidual, np.max(np.abs(residual)))
    final = self.state([self.endS])[:, 0]
    finalC = final[1:-1:2]
    return {
        'event_s': self.eventS, 'event_h': None if self.eventS is None else
self.eventS/3600,
        'end_s': self.endS, 'elapsed_s': self.elapsedS,
        'end_time_convention': 'ceil(critical_event_s)+1: conservative post-crossing
verification second; not claimed earliest integer second',
        'max_mass_balance_abs_kg_per_kg': float(maximumMassResidual),
        'min_C': float(minimumC), 'max_C': float(maximumC),
        'min_T_K': float(minimumT), 'max_T_K': float(maximumT),
        'min_D': float(minimumD), 'max_D': float(maximumD),
        'positive_properties': bool(minimumProperty > 0),
        'max_radial_C_increase': float(maximumRadialIncrease),
        'final_max_C': float(finalC.max()), 'final_surface_C': float(finalC[-1]),
        'strictly_dry_at_end': bool(finalC.max() < .15),
        'radius_end_m': float(self.model.radius(self.endS)),
        'radius_extrapolation_used': bool(self.model.settings.shrink and self.endS >
259200),
        'rhs_evaluations': self.model.evaluations,
        'accepted_time_points': sum(len(p.t) for p in self.pieces),
        'nfev': sum(p.nfev for p in self.pieces),
        'njev': sum(p.njev for p in self.pieces), 'nlu': sum(p.nlu for p in
self.pieces),
        'solver_success': all(p.success for p in self.pieces),
        'dense_storage': self.model.settings.denseStorage,
        'dense_coefficient_bytes': 0 if self.cache is None else
self.cache.bytesWritten,
        'dense_polynomial_count': 0 if self.cache is None else
self.cache.polynomialCount,
    }

```

```

def solveCase(settings: Settings) -> Run:
    started = time.perf_counter()
    if settings.denseStorage not in ('memory', 'disk'):
        raise ValueError('Dense storage must be memory or disk')
    if settings.denseStorage == 'disk' and settings.method != 'BDF':
        raise ValueError('Exact disk dense storage currently supports BDF only')
    cache = diskDense.DenseCache(ROOT) if settings.denseStorage == 'disk' else None
    try:
        return solveCaseImpl(settings, cache, started)
    except BaseException as error:
        if cache is not None:
            try:
                cache.close()
            except BaseException as cleanup_error:
                error.add_note('Private cache cleanup also failed: '+repr(cleanup_error))
        raise

def solveCaseImpl(settings, cache, started):
    model = RadialModel(settings)
    method = diskDense.DiskBDF if cache is not None else settings.method
    jacobianOptions = ({'jac': lambda t, y: analyticJacobian.jacobian(model, t, y)}
        if settings.jacobianMode == 'analytic' else {'jac_sparsity': model.jacPattern})
    if cache is not None:
        jacobianOptions['dense_cache'] = cache

    def dryEvent(t, y):
        return float(np.max(y[1:-1:2]) - .15)
    dryEvent.terminal, dryEvent.direction = True, -1
    atol = np.empty(2 * model.n + 1)
    atol[1:-1:2], atol[1:-1] = settings.atolTemperature, settings.atolMoisture,
    settings.atolMoisture
    horizon = 1800. if settings.question == 'Q1' else settings.horizonH * 3600.
    pieces, initial, eventS = [], model.initial(), None

    endpoints = [0., min(14400., horizon)]
    if horizon > 14400.:
        endpoints.append(horizon)
    for left, right in zip(endpoints[:-1], endpoints[1:]):
        piece = solve_ivp(model.rhs, (left, right), initial, method=method,
            rtol=settings.rtol, atol=atol, **jacobianOptions,
            max_step=settings.earlyMaxStepS if left < 14400. else settings.maxStepS,
            events=None if settings.question == 'Q1' else dryEvent, dense_output=True)
        if cache is not None:
            diskDense.alignBdfSegments(piece)
        pieces.append(piece)
        if not piece.success:
            raise RuntimeError(piece.message)
        initial = piece.y[:, -1].copy()
        if cache is not None:
            piece.y = cache.storeAccepted(piece.y)
        if piece.t_events is not None and len(piece.t_events[0]):
            eventS = float(piece.t_events[0][0])

    end = float(np.ceil(eventS) + 1)
    tail = solve_ivp(model.rhs, (eventS, end), initial, method=method,
        rtol=settings.rtol, atol=atol, **jacobianOptions,
        max_step=1., dense_output=True)
    if cache is not None:
        diskDense.alignBdfSegments(tail)
    if not tail.success:
        raise RuntimeError(tail.message)

```

```

        if cache is not None:
            tail.y = cache.storeAccepted(tail.y)
            pieces.append(tail)
            break
run = Run(model, pieces, time.perf_counter() - started, eventS, cache)
diagnostic = run.diagnostics()
if diagnostic['min_C'] < -1e-8 or not diagnostic['positive_properties']:
    raise FloatingPointError('Physical range/positive property check failed')
if diagnostic['max_mass_balance_abs_kg_per_kg'] > 1e-6:
    raise FloatingPointError('Dry-basis mass balance failed')
return run

def fileRecord(path):
    p = Path(path)
    return {'path': p.relative_to(ROOT).as_posix(), 'bytes': p.stat().st_size,
            'sha256': hashlib.sha256(p.read_bytes()).hexdigest(), 'exists': True}

def saveRun(run: Run, directory: Path):
    codeRecord = fileRecord(Path(__file__))
    if codeRecord['sha256'] != LOADED_CODE_SHA256:
        raise RuntimeError('Solver file changed after import; restart to obtain valid
provenance')
    jacobianRecord = fileRecord(Path(analyticJacobian.__file__))
    if jacobianRecord['sha256'] != analyticJacobian.LOADED_CODE_SHA256:
        raise RuntimeError('Jacobian file changed after import; restart to obtain valid
provenance')
    storageRecord = fileRecord(Path(diskDense.__file__))
    if storageRecord['sha256'] != diskDense.LOADED_CODE_SHA256:
        raise RuntimeError('Dense storage file changed after import; restart for valid
provenance')
    for record in run.model.inputRecords:
        if fileRecord(ROOT/record['path'])['sha256'] != record['sha256']:
            raise RuntimeError('Input changed after being loaded; retain failure and
rerun')
    directory.mkdir(parents=True, exist_ok=True)
    times = np.unique(np.r_[np.arange(0., run.endS, 60.),
                           [t for t in [100., 300., 600., 900., 1200., 1500., 1800., 3600., 5400.,
7200., 9000., 10800.] if t <= run.endS],
                           run.endS, [] if run.eventS is None else [run.eventS]])
    x = np.linspace(0., 1., 21)
    temperature, moisture, means, losses = [], [], [], []
    for first in range(0, len(times), 128):
        blockTimes = times[first:first+128]
        Tb, Cb = run.fields(blockTimes, materialX=x)
        raw = run.state(blockTimes)
        temperature.append(Tb); moisture.append(Cb)
        means.append(2*run.model.w@raw[1:-1:2]); losses.append(raw[-1].copy())
    T, C = np.vstack(temperature), np.vstack(moisture)
    np.savez_compressed(directory/'sampled_solution.npz', times_s=times, material_x=x,
                        T_K=T, C=C, radius_m=run.model.radius(times), mean_C=np.concatenate(means),
                        cumulative_loss=np.concatenate(losses))
    summary = {'settings': asdict(run.model.settings), 'diagnostics': run.diagnostics(),
              'code': codeRecord,
              'jacobian_code': jacobianRecord,
              'dense_storage_code': storageRecord,
              'inputs': run.model.inputRecords,
              'human_review_status': 'pending', 'gui_reproduced': False}
    (directory/'summary.json').write_text(json.dumps(summary, ensure_ascii=False,
indent=2), encoding='utf-8')
    return summary

```

D.7 analyticJacobian.py

路径: 03_程序代码/analyticJacobian.py

用途: 非线性径向模型的解析稀疏 Jacobian 及有限差分自检

使用范围: 正式采用: 生产求解、导出或数值参照

语言: Python; 源文件共 315 行。

SHA256: e52fd4c4ed7a11e7e6ac17d13c7e1de90b79d08b5e563f2e85248ec3afb17cdd

```
from __future__ import annotations

import argparse
from dataclasses import asdict
from datetime import datetime, timezone
import hashlib
import json
from pathlib import Path
import sys
import time
import warnings

import numpy as np
from scipy.integrate import solve_ivp
from scipy.sparse import coo_matrix
from scipy.special import expi

LOADED_CODE_SHA256 = hashlib.sha256(Path(__file__).read_bytes()).hexdigest()

def harmonicPartials(left, right):

    total = left + right
    tiny = np.finfo(float).tiny
    denominator = np.maximum(total, tiny)
    ordinary = total > tiny
    dl = np.where(ordinary, 2 * (right / denominator) ** 2, 2 * right / denominator)
    dr = np.where(ordinary, 2 * (left / denominator) ** 2, 2 * left / denominator)
    return dl, dr

def jacobian(model, t, y):

    state = np.asarray(y, dtype=float)
    n = model.n
    if state.ndim != 1 or state.size != 2*n + 1:
        raise ValueError("Jacobian requires interleaved 1D state of length 2*n+1")
    s = model.settings
    T, C = state[:-1:2], state[1:-1:2]
    cc = np.maximum(C, 1e-12)
    active = (C > 1e-12).astype(float)
    rho, cp, k, D = model.properties(T, C)
    if s.question == 'Q1':
        a, d0, tempConstant = .89, 7e-9, 0.
        rhoC, cpC, kC = np.zeros(n), np.zeros(n), np.zeros(n)
    elif s.question in ('Q2', 'Q3', 'Q23'):
        a, d0, tempConstant = .45, 2.4e-3, 3850.
        rhoC = np.full(n, 128.) * active
```

```

        cpC = 2736. / (1. + cc)**2 * active
        kC = .38 / (1. + cc)**2 * active
    elif s.question == 'Q4':
        a, d0, tempConstant = .30, 4.2e-4, 3850.
        rhoC = np.full(n, 90.) * active
        cpC = 2150. / (1. + cc)**2 * active
        kC = .20 / (1. + cc)**2 * active
    else:
        raise ValueError(s.question)
    if s.constantThermal:
        rhoC, cpC, kC = np.zeros(n), np.zeros(n), np.zeros(n)
    capacity = rho * cp
    capacityC = rhoC * cp + rho * cpC
    if s.constantD is None:
        dC = D * (a / cc**2) * active
        dT = D * tempConstant / T**2
    else:
        dC, dT = np.zeros(n), np.zeros(n)

    radius = float(model.radius(t))
    if radius <= 0:
        raise ValueError("Nonpositive radius")
    tair, ceq = model.environment(t)
    geom = model.internalFaces / model.dx
    deltaT, deltaC = np.diff(T), np.diff(C)

    kHarm = model.harmonic(k)
    khL, khR = harmonicPartials(k[:-1], k[1:])

    heatDeriv = np.column_stack((
        -geom * kHarm,
        geom * khL * kC[:-1] * deltaT,
        geom * kHarm,
        geom * khR * kC[1:] * deltaT,
    ))
    heatG = np.zeros(n + 1)
    heatG[1:-1] = geom * kHarm * deltaT
    heatG[-1] = -s.h * radius * (T[-1] - tair)

    if s.faceScheme == 'harmonic' or s.constantD is not None:
        dh = model.harmonic(D)
        dhL, dhR = harmonicPartials(D[:-1], D[1:])
        waterDeriv = np.column_stack((
            geom * dhL * dT[:-1] * deltaC,
            geom * (dhL * dC[:-1] * deltaC - dh),
            geom * dhR * dT[1:] * deltaC,
            geom * (dhR * dC[1:] * deltaC + dh),
        ))
    elif s.faceScheme == 'kirchhoff':
        primitive = cc * np.exp(-a / cc) + a * expi(-a / cc)
        primitiveDelta = np.diff(primitive)
        meanC = (cc[:-1] + cc[1:]) / 2.
        meanT = (T[:-1] + T[1:]) / 2.
        deltaCc = np.diff(cc)
        small = np.abs(deltaCc) < 1e-7 * np.maximum(meanC, 1e-3)
        expLeft, expRight = np.exp(-a/cc[:-1]), np.exp(-a/cc[1:])
        primitiveL = -expLeft * active[:-1]
        primitiveR = expRight * active[1:]

        if np.any(small):
            fMid = np.exp(-a / meanC[small])
            fPrimeMid = fMid * a / meanC[small]**2

```

```

        primitiveDelta[small] = fMid * deltaCc[small]
        primitiveL[small] = (0.5*fPrimeMid*deltaCc[small]-fMid)*active[:-1][small]
        primitiveR[small] = (0.5*fPrimeMid*deltaCc[small]+fMid)*active[1:][small]
        temperatureFactor = np.exp(-tempConstant / meanT)
        coefficient = geom * d0 * temperatureFactor
        waterG = coefficient * primitiveDelta
        waterT = waterG * tempConstant / (2 * meanT**2)
        waterDeriv = np.column_stack((waterT, coefficient*primitiveL,
                                      waterT, coefficient*primitiveR))
    else:
        raise ValueError('Unknown nonlinear face scheme')

    heatScale = 1. / (radius**2 * model.w * capacity)
    waterScale = 1. / (radius**2 * model.w)
    surfaceHeatC = 0.
    if s.surfaceLatentFraction:
        rhoD = model.rhoD0 * (.02 / radius)**2
        surfaceHeatC = -radius*s.surfaceLatentFraction*s.latentJKg*rhoD*s.beta
        heatG[-1] += surfaceHeatC * (C[-1] - ceq)
    tempDerivative = np.diff(heatG) * heatScale

    faceIndex = np.arange(n-1)
    faceColumns = np.column_stack((2*faceIndex, 2*faceIndex+1,
                                   2*faceIndex+2, 2*faceIndex+3)).ravel()

    rows, columns, values = [], [], []

    def addFace(rowIndex, derivatives, factor):
        rows.append(np.repeat(rowIndex, 4))
        columns.append(faceColumns)
        values.append((derivatives * factor[:, None]).ravel())

    addFace(2*faceIndex, heatDeriv, heatScale[:-1])
    addFace(2*faceIndex+2, heatDeriv, -heatScale[1:])
    addFace(2*faceIndex+1, waterDeriv, waterScale[:-1])
    addFace(2*faceIndex+3, waterDeriv, -waterScale[1:])
    cellIndex = np.arange(n)
    rows.append(2*cellIndex)
    columns.append(2*cellIndex+1)

    values.append(-tempDerivative * capacityC / capacity)
    rows.append(np.array([2*n-2, 2*n-2, 2*n-1, 2*n]))
    columns.append(np.array([2*n-2, 2*n-1, 2*n-1, 2*n-1]))
    values.append(np.array([-s.h*radius*heatScale[-1],
                            surfaceHeatC*heatScale[-1],
                            -s.beta*radius*waterScale[-1], 2*s.beta/radius]))

    matrix = coo_matrix((np.concatenate(values),
                        (np.concatenate(rows), np.concatenate(columns))),
                        shape=(2*n+1, 2*n+1)).tocsc()
    matrix.sum_duplicates()
    matrix.eliminate_zeros()
    return matrix

def selfTest(outputDirectory):
    import scipy
    from dryingCore import ROOT, Settings, RadialModel, LOADED_CODE_SHA256 as
    CORE_LOADED_CODE_SHA256

    started = time.perf_counter()
    rng = np.random.default_rng(20260910)
    relativeTolerance, absoluteTolerance = 5e-6, 5e-10

```

```

cases = []
for question in ['Q1', 'Q23', 'Q4']:
    for scheme in ['harmonic', 'kirchhoff']:
        for latent in [0., 1.]:
            cases.append(Settings(question=question, intervals=8, faceScheme=scheme,
                                shrink=(question=='Q4'),
                                surfaceLatentFraction=latent))
for question in ['Q1', 'Q23', 'Q4']:
    for constantDiffusivity, constantThermal in [(2e-9, False), (None, True), (2e-9,
True)]:
        cases.append(Settings(question=question, intervals=8, faceScheme='kirchhoff',
                                shrink=(question=='Q4'), constantD=constantDiffusivity,
                                constantThermal=constantThermal,
                                surfaceLatentFraction=1.))
records, failures = [], []
for settings in cases:
    model = RadialModel(settings)
    x = model.x
    states = {
        'initial': (0., model.initial()),
        'nonuniform': (18000., model.initial()),
        'late_dry': (150000., model.initial()),
        'near_uniform_small_branch': (14401., model.initial()),
    }
    states['nonuniform'][1][:1:-1:2] = 303. + 17.*x**2
    states['nonuniform'][1][1:-1:2] = 2.4 - 2.0*x**2
    states['late_dry'][1][:1:-1:2] = 321. + 2.0*x**2
    states['late_dry'][1][1:-1:2] = .175 - .115*x**2
    states['near_uniform_small_branch'][1][:1:-1:2] = 303. + 17.*x**2
    states['near_uniform_small_branch'][1][1:-1:2] = .15 + 1e-10*x
    for stateName, (t, y) in states.items():
        with warnings.catch_warnings(record=True) as captured:
            warnings.simplefilter('always')
            matrix = jacobian(model, t, y)
            vectors = []
            for _ in range(4):
                direction = rng.normal(size=y.size)
                direction[:1:-1:2] *= 1.0
                direction[1:-1:2] *= .02
                direction[-1] = .3
                vectors.append(direction)
            directional = []
            for direction in vectors:
                predicted = matrix @ direction
                steps = []
                for step in [1e-4, 3e-5, 1e-5]:
                    finiteDifference = (model.rhs(t, y+step*direction)-
                                        model.rhs(t, y-step*direction))/(2*step)
                    absoluteError = float(np.max(np.abs(predicted-finiteDifference)))
                    scale = max(float(np.max(np.abs(predicted))),
                                float(np.max(np.abs(finiteDifference))), 1e-30)
                    scaled = float(np.max(np.abs(predicted-finiteDifference)/
                                            (absoluteTolerance + relativeTolerance*np.maximum(
                                                np.abs(predicted), np.abs(finiteDifference)))))
                    steps.append({'step':step, 'max_abs_error':absoluteError,
                                'relative_inf_error':absoluteError/scale,
                                'max_component_tolerance_ratio':scaled})
                best = min(steps, key=lambda
item:item['max_component_tolerance_ratio'])
                directional.append({'all_step_errors':steps, 'best':best})
            passive = np.zeros(y.size); passive[-1] = 1.
            passiveAnalytic = float(np.max(np.abs(matrix @ passive)))
            passiveFd = float(np.max(np.abs(model.rhs(t, y+passive)-model.rhs(t,y-

```

```

passive))))
    massWeights = np.zeros(y.size)
    massWeights[1:-1:2] = 2*model.w
    massWeights[-1] = 1.
    conservation = float(np.max(np.abs(np.asarray(massWeights @ matrix))))
    warningMessages = [str(w.message) for w in captured]
    passed = (all(step['max_component_tolerance_ratio'] <= 1
        for d in directional for step in d['all_step_errors'])
        and passiveAnalytic == 0 and passiveFd == 0 and conservation < 1e-11
        and np.isfinite(matrix.data).all() and not warningMessages)
    record = {'settings':asdict(settings), 'state':stateName, 'time_s':t,
        'status':'PASS' if passed else 'FAIL', 'matrix_shape':matrix.shape,
        'matrix_nnz':matrix.nnz, 'directional_checks':directional,
        'passive_column_analytic_abs':passiveAnalytic,
        'passive_column_finite_difference_abs':passiveFd,
        'mass_balance_derivative_abs':conservation,
    'warnings':warningMessages}
    records.append(record)
    if not passed:

failures.append(f"{settings.question}/{settings.faceScheme}/latent={settings.surfaceLatentF
raction}/{stateName}/constant_D={settings.constantD}/
constant_thermal={settings.constantThermal}")
    smokeRecords = []
    for scheme in ['harmonic', 'kirchhoff']:
        for question in ['Q23', 'Q4']:
            settings = Settings(question=question, intervals=8, faceScheme=scheme,
                shrink=(question=='Q4'), surfaceLatentFraction=1.)
            model = RadialModel(settings)
            with warnings.catch_warnings(record=True) as captured:
                warnings.simplefilter('always')
                result = solve_ivp(model.rhs, (0., 2.), model.initial(), method='BDF',
                    jac=lambda t,y:jacobian(model,t,y), rtol=1e-10,
                    atol=1e-12, max_step=.2)
            msgs = [str(w.message) for w in captured]
            passed = bool(result.success and np.isfinite(result.y).all() and not msgs)
            smokeRecords.append({'question':question, 'face_scheme':scheme, 'interval_s':
[0,2],
                'status':'PASS' if passed else 'FAIL', 'warnings':msgs,
                'nfev':result.nfev, 'njev':result.njev, 'nlu':result.nlu,
                'purpose':'Short Jacobian/BDF wiring check, not
production accuracy'})
            if not passed:
                failures.append(f"BDF_smoke/{question}/{scheme}")
            currentSourceHash = hashlib.sha256(Path(__file__).read_bytes()).hexdigest()
            currentCoreHash = hashlib.sha256((ROOT/'dryingCore.py').read_bytes()).hexdigest()
            if currentSourceHash != LOADED_CODE_SHA256 or currentCoreHash !=
CORE_LOADED_CODE_SHA256:
                failures.append('Source file changed after module load during validation')
            report = {'schema_version':'1.0', 'created_utc':datetime.now(timezone.utc).isoformat(),
                'status':'PASS' if not failures else 'FAIL', 'failures':failures,
                'source_sha256':LOADED_CODE_SHA256,
                'core_sha256':CORE_LOADED_CODE_SHA256,
                'source_hash_policy':'SHA-256 frozen when each module is imported; files
checked unchanged before report save.',
                'runtime':
{'python':sys.version, 'executable':sys.executable, 'numpy':np.__version__, 'scipy':scipy.__ve
rsion__},
                'seed':20260910, 'relative_component_tolerance':relativeTolerance,
                'absolute_component_tolerance':absoluteTolerance,
                'finite_difference_policy':'Central directional differences, three decreasing
steps; every step must satisfy the mixed absolute/relative component tolerance. Best
comparison is supplementary only.',

```

```

        'case_state_count':len(records), 'direction_count':4*len(records),
        'checks':records,'short_BDF_checks':smokeRecords,
        'elapsed_s':time.perf_counter()-started,
        'limitations':['No full production run or grid convergence performed here.',
            'Tiny denominator/underflow extensions at nonphysical Newton
probes are not calibrated physical data.',
            'Only this module and the short tests use analytic Jacobian
until main agent hooks core.',
            'Visual Studio reproduction and human review remain
pending.']]
    outputDirectory.mkdir(parents=True, exist_ok=True)
    (outputDirectory/'analytic_jacobian_selftest.json').write_text(
        json.dumps(report,ensure_ascii=False,indent=2),encoding='utf-8')
    brief = {key:report[key] for key in
['status','failures','case_state_count','direction_count','elapsed_s']}
    brief['max_best_component_tolerance_ratio'] = max(
        d['best']['max_component_tolerance_ratio'] for r in records for d in
r['directional_checks'])
    brief['max_all_steps_component_tolerance_ratio'] = max(
        step['max_component_tolerance_ratio'] for r in records
        for d in r['directional_checks'] for step in d['all_step_errors'])
    brief['max_mass_balance_derivative_abs'] = max(r['mass_balance_derivative_abs'] for r
in records)
    brief['warning_count'] = sum(len(r['warnings']) for r in records+smokeRecords)
    print(json.dumps(brief,ensure_ascii=False,indent=2))
    return 0 if not failures else 1

if __name__ == '__main__':
    parser = argparse.ArgumentParser()
    parser.add_argument('--self-test', action='store_true')
    parser.add_argument('--output-dir', type=Path,
        default=Path(__file__).resolve().parent/'results/jacobianValidation')
    arguments = parser.parse_args()
    if not arguments.self_test:
        parser.error('Use --self-test, or import jacobian(model,t,y) from this module.')
    raise SystemExit(selfTest(arguments.output_dir))

```

D.8 diskDense.py

路径: 03_程序代码/diskDense.py

用途: 原精度 BDF 稠密多项式磁盘缓存、查询及释放

使用范围: 正式采用: 生产求解、导出或数值参照

语言: Python; 源文件共 114 行。

SHA256: cd57ab4add0e76ba7e282eba3884e8207ef0db2ec895828f8a84e0de08e05380

```
from __future__ import annotations
import hashlib
from pathlib import Path
import tempfile
import numpy as np
from scipy.integrate import OdeSolution
from scipy.integrate._ivp.bdf import BDF, BdfDenseOutput
from scipy.integrate._ivp.base import DenseOutput

LOADED_CODE_SHA256 = hashlib.sha256(Path(__file__).read_bytes()).hexdigest()

class DenseCache:
    def __init__(self, root):
        self.parent = (Path(root)/'tmp/cache/solver_runs').resolve()
        if not self.parent.is_relative_to(Path(root).resolve()):
            raise ValueError('Private solver cache must remain in this project')
        self.parent.mkdir(parents=True, exist_ok=True)
        self.directory = Path(tempfile.mkdtemp(prefix='bdf_', dir=self.parent)).resolve()
        try:
            self.handle = (self.directory/'polynomials.bin').open('w+b', buffering=0)
        except BaseException as error:
            try:
                self.directory.rmdir()
            except OSError as cleanup_error:
                error.add_note('Private cache directory cleanup failed: '
                               '+repr(cleanup_error)')
            raise
        self.bytesWritten = 0
        self.polynomialCount = 0
        self.acceptedArrays = []
        self.closed = False

    def append(self, values):
        if self.closed:
            raise RuntimeError('Dense solution cache is closed')
        array = np.ascontiguousarray(values, dtype=np.float64)
        offset = self.bytesWritten
        self.handle.seek(offset)
        array.tofile(self.handle)
        self.bytesWritten += array.nbytes
        self.polynomialCount += 1
        return offset, array.shape

    def read(self, offset, shape):
        if self.closed:
            raise RuntimeError('Dense solution cache is closed')
        count = int(np.prod(shape))
        self.handle.seek(offset)
        values = np.fromfile(self.handle, dtype=np.float64, count=count)
```

```

        if values.size != count:
            raise IOError('Incomplete BDF coefficient cache')
        return values.reshape(shape)

    def storeAccepted(self, values):
        path = self.directory/f'accepted_{len(self.acceptedArrays):03d}.numpy'
        mapped = np.lib.format.open_mmap(path, mode='w+', dtype=values.dtype,
shape=values.shape)

        self.acceptedArrays.append(mapped)
        mapped[:] = values
        mapped.flush()
        return mapped

    def close(self):
        if self.closed:
            return
        self.handle.close()
        for array in self.acceptedArrays:
            array._mmap.close()
        self.acceptedArrays.clear()

        if self.directory.parent != self.parent or not
self.directory.name.startswith('bdf_'):
            raise RuntimeError('Unexpected private cache path; cleanup refused')
        for path in self.directory.iterdir():
            if not path.is_file() or path.is_symlink():
                raise RuntimeError('Unexpected cache entry; cleanup refused')
            path.unlink()
        self.directory.rmdir()
        self.closed = True

class FileBdfDenseOutput(DenseOutput):
    def __init__(self, original, cache):
        super().__init__(original.t_old, original.t)
        self.order = original.order
        self.t_shift = original.t_shift.copy()
        self.denom = original.denom.copy()
        self.cache = cache
        self.offset, self.shape = cache.append(original.D)

    def _call_impl(self, t):

        dense = object.__new__(BdfDenseOutput)
        dense.D = self.cache.read(self.offset, self.shape)
        dense.t_shift, dense.denom = self.t_shift, self.denom
        return dense._call_impl(t)

class DiskBDF(BDF):
    def __init__(self, *args, dense_cache, **kwargs):
        self.dense_cache = dense_cache
        super().__init__(*args, **kwargs)

    def _dense_output_impl(self):
        return FileBdfDenseOutput(super()._dense_output_impl(), self.dense_cache)

```

```
def alignBdfSegments(result):  
  
    if result.sol is not None:  
        result.sol = OdeSolution(result.sol.ts, result.sol.interpolants, alt_segment=True)
```

D.9 exportOutputs.py

路径: 03_程序代码/exportOutputs.py

用途: 按题目模板生成四问工作簿、原精度归档、正文表, 并进行完整回读校验

使用范围: 正式采用: 生产求解、导出或数值参照

语言: Python; 源文件共 720 行。

SHA256: b7daf45a89dd54a4a5df26e47e2b07cbe1b145656234d53e7d20c3022112d5d9

```
from __future__ import annotations

import argparse
import csv
from dataclasses import asdict
from datetime import datetime, timezone
import gzip
import hashlib
import inspect
import json
import math
from pathlib import Path
import re
import sys
import tempfile
import time

import numpy as np
from openpyxl import Workbook, load_workbook
from openpyxl.cell import WriteOnlyCell
from openpyxl.styles import Alignment, Font, PatternFill
from openpyxl.utils import get_column_letter

ROOT = Path(__file__).resolve().parent
SOURCE = Path(__file__).resolve()
LOADED_EXPORTER_SHA256 = hashlib.sha256(SOURCE.read_bytes()).hexdigest()
TEMPLATE_DIR = ROOT / 'inputs/templates'
RADII_M = np.arange(21, dtype=float)*0.001
RADII_CM = [i/10 for i in range(21)]
NUMBER_FORMAT = '0.0000'
SIZE_LIMIT_BYTES = 20_000_000

def sha256File(path):
    h=hashlib.sha256()
    with Path(path).open('rb') as fh:
        for block in iter(lambda:fh.read(1024*1024),b''):
            h.update(block)
    return h.hexdigest()

def exportFileRecord(path, role=None):
    path=Path(path).resolve()
    result={'path':path.relative_to(ROOT).as_posix(),'absolute_path':str(path),
           'bytes':path.stat().st_size,'sha256':sha256File(path),'exists':True}
    if role:
        result['role']=role
    return result
```

```

def writeJson(path,value):
    path.write_text(json.dumps(value,ensure_ascii=False,indent=2,allow_nan=False)
+'\\n',encoding='utf-8')

def normalizeQuestionId(questionId):
    value=str(questionId).upper()
    if value in ['1','2','3','4']:
        value='Q'+value
    if value not in ['Q1','Q2','Q3','Q4']:
        raise ValueError('Export requires Q1, Q2, Q3 or Q4')
    return value

def outputTimes(questionId,end,event):
    if questionId=='Q1':
        if end<1800-1e-7:
            raise ValueError('Q1 requires a solved interval covering 0..1800s')
        return np.arange(1801,dtype=float)
    if questionId=='Q2':
        if abs(end-round(end))>1e-7:
            raise ValueError('Q2 needs an integer post-crossing verification endpoint')
        return np.arange(int(round(end))+1,dtype=float)
    if event is None:
        raise ValueError('Q3/Q4 final export requires an actual critical drying event')
    if not 0<=event<=end:
        raise ValueError('Critical event must lie inside the solved interval')
    return np.unique(np.r_[np.arange(math.floor(end/60)+1,dtype=float)*60,event,end])

def roundFourDecimals(x):
    if x is None:
        return None
    if not math.isfinite(float(x)):
        raise ValueError('A non-finite value cannot be exported as a numerical result')
    return float(round(float(x),4))

def loadTemplate(questionId):
    path=TEMPLATE_DIR/f'result{questionId[-1]}.xlsx'
    wb=load_workbook(path,read_only=True,data_only=False)
    try:
        header=next(wb.worksheets[0].iter_rows(values_only=True))
        names=wb.sheetnames
        return
    except:
        {'record':exportFileRecord(path,'original_result_template'),'time_header':header[0],
        'sheet_names':names,'surface_header':header[-1] if questionId=='Q4' else
        None}
    finally:
        wb.close()

def runProvenance(run):
    module=sys.modules[run.model.__class__.__module__]
    core=Path(inspect.getfile(run.model.__class__)).resolve()
    loaded=getattr(module,'LOADED_CODE_SHA256',None)
    if loaded is not None and loaded!=sha256File(core):
        raise RuntimeError('Core file changed after this Run implementation was imported')
    if sha256File(SOURCE)!=LOADED_EXPORTER_SHA256:
        raise RuntimeError('Exporter changed after import; restart for consistent
provenance')
    inputRecords=list(getattr(run.model,'inputRecords',[]))
    for rec in inputRecords:

```

```

        if sha256File(ROOT/rec['path'])!=rec['sha256']:
            raise RuntimeError('Observed input changed after solve: '+rec['path'])
    return
{'core':exportFileRecord(core,'solver_source'),'exporter':exportFileRecord(SOURCE,'exporter_source'),
    'inputs':inputRecords,'settings':asdict(run.model.settings),
    'event_s':None if run.eventS is None else float(run.eventS),
    'end_s':float(run.endS),'source_kind':'live_Run_dense_solution',
    'per_second_values_are_not_interpolated_from_60s_npz':True}

def fieldBlock(run,times,qid):
    T,C=run.fields(times,radiiM=RADII_M)
    radius=np.asarray(run.model.radius(times),dtype=float)
    if radius.ndim==0:
        radius=np.full(len(times),radius)
    surfaceC=None
    if qid=='Q4':
        _,surface=run.fields(times,materialX=np.array([1.0]))
        surfaceC=surface[:,0]
        inside=RADII_M[None,:] <= radius[:,None]+1e-12

        nearBoundary=inside & (~np.isfinite(C))
        if np.any(nearBoundary):
            C=np.where(nearBoundary,surfaceC[:,None],C)
            T=np.where(inside,T,np.nan)
            C=np.where(inside,C,np.nan)
            if not np.all(np.isfinite(C[inside])) or not np.all(np.isfinite(surfaceC)):
                raise ValueError('Non-finite moisture inside Q4 domain')
    else:
        inside=np.ones_like(C,dtype=bool)
        if not np.all(np.isfinite(T)) or not np.all(np.isfinite(C)):
            raise ValueError('Non-finite output inside fixed domain')
    return T-273.15,C,radius,surfaceC,inside

def rawHeaders(qid):
    h=['time_s','radius_m']
    if qid in ['Q1','Q2']:
        h += [f'T_C_r_{i/10:.1f}_cm' for i in range(21)]
    h += [f'C_r_{i/10:.1f}_cm' for i in range(21)]
    if qid=='Q4':
        h += ['C_surface']
    return h

def rawRow(qid,t,T,C,R,surface,inside):
    row=[float(t),float(R)]
    if qid in ['Q1','Q2']:
        row+=list(map(float,T))
    row += [float(c) if valid else None for c,valid in zip(C,inside)]
    if qid=='Q4':
        row.append(float(surface))
    return row

def roundedSheets(qid,row):
    t,R=row[:2]
    if qid in ['Q1','Q2']:
        return {'温度':[roundFourDecimals(t)]+[roundFourDecimals(v) for v in row[2:23]],
                '水分浓度':[roundFourDecimals(t)]+[roundFourDecimals(v) for v in
row[23:44]]}
    if qid=='Q3':

```

```

        return {'Sheet1':[roundFourDecimals(t)]+[roundFourDecimals(v) for v in row[2:23]]}
    return {'Sheet1':[roundFourDecimals(t)]+[roundFourDecimals(v) for v in row[2:24]],
            '半径':[roundFourDecimals(t),roundFourDecimals(R*100)]}

def createSheet(wb,name,header):
    ws=wb.create_sheet(name)
    ws.freeze_panes='B2'
    ws.sheet_view.showGridLines=False
    ws.column_dimensions['A'].width=30
    for j in range(2,len(header)+1):
        ws.column_dimensions[get_column_letter(j)].width=13 if j<len(header) else 16
    ws.row_dimensions[1].height=34
    cells=[]
    for value in header:
        cell=WriteOnlyCell(ws,value=value)
        cell.font=Font(name='Arial',size=10,bold=True,color='FFFFFF')
        cell.fill=PatternFill(fill_type='solid',fgColor='334155')
        cell.alignment=Alignment(horizontal='center',vertical='center',wrap_text=True)
        if isinstance(value,(int,float)):
            cell.number_format='0.0'
        cells.append(cell)

    cache=ROOT/'tmp/cache/openpyxl_exports'
    cache.mkdir(parents=True,exist_ok=True)
    previousTempdir=tempfile.tempdir
    try:
        tempfile.tempdir=str(cache)
        ws.append(cells)
    finally:
        tempfile.tempdir=previousTempdir
    return ws

def appendNumeric(ws,row):
    cells=[]
    for value in row:
        if value is None:
            cells.append(None)
        else:
            cell=WriteOnlyCell(ws,value=value)
            cell.number_format=NUMBER_FORMAT
            cells.append(cell)
    ws.append(cells)

def paperTables(run,qid,directory):
    if qid=='Q1':
        ts=np.array([100,300,600,900,1200,1500,1800],dtype=float)
    elif qid=='Q2':
        ts=np.arange(1,7,dtype=float)*1800
        if run.endS<10800-1e-7:
            raise ValueError('Q2 paper tables require 3h of actual solution')
    else:
        from q3Model import completion
        reportedEnd=completion(run)['reported_time_s']
        ts=np.unique(np.r_[np.arange(21600.,reportedEnd+1e-8,21600.),reportedEnd])
    rs=np.array([0.,.005,.01,.015,.02])
    T,C=run.fields(ts,radiiM=rs)
    T=T-273.15
    R=np.asarray(run.model.radius(ts))
    headers=['时间/s' if qid=='Q1' else '时间/h']+[0,0.5,1,1.5,2]

```

```

if qid=='Q4':
    headers += ['药材表面']
    _,Cs=run.fields(ts,materialX=[1.])
outputs=[]
fields=[('temperature',T),('moisture',C)] if qid in ['Q1','Q2'] else [('moisture',C)]
for kind,values in fields:
    path=directory/f'{qid.lower()}_paper_{kind}.csv'
    with path.open('w',encoding='utf-8-sig',newline='') as fh:
        w=csv.writer(fh);w.writerow(headers)
        for i,t in enumerate(ts):
            row=[f'{t if qid=="Q1" else t/3600:.4f}']
            for j,r in enumerate(rs):
                outside=qid=='Q4' and r>R[i]+1e-12
                value=values[i,j]
                if not outside and not np.isfinite(value) and qid=='Q4' and abs(r-
R[i])<=1e-12:
                    value=Cs[i,0]
                    row.append('' if outside else f'{float(value):.4f}')
                if qid=='Q4':
                    row += [f'{Cs[i,0]:.4f}']
                w.writerow(row)
            outputs.append(exportFileRecord(path,'paper_table'))
if qid=='Q4':
    path=directory/'q4_paper_radius.csv'
    with path.open('w',encoding='utf-8-sig',newline='') as fh:
        w=csv.writer(fh);w.writerow(['时间/h','表面半径/cm'])
        w.writerows([f'{t/3600:.4f}',f'{r*100:.4f}'] for t,r in zip(ts,R))
    outputs.append(exportFileRecord(path,'paper_surface_coordinates'))
return outputs

```

```

def validatePaperTables(manifest,run):

```

```

    qid=manifest['question_id']
    if run is None:
        return {'status':'NOT_REQUESTED','checked_cells':0,'files':[],
                'scope':'Paper CSV hashes only; no independent live Run available.'},[]
    errors=[];records=manifest['paper_tables'];completionInfo=None
    if qid=='Q1':
        times=np.asarray([100.,300.,600.,900.,1200.,1500.,1800.])
        expectedKinds=['temperature','moisture']
    elif qid=='Q2':
        times=1800.*np.arange(1,7)
        expectedKinds=['temperature','moisture']
    else:
        from q3Model import completion
        completionInfo=completion(run)
        finalS=float(completionInfo['reported_time_s'])

        regular=[21600.*k for k in range(1,math.floor(finalS/21600.)+1)
                  if 21600.*k<finalS-1e-8]
        times=np.asarray([*regular,finalS],dtype=float)
        expectedKinds=['moisture']+(['radius'] if qid=='Q4' else [])
        stored=manifest.get('paper_table_time_contract',{})
        for key in ['reported_time_s','reported_drying_time_h']:
            if key not in stored or not numericEqual(stored[key],completionInfo[key]):
                errors.append('Paper CSV report-time metadata differs from independent
completion: '+key)
            if not
numericEqual(stored.get('reported_time_h'),completionInfo['reported_drying_time_h']):
                errors.append('Paper CSV reported_time_h alias differs from independent

```

```

completion')
    if times[-1]>run.endS+1e-7:
        errors.append('Paper CSV contract exceeds the live solution horizon')
        return {'status': 'FAIL', 'checked_cells': 0, 'files': []}, errors
    expectedNames={f'{qid.lower()}_paper_{kind}.csv': kind for kind in expectedKinds}
    actualNames=[Path(r['path']).name for r in records]
    if set(actualNames)!=set(expectedNames) or len(actualNames)!=len(expectedNames):
        errors.append('Missing, duplicate, or unexpected paper CSV artifact')
    files=[];totalCells=0
    for rec in records:
        path=ROOT/rec['path'];kind=expectedNames.get(path.name)
        if kind is None:continue
        beforeErrors=len(errors)
        with path.open('r',encoding='utf-8-sig',newline='') as fh:
            reader=csv.reader(fh);header=next(reader,None);rows=list(reader)
        desiredTime='时间/s' if qid=='Q1' else '时间/h'
        if kind=='radius':
            correctHeader=['时间/h','表面半径/cm']
            if header!=correctHeader:
                errors.append(path.name+' : radius/time units or header mismatch')
                continue
            radii=None
            expected=np.asarray(run.model.radius(times),dtype=float).reshape(-1,1)*100.
            inside=np.ones_like(expected,dtype=bool)
            unit='cm';surfaceColumn=None
        else:
            wantedColumns=7 if qid=='Q4' else 6
            if header is None or len(header)!=wantedColumns or header[0]!=desiredTime:
                errors.append(path.name+' : time/column header mismatch');continue
            try:
                radii=np.asarray([float(v)*.01 for v in header[1:6]])
            except ValueError:
                errors.append(path.name+' : nonnumeric physical-radius header');continue
            if not np.all(np.isfinite(radii)) or not np.allclose(radii,
[0.,.005,.01,.015,.02],rtol=0,atol=1e-12):
                errors.append(path.name+' : physical-radius grid must be 0,0.5,1,1.5,2
cm');continue
            if qid=='Q4' and header[-1]!='药材表面':
                errors.append(path.name+' : distinct material surface column
missing');continue
            temperature,moisture=run.fields(times,radiiM=radii)
            expected=np.asarray(temperature)-273.15 if kind=='temperature' else
np.asarray(moisture)
            inside=np.ones_like(expected,dtype=bool);surfaceColumn=None
            if qid=='Q4':
                physicalR=np.asarray(run.model.radius(times),dtype=float)
                inside=radii[None,:]<=physicalR[:,None]+1e-12
                _,surfaceValues=run.fields(times,materialX=np.asarray([1.]))

                coincidence=np.abs(radii[None,:]-physicalR[:,None])<=1e-12
                expected=np.where(coincidence &
~np.isfinite(expected),surfaceValues,expected)
                expected=np.column_stack([expected,surfaceValues[:,0]])
                inside=np.column_stack([inside,np.ones(len(times),dtype=bool)])
                surfaceColumn=7
                unit='degC' if kind=='temperature' else 'kg/kg'
            if len(rows)!=len(times):
                errors.append(f'{path.name}: expected {len(times)} data rows, read
{len(rows)}')
            checked=0;blanks=0;maxFieldError=0.;maxTimeError=0.
            for i,(row,t) in enumerate(zip(rows,times)):
                if len(row)!=len(header):

```

```

        errors.append(f'{path.name}: column count mismatch at row {i+2}');continue
    timeValue=float(t if qid=='Q1' else t/3600.)
    if row[0]!=format(timeValue, '.4f'):
        errors.append(f'{path.name}: time label mismatch at row {i+2}; query time
is {t:.17g}s')
    else:maxTimeError=max(maxTimeError,abs(float(row[0])-timeValue))
    checked+=1
    for j,cell in enumerate(row[1:]):
        checked+=1
        if not inside[i,j]:
            blanks+=1
            if cell!='':errors.append(f'{path.name}: domain outside must be blank
at row {i+2}, column {j+2}')
            continue
        value=float(expected[i,j])
        if not math.isfinite(value):
            errors.append(f'{path.name}: live value is not finite inside
material');continue
        if not re.fullmatch(r'-?\d+\.\d{4}',cell):
            errors.append(f'{path.name}: finite four-decimal number required at row
{i+2}, column {j+2}')
            continue
        actual=float(cell)
        if not math.isfinite(actual) or abs(actual-
float(format(value, '.4f')))>1e-10:
            errors.append(f'{path.name}: independently queried value mismatch at
row {i+2}, column {j+2}')
            maxFieldError=max(maxFieldError,abs(actual-value))
        totalCells+=checked
        files.append({'artifact':exportFileRecord(path),'kind':kind,'status':'PASS' if
len(errors)==beforeErrors else 'FAIL',

'header':header,'data_rows':len(rows),'expected_data_rows':len(times),'checked_cells':check
ed,

        'source_query_times_s':times.tolist(),'display_time_unit':'s' if qid=='Q1' else
'h',

        'max_display_time_rounding_error':maxTimeError,'field_unit':unit,

'max_field_rounding_error_from_live_value':maxFieldError,'outside_domain_blank_cells':blank
s,

        'physical_radii_m':None if radii is None else radii.tolist(),
        'surface_column_1based':surfaceColumn,'surface_query':'material_x=1; never
fixed 2 cm' if surfaceColumn else None})
    return {'status':'FAIL' if errors else 'PASS','checked_cells':totalCells,'files':files,

'completion':completionInfo,'completion_source':exportFileRecord(SOURCE.with_name('q3Model.
py')) if completionInfo else None,

        'scope':'Every paper CSV cell independently re-evaluated from live Run and fixed
question contracts; no generation arrays or 60s archives reused.',

        'rounding_limit':'CSV stores four decimals; matching rounded values cannot
distinguish sub-rounding perturbations of the original source value.'},errors

def exportQuestion(run,questionId,outputDir,*,chunkRows=1000,overwrite=False):

    started=time.perf_counter()
    qid=normalizeQuestionId(questionId)
    directory=Path(outputDir).resolve()
    if not directory.is_relative_to(ROOT) or directory.is_relative_to(ROOT/'inputs'):
        raise ValueError('Outputs must be inside the extracted code package, outside
inputs')
    if not 1<=int(chunkRows)<=2000:

```

```

        raise ValueError('chunk_rows must be between 1 and 2000')
    directory.mkdir(parents=True,exist_ok=True)
    stem='result'+qid[-1]
    bookPath=directory/(stem+'.xlsx')
    rawPath=directory/(stem+'_unrounded.csv.gz')
    manifestPath=directory/(stem+'.export.json')
    if not overwrite and any(p.exists() for p in [bookPath,rawPath,manifestPath]):
        raise FileExistsError('Export target already exists; use a new output version')
    provenance=runProvenance(run)
    allowed={'Q1':{'Q1'}, 'Q2':{'Q2', 'Q23'}, 'Q3':{'Q2', 'Q3', 'Q23'}, 'Q4':{'Q4'}}
    if run.model.settings.question not in allowed[qid]:
        raise ValueError('Run physics/question do not match the requested output question')
    if qid=='Q4' and not run.model.settings.shrink:
        raise ValueError('Formal Q4 output requires the shrinking-domain Run, not its
fixed-radius control')
    template=loadTemplate(qid)
    times=outputTimes(qid,run.endS,run.eventS)
    if len(times)+1>1_048_576:
        raise ValueError('Required full time grid exceeds the XLSX worksheet row limit; do
not truncate')
    workbook=Workbook(write_only=True)
    workbook.properties.creator=''
    workbook.properties.lastModifiedBy=''
    workbook.properties.title=f'问题{qid[-1]}结果'
    headers={name:[template['time_header']] + RADII_CM for name in template['sheet_names']}
    if qid=='Q4':
        headers['Sheet1'].append(template['surface_header'])
        headers['半径']=['时间/s', '药材表面半径/cm']
    sheets={name:createSheet(workbook,name,header) for name,header in headers.items()}
    counts={name:
{'data_rows':0, 'total_rows':1, 'columns':len(header), 'outside_domain_blank_cells':0,
    'first_time_s':None, 'last_time_s':None, 'header':header} for name,header
in headers.items()}
    samplesIdx=set(np.linspace(0,len(times)-1,min(13,len(times)),dtype=int).tolist())
    samplesIdx.update([0,min(1,len(times)-1),len(times)-1])
    if run.eventS is not None:
        samplesIdx.update(np.flatnonzero(times==run.eventS).tolist())
    samples=[]
    with gzip.open(rawPath,'wt',encoding='utf-8',newline='',compresslevel=6) as fh:
        rawWriter=csv.writer(fh);rawWriter.writerow(rawHeaders(qid))
        for start in range(0,len(times),int(chunkRows)):
            ts=times[start:start+int(chunkRows)]
            T,C,R,Cs,inside=fieldBlock(run,ts,qid)
            for i,t in enumerate(ts):
                raw=rawRow(qid,t,T[i],C[i],R[i],None if Cs is None else Cs[i],inside[i])
                rawWriter.writerow(['' if v is None else format(v,'.17g') for v in raw])
                values=roundedSheets(qid,raw)
                for name,row in values.items():
                    appendNumeric(sheets[name],row)
                    count=counts[name]
                    count['data_rows']+=1;count['total_rows']+=1
                    count['outside_domain_blank_cells']+=sum(v is None for v in row)
                    if count['first_time_s'] is None:
                        count['first_time_s']=row[0]
                    count['last_time_s']=row[0]
                if start+i in samplesIdx:
                    samples.append({'data_row_index':start+i, 'source_time_s':float(t),
                        'raw_values':raw, 'rounded_sheets':values})
    workbook.save(bookPath)
    paperRecords=paperTables(run,qid,directory)
    paperTimeContract={'unit':'s' if qid=='Q1' else 'h', 'stored_decimals':4,
        'query_uses_unrounded_seconds':True}
    if qid in ['Q3','Q4']:

```

```

from q3Model import completion
completed=completion(run)
paperTimeContract.update({key:completed[key] for key in

['reported_time_s','reported_drying_time_h','critical_event_s',

'max_C_at_reported_time','conservative_post_verification_s']})
    paperTimeContract['reported_time_h']=completed['reported_drying_time_h']
    paperTimeContract['regular_step_s']=21600.
    paperTimeContract['endpoint_convention']='6-hour samples plus upward-reported
strict-drying time; distinct from workbook post-verification endpoint'
    endProvenance=runProvenance(run)
    if endProvenance!=provenance or sha256File(ROOT/template['record']['path'])!
=template['record']['sha256']:
        raise RuntimeError('Source, settings or template changed during export')
    bookRecord=exportFileRecord(bookPath,'contest_result_workbook')
    archiveRecord=exportFileRecord(rawPath,'internal_unrounded_output_grid_archive')
    manifest={
        'schema_version':'1.0','question_id':qid,'generated_by':'exportOutputs.py',

'generated_at':datetime.now(timezone.utc).isoformat(),'status':'EXPORTED_PENDING_STREAM_REA
DBACK',

'workbook':bookRecord,'unrounded_archive':archiveRecord,'paper_tables':paperRecords,
    'paper_table_time_contract':paperTimeContract,
    'provenance':provenance,'template':template['record'],'sheets':counts,
    'time_grid':{'start_s':0.,'end_s':float(times[-1]),'regular_step_s':1 if qid in
['Q1','Q2'] else 60,
        'critical_event_s':provenance['event_s'] if qid in ['Q3','Q4'] else
None,
        'include_post_verification_endpoint':qid!='Q1',

'count':len(times),'rounding_duplicate_time_count':int(np.sum(np.diff(np.round(times,4))==0
))),
    'radius_grid':
{'fixed_radii_cm':RADII_CM,'surface_column':qid=='Q4','surface_coordinate_sheet':'半径' if
qid=='Q4' else None},
    'rounding':{'decimal_places':4,'storage':'numeric rounded to four
decimals','number_format':NUMBER_FORMAT,
        'raw_archive':'17 significant digits, source times in seconds and
radius in metres',
        'threshold_judgement':'use original full precision event and max C,
never rounded 0.1500'},
    'domain_rule':{'outside_values':'blank (None); never
zero','comparison_tolerance_m':1e-12,
        'Q4_surface_is_not_fixed_2cm':True},
    'samples':samples,'elapsed_s':time.perf_counter()-started,
    'size':{'workbook_bytes':bookRecord['bytes'],'workbook_MiB':bookRecord['bytes']/
2**20,

'unrounded_archive_bytes':archiveRecord['bytes'],'warning_limit_bytes':SIZE_LIMIT_BYTES,
        'workbook_exceeds_20_decimal_MB':bookRecord['bytes']>SIZE_LIMIT_BYTES,
        'archive_is_internal_not_automatically_in_submission_zip':True,
        'complete_support_archive_still_requires_actual_size_check':True,
        'no_required_time_or_space_values_removed':True},
    'validation_required':'validate_exports([record], runs={question_id: run}); full
workbook/archive comparison, live Run workbook samples, and every paper CSV cell
independently queried from live Run',
        'visual_studio_gui':'pending','human_review':'pending',
    }
    writeJson(manifestPath,manifest)
    return
{'question_id':qid,'workbook_path':str(bookPath),'manifest_path':str(manifestPath),

```

```

        'artifacts':[bookRecord,archiveRecord]+paperRecords+
[exportFileRecord(manifestPath,'export_manifest')],
        'sheets':counts,'size':manifest['size'],'validation_status':'pending'}

def resolveManifestPath(item):
    if isinstance(item,dict):
        return Path(item['manifest_path'])
    path=Path(item)
    return path.with_suffix('.export.json') if path.suffix=='.xlsx' else path

def numericEqual(a,b,tolerance=1e-10):
    return a is None and b is None or (a is not None and b is not None and
        isinstance(a,(int,float)) and not isinstance(a,bool) and math.isfinite(float(a))
and abs(float(a)-float(b))<=tolerance)

def validateExports(paths,runs=None):

    if isinstance(paths,(str,Path,dict)):
        paths=[paths]
        runs={} if runs is None else runs
        results=[]
        for item in paths:
            started=time.perf_counter()
            manifestPath=resolveManifestPath(item).resolve()
            manifest=json.loads(manifestPath.read_text(encoding='utf-8'))
            qid=manifest['question_id'];errors=[];checkedCells=0;checkedSheets={}
            for rec in
[manifest['workbook'],manifest['unrounded_archive'],manifest['template']]
+manifest['paper_tables']:
                if not (ROOT/rec['path']).exists() or sha256File(ROOT/rec['path'])!
=rec['sha256']:
                    errors.append('Hash mismatch or missing artifact: '+rec['path'])
            if errors:
                raise RuntimeError('; '.join(errors))
            times=outputTimes(qid,manifest['provenance']['end_s'],manifest['provenance']
['event_s'])
            wb=load_workbook(ROOT/manifest['workbook']['path'],read_only=True,data_only=False)
            if wb.sheetnames!=list(manifest['sheets']):
                errors.append('Sheet names/order differ from export contract')
            try:
                for name,spec in manifest['sheets'].items():
                    ws=wb[name]
                    rows=ws.iter_rows(min_row=1,max_col=spec['columns'])
                    header=[cell.value for cell in next(rows)]
                    if header!=spec['header']:
                        errors.append(name+': header/physical radius grid mismatch')
                    count=0;blanks=0;first=None;last=None;maxRoundingError=0.
                    with gzip.open(ROOT/manifest['unrounded_archive']
['path'],'rt',encoding='utf-8',newline='') as rawfh:
                        sourceRows=csv.reader(rawfh)
                        if next(sourceRows)!=rawHeaders(qid):
                            errors.append('Unrounded archive schema mismatch')
                        for index,sourceRow in enumerate(sourceRows):
                            try:
                                cells=next(rows)
                            except StopIteration:
                                errors.append(name+': workbook ends before raw archive');break
                            raw=[None if v=='' else float(v) for v in sourceRow]
                            if index>=len(times) or abs(raw[0]-times[index])>1e-8:

```

```

        errors.append(name+' : source time schedule mismatch at
'+str(index))
        if qid=='Q4':
            for j,r in enumerate(RADII_M):
                outside=r>raw[1]+1e-12
                if (raw[2+j] is None)!=outside:
                    errors.append(name+' : incorrect moving-domain mask at
'+str(index))
            elif any(v is None for v in raw):
                errors.append(name+' : unexpected blank in fixed domain')
                expected=roundedSheets(qid,raw)[name]
                for j,(cell,value) in enumerate(zip(cells,expected)):
                    if not numericEqual(cell.value,value):
                        errors.append(f'{name}: value mismatch at row {index+2},
column {j+1}')
                    if value is not None:
                        if cell.number_format!=NUMBER_FORMAT:
                            errors.append(f'{name}: four-decimal format missing at
row {index+2}, column {j+1}')
                        if not math.isfinite(float(cell.value)) or
abs(float(cell.value)*1e4-round(float(cell.value)*1e4))>1e-5:
                            errors.append(f'{name}: non-finite or non-four-decimal
stored number')
                maxRoundingError=max(maxRoundingError,abs(float(cell.value)-value))
            else:
                blanks+=1
                checkedCells+=len(cells);count+=1
                if first is None:first=cells[0].value
                last=cells[-1].value
                if len(errors)>30:
                    raise RuntimeError('Export validation failed: '+';
'.join(errors[:30]))
                if next(rows,None) is not None:
                    errors.append(name+' : workbook has uncontracted extra rows')
                if count!=spec['data_rows'] or count!=len(times):
                    errors.append(name+' : row count differs from time grid')
                if blanks!=spec['outside_domain_blank_cells']:
                    errors.append(name+' : blank count mismatch')

checkedSheets[name]={'data_rows':count,'total_rows':count+1,'first_time_s':first,
'last_time_s':last,'outside_domain_blank_cells':blanks,'value_comparison_max_abs_error':max
RoundingError}
    finally:
        wb.close()
    run=runs.get(qid)
    live={'status':'NOT_REQUESTED','samples':0,'max_abs_unrounded_difference':None}
    if run is not None:
        current=runProvenance(run)
        if current!=manifest['provenance']:
            errors.append('Live Run provenance differs from the exported Run')
        samples=manifest['samples'];ts=np.array([s['source_time_s'] for s in samples])
        T,C,R,Cs,inside=fieldBlock(run,ts,qid)
        difference=0.
        for i,sample in enumerate(samples):
            actual=rawRow(qid,ts[i],T[i],C[i],R[i],None if Cs is None else
Cs[i],inside[i])
            for a,b in zip(actual,sample['raw_values']):
                if a is None or b is None:
                    if not (a is None and b is None):errors.append('Live Run sample
domain mismatch')
            else:

```

```

        difference=max(difference,abs(a-b))
    if difference>1e-10:
        errors.append('Independent live Run sample mismatch exceeds 1e-10')
    live={'status':'PASS' if difference<=1e-10 else 'FAIL','samples':len(samples),
        'max_abs_unrounded_difference':difference}
    paperValidation,paperErrors=validatePaperTables(manifest,run)
    errors.extend(paperErrors)
    report={'schema_version':'1.0','question_id':qid,'generated_by':'exportOutputs.py',
        'generated_at':datetime.now(timezone.utc).isoformat(),
        'status':'FAIL' if errors else 'PASS','fully_verified_with_live_Run':not
errors and run is not None,

'checked_cells':checkedCells,'sheets':checkedSheets,'live_Run_sample_validation':live,
    'paper_CSV_live_Run_validation':paperValidation,
    'errors':errors,'elapsed_s':time.perf_counter()-started,

'workbook':manifest['workbook'],'export_manifest':exportFileRecord(manifestPath),
    'scope':'Full workbook/raw-archive validation; independent workbook live
Run samples and every paper CSV time/position/value/surface/radius cell when live Run is
provided.',
        'four_decimals_are_storage_and_display_not_physical_accuracy_claim':True,

'visual_render':'pending','visual_studio_gui':'pending','human_review':'pending'}

reportPath=manifestPath.with_name(manifestPath.name.replace('.export.json','.validation.js
n'))

writeJson(reportPath,report);report['report_path']=str(reportPath);results.append(report)
    if errors:
        raise RuntimeError('Export validation failed: '+'; '.join(errors[:30]))
    return {'status':'PASS' if all(r['status']=='PASS' for r in results) else 'FAIL',
        'fully_verified_with_live_Run':all(r['fully_verified_with_live_Run'] for r in
results),'exports':results}

def paperCsvSelfcheck(directory):

    from dryingCore import Settings,solveCase
    import copy
    directory=Path(directory).resolve()
    directory.mkdir(parents=True,exist_ok=True)
    summaries=[];negativeChecks=[]
    for qid in ['Q1','Q2','Q3','Q4']:
        settings={'question':'Q23' if qid in ['Q2','Q3'] else qid,
            'intervals':40,'faceScheme':'kirchhoff','shrink':qid=='Q4',
            'rtol':1e-8,'atolTemperature':1e-8,'atolMoisture':1e-10,
            'earlyMaxStepS':5.}
        if qid=='Q2':
            settings.update(constantD=2e-8,beta=8e-6,horizonH=24.)
            print('PAPER_CSV_SELFHECK '+qid+' : actual N40 solve',flush=True)
            run=solveCase(Settings(**settings))
            try:
                if qid=='Q2' and (run.eventS is None or run.endS<10800.):
                    raise RuntimeError('Accelerated export test must still cover 3h and a real
drying event')
                exported=exportQuestion(run,qid,directory/qid)
                checked=validateExports([exported],runs={qid:run})

manifest=json.loads(Path(exported['manifest_path']).read_text(encoding='utf-8'))
        defects={'Q1':['temperature_value'],'Q2':['moisture_value'],
            'Q3':['last_time_label'],'Q4':
['outside_nonblank','surface_missing','radius_value']}[qid]
        for defect in defects:

```

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        changed=copy.deepcopy(manifest)
        kind='temperature' if defect=='temperature_value' else 'radius' if
defect=='radius_value' else 'moisture'
        selected=next(r for r in changed['paper_tables'] if
Path(r['path']).name==f'{qid.lower()}_paper_{kind}.csv')
        originalPath=ROOT/selected['path']
        with originalPath.open('r',encoding='utf-8-sig',newline='') as
fh:rows=list(csv.reader(fh))
        if defect in ['temperature_value','moisture_value']:
            rows[1][1]=format(float(rows[1][1])+.01, '.4f')
        elif defect=='last_time_label':
            rows[-1][0]=format(float(rows[-1][0])-.001, '.4f')
        elif defect=='outside_nonblank':
            ri,ci=next((i,j) for i in range(1,len(rows)) for j in range(1,6) if
rows[i][j]=='')
            rows[ri][ci]='0.0000'
        elif defect=='surface_missing':rows[-1][-1]=''
        elif defect=='radius_value':rows[-1][1]=format(float(rows[-1][1])
+.01, '.4f')
        target=directory/'deliberate_defects'/defect/originalPath.name
        target.parent.mkdir(parents=True,exist_ok=True)
        with target.open('w',encoding='utf-8-sig',newline='') as
fh:csv.writer(fh).writerows(rows)
        selected.update(exportFileRecord(target,selected.get('role')))
        result,errors=validatePaperTables(changed,run)
        if not errors or result['status']!='FAIL':
            raise AssertionError('Independent paper validator missed deliberate
defect: '+defect)

negativeChecks.append({'question_id':qid,'defect':defect,'correctly_rejected':True,
'errors':errors,'artifact':exportFileRecord(target)})

summaries.append({'question_id':qid,'settings':asdict(run.model.settings),'export':exported
,
                    'validation':checked,'event_s':run.eventS,'end_s':run.endS,
                    'Q2_is_accelerated_constant_D_export_test_only':qid=='Q2'})
    print(json.dumps({'question_id':qid,'status':'PASS','end_s':run.endS,
        'paper_cells':checked['exports'][0]['paper_CSV_live_Run_validation']
['checked_cells']},ensure_ascii=False),flush=True)
    finally:
        run.close()
    report={'status':'PASS','generated_at':datetime.now(timezone.utc).isoformat(),
        'scope':'N40 exporter fidelity only. Q2 uses constant_D=2e-8 and beta=8e-6
solely to shorten the real full-event export test; these are not formal physical results.
Q1/Q3/Q4 use their nominal physics.',
        'source':exportFileRecord(SOURCE),'exports':summaries,'negative_checks':negativeChecks,
        'negative_test_scope':'Copied paper CSVs only; six deliberate defects must be
rejected even after their recorded hashes are updated.',
        'visual_studio_gui':'pending','human_review':'pending'}
    writeJson(directory/'paper_csv_selfcheck_report.json',report)

print(json.dumps({'status':'PASS','questions':4,'deliberate_defects_rejected':len(negativeC
hecks),
'report':str(directory/'paper_csv_selfcheck_report.json')},ensure_ascii=False),flush=True)
    return 0

def main():
    parser=argparse.ArgumentParser(description=__doc__)
    parser.add_argument('--selfcheck',action='store_true')

```

```

    parser.add_argument('--paper-selfcheck',action='store_true')
    parser.add_argument('--output-dir',default='results/exportSelfcheck')
    args=parser.parse_args()
    if not args.selfcheck and not args.paper_selfcheck:
        parser.error('Use the Python API with final Runs, or --selfcheck for a real Q1
validation export')
    if Path.cwd().resolve()!=ROOT:
        raise RuntimeError('Use the extracted code package as cwd')
    if args.paper_selfcheck:
        return paperCsvSelfcheck(ROOT/args.output_dir)
    from dryingCore import Settings,solveCase

    run=solveCase(Settings(question='Q1',intervals=40,faceScheme='kirchhoff',
rtol=1e-8,atolTemperature=1e-8,atolMoisture=1e-10,earlyMaxStepS=5.))
    record=exportQuestion(run,'Q1',ROOT/args.output_dir)
    checks=validateExports([record],runs={'Q1':run})
    output=Path(args.output_dir).resolve()
    note={'status':checks['status'],'scope':'Actual Q1 export selfcheck only, not final
production or numerical-accuracy acceptance',
        'settings':asdict(run.model.settings),'export':record,'validation':checks}
    writeJson(output/'selfcheck_summary.json',note)

    print(json.dumps({'status':checks['status'],'fully_verified_with_live_Run':checks['fully_ve
rified_with_live_Run'],
    'output':str(output),'size':record['size'],'sheets':record['sheets']},ensure_ascii=False,in
dent=2))
    return 0

if __name__=='__main__':
    raise SystemExit(main())

```

D.10 reference_data.py

路径: 03_程序代码/reference/reference_data.py

用途: final_v6a 冻结数值参照的 Python 纯数据常量

使用范围: 正式采用: 生产求解、导出或数值参照

语言: Python; 源文件共 216 行。

SHA256: 9ab09a806872ef171824920a661ac3fdd422f356834a7d4563beba02d4f9a94e

```
REFERENCE_DATA = {'sourceVersion': 'final_v6a',
  'purpose': 'Frozen numerical reference, not a new solve or measured physical data',
  'questions': {'Q1': {'diagnostics': {'event_s': None,
    'event_h': None,
    'end_s': 1800.0,
    'elapsed_s': 5.367319200002385,
    'end_time_convention': 'ceil(critical_event_s)+1: '
      'conservative post-crossing '
      'verification second; not
claimed '
      'earliest integer second',
    'max_mass_balance_abs_kg_per_kg':
5.773159728050814e-15,
    'min_C': 1.510234685362519,
    'max_C': 2.5500000000000001,
    'min_T_K': 301.15,
    'max_T_K': 309.9355602874471,
    'min_D': 3.882960068132186e-09,
    'max_D': 4.937655094173939e-09,
    'positive_properties': True,
    'max_radial_C_increase': 3.552713678800501e-15,
    'final_max_C': 2.549992403819304,
    'final_surface_C': 1.5102346853625193,
    'strictly_dry_at_end': False,
    'radius_end_m': 0.02,
    'radius_extrapolation_used': False,
    'rhs_evaluations': 3914,
    'accepted_time_points': 1743,
    'nfev': 3914,
    'njev': 68,
    'nlu': 431,
    'solver_success': True,
    'dense_storage': 'disk',
    'dense_coefficient_bytes': 463474752,
    'dense_polynomial_count': 1742},
'reproduction_samples': {'times_s': [0.0, 1800.0],
  'material_x': [0.0, 0.25, 0.5, 0.75, 1.0],
  'T_K': [[301.15, 301.15, 301.15, 301.15,
301.15],
          [306.7253448172471,
306.92200190803965,
307.51416078664806,
308.51212454467895,
309.9355602874471]],
  'C': [[2.55, 2.55, 2.55, 2.55, 2.55],
[2.549992403819304,
2.549723930554662,
2.5383005254029065,
2.3754698868518354,
1.5102346853625193]]}},
```



```

323.0053407098511,
323.02461224281694,
323.0601477811811,
323.1164123947858],
[323.15,
323.15,
323.15,
323.15,
323.14999999999999],
[323.15, 323.15, 323.15, 323.15,
323.15]],

'C': [[2.55, 2.55, 2.55, 2.55, 2.55],
[2.549934098712396,
2.5488838430623373,
2.5255580144516143,
2.3257039999977334,
1.648617763296951],
[1.7661771002440243,
1.716533013166519,
1.5701979936396047,
1.3332634188779369,
1.0081231118401026],
[0.14999989718225315,
0.14768651296460814,
0.1402086276135209,
0.12491406907205477,
0.0526594803435617],
[0.14999950103529133,
0.14768612746407359,
0.14020827567351044,
0.12491378167987972,
0.05265944808510597]]}],

'Q4': {'diagnostics': {'event_s': 183926.06923258994,
'event_h': 51.09057478683054,
'end_s': 183928.0,
'elapsed_s': 164.5545338999982,
'end_time_convention': 'ceil(critical_event_s)+1: '
'conservative post-crossing '
'verification second; not
claimed '
'earliest integer second',
'max_mass_balance_abs_kg_per_kg':
7.993605777301127e-15,

'min_C': 0.05261996970729463,
'max_C': 2.5500000000000025,
'min_T_K': 301.15,
'max_T_K': 323.2000976421016,
'min_D': 9.398366254447737e-12,
'max_D': 2.477122820067253e-09,
'positive_properties': True,
'max_radial_C_increased': 4.440892098500626e-15,
'final_max_C': 0.14999900856092843,
'final_surface_C': 0.052619969707294635,
'strictly_dry_at_end': True,
'radius_end_m': 0.012,
'radius_extrapolation_used': False,
'rhs_evaluations': 36934,
'accepted_time_points': 15996,
'nfev': 36934,
'njev': 737,
'nlu': 4099,
'solver_success': True,
'dense_storage': 'disk',

```

```

        'dense_coefficient_bytes': 8327788168,
        'dense_polynomial_count': 15993},
'completion': {'critical_event_s': 183926.06923258994,
               'critical_event_h': 51.09057478683054,
               'reported_drying_time_h': 51.0906,
               'reported_time_s': 183926.16,
               'max_C_at_reported_time': 0.14999995339076624,
               'slowest_material_coordinate': 0.0,
               'conservative_post_verification_s': 183928.0,
               'rounding_convention': 'upward on 0.0001 h grid,

followed by '
                                   'actual strict threshold
check',
                                   'interpretation': 'A conditional numerical event, not
a '
                                   'confidence bound on physical drying
time.'},
'reproduction_samples': {'times_s': [0.0,
                                     1800.0,
                                     10800.0,
                                     183926.16,
                                     183928.0],
                          'material_x': [0.0, 0.25, 0.5, 0.75, 1.0],
                          'T_K': [[301.15, 301.15, 301.15, 301.15,
                                     304.18781907151936,
                                     304.46728468564174,
                                     305.3224885001255,
                                     306.80252608201823,
                                     309.05980848026536],
                                    [322.97124085294246,
                                     322.9784486845918,
                                     323.0028781283762,
                                     323.0496413187221,
                                     323.12744861066466],
                                    [323.15, 323.15, 323.15, 323.15,
                                     323.15],
                                    [323.15, 323.15, 323.15, 323.15,
                                     323.15]],

301.15],

323.15],

323.15]],

'C': [[2.55, 2.55, 2.55, 2.55, 2.55],
      [2.55,
        2.5499999999995815,
        2.549998436080619,
        2.5309421443741127,
        1.1313425016195608],
      [2.4547252277379927,
        2.3815382147789257,
        2.112246351658363,
        1.5585872624122925,
        0.6843727284423174],
      [0.14999995339076624,
        0.14692749737405153,
        0.13718379398934455,
        0.11813224940015873,
        0.05262002809955053],
      [0.14999900856092843,
        0.14692658503537287,
        0.13718298178958888,
        0.11813162008386399,
        0.052619969707294635]]]]]]}

```

D.11 runLogged.ps1

路径: 03_程序代码/runLogged.ps1

用途: 独立入口的 Windows 辅助监督与真实退出码留存

使用范围: 复现辅助入口

语言: PowerShell; 源文件共 40 行。

SHA256: ad9888888a757e07ec9be147ffecaac7a4e58347093f664d7894abd4323c4f17

```
param(
    [string]$Python = 'python',
    [ValidateSet('quick', 'final')][string]$Profile = 'quick',
    [string]$RunId = ''
)
$ErrorActionPreference = 'Stop'
if ([string]::IsNullOrEmpty($RunId)) {
    $RunId = $Profile + '_' + [DateTime]::UtcNow.ToString('yyyyMMddTHHmssZ')
}
if ($RunId -notmatch '^[A-Za-z0-9][A-Za-z0-9_-]{0,79}$') {
    throw 'Invalid run identifier.'
}
$packageDirectory = $PSScriptRoot
$entryPath = Join-Path $packageDirectory 'runDelivery.py'
$resultDirectory = Join-Path (Join-Path $packageDirectory 'results') $RunId
if (Test-Path -LiteralPath $resultDirectory) {
    throw 'The run directory already exists; use a new run identifier.'
}
$startedUtc = [DateTime]::UtcNow.ToString('o')
$timer = [Diagnostics.Stopwatch]::StartNew()
$entryHash = (Get-FileHash -LiteralPath $entryPath -Algorithm
SHA256).Hash.ToLowerInvariant()
& $Python -X utf8 -B $entryPath --profile $Profile --run-id $RunId
$processExit = $LASTEXITCODE
$timer.Stop()
if (Test-Path -LiteralPath $resultDirectory) {
    $record = [ordered]@{
        runId = $RunId
        profile = $Profile
        startedAtUtc = $startedUtc
        finishedAtUtc = [DateTime]::UtcNow.ToString('o')
        elapsedSeconds = $timer.Elapsed.TotalSeconds
        actualExitCode = $processExit
        entrySha256AtStart = $entryHash
        entrySha256AtEnd = (Get-FileHash -LiteralPath $entryPath -Algorithm
SHA256).Hash.ToLowerInvariant()
        visualStudioGui = 'NOT_OBSERVED_BY_THIS_SCRIPT'
        humanReview = 'PENDING_USER_REVIEW'
    }
    $record | ConvertTo-Json -Depth 5 | Set-Content -LiteralPath (Join-Path
$resultDirectory 'processExit.json') -Encoding UTF8
}
exit $processExit
```

D.12 compareMatlab.mjs

路径: 05_数值检验与实验/MATLAB 对照证据/compareMatlab.mjs

用途: 原 MATLAB 与 Python 共同时间/材料坐标样点、事件的对照程序; Node.js 自带库

使用范围: 保留用于追溯的未通过、受限或未执行检验

语言: JavaScript; 源文件共 199 行。

SHA256: 967fd3cb232b7e0f775728035ea51783fd1f5cd2e61e8533354e939151491d59

```
import fs from 'node:fs';
import crypto from 'node:crypto';
import path from 'node:path';
import { fileURLToPath } from 'node:url';
import { spawnSync } from 'node:child_process';

const scriptPath = fileURLToPath(import.meta.url);
const projectRoot = path.resolve(path.dirname(scriptPath), '../..');
const matlabRoot = path.join(projectRoot, 'paper_output/qa/matlab_crosscheck_20260911');
const pythonRoot = path.join(projectRoot, 'paper_output/results/code_delivery/camel_audit_v1');
const outputJson = path.join(matlabRoot, 'comparison-python.json');
const outputMarkdown = path.join(matlabRoot, 'comparison-python.md');

const thresholds = Object.freeze({ temperatureK: 1e-4, moistureDryBasis: 1e-6, eventSec: 0.01 });
const scriptRecord = fileRecord(scriptPath);

function now() { return new Date().toISOString(); }
function digest(bytes) { return crypto.createHash('sha256').update(bytes).digest('hex'); }
function fileRecord(filePath) {
  const bytes = fs.readFileSync(filePath);
  return { path: path.relative(projectRoot, filePath).split(path.sep).join('/'),
    bytes: bytes.length, sha256: digest(bytes) };
}
function readJson(filePath) { return JSON.parse(fs.readFileSync(filePath, 'utf8').replace(/\uFFFF/, '')); }
function writeJson(filePath, value) { fs.writeFileSync(filePath, JSON.stringify(value, null, 2) + '\n', 'utf8'); }
function assert(condition, message) { if (!condition) throw new Error(message); }
function finite(value, label) { assert(typeof value === 'number' && Number.isFinite(value), `Nonfinite ${label}`); return value; }

function compareWorker() {
  const startedAtUtc = now();
  const startTime = process.hrtime.bigint();
  const inputPaths = [path.join(matlabRoot, 'runSummary.json'), path.join(matlabRoot, 'runLog.txt'),
    path.join(pythonRoot, 'launch.json'), path.join(pythonRoot, 'runResult.json'),
    path.join(pythonRoot, 'artifactManifest.json')];
  for (const question of ['Q1', 'Q23', 'Q4']) {
    inputPaths.push(path.join(matlabRoot, `${question}_crossCheck.json`),
      path.join(pythonRoot, question, 'summary.json'), path.join(pythonRoot, question, 'sampled_solution.npz'));
  }
  const matlabRun = readJson(inputPaths[0]);
  const pythonLaunch = readJson(path.join(pythonRoot, 'launch.json'));
  const pythonRun = readJson(path.join(pythonRoot, 'runResult.json'));
  const noMatlabException = matlabRun.exception == null ||
(Array.isArray(matlabRun.exception) && matlabRun.exception.length === 0);
```

```

    assert(matlabRun.status === 'COMPLETED_NUMERICAL_CHECKS' && noMatlabException, 'MATLAB
run is incomplete');
    assert(pythonRun.status === 'PASS', 'Python baseline is incomplete');
    const provenanceChecks = [];

    function verifyDeclared(record, label) {
        const resolved = path.isAbsolute(record.path) ? record.path : path.join(projectRoot,
record.path);
        const actual = fileRecord(resolved);
        assert(actual.sha256 === record.sha256, `Declared hash differs: ${label}`);
        inputPaths.push(resolved);
        provenanceChecks.push({ label, actual, matchedRecordedHash: true });
    }
    verifyDeclared({ path: matlabRun.sourcePath, sha256: matlabRun.sourceSha256 }, 'MATLAB
source');
    for (const record of matlabRun.inputs) verifyDeclared(record, 'MATLAB observed input');
    for (const question of ['Q1', 'Q23', 'Q4']) {
        const baseline = readJson(path.join(pythonRoot, question, 'summary.json'));
        for (const key of ['code', 'jacobian_code', 'dense_storage_code'])
            verifyDeclared(baseline[key], `${question}/${key}`);
        for (const record of baseline.inputs) verifyDeclared(record, `${question}/observed
input`);
    }
    const uniqueInputPaths = [...new Set(inputPaths.map(item => path.resolve(item)))]
    const inputRecords = uniqueInputPaths.map(fileRecord);
    const questions = [];
    const failures = [];
    for (const question of ['Q1', 'Q23', 'Q4']) {
        const matlab = readJson(path.join(matlabRoot, `${question}_crossCheck.json`));
        const python = readJson(path.join(pythonRoot, question, 'summary.json'));
        const sample = python.crossLanguageSamples;
        assert(matlab.question === question && python.settings.question === question, `${
question}: question mismatch`);
        assert(matlab.intervalCount === 40 && python.settings.intervals === 40, `${question}:
require N40`);
        const settingsPairs = [
            ['rtol', 'relativeTolerance'], ['atolTemperature', 'temperatureAbsoluteTolerance'],
            ['atolMoisture', 'moistureAbsoluteTolerance'], ['earlyMaxStepS', 'earlyMaxStepSec'],
            ['maxStepS', 'lateMaxStepSec'], ['h', 'heatTransferCoefficient'], ['beta',
'massTransferCoefficient']];
        for (const [pythonKey, matlabKey] of settingsPairs) {
            assert(python.settings[pythonKey] === matlabRun.settings[matlabKey], `${question}: $
{pythonKey} differs`);
        }
        assert(python.settings.faceScheme.toLowerCase() ===
matlabRun.settings.waterFaceScheme.toLowerCase(), `${question}: face scheme differs`);
        assert(matlab.sampleTimesSec.length === matlab.sampleTypes.length, `${question}: sample
types malformed`);
        const times = question === 'Q1' ? [0, 1800] : [0, 1800, 10800];
        if (question !== 'Q1') {
            assert(Number.isInteger(matlab.endSec) && matlab.endSec === python.diagnostics.end_s,
`${question}: no common post-event integer second`);
            const postIndex = matlab.sampleTimesSec.indexOf(matlab.endSec);
            assert(postIndex >= 0 && matlab.sampleTypes[postIndex] === 'postEventIntegerSecond',
`${question}: endpoint is not postEventIntegerSecond`);
            times.push(matlab.endSec);
        }
        const coordinates = [0, 0.25, 0.5, 0.75, 1];
        const pairs = [];
        const maxima = { temperatureK: null, moistureDryBasis: null };
        for (const timeSec of times) {
            const matlabTimeIndex = matlab.sampleTimesSec.indexOf(timeSec);
            const pythonTimeIndex = sample.timesS.indexOf(timeSec);

```

```

    assert(matlabTimeIndex >= 0 && pythonTimeIndex >= 0, `${question}: missing exact
common time ${timeSec}`);
    for (const materialX of coordinates) {
        const matlabXIndex = matlab.sampleMaterialX.indexOf(materialX);
        const pythonXIndex = sample.materialX.indexOf(materialX);
        assert(matlabXIndex >= 0 && pythonXIndex >= 0, `${question}: missing exact common
coordinate ${materialX}`);
        const item = { timeSec, materialX, matlabTimeIndexZeroBased: matlabTimeIndex,
pythonTimeIndexZeroBased: pythonTimeIndex,
        matlabXIndexZeroBased: matlabXIndex, pythonXIndexZeroBased: pythonXIndex };
        for (const field of ['temperatureK', 'moistureDryBasis']) {
            const matlabValue = finite(matlab[field][matlabTimeIndex][matlabXIndex], `${
question}/${field}/MATLAB`);
            const pythonValue = finite(sample[field][pythonTimeIndex][pythonXIndex], `${
question}/${field}/Python`);
            const signedDifference = matlabValue - pythonValue;
            const absoluteDifference = Math.abs(signedDifference);
            const passed = absoluteDifference <= thresholds[field];
            item[field] = { matlabValue, pythonValue, signedDifferenceMatlabMinusPython:
signedDifference, absoluteDifference, passed };
            if (maxima[field] === null || absoluteDifference >
maxima[field].absoluteDifference) {
                maxima[field] = { timeSec, materialX, ...item[field] };
            }
            if (!passed) failures.push(`${question}/${field}/t=${timeSec}/x=${materialX}`);
        }
        pairs.push(item);
    }
}
let event = { applicable: false, reason: 'Q1 stops at 1800 s and has no drying
event' };
if (question !== 'Q1') {
    const matlabEventSec = finite(matlab.eventSec, `${question}/MATLAB event`);
    const pythonEventSec = finite(python.diagnostics.event_s, `${question}/Python
event`);
    const signedDifferenceSec = matlabEventSec - pythonEventSec;
    const absoluteDifferenceSec = Math.abs(signedDifferenceSec);
    const passed = absoluteDifferenceSec <= thresholds.eventSec;
    event = { applicable: true, matlabEventSec, pythonEventSec, signedDifferenceSec,
absoluteDifferenceSec, passed };
    if (!passed) failures.push(`${question}/event`);
}
questions.push({ question, status: failures.some(item => item.startsWith(question +
'/')) ? 'FAIL' : 'PASS',
    intervals: 40, commonTimesSec: times, commonMaterialX: coordinates,
matchedCoordinatePairs: pairs.length,
    maxima, event, comparisons: pairs,
    excludedPythonTimesSec: sample.timesS.filter(value => !times.includes(value)),
    excludedMatlabTimesSec: matlab.sampleTimesSec.filter(value => !
times.includes(value)),
    exclusionReason: 'Python strict reported time and MATLAB critical event are different
times; their field rows are not paired.',
    matlabWarnings: matlab.warnings, matlabWarningCapture: matlab.warningCapture,
pythonWarningCount: python.warningCount });
}
assert(JSON.stringify(inputRecords) === JSON.stringify(uniqueInputPaths.map(fileRecord)),
'Inputs changed during comparison');
assert(scriptRecord.sha256 === fileRecord(scriptPath).sha256, 'Comparison script changed
while running');
return { schemaVersion: 1, status: failures.length ? 'FAIL' : 'PASS', failures,
    startedAtUtc, finishedAtUtc: now(), elapsedSeconds: Number(process.hrtime.bigint() -
startTime) / 1e9,
    thresholds, thresholdPolicy: 'Fixed before reading results; not changed after observing

```

```

differences',
  matchingPolicy: 'Exact common time values and exact common material coordinates; no
row-index pairing or interpolation',
  differenceConvention: 'MATLAB minus Python', script: scriptRecord, inputRecords,
provenanceChecks,
  runtime: { nodeVersion: process.version, executable: process.execPath, platform:
process.platform, architecture: process.arch,
  matlabVersion: matlabRun.matlabVersion, matlabSolver: matlabRun.settings.solver,
  matlabJacobian: matlabRun.settings.jacobianMode, pythonVersion: pythonLaunch.python,
  pythonLibraries: pythonLaunch.libraries, pythonSolver: 'SciPy BDF', pythonJacobian:
'analytic sparse' },
  sourceRunTimes: { matlabStartedAtUtc: matlabRun.startedAtUtc, matlabFinishedAtUtc:
matlabRun.finishedAtUtc,
  pythonStartedAtUtc: pythonRun.startedAtUtc, pythonFinishedAtUtc:
pythonRun.finishedAtUtc },
  comparedFieldScalars: questions.reduce((total, item) => total + 2 *
item.matchedCoordinatePairs, 0),
  comparedEvents: 2, questions,
  scope: 'Independent N40 implementation agreement at recorded common times and five
material coordinates, plus continuous event times',
  limitations: ['NPZ files are hashed for provenance; numerical comparisons read
crossLanguageSamples in Python summary.json.',
  'No full-time/full-space error bound, physical calibration accuracy, production-grid
acceptance, or confidence interval is established.',
  'MATLAB GUI execution evidence is recorded separately by the main agent; this
comparison does not infer GUI completion.',
  'Human code review remains pending.'], guiEvidence: 'separate external record',
humanReview: 'pending' };
}

function markdown(report) {
  if (!report.questions?.length) {
    return `# MATLAB/Python N40 独立实现比较\n\n状态: **FAIL**. 监督进程实际读取的退出码: $
{report.processSupervision.workerActualExitCode}.\n\n比较器未生成可用数值报告:
\n\n\`\`\`text\n${report.processSupervision.workerStderr}\n\n\`\`\`\n\n没有放宽阈值或填充成功状
态.\n`;
  }
  const number = value => value === null || value === undefined ? '不适用' :
value.toExponential(12);
  const rows = report.questions.map(item => `| ${item.question} | $
{item.commonTimesSec.join(',')} | ${item.matchedCoordinatePairs} | $
{number(item.maxima.temperatureK.absoluteDifference)} | $
{number(item.maxima.moistureDryBasis.absoluteDifference)} | $
{number(item.event.absoluteDifferenceSec)} | ${item.status} |`);
  return `# MATLAB/Python N40 独立实现比较\n\n状态: **${report.status}**. 比较进程实际退出码:
**${report.processSupervision.workerActualExitCode}**, 由父 Node 监督进程读取.\n\n固定接纳阈
值为温度 ${thresholds.temperatureK} K、干基含水率 ${thresholds.moistureDryBasis} kg/kg、连续事
件时间 ${thresholds.eventSec} s; 未根据观察结果调整。全部比较按精确相同时间及材料坐标 x 配对, x 为
0、0.25、0.5、0.75、1.\n\n| 问题轨迹 | 共同时间 / s | 时间坐标配对数 | 最大温差 / K | 最大干基含
水率差 / kg/kg | 连续事件绝对差 / s | 结论 |\n|---|---|---|---|---|---|---|\n$
{rows.join('\n')}\n\n共比较 ${report.comparedFieldScalars} 个场标量和 $
{report.comparedEvents} 个连续事件。Q23 同时服务 Q2 与 Q3, 不另造第二份轨迹。Q1 没有完整干燥事
件.\n\nPython 的额外一行是严格四位报告时刻, MATLAB 的额外一行是其连续事件时刻, 两者不同; 这些场值
未按行相减。连续事件只单独比较秒。最大值所在的时间、坐标、原始数值和有符号差均保存在 comparison-
python.json.\n\n开始: ${report.startedAtUtc}; 结束: ${report.finishedAtUtc}; 比较耗时: $
{report.elapsedSeconds.toFixed(6)} s。Node ${report.runtime.nodeVersion}; MATLAB $
{report.runtime.matlabVersion}, ode15s 默认 NDF 与稀疏数值差分; Python 使用 SciPy BDF 和解析稀
疏 Jacobian.\n\n输入文件、声明的源码与观测数据哈希已经实际回读匹配, 并在比较结束时再次检查未改变。
Python NPZ 只作来源哈希记录, 比较数值取自 summary.json 的 crossLanguageSamples。比较脚本
SHA-256: \`${report.script.sha256}\`. 监督过程和两个输出文件的 SHA-256 见 comparison-python-
manifest.json, 避免报告文件自哈希的循环依赖.\n\n这仅是 N40 在共同记录时间和五个材料坐标的跨实现一
致性检查, 不能推广为连续全域误差上界、正式网格精度、物理预测精度或置信区间。MATLAB GUI 操作证据由主
代理另存, 本脚本不补写 GUI 成功; 用户人工代码审查仍待完成.\n`;
}

```

```

}

if (process.argv.includes('--worker')) {
  try {
    const report = compareWorker();
    process.stdout.write(JSON.stringify(report));
    process.exitCode = report.status === 'PASS' ? 0 : 1;
  } catch (error) {
    process.stderr.write(String(error.stack ?? error) + '\n');
    process.exitCode = 2;
  }
} else {
  const supervisorStartedAtUtc = now();
  const worker = spawnSync(process.execPath, [scriptPath, '--worker'], { cwd: projectRoot,
encoding: 'utf8', maxBuffer: 8 * 1024 * 1024 });
  const supervision = { supervisorStartedAtUtc, supervisorFinishedAtUtc: now(),
    executable: process.execPath, commandArguments: [scriptPath, '--worker'], cwd:
projectRoot,
    workerActualExitCode: worker.status, workerSignal: worker.signal, workerSpawnError:
worker.error ? String(worker.error) : null,
    workerStderr: worker.stderr, source: 'Node parent process spawnSync return status; not
a child prefilled success flag' };
  fs.appendFileSync(path.join(matlabRoot, 'comparison-python-process.jsonl'),
JSON.stringify({
    script: scriptRecord, ...supervision, stdoutSha256: digest(Buffer.from(worker.stdout ??
'', 'utf8')) }) + '\n', 'utf8'));
  const report = worker.stdout?.length ? JSON.parse(worker.stdout) : {
    schemaVersion: 1, status: 'FAIL', failures: ['Worker did not produce a numerical
report'],
    script: scriptRecord, thresholds, startedAtUtc: supervisorStartedAtUtc, finishedAtUtc:
now(),
    comparedFieldScalars: 0, comparedEvents: 0, questions: [], humanReview: 'pending' };
  report.processSupervision = supervision;
  if (worker.status !== 0) report.status = 'FAIL';
  writeJson(outputJson, report);
  fs.writeFileSync(outputMarkdown, markdown(report), 'utf8');
  writeJson(path.join(matlabRoot, 'comparison-python-manifest.json'), {
    status: report.status, createdAtUtc: now(), script: scriptRecord, processSupervision:
supervision,
    outputs: [fileRecord(outputJson), fileRecord(outputMarkdown)] });
  process.stdout.write(JSON.stringify({ status: report.status, workerActualExitCode:
worker.status,
    comparedFieldScalars: report.comparedFieldScalars, comparedEvents:
report.comparedEvents,
    maxima: report.questions.map(item => ({ question: item.question,
      temperatureK: item.maxima.temperatureK.absoluteDifference, moistureDryBasis:
item.maxima.moistureDryBasis.absoluteDifference,
      eventSec: item.event.absoluteDifferenceSec ?? null }))) }) + '\n');
  process.exitCode = worker.status ?? 2;
}

```

D.13 runCrossCheck.m

路径: 05_数值检验与实验/MATLAB 对照证据/runCrossCheck.m

用途: 实际使用的独立 MATLAB N40 热湿求解和共同样点导出

使用范围: 保留用于追溯的未通过、受限或未执行检验

语言: MATLAB; 源文件共 491 行。

SHA256: 0132b3bed4a6b43347deb0be102fd85bf077f6a116760f267037cd3e665874aa

```
function runSummary = runCrossCheck(outputDirectory)

    sourcePath = [mfilename('fullpath'), '.m'];
    projectRoot = fileparts(sourcePath);
    for parentIndex = 1:4
        projectRoot = fileparts(projectRoot);
    end
    if nargin < 1 || strlength(string(outputDirectory)) == 0
        runStamp = char(datetime('now', 'Format', 'yyyyMMdd_HHmmsSS'));
        outputDirectory = fullfile(projectRoot, 'paper_output', 'qa', ...
            ['matlab_crosscheck_', runStamp]);
    end
    outputDirectory = char(outputDirectory);

    outputFile = java.io.File(outputDirectory);
    projectRootFile = java.io.File(projectRoot);
    outputDirectory = char(outputFile.getCanonicalPath());
    canonicalRoot = char(projectRootFile.getCanonicalPath());
    if ~startsWith(lower(outputDirectory), [lower(canonicalRoot), filesep])
        error('CrossCheck:WorkspaceBoundary', '输出必须位于本次 2026CUMCM 子工作区。');
    end
    summaryPath = fullfile(outputDirectory, 'runSummary.json');
    if isfile(summaryPath)
        error('CrossCheck:ExistingRun', '该目录已有运行记录, 请使用新的输出目录以保留证据。');
    end
    if ~isfolder(outputDirectory)
        mkdir(outputDirectory);
    end
    logPath = fullfile(outputDirectory, 'runLog.txt');
    logFileId = fopen(logPath, 'w', 'n', 'UTF-8');
    if logFileId < 0
        error('CrossCheck:LogOpen', '无法创建运行日志。');
    end
    logCleanup = onCleanup(@() fclose(logFileId));
    runTimer = tic;
    runSummary = struct('schemaVersion', 1, 'status', 'RUNNING', ...
        'startedAtUtc', utcStamp(), 'finishedAtUtc', '', 'elapsedSec', [], ...
        'matlabVersion', version, 'matlabRelease', version('-release'), ...
        'computer', computer, 'sourcePath', sourcePath, ...
        'sourceSha256', sha256File(sourcePath), 'outputDirectory', outputDirectory, ...
        'guiReproduced', [], 'guiEvidenceStatus', 'external_observation_required', ...
        'humanReviewStatus', 'pending', 'caseResults', {}, 'exception', []);
    settings = struct('intervalCount', 40, 'nodeCount', 41, 'relativeTolerance', 1e-10, ...
        'temperatureAbsoluteTolerance', 1e-10, 'moistureAbsoluteTolerance', 1e-12, ...
        'earlyMaxStepSec', 2, 'lateMaxStepSec', 120, 'horizonSec', 240*3600, ...
        'initialTemperatureK', 301.15, 'initialMoisture', 2.55, ...
        'initialRadiusM', 0.02, 'fixedLengthM', 0.25, ...
        'heatTransferCoefficient', 25, 'massTransferCoefficient', 8e-7, ...
        'dryThreshold', 0.15, 'environmentSwitchSec', 14400, ...
```

```

        'tailTemperatureK', 323.15, 'tailEquilibriumMoisture', 0.05, ...
        'solver', 'ode15s_default_NDF', 'jacobianMode', 'JPattern_finite_difference', ...
        'waterFaceScheme', 'Kirchhoff', 'heatFaceScheme', 'harmonic', ...
        'coefficientFloor', 1e-12, 'latentHeatIncluded', false);
runSummary.settings = settings;
writeJson(summaryPath, runSummary);
logLine(logFileId, 'START %s | MATLAB %s | output=%s', ...
    runSummary.startedAtUtc, version, outputDirectory);
try
    environmentPath = fullfile(projectRoot, 'paper_output', 'data_cleaned',
'A_environment_observed.csv');
    radiusPath = fullfile(projectRoot, 'paper_output', 'data_cleaned',
'A_radius_observed.csv');
    inputData.environment = readtable(environmentPath, 'VariableNamingRule',
'preserve');
    inputData.radius = readtable(radiusPath, 'VariableNamingRule', 'preserve');
    validateInputs(inputData);
    runSummary.inputs = {fileRecord(environmentPath), fileRecord(radiusPath)};
    caseNames = {'Q1', 'Q23', 'Q4'};
    for caseIndex = 1:numel(caseNames)
        caseName = caseNames{caseIndex};
        logLine(logFileId, 'CASE_START %s at %s', caseName, utcStamp());
        caseTimer = tic;
        caseResult = solveSingleCase(caseName, inputData, settings, logFileId);
        caseResult.elapsedSec = toc(caseTimer);
        caseResult.finishedAtUtc = utcStamp();
        casePath = fullfile(outputDirectory, [caseName, '_crossCheck.json']);
        writeJson(casePath, caseResult);
        sampleTable = makeSampleTable(caseResult);
        writetable(sampleTable, fullfile(outputDirectory, [caseName, '_samples.csv']));
        if strcmp(caseName, 'Q23')

            sampleTable.question(:) = "Q2";
            writetable(sampleTable, fullfile(outputDirectory, 'Q2_samples.csv'));
            sampleTable.question(:) = "Q3";
            writetable(sampleTable, fullfile(outputDirectory, 'Q3_samples.csv'));
        end
        runSummary.caseResults{end+1} = caseResult;
        writeJson(summaryPath, runSummary);
        logLine(logFileId, 'CASE_END %s | elapsed=%.6f s | event=%.12g s |
massResidual=%.3g', ...
            caseName, caseResult.elapsedSec, scalarOrNaN(caseResult.eventSec), ...
            caseResult.diagnostics.maxMassBalanceAbs);
    end
    if ~strcmp(runSummary.sourceSha256, sha256File(sourcePath))
        error('CrossCheck:SourceChanged', '源码在运行中改变，本轮不能作为固定版本证据。');
    end
    for inputIndex = 1:numel(runSummary.inputs)
        record = runSummary.inputs{inputIndex};
        if ~strcmp(record.sha256, sha256File(record.path))
            error('CrossCheck:InputChanged', '输入在运行中改变: %s', record.path);
        end
    end
    runSummary.status = 'COMPLETED_NUMERICAL_CHECKS';
    runSummary.finishedAtUtc = utcStamp();
    runSummary.elapsedSec = toc(runTimer);
    writeJson(summaryPath, runSummary);
    logLine(logFileId, 'FINISH %s | elapsed=%.6f s | humanReview=pending', ...
        runSummary.finishedAtUtc, runSummary.elapsedSec);
catch runError
    runSummary.status = 'FAILED';
    runSummary.finishedAtUtc = utcStamp();
    runSummary.elapsedSec = toc(runTimer);

```

```

        runSummary.exception = struct('identifier', runError.identifier, ...
            'message', runError.message, 'report', getReport(runError, 'extended',
'hyperlinks', 'off'));
        writeJson(summaryPath, runSummary);
        logLine(logFileId, 'FAILED %s\n%s', runSummary.finishedAtUtc,
runSummary.exception.report);
        rethrow(runError);
    end
end

function caseResult = solveSingleCase(caseName, inputData, settings, logFileId)
    caseStartedAtUtc = utcStamp();
    nodeCount = settings.nodeCount;
    materialX = linspace(0, 1, nodeCount)';
    deltaX = 1/settings.intervalCount;
    faceX = [0; (materialX(1:end-1)+materialX(2:end))/2; 1];
    cellWeights = diff(faceX.^2)/2;
    internalFaceX = faceX(2:end-1);
    stateCount = 2*nodeCount+1;
    initialState = zeros(stateCount, 1);
    initialState(1:2:end-1) = settings.initialTemperatureK;
    initialState(2:2:end-1) = settings.initialMoisture;
    absoluteTolerance = repmat(settings.moistureAbsoluteTolerance, stateCount, 1);
    absoluteTolerance(1:2:end-1) = settings.temperatureAbsoluteTolerance;
    jacobianPattern = makeJacobianPattern(nodeCount);
    baseOptions = odeset('RelTol', settings.relativeTolerance, 'AbsTol', absoluteTolerance,
...
    'JPattern', jacobianPattern, 'Vectorized', 'off', 'Stats', 'off');
    if strcmp(caseName, 'Q1')
        horizonSec = 1800;
        dryDensityInitial = 820/(1+settings.initialMoisture);
    elseif strcmp(caseName, 'Q4')
        horizonSec = settings.horizonSec;
        dryDensityInitial = (760+90*settings.initialMoisture)/(1+settings.initialMoisture);
    else
        horizonSec = settings.horizonSec;
        dryDensityInitial =
(650+128*settings.initialMoisture)/(1+settings.initialMoisture);
    end
    dryMassKg = dryDensityInitial*pi*settings.initialRadiusM^2*settings.fixedLengthM;
    endPoints = unique([0, min(settings.environmentSwitchSec, horizonSec), horizonSec]);
    solutions = {};
    warningRecords = {};
    eventSec = [];
    rhsEvaluations = 0;
    for segmentIndex = 1:numel(endPoints)-1
        leftSec = endPoints(segmentIndex);
        rightSec = endPoints(segmentIndex+1);
        if leftSec < settings.environmentSwitchSec
            maxStepSec = settings.earlyMaxStepSec;
        else
            maxStepSec = settings.lateMaxStepSec;
        end
        segmentOptions = odeset(baseOptions, 'MaxStep', maxStepSec);
        if ~strcmp(caseName, 'Q1')
            segmentOptions = odeset(segmentOptions, 'Events', @dryEvent);
        end
        lastwarn('');
        segment = ode15s(@balanceRhs, [leftSec, rightSec], initialState, segmentOptions);
        [warningText, warningId] = lastwarn;
        recordWarning(warningText, warningId, leftSec, rightSec);
        solutions{end+1} = segment;
        hasEvent = isfield(segment, 'xe') && ~isempty(segment.xe);
    end
end

```

```

if hasEvent
    eventSec = segment.xe(end);
    initialState = segment.ye(:, end);
    tailEndSec = ceil(eventSec)+1;

    tailOptions = odeset(baseOptions, 'MaxStep', 1, 'Events', []);
    lastwarn('');
    tail = ode15s(@balanceRhs, [eventSec, tailEndSec], initialState, tailOptions);
    [warningText, warningId] = lastwarn;
    recordWarning(warningText, warningId, eventSec, tailEndSec);
    assertReached(tail, tailEndSec);
    solutions{end+1} = tail;
    break;
end
assertReached(segment, rightSec);
initialState = segment.y(:, end);
end
if ~strcmp(caseName, 'Q1') && isempty(eventSec)
    error('CrossCheck.NoDryEvent', '%s 在 240 小时内没有找到干燥事件。', caseName);
end
endSec = solutions{end}.x(end);
rawSampleTimes = [0, 1800, 10800];
rawSampleTimes = rawSampleTimes(rawSampleTimes <= endSec);
sampleTimesSec = unique([rawSampleTimes, eventSec, endSec]);
sampleStates = evaluatePieces(solutions, sampleTimesSec);
sampleX = [0, 0.25, 0.5, 0.75, 1];
temperatureK = interp1(materialX, sampleStates(1:2:end-1, :), sampleX, 'linear');
moisture = interp1(materialX, sampleStates(2:2:end-1, :), sampleX, 'linear');
sampleRadiusM = zeros(size(sampleTimesSec));
sampleTypes = strings(size(sampleTimesSec));
for sampleIndex = 1:numel(sampleTimesSec)
    sampleRadiusM(sampleIndex) = radiusAt(sampleTimesSec(sampleIndex));
    if sampleTimesSec(sampleIndex) == 0
        sampleTypes(sampleIndex) = "initial";
    elseif ~isempty(eventSec) && sampleTimesSec(sampleIndex) == eventSec
        sampleTypes(sampleIndex) = "criticalEvent";
    elseif ~isempty(eventSec) && sampleTimesSec(sampleIndex) == endSec
        sampleTypes(sampleIndex) = "postEventIntegerSecond";
    else
        sampleTypes(sampleIndex) = "fixedTime";
    end
end
end
diagnostic = struct('minMoisture', Inf, 'maxMoisture', -Inf, ...
    'minTemperatureK', Inf, 'maxTemperatureK', -Inf, 'minDiffusivity', Inf, ...
    'positiveProperties', true, 'maxMassBalanceAbs', 0, ...
    'maxRadialMoistureIncrease', -Inf, 'acceptedTimePoints', 0);
for pieceIndex = 1:numel(solutions)
    acceptedStates = solutions{pieceIndex}.y;
    acceptedTemperature = acceptedStates(1:2:end-1, :);
    acceptedMoisture = acceptedStates(2:2:end-1, :);
    [density, heatCapacity, conductivity, diffusivity] =
materialProperties(acceptedTemperature, acceptedMoisture, caseName);
    diagnostic.minMoisture = min(diagnostic.minMoisture, min(acceptedMoisture, [],
'all'));
    diagnostic.maxMoisture = max(diagnostic.maxMoisture, max(acceptedMoisture, [],
'all'));
    diagnostic.minTemperatureK = min(diagnostic.minTemperatureK,
min(acceptedTemperature, [], 'all'));
    diagnostic.maxTemperatureK = max(diagnostic.maxTemperatureK,
max(acceptedTemperature, [], 'all'));
    diagnostic.minDiffusivity = min(diagnostic.minDiffusivity, min(diffusivity, [],
'all'));
    diagnostic.positiveProperties = diagnostic.positiveProperties && ...

```

```

        all(density > 0 & heatCapacity > 0 & conductivity > 0 & diffusivity > 0,
'all');
        massResidual = 2*cellWeights'*acceptedMoisture+acceptedStates(end, :)-
settings.initialMoisture;
        diagnostic.maxMassBalanceAbs = max(diagnostic.maxMassBalanceAbs,
max(abs(massResidual)));
        diagnostic.maxRadialMoistureIncrease =
max(diagnostic.maxRadialMoistureIncrease, ...
        max(diff(acceptedMoisture, 1, 1), [], 'all'));
        diagnostic.acceptedTimePoints =
diagnostic.acceptedTimePoints+numel(solutions{pieceIndex}.x);
    end
    finalState = evaluatePieces(solutions, endSec);
    diagnostic.finalMaxMoisture = max(finalState(2:2:end-1));
    diagnostic.strictlyDryAtEnd = diagnostic.finalMaxMoisture < settings.dryThreshold;
    diagnostic.rhsEvaluations = rhsEvaluations;
    diagnostic.radiusExtrapolationUsed = strcmp(caseName, 'Q4') && endSec >
inputData.radius.time_s(end);
    diagnostic.checkScope = 'accepted ode15s states; not a continuous exact-solution error
bound';
    if diagnostic.minMoisture < -1e-8 || ~diagnostic.positiveProperties
        error('CrossCheck:PhysicalRange', '%s 的接受状态违反物性正值或水分范围。', caseName);
    end
    if diagnostic.maxMassBalanceAbs > 1e-6
        error('CrossCheck:MassBalance', '%s 的离散干基质量守恒残差超限。', caseName);
    end
    if ~strcmp(caseName, 'Q1') && ~diagnostic.strictlyDryAtEnd
        error('CrossCheck:StrictDryness', '%s 的事件后原精度 maxC 没有严格小于 0.15。',
caseName);
    end
    caseResult = struct('question', caseName, 'startedAtUtc', caseStartedAtUtc, ...
        'intervalCount', settings.intervalCount, 'eventSec', eventSec, ...
        'eventHours', eventSec/3600, 'endSec', endSec, 'dryMassKg', dryMassKg, ...
        'endTimeConvention', 'ceil(eventSec)+1; actual integration, not earliest integer-
second claim', ...
        'sampleTimesSec', sampleTimesSec, 'sampleTypes', sampleTypes, 'sampleMaterialX',
sampleX, ...
        'sampleRadiusM', sampleRadiusM, 'temperatureK', temperatureK, ...
        'temperatureC', temperatureK-273.15, 'moistureDryBasis', moisture, ...
        'fullMaterialX', materialX, 'fullStateBySample', sampleStates, ...
        'stateOrder', 'T0,C0,T1,C1,...,TN,CN,cumulativeDryBasisLoss', ...
        'meanMoisture', 2*cellWeights'*sampleStates(2:2:end-1, :), ...
        'cumulativeLoss', sampleStates(end, :), 'diagnostics', diagnostic, ...
        'warnings', {warningRecords}, 'warningCapture', 'last warning per segment; not all
warning messages', ...
        'humanReviewStatus', 'pending');

    function derivative = balanceRhs(timeSec, state)
        rhsEvaluations = rhsEvaluations+1;
        temperature = state(1:2:end-1);
        waterContent = state(2:2:end-1);
        [density, heatCapacity, conductivity, ~] = materialProperties(temperature,
waterContent, caseName);
        radiusM = radiusAt(timeSec);
        [airTemperatureK, equilibriumMoisture] = environmentAt(timeSec);
        heatFlux = zeros(nodeCount+1, 1);
        waterFlux = zeros(nodeCount+1, 1);
        faceConductivity = 2*conductivity(1:end-1).*conductivity(2:end) ./ ...
            max(conductivity(1:end-1)+conductivity(2:end), realmin);
        heatFlux(2:end-1) = internalFaceX.*faceConductivity.*diff(temperature)/deltaX;
        if strcmp(caseName, 'Q1')
            moistureExponent = 0.89;
            diffusionPrefactor = 7e-9;

```

```

        thermalFactor = ones(nodeCount-1, 1);
    elseif strcmp(caseName, 'Q4')
        moistureExponent = 0.30;
        diffusionPrefactor = 4.2e-4;
        thermalFactor = exp(-3850./((temperature(1:end-1)+temperature(2:end))/2));
    else
        moistureExponent = 0.45;
        diffusionPrefactor = 2.4e-3;
        thermalFactor = exp(-3850./((temperature(1:end-1)+temperature(2:end))/2));
    end
    positiveMoisture = max(waterContent, settings.coefficientFloor);

    kirchhoffPotential = positiveMoisture.*exp(-moistureExponent./positiveMoisture) ...
        -moistureExponent*expint(moistureExponent./positiveMoisture);
    potentialDifference = diff(kirchhoffPotential);
    meanFaceMoisture = (positiveMoisture(1:end-1)+positiveMoisture(2:end))/2;
    moistureDifference = diff(positiveMoisture);
    closePair = abs(moistureDifference) < 1e-7*max(meanFaceMoisture, 1e-3);
    potentialDifference(closePair) =
exp(-moistureExponent./meanFaceMoisture(closePair)) ...
        .*moistureDifference(closePair);

    waterFlux(2:end-1) =
internalFaceX.*diffusionPrefactor.*thermalFactor.*potentialDifference/deltaX;
    waterFlux(end) = -settings.massTransferCoefficient*radiusM*(waterContent(end)-
equilibriumMoisture);
    heatFlux(end) = -settings.heatTransferCoefficient*radiusM*(temperature(end)-
airTemperatureK);
    derivative = zeros(stateCount, 1);
    derivative(1:2:end-1) =
diff(heatFlux)./(radiusM^2*cellWeights.*density.*heatCapacity);

    derivative(2:2:end-1) = diff(waterFlux)./(radiusM^2*cellWeights);
    derivative(end) = 2*settings.massTransferCoefficient/radiusM*(waterContent(end)-
equilibriumMoisture);
end

function radiusM = radiusAt(timeSec)
    if strcmp(caseName, 'Q4')
        boundedTimeSec = min(max(timeSec, inputData.radius.time_s(1)),
inputData.radius.time_s(end));
        radiusM = interp1(inputData.radius.time_s, inputData.radius.radius_m,
boundedTimeSec, 'linear');
    else
        radiusM = settings.initialRadiusM;
    end
end

function [airTemperatureK, equilibriumMoisture] = environmentAt(timeSec)

    if timeSec > inputData.environment.time_s(end)
        airTemperatureK = settings.tailTemperatureK;
        equilibriumMoisture = settings.tailEquilibriumMoisture;
    else
        boundedTimeSec = max(timeSec, inputData.environment.time_s(1));
        airTemperatureK = interp1(inputData.environment.time_s,
inputData.environment.temperature_K, boundedTimeSec, 'linear');
        equilibriumMoisture = interp1(inputData.environment.time_s,
inputData.environment.air_moisture_kg_per_kg, boundedTimeSec, 'linear');
    end
end

function [eventValue, isTerminal, direction] = dryEvent(~, state)

```

```

        eventValue = max(state(2:2:end-1))-settings.dryThreshold;
        isTerminal = 1;
        direction = -1;
    end

    function recordWarning(warningText, warningId, leftSec, rightSec)
        if ~isempty(warningText)
            warningRecords{end+1} = struct('identifier', warningId, 'message', warningText,
            ...
            'segmentStartSec', leftSec, 'segmentTargetSec', rightSec);
            logLine(logFileId, 'WARNING %s [%s] %s', caseName, warningId, warningText);
        end
    end
end

function [density, heatCapacity, conductivity, diffusivity] =
materialProperties(temperatureK, waterContent, caseName)
    if any(~isfinite(temperatureK) | temperatureK <= 0, 'all') ||
any(~isfinite(waterContent), 'all')
        error('CrossCheck:InvalidState', '遇到非有限状态或非正绝对温度。');
    end

    positiveMoisture = max(waterContent, 1e-12);
    wetFraction = positiveMoisture./(1+positiveMoisture);
    switch caseName
        case 'Q1'
            density = 820*ones(size(waterContent));
            heatCapacity = 2600*ones(size(waterContent));
            conductivity = 0.36*ones(size(waterContent));
            diffusivity = 7e-9*exp(-0.89./positiveMoisture);
        case 'Q23'
            density = 650+128*positiveMoisture;
            heatCapacity = 1450+2736*wetFraction;
            conductivity = 0.21+0.38*wetFraction;
            diffusivity = 2.4e-3*exp(-0.45./positiveMoisture-3850./temperatureK);
        case 'Q4'
            density = 760+90*positiveMoisture;
            heatCapacity = 1850+2150*wetFraction;
            conductivity = 0.12+0.20*wetFraction;
            diffusivity = 4.2e-4*exp(-0.30./positiveMoisture-3850./temperatureK);
        otherwise
            error('CrossCheck:Question', '不支持的问题名: %s', caseName);
        end
    end
end

function jacobianPattern = makeJacobianPattern(nodeCount)
    stateCount = 2*nodeCount+1;
    jacobianPattern = sparse(stateCount, stateCount);
    for nodeIndex = 1:nodeCount
        rowIndices = 2*nodeIndex-1:2*nodeIndex;
        for neighborIndex = max(1, nodeIndex-1):min(nodeCount, nodeIndex+1)
            columnIndices = 2*neighborIndex-1:2*neighborIndex;
            jacobianPattern(rowIndices, columnIndices) = 1;
        end
    end
    jacobianPattern(end, 2*nodeCount) = 1;
end

function sampleStates = evaluatePieces(solutions, queryTimesSec)
    sampleStates = zeros(size(solutions{1}.y, 1), numel(queryTimesSec));
    for queryIndex = 1:numel(queryTimesSec)
        queryTimeSec = queryTimesSec(queryIndex);
        foundPiece = false;
    end
end

```

```

        for pieceIndex = 1:numel(solutions)
            piece = solutions{pieceIndex};
            if queryTimeSec >= piece.x(1) && queryTimeSec <= piece.x(end)
                sampleStates(:, queryIndex) = deval(piece, queryTimeSec);
                foundPiece = true;
                break;
            end
        end
        if ~foundPiece
            error('CrossCheck:DenseDomain', '请求的时刻 %.17g 不在已积分分段内。',
queryTimeSec);
        end
    end
end

function sampleTable = makeSampleTable(caseResult)
    sampleCount = numel(caseResult.sampleTimesSec);
    radialCount = numel(caseResult.sampleMaterialX);
    question = repmat(string(caseResult.question), sampleCount*radialCount, 1);
    sampleType = repelem(caseResult.sampleTypes(:), radialCount);
    timeSec = repelem(caseResult.sampleTimesSec(:), radialCount);
    materialX = repmat(caseResult.sampleMaterialX(:), sampleCount, 1);
    currentRadiusM = repelem(caseResult.sampleRadiusM(:), radialCount);
    radiusM = currentRadiusM.*materialX;
    temperatureK = reshape(caseResult.temperatureK', [], 1);
    temperatureC = temperatureK-273.15;
    moistureDryBasis = reshape(caseResult.moistureDryBasis', [], 1);
    sampleTable = table(question, sampleType, timeSec, materialX, radiusM,
currentRadiusM, ...
        temperatureK, temperatureC, moistureDryBasis);
end

function validateInputs(inputData)
    environmentColumns = {'time_s', 'temperature_K', 'air_moisture_kg_per_kg'};
    radiusColumns = {'time_s', 'radius_m'};
    if ~all(ismember(environmentColumns, inputData.environment.Properties.VariableNames))
|| ...
        ~all(ismember(radiusColumns, inputData.radius.Properties.VariableNames))
        error('CrossCheck:InputColumns', '清洗 CSV 字段不符合正式输入接口。');
    end
    environmentValues = inputData.environment{:, environmentColumns};
    radiusValues = inputData.radius{:, radiusColumns};
    if any(~isfinite(environmentValues), 'all') || any(~isfinite(radiusValues), 'all')
|| ...
        any(diff(environmentValues(:, 1)) <= 0) || any(diff(radiusValues(:, 1)) <= 0)
|| ...
        environmentValues(1, 1) ~= 0 || radiusValues(1, 1) ~= 0 || ...
        environmentValues(end, 1) ~= 14400 || radiusValues(end, 1) ~= 259200 || ...
        any(environmentValues(:, 2) <= 0) || any(environmentValues(:, 3) < 0) || ...
        any(radiusValues(:, 2) <= 0) || abs(radiusValues(1, 2)-0.02) > 1e-14
        error('CrossCheck:InputAudit', '时间、数值范围或初始半径审计未通过。');
    end
end

function assertReached(solution, requestedEndSec)
    if abs(solution.x(end)-requestedEndSec) > 1e-7
        error('CrossCheck:SolverStopped', 'ode15s 实际终点 %.17g 未到请求终点 %.17g。',
solution.x(end), requestedEndSec);
    end
    if any(~isfinite(solution.y), 'all')
        error('CrossCheck:NonfiniteSolution', 'ode15s 返回非有限接受状态。');
    end
end

```

```

function record = fileRecord(filePath)
    fileInfo = dir(filePath);
    record = struct('path', filePath, 'bytes', fileInfo.bytes, 'sha256',
sha256File(filePath));
end

function hashText = sha256File(filePath)
    fileId = fopen(filePath, 'r');
    if fileId < 0
        error('CrossCheck:HashRead', '无法读取待哈希文件: %s', filePath);
    end
    fileCleanup = onCleanup(@() fclose(fileId));
    fileBytes = fread(fileId, Inf, '*uint8');
    digest = java.security.MessageDigest.getInstance('SHA-256');
    digest.update(typecast(fileBytes, 'int8'));
    digestBytes = typecast(digest.digest(), 'uint8');
    hashText = lower(reshape(dec2hex(digestBytes, 2)', 1, []));
end

function writeJson(filePath, value)
    fileId = fopen(filePath, 'w', 'n', 'UTF-8');
    if fileId < 0
        error('CrossCheck:JsonWrite', '无法写入 JSON: %s', filePath);
    end
    fileCleanup = onCleanup(@() fclose(fileId));
    fprintf(fileId, '%s\n', jsonencode(value, 'PrettyPrint', true));
end

function logLine(fileId, messageFormat, varargin)
    logText = sprintf(messageFormat, varargin{:});
    fprintf('%s\n', logText);
    fprintf(fileId, '%s\n', logText);
end

function timeText = utcStamp()
    timeText = char(datetime('now', 'TimeZone', 'UTC', 'Format', 'yyyy-MM-
dd''T''HH:mm:ss.SSS''Z''));
end

function value = scalarOrNaN(optionalValue)
    if isempty(optionalValue)
        value = NaN;
    else
        value = optionalValue;
    end
end

```

D.14 analytic_jacobian.py

路径: 05_数值检验与实验/Python 检验源码/paper_output/code/modeling/analytic_jacobian.py

用途: 解析稀疏 Jacobian 与有限差分自检; 原求解器依赖

使用范围: 检验与历史诊断

语言: Python; 源文件共 311 行。

SHA256: 43960b47f92f34770b1be3eebba9fed8c355d8819ad1dd030a6f9803110b33

```
from __future__ import annotations

import argparse
from dataclasses import asdict
from datetime import datetime, timezone
import hashlib
import json
from pathlib import Path
import sys
import time
import warnings

import numpy as np
from scipy.integrate import solve_ivp
from scipy.sparse import coo_matrix
from scipy.special import expi

LOADED_CODE_SHA256 = hashlib.sha256(Path(__file__).read_bytes()).hexdigest()

def _harmonic_partials(left, right):

    total = left + right
    tiny = np.finfo(float).tiny
    denominator = np.maximum(total, tiny)
    ordinary = total > tiny
    dl = np.where(ordinary, 2 * (right / denominator) ** 2, 2 * right / denominator)
    dr = np.where(ordinary, 2 * (left / denominator) ** 2, 2 * left / denominator)
    return dl, dr

def jacobian(model, t, y):

    state = np.asarray(y, dtype=float)
    n = model.n
    if state.ndim != 1 or state.size != 2*n + 1:
        raise ValueError("Jacobian requires interleaved 1D state of length 2*n+1")
    s = model.settings
    T, C = state[:-1:2], state[1:-1:2]
    cc = np.maximum(C, 1e-12)
    active = (C > 1e-12).astype(float)
    rho, cp, k, D = model.properties(T, C)
    if s.question == 'Q1':
        a, d0, temp_constant = .89, 7e-9, 0.
        rho_c, cp_c, k_c = np.zeros(n), np.zeros(n), np.zeros(n)
    elif s.question in ('Q2', 'Q3', 'Q23'):
        a, d0, temp_constant = .45, 2.4e-3, 3850.
        rho_c = np.full(n, 128.) * active
```

```

        cp_c = 2736. / (1. + cc)**2 * active
        k_c = .38 / (1. + cc)**2 * active
    elif s.question == 'Q4':
        a, d0, temp_constant = .30, 4.2e-4, 3850.
        rho_c = np.full(n, 90.) * active
        cp_c = 2150. / (1. + cc)**2 * active
        k_c = .20 / (1. + cc)**2 * active
    else:
        raise ValueError(s.question)
    if s.constant_thermal:
        rho_c, cp_c, k_c = np.zeros(n), np.zeros(n), np.zeros(n)
    capacity = rho * cp
    capacity_c = rho_c * cp + rho * cp_c
    if s.constant_D is None:
        d_c = D * (a / cc**2) * active
        d_t = D * temp_constant / T**2
    else:
        d_c, d_t = np.zeros(n), np.zeros(n)

    radius = float(model.radius(t))
    if radius <= 0:
        raise ValueError("Nonpositive radius")
    tair, ceq = model.environment(t)
    geom = model.internal_faces / model.dx
    delta_t, delta_c = np.diff(T), np.diff(C)

    k_harm = model.harmonic(k)
    kh_l, kh_r = _harmonicpartials(k[:-1], k[1:])
    heat_deriv = np.column_stack((
        -geom * k_harm,
        geom * kh_l * k_c[:-1] * delta_t,
        geom * k_harm,
        geom * kh_r * k_c[1:] * delta_t,
    ))
    heat_g = np.zeros(n + 1)
    heat_g[1:-1] = geom * k_harm * delta_t
    heat_g[-1] = -s.h * radius * (T[-1] - tair)

    if s.face_scheme == 'harmonic' or s.constant_D is not None:
        dh = model.harmonic(D)
        dh_l, dh_r = _harmonicpartials(D[:-1], D[1:])
        water_deriv = np.column_stack((
            geom * dh_l * d_t[:-1] * delta_c,
            geom * (dh_l * d_c[:-1] * delta_c - dh),
            geom * dh_r * d_t[1:] * delta_c,
            geom * (dh_r * d_c[1:] * delta_c + dh),
        ))
    elif s.face_scheme == 'kirchhoff':
        primitive = cc * np.exp(-a / cc) + a * expi(-a / cc)
        primitive_delta = np.diff(primitive)
        mean_c = (cc[:-1] + cc[1:]) / 2.
        mean_t = (T[:-1] + T[1:]) / 2.
        delta_cc = np.diff(cc)
        small = np.abs(delta_cc) < 1e-7 * np.maximum(mean_c, 1e-3)
        exp_left, exp_right = np.exp(-a/cc[:-1]), np.exp(-a/cc[1:])
        primitive_l = -exp_left * active[:-1]
        primitive_r = exp_right * active[1:]
        if np.any(small):
            f_mid = np.exp(-a / mean_c[small])
            f_prime_mid = f_mid * a / mean_c[small]**2
            primitive_delta[small] = f_mid * delta_cc[small]
            primitive_l[small] = (0.5*f_prime_mid*delta_cc[small]-f_mid)*active[:-1][small]

```

```

        primitive_r[small] = (0.5*f_prime_mid*delta_cc[small]+f_mid)*active[1:][small]
    temperature_factor = np.exp(-temp_constant / mean_t)
    coefficient = geom * d0 * temperature_factor
    water_g = coefficient * primitive_delta
    water_t = water_g * temp_constant / (2 * mean_t**2)
    water_deriv = np.column_stack((water_t, coefficient*primitive_l,
                                   water_t, coefficient*primitive_r))
else:
    raise ValueError('Unknown nonlinear face scheme')

heat_scale = 1. / (radius**2 * model.w * capacity)
water_scale = 1. / (radius**2 * model.w)
surface_heat_c = 0.
if s.surface_latent_fraction:
    rho_d = model.rho_d0 * (.02 / radius)**2
    surface_heat_c = -radius*s.surface_latent_fraction*s.latent_J_kg*rho_d*s.beta
    heat_g[-1] += surface_heat_c * (C[-1] - ceq)
temp_derivative = np.diff(heat_g) * heat_scale

face_index = np.arange(n-1)
face_columns = np.column_stack((2*face_index, 2*face_index+1,
                                2*face_index+2, 2*face_index+3)).ravel()
rows, columns, values = [], [], []

def add_face(row_index, derivatives, factor):
    rows.append(np.repeat(row_index, 4))
    columns.append(face_columns)
    values.append((derivatives * factor[:, None]).ravel())

add_face(2*face_index, heat_deriv, heat_scale[:-1])
add_face(2*face_index+2, heat_deriv, -heat_scale[1:])
add_face(2*face_index+1, water_deriv, water_scale[:-1])
add_face(2*face_index+3, water_deriv, -water_scale[1:])
cell_index = np.arange(n)
rows.append(2*cell_index)
columns.append(2*cell_index+1)
values.append(-temp_derivative * capacity_c / capacity)
rows.append(np.array([2*n-2, 2*n-2, 2*n-1, 2*n]))
columns.append(np.array([2*n-2, 2*n-1, 2*n-1, 2*n-1]))
values.append(np.array([-s.h*radius*heat_scale[-1],
                        surface_heat_c*heat_scale[-1],
                        -s.beta*radius*water_scale[-1], 2*s.beta/radius]))
matrix = coo_matrix((np.concatenate(values),
                    (np.concatenate(rows), np.concatenate(columns))),
                    shape=(2*n+1, 2*n+1)).tocsc()
matrix.sum_duplicates()
matrix.eliminate_zeros()
return matrix

def _self_test(output_directory):
    import scipy
    from drying_core import ROOT, Settings, RadialModel, LOADED_CODE_SHA256 as
    CORE_LOADED_CODE_SHA256

    started = time.perf_counter()
    rng = np.random.default_rng(20260910)
    relative_tolerance, absolute_tolerance = 5e-6, 5e-10
    cases = []
    for question in ['Q1', 'Q23', 'Q4']:
        for scheme in ['harmonic', 'kirchhoff']:
            for latent in [0., 1.]:

```

```

        cases.append(Settings(question=question, intervals=8, face_scheme=scheme,
                               shrink=(question=='Q4'),
surface_latent_fraction=latent))
    for question in ['Q1', 'Q23', 'Q4']:
        for constant_d, constant_thermal in [(2e-9, False), (None, True), (2e-9, True)]:
            cases.append(Settings(question=question, intervals=8, face_scheme='kirchhoff',
                                   shrink=(question=='Q4'), constant_D=constant_d,
                                   constant_thermal=constant_thermal,
surface_latent_fraction=1.))
    records, failures = [], []
    for settings in cases:
        model = RadialModel(settings)
        x = model.x
        states = {
            'initial': (0., model.initial()),
            'nonuniform': (18000., model.initial()),
            'late_dry': (150000., model.initial()),
            'near_uniform_small_branch': (14401., model.initial()),
        }
        states['nonuniform'][1][:-1:2] = 303. + 17.*x**2
        states['nonuniform'][1][1:-1:2] = 2.4 - 2.0*x**2
        states['late_dry'][1][:-1:2] = 321. + 2.0*x**2
        states['late_dry'][1][1:-1:2] = .175 - .115*x**2
        states['near_uniform_small_branch'][1][:-1:2] = 303. + 17.*x**2
        states['near_uniform_small_branch'][1][1:-1:2] = .15 + 1e-10*x
        for state_name, (t, y) in states.items():
            with warnings.catch_warnings(record=True) as captured:
                warnings.simplefilter('always')
                matrix = jacobian(model, t, y)
                vectors = []
                for _ in range(4):
                    direction = rng.normal(size=y.size)
                    direction[:-1:2] *= 1.0
                    direction[1:-1:2] *= .02
                    direction[-1] = .3
                    vectors.append(direction)
                directional = []
                for direction in vectors:
                    predicted = matrix @ direction
                    steps = []
                    for step in [1e-4, 3e-5, 1e-5]:
                        finite_difference = (model.rhs(t, y+step*direction)-
                                              model.rhs(t, y-step*direction))/(2*step)
                        absolute_error = float(np.max(np.abs(predicted-finite_difference)))
                        scale = max(float(np.max(np.abs(predicted))),
                                    float(np.max(np.abs(finite_difference))), 1e-30)
                        scaled = float(np.max(np.abs(predicted-finite_difference)/
                                                  (absolute_tolerance + relative_tolerance*np.maximum(
                                                      np.abs(predicted), np.abs(finite_difference))))))
                        steps.append({'step':step, 'max_abs_error':absolute_error,
                                     'relative_inf_error':absolute_error/scale,
                                     'max_component_tolerance_ratio':scaled})
                    best = min(steps, key=lambda
item:item['max_component_tolerance_ratio'])
                    directional.append({'all_step_errors':steps, 'best':best})
                passive = np.zeros(y.size); passive[-1] = 1.
                passive_analytic = float(np.max(np.abs(matrix @ passive)))
                passive_fd = float(np.max(np.abs(model.rhs(t, y+passive)-model.rhs(t,y-
passive))))
                mass_weights = np.zeros(y.size)
                mass_weights[1:-1:2] = 2*model.w
                mass_weights[-1] = 1.
                conservation = float(np.max(np.abs(np.asarray(mass_weights @ matrix))))

```

```

warning_messages = [str(w.message) for w in captured]
passed = (all(step['max_component_tolerance_ratio'] <= 1
               for d in directional for step in d['all_step_errors'])
          and passive_analytic == 0 and passive_fd == 0 and conservation <
1e-11
          and np.isfinite(matrix.data).all() and not warning_messages)
record = {'settings':asdict(settings), 'state':state_name, 'time_s':t,
          'status':'PASS' if passed else 'FAIL', 'matrix_shape':matrix.shape,
          'matrix_nnz':matrix.nnz, 'directional_checks':directional,
          'passive_column_analytic_abs':passive_analytic,
          'passive_column_finite_difference_abs':passive_fd,
          'mass_balance_derivative_abs':conservation,
'warnings':warning_messages}
records.append(record)
if not passed:

failures.append(f"{settings.question}/{settings.face_scheme}/latent={settings.surface_latent_
fraction}/{state_name}/constant_D={settings.constant_D}/
constant_thermal={settings.constant_thermal}")
smoke_records = []
for scheme in ['harmonic', 'kirchhoff']:
    for question in ['Q23', 'Q4']:
        settings = Settings(question=question, intervals=8, face_scheme=scheme,
                             shrink=(question=='Q4'), surface_latent_fraction=1.)
        model = RadialModel(settings)
        with warnings.catch_warnings(record=True) as captured:
            warnings.simplefilter('always')
            result = solve_ivp(model.rhs, (0., 2.), model.initial(), method='BDF',
                               jac=lambda t,y:jacobian(model,t,y), rtol=1e-10,
                               atol=1e-12, max_step=.2)
        msgs = [str(w.message) for w in captured]
        passed = bool(result.success and np.isfinite(result.y).all() and not msgs)
        smoke_records.append({'question':question, 'face_scheme':scheme, 'interval_s':
[0,2],
                             'status':'PASS' if passed else 'FAIL', 'warnings':msgs,
                             'nfev':result.nfev, 'njev':result.njev, 'nlu':result.nlu,
                             'purpose':'Short Jacobian/BDF wiring check, not
production accuracy'})
        if not passed:
            failures.append(f"BDF_smoke/{question}/{scheme}")
            current_source_hash = hashlib.sha256(Path(__file__).read_bytes()).hexdigest()
            current_core_hash =
hashlib.sha256((ROOT/'paper_output/code/modeling/drying_core.py').read_bytes()).hexdigest()
            if current_source_hash != LOADED_CODE_SHA256 or current_core_hash !=
CORE_LOADED_CODE_SHA256:
                failures.append('Source file changed after module load during validation')
            report = {'schema_version':'1.0', 'created_utc':datetime.now(timezone.utc).isoformat(),
                    'status':'PASS' if not failures else 'FAIL', 'failures':failures,
                    'source_sha256':LOADED_CODE_SHA256,
                    'core_sha256':CORE_LOADED_CODE_SHA256,
                    'source_hash_policy':'SHA-256 frozen when each module is imported; files
checked unchanged before report save.',
                    'runtime':
{'python':sys.version, 'executable':sys.executable, 'numpy':np.__version__, 'scipy':scipy.__ve
rsion__},
                    'seed':20260910, 'relative_component_tolerance':relative_tolerance,
                    'absolute_component_tolerance':absolute_tolerance,
                    'finite_difference_policy':'Central directional differences, three decreasing
steps; every step must satisfy the mixed absolute/relative component tolerance. Best
comparison is supplementary only.',
                    'case_state_count':len(records), 'direction_count':4*len(records),
                    'checks':records, 'short_BDF_checks':smoke_records,
                    'elapsed_s':time.perf_counter()-started,

```

```

        'limitations':['No full production run or grid convergence performed here.',
                        'Tiny denominator/underflow extensions at nonphysical Newton
probes are not calibrated physical data.',
                        'Only this module and the short tests use analytic Jacobian
until main agent hooks core.',
                        'Visual Studio reproduction and human review remain
pending.']]
    output_directory.mkdir(parents=True, exist_ok=True)
    (output_directory/'analytic_jacobian_selftest.json').write_text(
        json.dumps(report,ensure_ascii=False,indent=2),encoding='utf-8')
    brief = {key:report[key] for key in
['status','failures','case_state_count','direction_count','elapsed_s']}
    brief['max_best_component_tolerance_ratio'] = max(
        d['best']['max_component_tolerance_ratio'] for r in records for d in
r['directional_checks'])
    brief['max_all_steps_component_tolerance_ratio'] = max(
        step['max_component_tolerance_ratio'] for r in records
        for d in r['directional_checks'] for step in d['all_step_errors'])
    brief['max_mass_balance_derivative_abs'] = max(r['mass_balance_derivative_abs'] for r
in records)
    brief['warning_count'] = sum(len(r['warnings']) for r in records+smoke_records)
    print(json.dumps(brief,ensure_ascii=False,indent=2))
    return 0 if not failures else 1

if __name__ == '__main__':
    parser = argparse.ArgumentParser()
    parser.add_argument('--self-test', action='store_true')
    parser.add_argument('--output-dir', type=Path,
        default=Path(__file__).resolve().parents[2]/'results/jacobian_validation')
    arguments = parser.parse_args()
    if not arguments.self_test:
        parser.error('Use --self-test, or import jacobian(model,t,y) from this module.')
    raise SystemExit(_self_test(arguments.output_dir))

```

D.15 axisymmetric_check.py

路径: 05_数值检验与实验/Python 检验源码/paper_output/code/modeling/axisymmetric_check.py

用途: 轴对称二维含端面情景的适用性检查源代码

使用范围: 检验与历史诊断

语言: Python; 源文件共 283 行。

SHA256: 666db958b1c300db3adc207e8069347540af1f75e41c4d984b255004d85105e9

```
from __future__ import annotations

import argparse
from dataclasses import asdict
from datetime import datetime, timezone
import hashlib
import json
import os
from pathlib import Path
import time
import traceback

for _name in ("OMP_NUM_THREADS", "OPENBLAS_NUM_THREADS", "MKL_NUM_THREADS"):
    os.environ.setdefault(_name, "1")

import numpy as np
from scipy.integrate import solve_ivp
from scipy.sparse import coo_matrix
from scipy.special import expi

from drying_core import ROOT, C0, T0, R0, LENGTH, RadialModel, Settings, solve_case

CODE_PATH = Path(__file__).resolve()
CORE_PATH = CODE_PATH.with_name("drying_core.py")
CODE_HASH = hashlib.sha256(CODE_PATH.read_bytes()).hexdigest()
CORE_HASH = hashlib.sha256(CORE_PATH.read_bytes()).hexdigest()

def harmonic_pair(a, b):
    return 2 * a * b / np.maximum(a + b, np.finfo(float).tiny)

def concentration_potential(c):
    cc = np.maximum(c, 1e-12)
    return cc * np.exp(-0.45 / cc) + 0.45 * expi(-0.45 / cc)

class AxisymmetricModel:
    def __init__(self, nr=30, nz=30, end_faces=True):
        self.nr, self.nz = nr, nz
        self.shape = (nr + 1, nz + 1)
        self.nodes = (nr + 1) * (nz + 1)
        self.settings = Settings(question="Q23", intervals=nr,
                                  face_scheme="kirchhoff", horizon_h=100.)
        self.reference = RadialModel(self.settings)
        self.x, self.wr = self.reference.x, self.reference.w
        self.xr_faces = self.reference.internal_faces
        self.y = np.linspace(0., 1., nz + 1)
        faces_z = np.r_[0., (self.y[:-1] + self.y[1:]) / 2., 1.]
```

```

self.wz = np.diff(faces_z)
self.dx, self.dy = 1. / nr, 1. / nz
self.half_length = LENGTH / 2.
self.end_faces = bool(end_faces)
self.mean_weights = 2 * self.wr[:, None] * self.wz[None, :]
self.calls = 0
self.started = 0.
self.budget_s = 900.
self.jac_pattern = self._sparsity()

def _sparsity(self):
    idx = np.arange(self.nodes).reshape(self.shape)
    pairs = [(idx.ravel(), idx.ravel())]
    for a, b in [(idx[:-1], idx[1:]), (idx[:, :-1], idx[:, 1:]):
        pairs += [(a.ravel(), b.ravel()), (b.ravel(), a.ravel())]
    rows, cols = [], []
    for a, b in pairs:
        for i in (0, 1):
            for j in (0, 1):
                rows.extend((2 * a + i).tolist())
                cols.extend((2 * b + j).tolist())
    surface = np.unique(np.r_[idx[-1, :], idx[:, -1]])
    rows.extend([2 * self.nodes] * len(surface))
    cols.extend((2 * surface + 1).tolist())
    size = 2 * self.nodes + 1
    return coo_matrix((np.ones(len(rows)), (rows, cols)),
                      shape=(size, size)).tocsr()

def initial(self):
    out = np.empty(2 * self.nodes + 1)
    out[:-1:2], out[1:-1:2], out[-1] = T0, C0, 0.
    return out

def fields(self, state):
    return state[:-1:2].reshape(self.shape), state[1:-1:2].reshape(self.shape)

@staticmethod
def face_water(Ta, Tb, Ca, Cb, Fa, Fb):
    cc_a, cc_b = np.maximum(Ca, 1e-12), np.maximum(Cb, 1e-12)
    dc = cc_b - cc_a
    dc_potential = Fb - Fa
    small = np.abs(dc) < 1e-7 * np.maximum((cc_a + cc_b) / 2, 1e-3)
    dc_potential = np.where(small, np.exp(-.45 / ((cc_a + cc_b) / 2)) * dc,
                           dc_potential)
    return 2.4e-3 * np.exp(-3850 / ((Ta + Tb) / 2)) * dc_potential

def rhs(self, t, state):
    self.calls += 1
    if self.calls % 50 == 0 and time.perf_counter() - self.started > self.budget_s:
        raise TimeoutError("Axisymmetric per-run CPU/wall budget reached")
    T, C = self.fields(state)
    rho, cp, k, _ = self.reference.properties(T, C)
    tair, ceq = self.reference.environment(t)
    F = concentration_potential(C)
    gr_t = np.zeros((self.nr + 2, self.nz + 1))
    gr_c = np.zeros_like(gr_t)
    gz_t = np.zeros((self.nr + 1, self.nz + 2))
    gz_c = np.zeros_like(gz_t)
    gr_t[1:-1] = (self.xr_faces[:, None] *
                  harmonic_pair(k[:-1], k[1:]) * np.diff(T, axis=0) / self.dx)
    gr_c[1:-1] = (self.xr_faces[:, None] *
                  self.face_water(T[:-1], T[1:], C[:-1], C[1:], F[:-1], F[1:]) / self.dx)
    gz_t[:, 1:-1] = harmonic_pair(k[:, :-1], k[:, 1:]) * np.diff(T, axis=1) / self.dy

```

```

gz_c[:, 1:-1] = self.face_water(T[:, :-1], T[:, 1:], C[:, :-1], C[:, 1:],
                                F[:, :-1], F[:, 1:]) / self.dy
gr_t[-1] = -self.settings.h * R0 * (T[-1] - tair)
gr_c[-1] = -self.settings.beta * R0 * (C[-1] - ceq)
if self.end_faces:
    gz_t[:, -1] = -self.settings.h * self.half_length * (T[:, -1] - tair)
    gz_c[:, -1] = -self.settings.beta * self.half_length * (C[:, -1] - ceq)
heat_div = (np.diff(gr_t, axis=0) / (R0**2 * self.wr[:, None]) +
            np.diff(gz_t, axis=1) / (self.half_length**2 * self.wz[None, :]))
mass_div = (np.diff(gr_c, axis=0) / (R0**2 * self.wr[:, None]) +
            np.diff(gz_c, axis=1) / (self.half_length**2 * self.wz[None, :]))
derivative = np.empty_like(state)
derivative[:-1:2] = (heat_div / (rho * cp)).ravel()
derivative[1:-1:2] = mass_div.ravel()
lateral_loss = 2 * self.settings.beta / R0 * (self.wz @ (C[-1] - ceq))
end_loss = (self.settings.beta / self.half_length *
            (2 * self.wr @ (C[:, -1] - ceq))) if self.end_faces else 0.
derivative[-1] = lateral_loss + end_loss
return derivative

class AxisymmetricRun:
    def __init__(self, model, pieces, elapsed_s, event):
        self.model, self.pieces, self.elapsed_s, self.event_s = model, pieces, elapsed_s, event
        self.end_s = float(pieces[-1].t[-1])

    def state(self, times):
        tt = np.atleast_1d(np.asarray(times, float))
        result = np.empty((2 * self.model.nodes + 1, len(tt)))
        pending = np.ones(len(tt), bool)
        for p in self.pieces:
            take = pending & (tt >= p.t[0] - 1e-8) & (tt <= p.t[-1] + 1e-8)
            if np.any(take):
                result[:, take] = p.sol(tt[take])
                pending[take] = False
        if pending.any():
            raise ValueError("Requested time outside computed axisymmetric solution")
        return result

    def diagnostics(self):
        m = self.model
        mass_error, cmin, cmax, tmin, tmax, axial_spread = 0., C0, C0, T0, T0, 0.
        for p in self.pieces:
            T, C = p.y[:-1:2], p.y[1:-1:2]
            means = m.mean_weights.ravel() @ C
            mass_error = max(mass_error, float(np.max(np.abs(means + p.y[-1] - C0))))
            cmin, cmax = min(cmin, float(C.min())), max(cmax, float(C.max()))
            tmin, tmax = min(tmin, float(T.min())), max(tmax, float(T.max()))
            CC = C.reshape(*m.shape, -1)
            axial_spread = max(axial_spread, float(np.max(np.ptp(CC, axis=1))))
        event_state = self.state([self.event_s if self.event_s is not None else
self.end_s])[ :, 0]
        _, finalC = m.fields(event_state)
        index = np.unravel_index(np.argmax(finalC), m.shape)
        result = {"event_s":self.event_s,
            "event_h":None if self.event_s is None else self.event_s/3600.,
            "end_s":self.end_s, "elapsed_s":self.elapsed_s,
            "max_mass_residual_kg_per_kg":mass_error, "min_C":cmin, "max_C":cmax,
            "min_T_K":tmin, "max_T_K":tmax, "max_axial_C_spread":axial_spread,
            "slowest_point_at_event_m":{"r":float(m.x[index[0]]*R0),
            "z_from_midplane":float(m.y[index[1]]*m.half_length)},

```

```

        "max_C_at_event":float(finalC.max()),
        "mean_C_at_event":float(np.sum(m.mean_weights*finalC)),
        "nfev":sum(p.nfev for p in self.pieces), "njev":sum(p.njev for p in
self.pieces),
        "nlu":sum(p.nlu for p in self.pieces),
        "rhs_calls_including_jacobian":m.calls,
        "accepted_time_points":sum(len(p.t) for p in self.pieces),
        "solver_success":all(p.success for p in self.pieces)}
    if mass_error > 1e-6 or cmin < -1e-8:
        raise FloatingPointError(f"Axisymmetric physical check failed: {result}")
    return result

def solve_axisymmetric(nr=30, nz=30, end_faces=True, budget_s=600.):
    started = time.perf_counter()
    model = AxisymmetricModel(nr, nz, end_faces)
    model.started, model.budget_s = started, budget_s
    def event(t, y):
        return np.max(y[1:-1:2]) - .15
    event.terminal, event.direction = True, -1
    atol = np.empty(2*model.nodes+1)
    atol[:-1:2], atol[1:-1:2], atol[-1] = 1e-7, 1e-9, 1e-9
    pieces, initial, event_s = [], model.initial(), None
    for left, right in [(0.,14400.), (14400.,360000.)]:
        p = solve_ivp(model.rhs, (left,right), initial, method="BDF",
            rtol=1e-7, atol=atol, max_step=30. if left==0 else 600.,
            jac_sparsity=model.jac_pattern, events=event, dense_output=True)
        pieces.append(p)
        if not p.success:
            raise RuntimeError(p.message)
        initial = p.y[:, -1]
        if len(p.t_events[0]):
            event_s = float(p.t_events[0][0])
            break
    return AxisymmetricRun(model,pieces,time.perf_counter()-started,event_s)

def compare(run, radial):
    end=min(run.end_s,radial.end_s)
    times=np.unique(np.r_[np.arange(0.,end,21600.),1800.,10800.,end])
    times=times[times<=end]
    state2=run.state(times)
    T2=state2[:-1:2].reshape(*run.model.shape,-1)
    C2=state2[1:-1:2].reshape(*run.model.shape,-1)
    state1=radial.state(times)
    T1,C1=state1[:-1:2],state1[1:-1:2]
    mean2=run.model.mean_weights.ravel()@state2[1:-1:2]
    mean1=2*radial.model.w@C1
    delta_t=None if run.event_s is None or radial.event_s is None else (run.event_s-
radial.event_s)/3600
    comparison={"same_radial_intervals":run.model.nr,
        "one_dimensional_event_h":None if radial.event_s is None else radial.event_s/3600,
        "axisymmetric_event_h":None if run.event_s is None else run.event_s/3600,
        "event_difference_h_2d_minus_1d":delta_t,
        "event_relative_difference":None if delta_t is None else
delta_t/(radial.event_s/3600),
        "max_midplane_C_difference":float(np.max(np.abs(C2[:,0,:]-C1))),
        "max_midplane_T_difference_K":float(np.max(np.abs(T2[:,0,:]-T1))),
        "max_mean_C_difference":float(np.max(np.abs(mean2-mean1))),
        "times_s":times.tolist(),
        "mean_C_2d":mean2.tolist(),"mean_C_1d":mean1.tolist(),
        "center_C_2d":C2[0,0].tolist(),"center_C_1d":C1[0].tolist(),
        "center_T_K_2d":T2[0,0].tolist(),"center_T_K_1d":T1[0].tolist()}

```

```

sample={"times_s":times,"x":run.model.x,"y":run.model.y,
        "T_K_2d":T2,"C_2d":C2,"T_K_1d":T1,"C_1d":C1,
        "mean_C_2d":mean2,"mean_C_1d":mean1}
return comparison,sample

def main():
    p=argparse.ArgumentParser()
    p.add_argument("--nr",type=int,default=30)
    p.add_argument("--nz",type=int,default=30)
    p.add_argument("--end-faces",choices=["on","off"],default="on")
    p.add_argument("--budget-s",type=float,default=600.)
    p.add_argument("--tag",default="")
    a=p.parse_args()
    folder=ROOT/"paper_output/results/axisymmetric_check"/f"Nr{a.nr}_Nz{a.nz}
_end_{a.end_faces}_{a.tag}"
    if folder.exists():
        raise FileExistsError(f"Refusing to overwrite {folder}")
    folder.mkdir(parents=True)
    start=datetime.now(timezone.utc).isoformat()
    print(f"START axisymmetric {folder.name} {start}",flush=True)
    try:
        run=solve_axisymmetric(a.nr,a.nz,a.end_faces=="on",a.budget_s)
        diagnostic=run.diagnostics()
        print(json.dumps({"axisymmetric_complete":diagnostic}),flush=True)

    radial=solve_case(Settings(question="Q23",intervals=a.nr,face_scheme="kirchhoff",horizon_h=
100.))
        comparison,samples=compare(run,radial)
        np.savez_compressed(folder/"comparison_fields.npz",**samples)
        if hashlib.sha256(CODE_PATH.read_bytes()).hexdigest()!=CODE_HASH:
            raise RuntimeError("axisymmetric source changed during execution")
        if hashlib.sha256(CORE_PATH.read_bytes()).hexdigest()!=CORE_HASH:
            raise RuntimeError("shared core source changed during execution")

    summary={"started_at_utc":start,"finished_at_utc":datetime.now(timezone.utc).isoformat(),
            "settings":{"nr":a.nr,"nz":a.nz,"end_faces":a.end_faces=="on",
                        "R_m":R0,"half_length_m":LENGTH/2,
                        "reference":asdict(run.model.settings)},
            "diagnostics":diagnostic,"one_dimensional_diagnostics":radial.diagnostics(),
            "comparison":comparison,
            "provenance":{"axisymmetric_sha256":CODE_HASH,"core_sha256":CORE_HASH,
                        "inputs":run.model.reference.input_records},
            "gui_reproduced":False,"human_review_status":"pending",
            "limits":"Same-physics numerical comparison, not experimental validation.
Grid/time convergence evaluated separately."}

    (folder/"summary.json").write_text(json.dumps(summary,ensure_ascii=False,indent=2),encoding
="utf-8")
        print(json.dumps({"folder":str(folder),"comparison":comparison}),flush=True)
    except Exception:
        (folder/"failure.txt").write_text(traceback.format_exc(),encoding="utf-8")
        raise

if __name__=="__main__":
    main()

```

D.16 compare_analytic.py

路径: 05_数值检验与实验/Python 检验源码/paper_output/code/modeling/compare_analytic.py

用途: 有限体积结果与独立 Bessel 基准的数值对照

使用范围: 保留用于追溯的未通过、受限或未执行检验

语言: Python; 源文件共 234 行。

SHA256: 18dbb8ede83416eeb7d5d08b9a5d14131937f22c4d44ffe304a5b36180364f8a

```
from __future__ import annotations

import csv
import hashlib
import json
import os
from dataclasses import asdict
from datetime import datetime, timezone
from pathlib import Path
import platform
import sys
import time

import numpy as np

from drying_core import Settings, solve_case, load_inputs, LOADED_CODE_SHA256
from validate_bessel import bessel_temperature, bessel_moisture

ROOT=Path(__file__).resolve().parents[3]
OUT=ROOT/'paper_output/results/analytic_comparison'
SOURCE=Path(__file__).resolve()
CORE=SOURCE.with_name('drying_core.py')
REFERENCE=SOURCE.with_name('validate_bessel.py')

def sha(path):
    return hashlib.sha256(Path(path).read_bytes()).hexdigest()

def file_record(path):
    p=Path(path)
    return {'path':p.relative_to(ROOT).as_posix(),'bytes':p.stat().st_size,'sha256':sha(p)}

def write_json(path,data):
    path.write_text(json.dumps(data,ensure_ascii=False,indent=2,allow_nan=False)
+'\\n',encoding='utf-8')

def norm_summary(actual,reference,times,radii):
    err=actual-reference
    idx=np.unravel_index(np.argmax(np.abs(err)),err.shape)
    return {
        'max_abs_error':float(np.max(np.abs(err))),
        'rms_error':float(np.sqrt(np.mean(err**2))),
        'mean_signed_error':float(np.mean(err)),
        'max_error_time_s':float(times[idx[0]]),
        'max_error_radius_m':float(radii[idx[1]]),
        'error_at_max_location':float(err[idx]),
        'centre_max_abs_error':float(np.max(np.abs(err[:,0]))),
```

```

        'surface_max_abs_error':float(np.max(np.abs(err[:,-1]))),
        'finite_values':bool(np.all(np.isfinite(actual)) and
np.all(np.isfinite(reference))),
    }

def write_comparison_csv(path,times,radii,T,Tref,C,Cref):
    with path.open('w',encoding='utf-8',newline='') as fh:
        writer=csv.writer(fh)
        writer.writerow(['time_s','radius_m','fvm_temperature_K','analytic_temperature_K',
            'temperature_error_K','fvm_moisture_kg_per_kg',

'analytic_frozen_D_moisture_kg_per_kg','moisture_error_kg_per_kg'])
        for i,t in enumerate(times):
            for j,r in enumerate(radii):
                numeric=[t,r,T[i,j],Tref[i,j],T[i,j]-Tref[i,j],C[i,j]]
                values=[format(float(v),'.17g') for v in numeric]
                values += ['', ''] if Cref is None else
[format(Cref[i,j],'.17g'),format(C[i,j]-Cref[i,j],'.17g')]
                writer.writerow(values)

def main():
    started=time.perf_counter()
    if Path.cwd().resolve() != ROOT:
        raise RuntimeError('Run from the 2026CUMCM workspace root')
    OUT.mkdir(parents=True,exist_ok=True)
    source_records=[file_record(p) for p in [SOURCE,CORE,REFERENCE]]
    if sha(CORE) != LOADED_CODE_SHA256:
        raise RuntimeError('Solver file changed after import')
    env_,inputs=load_inputs(with_records=True)
    env_t,env_T,env_C=env['time_s'],env['temperature_K'],env['air_moisture_kg_per_kg']
    paper_times=np.array([100.,300.,600.,900.,1200.,1500.,1800.])
    paper_radii=np.array([0.,.005,.01,.015,.02])
    dense_times=np.arange(0.,1800.+1,10.)
    dense_radii=np.linspace(0.,.02,21)
    D=float(7e-9*np.exp(-.89/2.55))
    analytic={}
    for grid,ts,rs in [('paper',paper_times,paper_radii),
('dense',dense_times,dense_radii)]:
        exact_T=bessel_temperature(ts,rs,env_t,env_T,n_terms=240)
        exact_C=bessel_moisture(ts,rs,env_t,env_C,D,n_terms=240)
        exact_T_refined=bessel_temperature(ts,rs,env_t,env_T,n_terms=480)
        exact_C_refined=bessel_moisture(ts,rs,env_t,env_C,D,n_terms=480)
        analytic[grid]=(exact_T,exact_C,{
            'heat_240_480_max_difference_K':float(np.max(np.abs(exact_T-exact_T_refined))),
            'frozen_water_240_480_max_difference_kg_per_kg':float(np.max(np.abs(exact_C-
exact_C_refined))))
        case_results=[]
        plot_payload=[]
        for mode,N in [('nonlinear_Q1',100),('nonlinear_Q1',200),('nonlinear_Q1',400),
('frozen_D',100),('frozen_D',400),('frozen_D',800)]:
            name=f'{mode}_N{N}'
            setting=Settings(question='Q1',intervals=N,rtol=1e-10,
                atol_temperature=1e-10,atol_moisture=1e-11,
                early_max_step_s=2.,max_step_s=2.,
                face_scheme='kirchhoff',constant_D=D if mode=='frozen_D' else None)

print(json.dumps({'event':'starting_case','case':name,'elapsed_total_s':time.perf_counter()
-started}),flush=True)
        run=solve_case(setting)
        for rec in source_records+inputs:
            if sha(ROOT/rec['path']) != rec['sha256']:

```

```

        raise RuntimeError('Source/input changed during comparison: '+rec['path'])
    case={'case_id':name,'settings':asdict(setting),'diagnostics':run.diagnostics(),
        'comparison_scope':'temperature only; nonlinear C has no Bessel reference' if
mode=='nonlinear_Q1'
        else 'constant-property heat and frozen-D linear
moisture',
        'comparisons':{},'output_artifacts':[]}
    for grid,ts,rs in [(['paper',paper_times,paper_radii),
('dense',dense_times,dense_radii)]:
        T,C=run.fields(ts,radii_m=rs)
        Tref,Cref,truncation=analytic[grid]
        if mode!='frozen_D':
            Cref=None

comparisons={'temperature':norm_summary(T,Tref,ts,rs),'analytic_truncation_sensitivity':tru
ncation}
    if Cref is not None:
        comparisons['frozen_D_moisture']=norm_summary(C,Cref,ts,rs)
    case['comparisons'][grid]=comparisons
    path=OUT/f'{name}_{grid}.csv'
    write_comparison_csv(path,ts,rs,T,Tref,C,Cref)
    case['output_artifacts'].append(file_record(path))
    if grid=='dense':
        plot_payload.append((mode,N,ts,np.max(np.abs(T-Tref),axis=1),
            None if Cref is None else np.max(np.abs(C-
Cref),axis=1)))

    node_T,node_C=run.fields(paper_times)
    node_r=run.model.x*.02
    edges=np.r_[0.,(node_r[1:]+node_r[:-1])/2.,.02]

mean_T=bessel_temperature(paper_times,node_r,env_t,env_T,n_terms=240,cell_edges_m=edges)
point_T=bessel_temperature(paper_times,node_r,env_t,env_T,n_terms=240)
case['representation_diagnostic']={
    'primary_interpretation':'collocated nodal values with surrounding dual
control-volume balances',
    'strict_cell_average_interpretation_is_not_primary':True,
    'heat_node_max_abs_error_K':float(np.max(np.abs(node_T-point_T))),
    'heat_vs_annular_average_max_abs_error_K':float(np.max(np.abs(node_T-mean_T))),

'analytic_point_vs_annular_average_max_difference_K':float(np.max(np.abs(point_T-mean_T))),
    'note':'Endpoint dual cells are one-sided, so point-vs-average differences can
be first order; they are not physical model error.'}
    if mode=='frozen_D':
        early_times=np.array([1.,2.,5.,10.,20.,30.,60.])
        early_radii=np.array([0.,.019,.02])
        early_T,early_C=run.fields(early_times,radii_m=early_radii)
        early_Tref=bessel_temperature(early_times,early_radii,env_t,env_T,n_terms=480)
        early_Cref=bessel_moisture(early_times,early_radii,env_t,env_C,D,n_terms=480)

early_Cref240=bessel_moisture(early_times,early_radii,env_t,env_C,D,n_terms=240)
    case['early_boundary_layer']={
        'times_s':early_times.tolist(),'radii_m':early_radii.tolist(),
        'moisture':norm_summary(early_C,early_Cref,early_times,early_radii),
        'surface_errors_kg_per_kg':(early_C[:-1]-early_Cref[:-1]).tolist(),
        'analytic_240_480_max_difference':float(np.max(np.abs(early_Cref240-
early_Cref)))}
    early_path=OUT/f'{name}_early.csv'

write_comparison_csv(early_path,early_times,early_radii,early_T,early_Tref,early_C,early_Cr
ef)
    case['output_artifacts'].append(file_record(early_path))

```



```

        'scope':'Numerical verification against the selected constant-coefficient PDE; no
internal experimental accuracy claim.',
        'point_value_comparison':'Nodal-dual state is compared to exact point values at the
same physical radii. Separate representation diagnostics quantify strict annular
averages.',
        'primary_space_time_samples':
{'paper_times_s':paper_times.tolist(), 'paper_radii_m':paper_radii.tolist(),
        'dense_time_step_s':10., 'dense_time_range_s':
[0.,1800.],
        'dense_radius_step_m':.001, 'dense_radius_range_m':
[0.,.02]},
        'nonlinear_moisture_note':'The nonlinear Q1 C solution is stored for provenance
only, not compared with constant-D Bessel or labeled analytic truth.',
        'constant_diffusion_coefficient_m2_s':D,
        'core_static_review':{
            'kirchhoff_primitive':'F(C)=C*exp(-a/C)+a*Ei(-a/C), so F_prime(C)=exp(-a/C)',
            'water_face_flux':'Internal face radius times D0*exp(-3850/T_face)*(F(C_right)-
F(C_left))/dx, with no thermal factor in the differenced potential.',
            'Q1_temperature_factor':'Exactly one, since D1 has no temperature dependence.',
            'no_spurious_soret_term':True,
            'internal_flux_cancellation':'Each face occurs with opposite signs in adjacent
control-volume balances; weighted sum leaves only the surface flux.',
            'density_interpretation':'rho*cp is effective thermal capacity; independent
dry-matter weighting is used for conserved moisture.',
            'scope_limitation':'Static consistency and linear benchmark do not alone
validate all nonlinear long-time or shrinkage scenarios.'},
        'convergence':convergence, 'cases':case_results,
        'total_elapsed_s':time.perf_counter()-started,
        'visual_studio_gui':'pending', 'human_review':'pending',
        'execution_provenance':{'run_command':sys.executable+' -B '+' '.join(sys.argv),
            'run_exit_code':0, 'python_version':platform.python_version(),
            'source_files':source_records, 'input_artifacts':inputs,
            'output_artifacts':[file_record(p) for p in sorted(OUT.glob('*.csv'))]+
[file_record(plot_path)]},
        }
    write_json(OUT/'analytic_comparison_report.json',report)

print(json.dumps({'status':report['status'], 'convergence':convergence, 'total_elapsed_s':rep
ort['total_elapsed_s']}, indent=2), flush=True)
    return 0

if __name__=='__main__':
    raise SystemExit(main())

```

D.17 disk_dense.py

路径: 05_数值检验与实验/Python 检验源码/paper_output/code/modeling/disk_dense.py

用途: BDF 稠密多项式磁盘保存和查询; 原求解器依赖

使用范围: 检验与历史诊断

语言: Python; 源文件共 111 行。

SHA256: f8ade59c2f45166339613af0dc95666d8b240d45ef238ef91f5586f41e368a3a

```
from __future__ import annotations
import hashlib
from pathlib import Path
import tempfile
import numpy as np
from scipy.integrate import OdeSolution
from scipy.integrate._ivp.bdf import BDF, BdfDenseOutput
from scipy.integrate._ivp.base import DenseOutput

LOADED_CODE_SHA256 = hashlib.sha256(Path(__file__).read_bytes()).hexdigest()

class DenseCache:
    def __init__(self, root):
        self.parent = (Path(root)/'tmp/cache/solver_runs').resolve()
        if not self.parent.is_relative_to(Path(root).resolve()):
            raise ValueError('Private solver cache must remain in this project')
        self.parent.mkdir(parents=True, exist_ok=True)
        self.directory = Path(tempfile.mkdtemp(prefix='bdf_', dir=self.parent)).resolve()
        try:
            self.handle = (self.directory/'polynomials.bin').open('w+b', buffering=0)
        except BaseException as error:
            try:
                self.directory.rmdir()
            except OSError as cleanup_error:
                error.add_note('Private cache directory cleanup failed: '
                               '+repr(cleanup_error)')
            raise
        self.bytes_written = 0
        self.polynomial_count = 0
        self.accepted_arrays = []
        self.closed = False

    def append(self, values):
        if self.closed:
            raise RuntimeError('Dense solution cache is closed')
        array = np.ascontiguousarray(values, dtype=np.float64)
        offset = self.bytes_written
        self.handle.seek(offset)
        array.tofile(self.handle)
        self.bytes_written += array.nbytes
        self.polynomial_count += 1
        return offset, array.shape

    def read(self, offset, shape):
        if self.closed:
            raise RuntimeError('Dense solution cache is closed')
        count = int(np.prod(shape))
        self.handle.seek(offset)
        values = np.fromfile(self.handle, dtype=np.float64, count=count)
```

```

        if values.size != count:
            raise IOError('Incomplete BDF coefficient cache')
        return values.reshape(shape)

    def store_accepted(self, values):
        path = self.directory/f'accepted_{len(self.accepted_arrays):03d}.npy'
        mapped = np.lib.format.open_mmap(path, mode='w+', dtype=values.dtype,
        shape=values.shape)

        self.accepted_arrays.append(mapped)
        mapped[:] = values
        mapped.flush()
        return mapped

    def close(self):
        if self.closed:
            return
        self.handle.close()
        for array in self.accepted_arrays:
            array._mmap.close()
        self.accepted_arrays.clear()

        if self.directory.parent != self.parent or not
self.directory.name.startswith('bdf_'):
            raise RuntimeError('Unexpected private cache path; cleanup refused')
        for path in self.directory.iterdir():
            if not path.is_file() or path.is_symlink():
                raise RuntimeError('Unexpected cache entry; cleanup refused')
            path.unlink()
        self.directory.rmdir()
        self.closed = True

class FileBdfDenseOutput(DenseOutput):
    def __init__(self, original, cache):
        super().__init__(original.t_old, original.t)
        self.order = original.order
        self.t_shift = original.t_shift.copy()
        self.denom = original.denom.copy()
        self.cache = cache
        self.offset, self.shape = cache.append(original.D)

    def _call_impl(self, t):

        dense = object.__new__(BdfDenseOutput)
        dense.D = self.cache.read(self.offset, self.shape)
        dense.t_shift, dense.denom = self.t_shift, self.denom
        return dense._call_impl(t)

class DiskBDF(BDF):
    def __init__(self, *args, dense_cache, **kwargs):
        self.dense_cache = dense_cache
        super().__init__(*args, **kwargs)

    def _dense_output_impl(self):
        return FileBdfDenseOutput(super()._dense_output_impl(), self.dense_cache)

```

```
def align_bdf_segments(result):  
  
    if result.sol is not None:  
        result.sol = OdeSolution(result.sol.ts, result.sol.interpolants, alt_segment=True)
```

D.18 drying_core.py

路径: 05_数值检验与实验/Python 检验源码/paper_output/code/modeling/drying_core.py

用途: 原径向热湿方程、材料坐标、Kirchhoff 水通量与求解器; 检验共同依赖

使用范围: 检验与历史诊断

语言: Python; 源文件共 392 行。

SHA256: dedfe48af60c43d41faf11ba66d8e767f2ed5ea9ecd0f6f9cf9a86f4467b6ebf

```
from __future__ import annotations

from dataclasses import asdict, dataclass
from pathlib import Path
import hashlib
import io
import json
import time

import numpy as np
from scipy.integrate import solve_ivp
from scipy.sparse import lil_matrix
from scipy.special import expi
import analytic_jacobian
import disk_dense

ROOT = Path(__file__).resolve().parents[3]
LOADED_CODE_SHA256 = hashlib.sha256(Path(__file__).read_bytes()).hexdigest()
C0, T0, R0, LENGTH = 2.55, 301.15, 0.02, 0.25

def load_inputs(with_records=False):
    arrays, records = [], []
    for name in ['A_environment_observed.csv', 'A_radius_observed.csv']:
        path = ROOT / 'paper_output/data_cleaned' / name
        content = path.read_bytes()
        arrays.append(np.genfromtxt(io.StringIO(content.decode('utf-8-sig')),
                                   delimiter=',', names=True))
        records.append({'path':path.relative_to(ROOT).as_posix(), 'bytes':len(content),
                       'sha256':hashlib.sha256(content).hexdigest(), 'exists':True})
    return (*arrays, records) if with_records else tuple(arrays)

@dataclass(frozen=True)
class Settings:
    question: str = 'Q23'
    intervals: int = 100
    rtol: float = 1e-7
    atol_temperature: float = 1e-7
    atol_moisture: float = 1e-9
    max_step_s: float = 600.0
    early_max_step_s: float = 30.0
    horizon_h: float = 240.0
    shrink: bool = False
    boundary_extension: str = 'nominal'
    tail_temperature_C: float = 50.0
    tail_equilibrium: float = 0.05
    h: float = 25.0
    beta: float = 8e-7
    equilibrium_scale: float = 1.0
```

```

surface_latent_fraction: float = 0.0
latent_J_kg: float = 2.4e6
constant_D: float | None = None
constant_thermal: bool = False
method: str = 'BDF'
face_scheme: str = 'harmonic'
jacobian_mode: str = 'analytic'
dense_storage: str = 'disk'

class RadialModel:
    def __init__(self, settings: Settings):
        self.settings = settings
        if settings.intervals < 2 or settings.jacobian_mode not in ('analytic',
'finite_difference'):
            raise ValueError('At least two intervals and a supported Jacobian mode are
required')
        self.env, self.rad, self.input_records = load_inputs(with_records=True)
        self.x = np.linspace(0., 1., settings.intervals + 1)
        self.dx = 1. / settings.intervals
        faces = np.r_[0., (self.x[1:] + self.x[:-1]) / 2., 1.]
        self.w = np.diff(faces ** 2) / 2.
        self.internal_faces = faces[1:-1]
        self.n = len(self.x)
        self.rho_d0 = (760 + 90 * C0) / (1 + C0) if settings.question == 'Q4' else (
            820. / (1 + C0) if settings.question == 'Q1' else (650 + 128 * C0) / (1 + C0))
        self evaluations = 0
        self.jac_pattern = self._sparsity()

    def radius(self, t):
        if self.settings.shrink:
            return np.interp(t, self.rad['time_s'], self.rad['radius_m'])
        return np.asarray(t) * 0. + R0

    def environment(self, t):
        s = self.settings
        tair = np.interp(t, self.env['time_s'], self.env['temperature_K'])
        ceq = np.interp(t, self.env['time_s'], self.env['air_moisture_kg_per_kg'])
        after = np.asarray(t) > self.env['time_s'][-1]
        if s.boundary_extension == 'nominal':
            tair = np.where(after, s.tail_temperature_C + 273.15, tair)
            ceq = np.where(after, s.tail_equilibrium, ceq)
        elif s.boundary_extension == 'tail_mean':
            tail = self.env['time_s'] >= 10800
            tair = np.where(after, self.env['temperature_K'][tail].mean(), tair)
            ceq = np.where(after, self.env['air_moisture_kg_per_kg'][tail].mean(), ceq)
        elif s.boundary_extension != 'last':
            raise ValueError('Unknown boundary extension')
        return tair, ceq * s.equilibrium_scale

    def properties(self, T, C):
        s = self.settings
        positive_C = np.maximum(C, 1e-12)
        if np.any(T <= 0):
            raise FloatingPointError('Nonpositive absolute temperature')
        wet = positive_C / (1. + positive_C)
        if s.question == 'Q1':
            rho = np.full_like(C, 820.)
            cp = np.full_like(C, 2600.)
            k = np.full_like(C, .36)
            D = 7e-9 * np.exp(-.89 / positive_C)
        elif s.question in ('Q2', 'Q3', 'Q23'):
            rho, cp, k = 650 + 128 * positive_C, 1450 + 2736 * wet, .21 + .38 * wet

```

```

        D = 2.4e-3 * np.exp(-.45 / positive_C - 3850 / T)
    elif s.question == 'Q4':
        rho, cp, k = 760 + 90 * positive_C, 1850 + 2150 * wet, .12 + .20 * wet
        D = 4.2e-4 * np.exp(-.30 / positive_C - 3850 / T)
    else:
        raise ValueError(s.question)
    if s.constant_D is not None:
        D = np.full_like(C, s.constant_D)
    if s.constant_thermal:
        rho, cp, k = np.full_like(C, 820.), np.full_like(C, 2600.), np.full_like(C,
.36)
        return rho, cp, k, D

    @staticmethod
    def harmonic(a):
        return 2 * a[:-1] * a[1:] / np.maximum(a[:-1] + a[1:], np.finfo(float).tiny)

    def _sparsity(self):
        p = lil_matrix((2 * self.n + 1, 2 * self.n + 1), dtype=int)
        for i in range(self.n):
            for j in range(max(0, i - 1), min(self.n, i + 2)):
                p[2*i+2, 2*j+2] = 1
        p[-1, 2 * (self.n - 1) + 1] = 1
        return p.tocsr()

    def water_internal_flux(self, T, C, D):
        if self.settings.face_scheme == 'harmonic' or self.settings.constant_D is not None:
            return self.internal_faces * self.harmonic(D) * np.diff(C) / self.dx
        if self.settings.face_scheme != 'kirchhoff':
            raise ValueError('Unknown nonlinear face scheme')
        a, D0 = {'Q1':(.89,7e-9), 'Q23':(.45,2.4e-3), 'Q2':(.45,2.4e-3),
                'Q3':(.45,2.4e-3), 'Q4':(.30,4.2e-4)}[self.settings.question]
        cc = np.maximum(C, 1e-12)
        potential = cc * np.exp(-a/cc) + a * expi(-a/cc)
        difference = np.diff(potential)
        small = np.abs(np.diff(cc)) < 1e-7 * np.maximum((cc[:-1]+cc[1:])/2, 1e-3)
        difference[small] = (np.exp(-a/((cc[:-1][small]+cc[1:][small])/2)) * np.diff(cc)
[small])
        thermal_factor = 1. if self.settings.question == 'Q1' else
np.exp(-3850/((T[:-1]+T[1:])/2))

        return self.internal_faces * D0 * thermal_factor * difference / self.dx

    def rhs(self, t, state):
        self.evaluations += 1
        T, C = state[:-1:2], state[1:-1:2]
        rho, cp, k, D = self.properties(T, C)
        radius = float(self.radius(t))
        tair, ceq = self.environment(t)
        heat_g = np.zeros(self.n + 1)
        water_g = np.zeros(self.n + 1)
        heat_g[1:-1] = self.internal_faces * self.harmonic(k) * np.diff(T) / self.dx
        water_g[1:-1] = self.water_internal_flux(T, C, D)
        water_g[-1] = -self.settings.beta * radius * (C[-1] - ceq)
        heat_g[-1] = -self.settings.h * radius * (T[-1] - tair)
        if self.settings.surface_latent_fraction:
            rho_d = self.rho_d0 * (R0 / radius) ** 2
            j_evap = rho_d * self.settings.beta * (C[-1] - ceq)
            heat_g[-1] -= (radius * self.settings.surface_latent_fraction *
                self.settings.latent_J_kg * j_evap)
        derivative = np.empty_like(state)

```

```

        derivative[:-1:2] = np.diff(heat_g) / (radius ** 2 * self.w * rho * cp)
        derivative[1:-1:2] = np.diff(water_g) / (radius ** 2 * self.w)
        derivative[-1] = 2 * self.settings.beta / radius * (C[-1] - ceq)
        return derivative

    def initial(self):
        state = np.empty(2 * self.n + 1)
        state[:-1:2], state[1:-1:2], state[-1] = T0, C0, 0.
        return state

class Run:
    def __init__(self, model, pieces, elapsed, event_s, cache=None):
        self.model, self.pieces = model, pieces
        self.cache = cache
        self.elapsed_s, self.event_s = elapsed, event_s
        self.end_s = float(pieces[-1].t[-1])

    def close(self):
        self.pieces.clear()
        if self.cache is not None:
            self.cache.close()

    def state(self, times):
        tt = np.atleast_1d(np.asarray(times, dtype=float))
        if np.min(tt) < -1e-10 or np.max(tt) > self.end_s + 1e-7:
            raise ValueError('Requested time outside solved interval')
        out = np.empty((2 * self.model.n + 1, len(tt)))
        remaining = np.ones(len(tt), dtype=bool)
        for result in self.pieces:
            select = remaining & (tt >= result.t[0]-1e-7) & (tt <= result.t[-1]+1e-7)
            if np.any(select):
                out[:, select] = result.sol(tt[select])
                remaining[select] = False
        if np.any(remaining):
            raise RuntimeError('Missing dense solution segment')
        return out

    def fields(self, times, radii_m=None, material_x=None):
        tt = np.atleast_1d(np.asarray(times, dtype=float))
        state = self.state(tt)
        Ts, Cs = state[:-1:2].T, state[1:-1:2].T
        if material_x is not None:
            points = np.asarray(material_x, dtype=float)
            if np.any(~np.isfinite(points)) or np.any((points < 0.) | (points > 1.)):
                raise ValueError('Material coordinates must be finite and within [0,1]')
            return np.array([np.interp(points, self.model.x, row) for row in Ts]),
np.array([
            np.interp(points, self.model.x, row) for row in Cs])
        if radii_m is None:
            return Ts, Cs
        radial = np.asarray(radii_m)
        Tout, Cout = [], []
        for i, t in enumerate(tt):
            xx = radial / self.model.radius(t)
            Tout.append(np.interp(xx, self.model.x, Ts[i], left=np.nan, right=np.nan))
            Cout.append(np.interp(xx, self.model.x, Cs[i], left=np.nan, right=np.nan))
        return np.asarray(Tout), np.asarray(Cout)

    def diagnostics(self):

```

```

minimum_C = minimum_T = minimum_D = minimum_property = np.inf
maximum_C = maximum_T = maximum_D = maximum_radial_increase = -np.inf
maximum_mass_residual = 0.
for piece in self.pieces:
    for first in range(0, len(piece.t), 256):
        raw = piece.y[:, first:first+256]
        T, C = raw[:-1:2], raw[1:-1:2]
        residual = 2*self.model.w@C + raw[-1] - C0
        rho, cp, k, D = self.model.properties(T.ravel(), C.ravel())
        minimum_C, maximum_C = min(minimum_C, C.min()), max(maximum_C, C.max())
        minimum_T, maximum_T = min(minimum_T, T.min()), max(maximum_T, T.max())
        minimum_D = min(minimum_D, D.min())
        maximum_D = max(maximum_D, D.max())
        minimum_property = min(minimum_property, rho.min(), cp.min(), k.min(), D.min())
        maximum_radial_increase =
max(maximum_radial_increase, np.max(np.diff(C, axis=0)))
        maximum_mass_residual = max(maximum_mass_residual, np.max(np.abs(residual)))
    final = self.state([self.end_s])[0]
    finalC = final[1:-1:2]
    return {
        'event_s': self.event_s, 'event_h': None if self.event_s is None else
self.event_s/3600,
        'end_s': self.end_s, 'elapsed_s': self.elapsed_s,
        'end_time_convention': 'ceil(critical_event_s)+1: conservative post-crossing
verification second; not claimed earliest integer second',
        'max_mass_balance_abs_kg_per_kg': float(maximum_mass_residual),
        'min_C': float(minimum_C), 'max_C': float(maximum_C),
        'min_T_K': float(minimum_T), 'max_T_K': float(maximum_T),
        'min_D': float(minimum_D), 'max_D': float(maximum_D),
        'positive_properties': bool(minimum_property > 0),
        'max_radial_C_increase': float(maximum_radial_increase),
        'final_max_C': float(finalC.max()), 'final_surface_C': float(finalC[-1]),
        'strictly_dry_at_end': bool(finalC.max() < .15),
        'radius_end_m': float(self.model.radius(self.end_s)),
        'radius_extrapolation_used': bool(self.model.settings.shrink and self.end_s >
259200),
        'rhs_evaluations': self.model.evaluations,
        'accepted_time_points': sum(len(p.t) for p in self.pieces),
        'nfev': sum(p.nfev for p in self.pieces),
        'njev': sum(p.njev for p in self.pieces), 'nlu': sum(p.nlu for p in
self.pieces),
        'solver_success': all(p.success for p in self.pieces),
        'dense_storage': self.model.settings.dense_storage,
        'dense_coefficient_bytes': 0 if self.cache is None else
self.cache.bytes_written,
        'dense_polynomial_count': 0 if self.cache is None else
self.cache.polynomial_count,
    }

def solve_case(settings: Settings) -> Run:
    started = time.perf_counter()
    if settings.dense_storage not in ('memory', 'disk'):
        raise ValueError('Dense storage must be memory or disk')
    if settings.dense_storage == 'disk' and settings.method != 'BDF':
        raise ValueError('Exact disk dense storage currently supports BDF only')
    cache = disk_dense.DenseCache(ROOT) if settings.dense_storage == 'disk' else None
    try:
        return _solve_case_impl(settings, cache, started)
    except BaseException as error:
        if cache is not None:
            try:
                cache.close()

```

```

        except BaseException as cleanup_error:
            error.add_note('Private cache cleanup also failed: '+repr(cleanup_error))
    raise

def _solve_case_impl(settings, cache, started):
    model = RadialModel(settings)
    method = disk_dense.DiskBDF if cache is not None else settings.method
    jacobian_options = ({'jac': lambda t, y: analytic_jacobian.jacobian(model, t, y)}
                        if settings.jacobian_mode == 'analytic' else {'jac_sparsity': model.jac_pattern})
    if cache is not None:
        jacobian_options['dense_cache'] = cache
    def dry_event(t, y):
        return float(np.max(y[1:-1:2]) - .15)
    dry_event.terminal, dry_event.direction = True, -1
    atol = np.empty(2 * model.n + 1)
    atol[:-1:2], atol[1:-1:2], atol[-1] = settings.atol_temperature,
    settings.atol_moisture, settings.atol_moisture
    horizon = 1800. if settings.question == 'Q1' else settings.horizon_h * 3600.
    pieces, initial, event_s = [], model.initial(), None

    endpoints = [0., min(14400., horizon)]
    if horizon > 14400.:
        endpoints.append(horizon)
    for left, right in zip(endpoints[:-1], endpoints[1:]):
        piece = solve_ivp(model.rhs, (left, right), initial, method=method,
                        rtol=settings.rtol, atol=atol, **jacobian_options,
                        max_step=settings.early_max_step_s if left < 14400. else settings.max_step_s,
                        events=None if settings.question == 'Q1' else dry_event, dense_output=True)
        if cache is not None:
            disk_dense.align_bdf_segments(piece)
        pieces.append(piece)
        if not piece.success:
            raise RuntimeError(piece.message)
        initial = piece.y[:, -1].copy()
        if cache is not None:
            piece.y = cache.store_accepted(piece.y)
        if piece.t_events is not None and len(piece.t_events[0]):
            event_s = float(piece.t_events[0][0])

        end = float(np.ceil(event_s) + 1)
        tail = solve_ivp(model.rhs, (event_s, end), initial, method=method,
                        rtol=settings.rtol, atol=atol, **jacobian_options,
                        max_step=1., dense_output=True)
        if cache is not None:
            disk_dense.align_bdf_segments(tail)
        if not tail.success:
            raise RuntimeError(tail.message)
        if cache is not None:
            tail.y = cache.store_accepted(tail.y)
        pieces.append(tail)
        break
    run = Run(model, pieces, time.perf_counter() - started, event_s, cache)
    diagnostic = run.diagnostics()
    if diagnostic['min_C'] < -1e-8 or not diagnostic['positive_properties']:
        raise FloatingPointError('Physical range/positive property check failed')
    if diagnostic['max_mass_balance_abs_kg_per_kg'] > 1e-6:
        raise FloatingPointError('Dry-basis mass balance failed')
    return run

def file_record(path):
    p = Path(path)

```

```

    return {'path': p.relative_to(ROOT).as_posix(), 'bytes': p.stat().st_size,
            'sha256': hashlib.sha256(p.read_bytes()).hexdigest(), 'exists': True}

def save_run(run: Run, directory: Path):
    code_record = file_record(Path(__file__))
    if code_record['sha256'] != LOADED_CODE_SHA256:
        raise RuntimeError('Solver file changed after import; restart to obtain valid
provenance')
    jacobian_record = file_record(Path(analytic_jacobian.__file__))
    if jacobian_record['sha256'] != analytic_jacobian.LOADED_CODE_SHA256:
        raise RuntimeError('Jacobian file changed after import; restart to obtain valid
provenance')
    storage_record = file_record(Path(disk_dense.__file__))
    if storage_record['sha256'] != disk_dense.LOADED_CODE_SHA256:
        raise RuntimeError('Dense storage file changed after import; restart for valid
provenance')
    for record in run.model.input_records:
        if file_record(ROOT/record['path'])['sha256'] != record['sha256']:
            raise RuntimeError('Input changed after being loaded; retain failure and
rerun')
    directory.mkdir(parents=True, exist_ok=True)
    times = np.unique(np.r_[np.arange(0., run.end_s, 60.),
        [t for t in [100., 300., 600., 900., 1200., 1500., 1800., 3600., 5400.,
7200., 9000., 10800.] if t <= run.end_s],
        run.end_s, [] if run.event_s is None else [run.event_s]])
    x = np.linspace(0., 1., 21)
    temperature, moisture, means, losses = [], [], [], []
    for first in range(0, len(times), 128):
        block_times = times[first:first+128]
        Tb, Cb = run.fields(block_times, material_x=x)
        raw = run.state(block_times)
        temperature.append(Tb); moisture.append(Cb)
        means.append(2*run.model.w@raw[1:-1:2]); losses.append(raw[-1].copy())
    T, C = np.vstack(temperature), np.vstack(moisture)
    np.savez_compressed(directory/'sampled_solution.npz', times_s=times, material_x=x,
        T_K=T, C=C, radius_m=run.model.radius(times), mean_C=np.concatenate(means),
        cumulative_loss=np.concatenate(losses))
    summary = {'settings': asdict(run.model.settings), 'diagnostics': run.diagnostics(),
        'code': code_record,
        'jacobian_code': jacobian_record,
        'dense_storage_code': storage_record,
        'inputs': run.model.input_records,
        'human_review_status': 'pending', 'gui_reproduced': False}
    (directory/'summary.json').write_text(json.dumps(summary, ensure_ascii=False,
indent=2), encoding='utf-8')
    return summary

```

D.19 export_outputs.py

路径: 05_数值检验与实验/Python 检验源码/paper_output/code/modeling/export_outputs.py

用途: run_modeling 动态加载的原结果导出/回读模块; 依赖补齐

使用范围: 检验与历史诊断

语言: Python; 源文件共 720 行。

SHA256: 1b9c358f5219bf522ac9f5fe7a4f62b0395c419127c614103957643d98b4f7e2

```
from __future__ import annotations

import argparse
import csv
from dataclasses import asdict
from datetime import datetime, timezone
import gzip
import hashlib
import inspect
import json
import math
from pathlib import Path
import re
import sys
import tempfile
import time

import numpy as np
from openpyxl import Workbook, load_workbook
from openpyxl.cell import WriteOnlyCell
from openpyxl.styles import Alignment, Font, PatternFill
from openpyxl.utils import get_column_letter

ROOT = Path(__file__).resolve().parents[3]
SOURCE = Path(__file__).resolve()
LOADED_EXPORTER_SHA256 = hashlib.sha256(SOURCE.read_bytes()).hexdigest()
TEMPLATE_DIR = ROOT / 'problem_files/CUMCM2026Problems/A 题/附件/附件 3'
RADII_M = np.arange(21, dtype=float)*0.001
RADII_CM = [i/10 for i in range(21)]
NUMBER_FORMAT = '0.0000'
SIZE_LIMIT_BYTES = 20_000_000

def _sha(path):
    h=hashlib.sha256()
    with Path(path).open('rb') as fh:
        for block in iter(lambda:fh.read(1024*1024),b''):
            h.update(block)
    return h.hexdigest()

def _record(path, role=None):
    path=Path(path).resolve()
    result={'path':path.relative_to(ROOT).as_posix(),'absolute_path':str(path),
            'bytes':path.stat().st_size,'sha256':_sha(path),'exists':True}
    if role:
        result['role']=role
    return result
```

```

def _json(path,value):
    path.write_text(json.dumps(value,ensure_ascii=False,indent=2,allow_nan=False)
+'\\n',encoding='utf-8')

def _qid(question_id):
    value=str(question_id).upper()
    if value in ['1','2','3','4']:
        value='Q'+value
    if value not in ['Q1','Q2','Q3','Q4']:
        raise ValueError('Export requires Q1, Q2, Q3 or Q4')
    return value

def _output_times(question_id,end,event):
    if question_id=='Q1':
        if end<1800-1e-7:
            raise ValueError('Q1 requires a solved interval covering 0..1800s')
        return np.arange(1801,dtype=float)
    if question_id=='Q2':
        if abs(end-round(end))>1e-7:
            raise ValueError('Q2 needs an integer post-crossing verification endpoint')
        return np.arange(int(round(end))+1,dtype=float)
    if event is None:
        raise ValueError('Q3/Q4 final export requires an actual critical drying event')
    if not 0<=event<=end:
        raise ValueError('Critical event must lie inside the solved interval')
    return np.unique(np.r_[np.arange(math.floor(end/60)+1,dtype=float)*60,event,end])

def _q4(x):
    if x is None:
        return None
    if not math.isfinite(float(x)):
        raise ValueError('A non-finite value cannot be exported as a numerical result')
    return float(round(float(x),4))

def _template(question_id):
    path=TEMPLATE_DIR/f'result{question_id[-1]}.xlsx'
    wb=load_workbook(path,read_only=True,data_only=False)
    try:
        header=next(wb.worksheets[0].iter_rows(values_only=True))
        names=wb.sheetnames
        return {'record':_record(path,'original_result_template'),'time_header':header[0],
                'sheet_names':names,'surface_header':header[-1] if question_id=='Q4' else
None}
    finally:
        wb.close()

def _run_provenance(run):
    module=sys.modules[run.model.__class__.__module__]
    core=Path(inspect.getfile(run.model.__class__)).resolve()
    loaded=getattr(module,'LOADED_CODE_SHA256',None)
    if loaded is not None and loaded!=_sha(core):
        raise RuntimeError('Core file changed after this Run implementation was imported')
    if _sha(SOURCE)!=LOADED_EXPORTER_SHA256:
        raise RuntimeError('Exporter changed after import; restart for consistent
provenance')
    input_records=list(getattr(run.model,'input_records',[]))
    for rec in input_records:
        if _sha(ROOT/rec['path'])!=rec['sha256']:

```

```

        raise RuntimeError('Observed input changed after solve: '+rec['path'])
    return
{'core':_record(core,'solver_source'),'exporter':_record(SOURCE,'exporter_source'),
 'inputs':input_records,'settings':asdict(run.model.settings),
 'event_s':None if run.event_s is None else float(run.event_s),
 'end_s':float(run.end_s),'source_kind':'live_Run_dense_solution',
 'per_second_values_are_not_interpolated_from_60s_npz':True}

def _field_block(run,times,qid):
    T,C=run.fields(times,radii_m=RADII_M)
    radius=np.asarray(run.model.radius(times),dtype=float)
    if radius.ndim==0:
        radius=np.full(len(times),radius)
    surface_C=None
    if qid=='Q4':
        _,surface=run.fields(times,material_x=np.array([1.0]))
        surface_C=surface[:,0]
        inside=RADII_M[None,:]<= radius[:,None]+1e-12

        near_boundary=inside & (~np.isfinite(C))
        if np.any(near_boundary):
            C=np.where(near_boundary,surface_C[:,None],C)
            T=np.where(inside,T,np.nan)
            C=np.where(inside,C,np.nan)
        if not np.all(np.isfinite(C[inside])) or not np.all(np.isfinite(surface_C)):
            raise ValueError('Non-finite moisture inside Q4 domain')
    else:
        inside=np.ones_like(C,dtype=bool)
        if not np.all(np.isfinite(T)) or not np.all(np.isfinite(C)):
            raise ValueError('Non-finite output inside fixed domain')
    return T-273.15,C,radius,surface_C,inside

def _raw_headers(qid):
    h=['time_s','radius_m']
    if qid in ['Q1','Q2']:
        h += [f'T_C_r_{i/10:.1f}_cm' for i in range(21)]
    h += [f'C_r_{i/10:.1f}_cm' for i in range(21)]
    if qid=='Q4':
        h += ['C_surface']
    return h

def _raw_row(qid,t,T,C,R,surface,inside):
    row=[float(t),float(R)]
    if qid in ['Q1','Q2']:
        row+=list(map(float,T))
    row += [float(c) if valid else None for c,valid in zip(C,inside)]
    if qid=='Q4':
        row.append(float(surface))
    return row

def _rounded_sheets(qid,row):
    t,R=row[:2]
    if qid in ['Q1','Q2']:
        return {'温度':[_q4(t)]+[_q4(v) for v in row[2:23]],
                '水分浓度':[_q4(t)]+[_q4(v) for v in row[23:44]]}
    if qid=='Q3':
        return {'Sheet1':[_q4(t)]+[_q4(v) for v in row[2:23]]}
    return {'Sheet1':[_q4(t)]+[_q4(v) for v in row[2:24]],
            '半径':[_q4(t),_q4(R*100)]}

```

```

def _sheet(wb,name,header):
    ws=wb.create_sheet(name)
    ws.freeze_panes='B2'
    ws.sheet_view.showGridLines=False
    ws.column_dimensions['A'].width=30
    for j in range(2,len(header)+1):
        ws.column_dimensions[get_column_letter(j)].width=13 if j<len(header) else 16
    ws.row_dimensions[1].height=34
    cells=[]
    for value in header:
        cell=WriteOnlyCell(ws,value=value)
        cell.font=Font(name='Arial',size=10,bold=True,color='FFFFFF')
        cell.fill=PatternFill(fill_type='solid',fgColor='334155')
        cell.alignment=Alignment(horizontal='center',vertical='center',wrap_text=True)
        if isinstance(value,(int,float)):
            cell.number_format='0.0'
        cells.append(cell)

    cache=ROOT/'tmp/cache/openpyxl_exports'
    cache.mkdir(parents=True,exist_ok=True)
    previous_tempdir=tempfile.tempdir
    try:
        tempfile.tempdir=str(cache)
        ws.append(cells)
    finally:
        tempfile.tempdir=previous_tempdir
    return ws

def _append_numeric(ws,row):
    cells=[]
    for value in row:
        if value is None:
            cells.append(None)
        else:
            cell=WriteOnlyCell(ws,value=value)
            cell.number_format=NUMBER_FORMAT
            cells.append(cell)
    ws.append(cells)

def _paper_tables(run,qid,directory):
    if qid=='Q1':
        ts=np.array([100,300,600,900,1200,1500,1800],dtype=float)
    elif qid=='Q2':
        ts=np.arange(1,7,dtype=float)*1800
        if run.end_s<10800-1e-7:
            raise ValueError('Q2 paper tables require 3h of actual solution')
    else:
        from q3_model import completion
        reported_end=completion(run)['reported_time_s']
        ts=np.unique(np.r_[np.arange(21600.,reported_end+1e-8,21600.),reported_end])
    rs=np.array([0.,.005,.01,.015,.02])
    T,C=run.fields(ts,radii_m=rs)
    T=T-273.15
    R=np.asarray(run.model.radius(ts))
    headers=['时间/s' if qid=='Q1' else '时间/h']+[0,0.5,1,1.5,2]
    if qid=='Q4':
        headers += ['药材表面']
        _,Cs=run.fields(ts,material_x=[1.])

```

```

outputs=[]
fields=[('temperature',T),('moisture',C)] if qid in ['Q1','Q2'] else [('moisture',C)]
for kind,values in fields:
    path=directory/f'{qid.lower()}_paper_{kind}.csv'
    with path.open('w',encoding='utf-8-sig',newline='') as fh:
        w=csv.writer(fh);w.writerow(headers)
        for i,t in enumerate(ts):
            row=[f'{t if qid=="Q1" else t/3600:.4f}']
            for j,r in enumerate(rs):
                outside=qid=="Q4" and r>R[i]+1e-12
                value=values[i,j]
                if not outside and not np.isfinite(value) and qid=="Q4" and abs(r-
R[i])<=1e-12:
                    value=Cs[i,0]
                    row.append(' if outside else f'{float(value):.4f}')
                if qid=="Q4":
                    row += [f'{Cs[i,0]:.4f}']
                w.writerow(row)
            outputs.append(_record(path,'paper_table'))
if qid=="Q4":
    path=directory/'q4_paper_radius.csv'
    with path.open('w',encoding='utf-8-sig',newline='') as fh:
        w=csv.writer(fh);w.writerow(['时间/h','表面半径/cm'])
        w.writerows([[f'{t/3600:.4f}',f'{r*100:.4f}'] for t,r in zip(ts,R)])
    outputs.append(_record(path,'paper_surface_coordinates'))
return outputs

```

```

def _validate_paper_tables(manifest,run):

```

```

    qid=manifest['question_id']
    if run is None:
        return {'status':'NOT_REQUESTED','checked_cells':0,'files':[],
                'scope':'Paper CSV hashes only; no independent live Run available.'},[]
    errors=[];records=manifest['paper_tables'];completion_info=None
    if qid=="Q1":
        times=np.asarray([100.,300.,600.,900.,1200.,1500.,1800.])
        expected_kinds=['temperature','moisture']
    elif qid=="Q2":
        times=1800.*np.arange(1,7)
        expected_kinds=['temperature','moisture']
    else:
        from q3_model import completion
        completion_info=completion(run)
        final_s=float(completion_info['reported_time_s'])

        regular=[21600.*k for k in range(1,math.floor(final_s/21600.)+1)
                  if 21600.*k<final_s-1e-8]
        times=np.asarray([*regular,final_s],dtype=float)
        expected_kinds=['moisture']+(['radius'] if qid=="Q4" else [])
        stored=manifest.get('paper_table_time_contract',{})
        for key in ['reported_time_s','reported_drying_time_h']:
            if key not in stored or not _numeric_equal(stored[key],completion_info[key]):
                errors.append('Paper CSV report-time metadata differs from independent
completion: '+key)
            if not
_numeric_equal(stored.get('reported_time_h'),completion_info['reported_drying_time_h']):
                errors.append('Paper CSV reported_time_h alias differs from independent
completion')
        if times[-1]>run.end_s+1e-7:
            errors.append('Paper CSV contract exceeds the live solution horizon')

```

```

        return {'status': 'FAIL', 'checked_cells': 0, 'files': []}, errors
    expected_names = {f'{qid.lower()}_paper_{kind}.csv': kind for kind in expected_kinds}
    actual_names = [Path(r['path']).name for r in records]
    if set(actual_names) != set(expected_names) or len(actual_names) != len(expected_names):
        errors.append('Missing, duplicate, or unexpected paper CSV artifact')
    files = []; total_cells = 0
    for rec in records:
        path = ROOT/rec['path']; kind = expected_names.get(path.name)
        if kind is None: continue
        before_errors = len(errors)
        with path.open('r', encoding='utf-8-sig', newline='') as fh:
            reader = csv.reader(fh); header = next(reader, None); rows = list(reader)
        desired_time = '时间/s' if qid == 'Q1' else '时间/h'
        if kind == 'radius':
            correct_header = ['时间/h', '表面半径/cm']
            if header != correct_header:
                errors.append(path.name + ': radius/time units or header mismatch')
                continue
            radii = None
            expected = np.asarray(run.model.radius(times), dtype=float).reshape(-1, 1)*100.
            inside = np.ones_like(expected, dtype=bool)
            unit = 'cm'; surface_column = None
        else:
            wanted_columns = 7 if qid == 'Q4' else 6
            if header is None or len(header) != wanted_columns or header[0] != desired_time:
                errors.append(path.name + ': time/column header mismatch'); continue
            try:
                radii = np.asarray([float(v)*.01 for v in header[1:6]])
            except ValueError:
                errors.append(path.name + ': nonnumeric physical-radius header'); continue
            if not np.all(np.isfinite(radii)) or not np.allclose(radii,
[0., .005, .01, .015, .02], rtol=0, atol=1e-12):
                errors.append(path.name + ': physical-radius grid must be 0,0.5,1,1.5,2
cm'); continue
            if qid == 'Q4' and header[-1] != '药材表面':
                errors.append(path.name + ': distinct material surface column
missing'); continue
            temperature, moisture = run.fields(times, radii_m=radii)
            expected = np.asarray(temperature)-273.15 if kind == 'temperature' else
np.asarray(moisture)
            inside = np.ones_like(expected, dtype=bool); surface_column = None
            if qid == 'Q4':
                physical_R = np.asarray(run.model.radius(times), dtype=float)
                inside = radii[None, :] <= physical_R[:, None]+1e-12
                _, surface_values = run.fields(times, material_x=np.asarray([1.]))

                coincidence = np.abs(radii[None, :]-physical_R[:, None])<=1e-12
                expected = np.where(coincidence &
~np.isfinite(expected), surface_values, expected)
                expected = np.column_stack([expected, surface_values[:, 0]])
                inside = np.column_stack([inside, np.ones(len(times), dtype=bool)])
                surface_column = 7
            unit = 'degC' if kind == 'temperature' else 'kg/kg'
            if len(rows) != len(times):
                errors.append(f'{path.name}: expected {len(times)} data rows, read
{len(rows)}')
            checked = 0; blanks = 0; max_field_error = 0.; max_time_error = 0.
            for i, (row, t) in enumerate(zip(rows, times)):
                if len(row) != len(header):
                    errors.append(f'{path.name}: column count mismatch at row {i+2}'); continue
                time_value = float(t if qid == 'Q1' else t/3600.)
                if row[0] != format(time_value, '.4f'):

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```

        errors.append(f'{path.name}: time label mismatch at row {i+2}; query time
is {t:.17g}s')
    else: max_time_error = max(max_time_error, abs(float(row[0]) - time_value))
    checked += 1
    for j, cell in enumerate(row[1:]):
        checked += 1
        if not inside[i, j]:
            blanks += 1
            if cell != '': errors.append(f'{path.name}: domain outside must be blank
at row {i+2}, column {j+2}')
            continue
        value = float(expected[i, j])
        if not math.isfinite(value):
            errors.append(f'{path.name}: live value is not finite inside
material'); continue
        if not re.fullmatch(r'-?\d+\.\d{4}', cell):
            errors.append(f'{path.name}: finite four-decimal number required at row
{i+2}, column {j+2}')
            continue
        actual = float(cell)
        if not math.isfinite(actual) or abs(actual -
float(format(value, '.4f'))) > 1e-10:
            errors.append(f'{path.name}: independently queried value mismatch at
row {i+2}, column {j+2}')
            max_field_error = max(max_field_error, abs(actual - value))
        total_cells += checked
        files.append({'artifact': _record(path), 'kind': kind, 'status': 'PASS' if
len(errors) == before_errors else 'FAIL',

'header': header, 'data_rows': len(rows), 'expected_data_rows': len(times), 'checked_cells': check
ed,

'source_query_times_s': times.tolist(), 'display_time_unit': 's' if qid == 'Q1' else
'h',

'max_display_time_rounding_error': max_time_error, 'field_unit': unit,

'max_field_rounding_error_from_live_value': max_field_error, 'outside_domain_blank_cells': bla
nks,

'physical_radii_m': None if radii is None else radii.tolist(),
'surface_column_1based': surface_column, 'surface_query': 'material_x=1; never
fixed 2 cm' if surface_column else None})
    return {'status': 'FAIL' if errors else
'PASS', 'checked_cells': total_cells, 'files': files,

'completion': completion_info, 'completion_source': _record(SOURCE.with_name('q3_model.py'))
if completion_info else None,

'scope': 'Every paper CSV cell independently re-evaluated from live Run and fixed
question contracts; no generation arrays or 60s archives reused.',

'rounding_limit': 'CSV stores four decimals; matching rounded values cannot
distinguish sub-rounding perturbations of the original source value.'}, errors

def export_question(run, question_id, output_dir, *, chunk_rows=1000, overwrite=False):

    started = time.perf_counter()
    qid = _qid(question_id)
    directory = Path(output_dir).resolve()
    if not directory.is_relative_to(ROOT) or
directory.is_relative_to(ROOT / 'problem_files'):
        raise ValueError('Outputs must be inside the competition workspace, outside
problem_files')
    if not 1 <= int(chunk_rows) <= 2000:
        raise ValueError('chunk_rows must be between 1 and 2000')

```

```

directory.mkdir(parents=True,exist_ok=True)
stem='result'+qid[-1]
book_path=directory/(stem+'.xlsx')
raw_path=directory/(stem+'_unrounded.csv.gz')
manifest_path=directory/(stem+'.export.json')
if not overwrite and any(p.exists() for p in [book_path,raw_path,manifest_path]):
    raise FileExistsError('Export target already exists; use a new output version')
provenance=_run_provenance(run)
allowed={'Q1':{'Q1'},'Q2':{'Q2','Q23'},'Q3':{'Q2','Q3','Q23'},'Q4':{'Q4'}}
if run.model.settings.question not in allowed[qid]:
    raise ValueError('Run physics/question do not match the requested output question')
if qid=='Q4' and not run.model.settings.shrink:
    raise ValueError('Formal Q4 output requires the shrinking-domain Run, not its
fixed-radius control')
template=_template(qid)
times=_output_times(qid,run.end_s,run.event_s)
if len(times)+1>1_048_576:
    raise ValueError('Required full time grid exceeds the XLSX worksheet row limit; do
not truncate')
workbook=Workbook(write_only=True)
workbook.properties.creator=''
workbook.properties.lastModifiedBy=''
workbook.properties.title=f'问题{qid[-1]}结果'
headers={name:[template['time_header']] + RADII_CM for name in template['sheet_names']}
if qid=='Q4':
    headers['Sheet1'].append(template['surface_header'])
    headers['半径']=['时间/s','药材表面半径/cm']
sheets={name:_sheet(workbook,name,header) for name,header in headers.items()}
counts={name:
{'data_rows':0,'total_rows':1,'columns':len(header),'outside_domain_blank_cells':0,
'first_time_s':None,'last_time_s':None,'header':header} for name,header
in headers.items()}
samples_idx=set(np.linspace(0,len(times)-1,min(13,len(times)),dtype=int).tolist())
samples_idx.update([0,min(1,len(times)-1),len(times)-1])
if run.event_s is not None:
    samples_idx.update(np.flatnonzero(times==run.event_s).tolist())
samples=[]
with gzip.open(raw_path,'wt',encoding='utf-8',newline='',compresslevel=6) as fh:
    raw_writer=csv.writer(fh);raw_writer.writerow(_raw_headers(qid))
    for start in range(0,len(times),int(chunk_rows)):
        ts=times[start:start+int(chunk_rows)]
        T,C,R,Cs,inside=_field_block(run,ts,qid)
        for i,t in enumerate(ts):
            raw=_raw_row(qid,t,T[i],C[i],R[i],None if Cs is None else Cs[i],inside[i])
            raw_writer.writerow(['' if v is None else format(v,'.17g') for v in raw])
            values=_rounded_sheets(qid,raw)
            for name,row in values.items():
                _append_numeric(sheets[name],row)
                count=counts[name]
                count['data_rows']+=1;count['total_rows']+=1
                count['outside_domain_blank_cells']+=sum(v is None for v in row)
                if count['first_time_s'] is None:
                    count['first_time_s']=row[0]
                count['last_time_s']=row[0]
            if start+i in samples_idx:
                samples.append({'data_row_index':start+i,'source_time_s':float(t),
'raw_values':raw,'rounded_sheets':values})
workbook.save(book_path)
paper_records=_paper_tables(run,qid,directory)
paper_time_contract={'unit':'s' if qid=='Q1' else 'h','stored_decimals':4,
'query_uses_unrounded_seconds':True}
if qid in ['Q3','Q4']:
    from q3_model import completion

```

```

        completed=completion(run)
        paper_time_contract.update({key:completed[key] for key in
['reported_time_s','reported_drying_time_h','critical_event_s',
'max_C_at_reported_time','conservative_post_verification_s']})
        paper_time_contract['reported_time_h']=completed['reported_drying_time_h']
        paper_time_contract['regular_step_s']=21600.
        paper_time_contract['endpoint_convention']='6-hour samples plus upward-reported
strict-drying time; distinct from workbook post-verification endpoint'
        end_provenance=_run_provenance(run)
        if end_provenance!=provenance or _sha(ROOT/template['record']['path'])!
=template['record']['sha256']:
            raise RuntimeError('Source, settings or template changed during export')
        book_record=_record(book_path,'contest_result_workbook')
        archive_record=_record(raw_path,'internal_unrounded_output_grid_archive')
        manifest={
            'schema_version':'1.0','question_id':qid,'generated_by':'paper_output/code/
modeling/export_outputs.py',

'generated_at':datetime.now(timezone.utc).isoformat(),'status':'EXPORTED_PENDING_STREAM_REA
DBACK',

'workbook':book_record,'unrounded_archive':archive_record,'paper_tables':paper_records,
    'paper_table_time_contract':paper_time_contract,
    'provenance':provenance,'template':template['record'],'sheets':counts,
    'time_grid':{'start_s':0.,'end_s':float(times[-1]),'regular_step_s':1 if qid in
['Q1','Q2'] else 60,
                'critical_event_s':provenance['event_s'] if qid in ['Q3','Q4'] else
None,
                'include_post_verification_endpoint':qid!='Q1',

'count':len(times),'rounding_duplicate_time_count':int(np.sum(np.diff(np.round(times,4))==0
))),
    'radius_grid':
{'fixed_radii_cm':RADII_CM,'surface_column':qid=='Q4','surface_coordinate_sheet':'半径' if
qid=='Q4' else None},
    'rounding':{'decimal_places':4,'storage':'numeric rounded to four
decimals','number_format':NUMBER_FORMAT,
                'raw_archive':'17 significant digits, source times in seconds and
radius in metres',
                'threshold_judgement':'use original full precision event and max C,
never rounded 0.1500'},
    'domain_rule':{'outside_values':'blank (None); never
zero','comparison_tolerance_m':1e-12,
                'Q4_surface_is_not_fixed_2cm':True},
    'samples':samples,'elapsed_s':time.perf_counter()-started,
    'size':{'workbook_bytes':book_record['bytes'],'workbook_MiB':book_record['bytes']/
2**20,

'unrounded_archive_bytes':archive_record['bytes'],'warning_limit_bytes':SIZE_LIMIT_BYTES,
    'workbook_exceeds_20_decimal_MB':book_record['bytes']>SIZE_LIMIT_BYTES,
    'archive_is_internal_not_automatically_in_submission_zip':True,
    'complete_support_archive_still_requires_actual_size_check':True,
    'no_required_time_or_space_values_removed':True},
    'validation_required':'validate_exports([record], runs={question_id: run}); full
workbook/archive comparison, live Run workbook samples, and every paper CSV cell
independently queried from live Run',
    'visual_studio_gui':'pending','human_review':'pending',
    }
    _json(manifest_path,manifest)
    return
{'question_id':qid,'workbook_path':str(book_path),'manifest_path':str(manifest_path),

```

```

        'artifacts':[book_record,archive_record]+paper_records+
[_record(manifest_path,'export_manifest')],
        'sheets':counts,'size':manifest['size'],'validation_status':'pending'}

def _manifest_path(item):
    if isinstance(item,dict):
        return Path(item['manifest_path'])
    path=Path(item)
    return path.with_suffix('.export.json') if path.suffix=='.xlsx' else path

def _numeric_equal(a,b,tolerance=1e-10):
    return a is None and b is None or (a is not None and b is not None and
        isinstance(a,(int,float)) and not isinstance(a,bool) and math.isfinite(float(a))
and abs(float(a)-float(b))<=tolerance)

def validate_exports(paths,runs=None):

    if isinstance(paths,(str,Path,dict)):
        paths=[paths]
        runs={} if runs is None else runs
        results=[]
        for item in paths:
            started=time.perf_counter()
            manifest_path=_manifest_path(item).resolve()
            manifest=json.loads(manifest_path.read_text(encoding='utf-8'))
            qid=manifest['question_id'];errors=[];checked_cells=0;checked_sheets={}
            for rec in
[manifest['workbook'],manifest['unrounded_archive'],manifest['template']]
+manifest['paper_tables']:
                if not (ROOT/rec['path']).exists() or _sha(ROOT/rec['path'])!=rec['sha256']:
                    errors.append('Hash mismatch or missing artifact: '+rec['path'])
            if errors:
                raise RuntimeError('; '.join(errors))
            times=_output_times(qid,manifest['provenance']['end_s'],manifest['provenance']
['event_s'])
            wb=load_workbook(ROOT/manifest['workbook']['path'],read_only=True,data_only=False)
            if wb.sheetnames!=list(manifest['sheets']):
                errors.append('Sheet names/order differ from export contract')
            try:
                for name,spec in manifest['sheets'].items():
                    ws=wb[name]
                    rows=ws.iter_rows(min_row=1,max_col=spec['columns'])
                    header=[cell.value for cell in next(rows)]
                    if header!=spec['header']:
                        errors.append(name+': header/physical radius grid mismatch')
                    count=0;blanks=0;first=None;last=None;max_rounding_error=0.
                    with gzip.open(ROOT/manifest['unrounded_archive']
['path'],'rt',encoding='utf-8',newline='') as rawfh:
                        source_rows=csv.reader(rawfh)
                        if next(source_rows)!=_raw_headers(qid):
                            errors.append('Unrounded archive schema mismatch')
                        for index,source_row in enumerate(source_rows):
                            try:
                                cells=next(rows)
                            except StopIteration:
                                errors.append(name+': workbook ends before raw archive');break
                            raw=[None if v=='' else float(v) for v in source_row]
                            if index>=len(times) or abs(raw[0]-times[index])>1e-8:
                                errors.append(name+': source time schedule mismatch at

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```

'+str(index))
        if qid=='Q4':
            for j,r in enumerate(RADII_M):
                outside=r>raw[1]+1e-12
                if (raw[2+j] is None)!=outside:
                    errors.append(name+': incorrect moving-domain mask at
'+str(index))

            elif any(v is None for v in raw):
                errors.append(name+': unexpected blank in fixed domain')
            expected=_rounded_sheets(qid,raw)[name]
            for j,(cell,value) in enumerate(zip(cells,expected)):
                if not _numeric_equal(cell.value,value):
                    errors.append(f'{name}: value mismatch at row {index+2},
column {j+1}')

                if value is not None:
                    if cell.number_format!=NUMBER_FORMAT:
                        errors.append(f'{name}: four-decimal format missing at
row {index+2}, column {j+1}')
                    if not math.isfinite(float(cell.value)) or
abs(float(cell.value)*1e4-round(float(cell.value)*1e4))>1e-5:
                        errors.append(f'{name}: non-finite or non-four-decimal
stored number')

            max_rounding_error=max(max_rounding_error,abs(float(cell.value)-value))
            else:
                blanks+=1
                checked_cells+=len(cells);count+=1
                if first is None:first=cells[0].value
                last=cells[0].value
                if len(errors)>30:
                    raise RuntimeError('Export validation failed: '+';
'.join(errors[:30]))

            if next(rows,None) is not None:
                errors.append(name+': workbook has uncontracted extra rows')
            if count!=spec['data_rows'] or count!=len(times):
                errors.append(name+': row count differs from time grid')
            if blanks!=spec['outside_domain_blank_cells']:
                errors.append(name+': blank count mismatch')

checked_sheets[name]={'data_rows':count,'total_rows':count+1,'first_time_s':first,
'last_time_s':last,'outside_domain_blank_cells':blanks,'value_comparison_max_abs_error':max
_rounding_error}
        finally:
            wb.close()
            run=runs.get(qid)
            live={'status':'NOT_REQUESTED','samples':0,'max_abs_unrounded_difference':None}
            if run is not None:
                current=_run_provenance(run)
                if current!=manifest['provenance']:
                    errors.append('Live Run provenance differs from the exported Run')
                samples=manifest['samples'];ts=np.array([s['source_time_s'] for s in samples])
                T,C,R,Cs,inside=_field_block(run,ts,qid)
                difference=0.
                for i,sample in enumerate(samples):
                    actual=_raw_row(qid,ts[i],T[i],C[i],R[i],None if Cs is None else
Cs[i],inside[i])
                    for a,b in zip(actual,sample['raw_values']):
                        if a is None or b is None:
                            if not (a is None and b is None):errors.append('Live Run sample
domain mismatch')
                        else:
                            difference=max(difference,abs(a-b))

```

```

        if difference>1e-10:
            errors.append('Independent live Run sample mismatch exceeds 1e-10')
            live={'status':'PASS' if difference<=1e-10 else 'FAIL','samples':len(samples),
                'max_abs_unrounded_difference':difference}
            paper_validation,paper_errors=_validate_paper_tables(manifest,run)
            errors.extend(paper_errors)
            report={'schema_version':'1.0','question_id':qid,'generated_by':'paper_output/
code/modeling/export_outputs.py',
                'generated_at':datetime.now(timezone.utc).isoformat(),
                'status':'FAIL' if errors else 'PASS','fully_verified_with_live_Run':not
errors and run is not None,

'checked_cells':checked_cells,'sheets':checked_sheets,'live_Run_sample_validation':live,
'paper_CSV_live_Run_validation':paper_validation,
'errors':errors,'elapsed_s':time.perf_counter()-started,
'workbook':manifest['workbook'],'export_manifest':_record(manifest_path),
'scope':'Full workbook/raw-archive validation; independent workbook live
Run samples and every paper CSV time/position/value/surface/radius cell when live Run is
provided.'},

'four_decimals_are_storage_and_display_not_physical_accuracy_claim':True,

'visual_render':'pending','visual_studio_gui':'pending','human_review':'pending'}

report_path=manifest_path.with_name(manifest_path.name.replace('.export.json','.validation.
json'))

_json(report_path,report);report['report_path']=str(report_path);results.append(report)
    if errors:
        raise RuntimeError('Export validation failed: '+'; '.join(errors[:30]))
    return {'status':'PASS' if all(r['status']=='PASS' for r in results) else 'FAIL',
        'fully_verified_with_live_Run':all(r['fully_verified_with_live_Run'] for r in
results),'exports':results}

def _paper_csv_selfcheck(directory):

    from drying_core import Settings,solve_case
    import copy
    directory=Path(directory).resolve()
    directory.mkdir(parents=True,exist_ok=True)
    summaries=[];negative_checks=[]
    for qid in ['Q1','Q2','Q3','Q4']:
        settings={'question':'Q23' if qid in ['Q2','Q3'] else qid,
            'intervals':40,'face_scheme':'kirchhoff','shrink':qid=='Q4',
            'rtol':1e-8,'atol_temperature':1e-8,'atol_moisture':1e-10,
            'early_max_step_s':5.}
        if qid=='Q2':
            settings.update(constant_D=2e-8,beta=8e-6,horizon_h=24.)
            print('PAPER_CSV_SELFHECK '+qid+': actual N40 solve',flush=True)
            run=solve_case(Settings(**settings))
            try:
                if qid=='Q2' and (run.event_s is None or run.end_s<10800.):
                    raise RuntimeError('Accelerated export test must still cover 3h and a real
drying event')
                exported=export_question(run,qid,directory/qid)
                checked=validate_exports([exported],runs={qid:run})

manifest=json.loads(Path(exported['manifest_path']).read_text(encoding='utf-8'))
                defects={'Q1':['temperature_value'],'Q2':['moisture_value'],
                    'Q3':['last_time_label'],'Q4':
['outside_nonblank','surface_missing','radius_value']}[qid]
                for defect in defects:
                    changed=copy.deepcopy(manifest)

```

```

        kind='temperature' if defect=='temperature_value' else 'radius' if
defect=='radius_value' else 'moisture'
        selected=next(r for r in changed['paper_tables'] if
Path(r['path']).name==f'{qid.lower()}_paper_{kind}.csv')
        original_path=ROOT/selected['path']
        with original_path.open('r',encoding='utf-8-sig',newline='') as
fh:rows=list(csv.reader(fh))
        if defect in ['temperature_value','moisture_value']:
            rows[1][1]=format(float(rows[1][1])+.01, '.4f')
        elif defect=='last_time_label':
            rows[-1][0]=format(float(rows[-1][0])-.001, '.4f')
        elif defect=='outside_nonblank':
            ri,ci=next((i,j) for i in range(1,len(rows)) for j in range(1,6) if
rows[i][j]=='')
            rows[ri][ci]='0.0000'
        elif defect=='surface_missing':rows[-1][-1]=''
        elif defect=='radius_value':rows[-1][1]=format(float(rows[-1][1])
+.01, '.4f')
        target=directory/'deliberate_defects'/defect/original_path.name
        target.parent.mkdir(parents=True,exist_ok=True)
        with target.open('w',encoding='utf-8-sig',newline='') as
fh:csv.writer(fh).writerows(rows)
        selected.update(_record(target,selected.get('role')))
        result,errors=_validate_paper_tables(changed,run)
        if not errors or result['status']!='FAIL':
            raise AssertionError('Independent paper validator missed deliberate
defect: '+defect)

negative_checks.append({'question_id':qid,'defect':defect,'correctly_rejected':True,
                        'errors':errors,'artifact':_record(target)})

summaries.append({'question_id':qid,'settings':asdict(run.model.settings),'export':exported
,
                  'validation':checked,'event_s':run.event_s,'end_s':run.end_s,
                  'Q2_is_accelerated_constant_D_export_test_only':qid=='Q2'})
print(json.dumps({'question_id':qid,'status':'PASS','end_s':run.end_s,
                  'paper_cells':checked['exports'][0]['paper_CSV_live_Run_validation']
['checked_cells']},ensure_ascii=False),flush=True)
finally:
    run.close()
    report={'status':'PASS','generated_at':datetime.now(timezone.utc).isoformat(),
            'scope':'N40 exporter fidelity only. Q2 uses constant_D=2e-8 and beta=8e-6
solely to shorten the real full-event export test; these are not formal physical results.
Q1/Q3/Q4 use their nominal physics.',
            'source':_record(SOURCE),'exports':summaries,'negative_checks':negative_checks,
            'negative_test_scope':'Copied paper CSVs only; six deliberate defects must be
rejected even after their recorded hashes are updated.',
            'visual_studio_gui':'pending','human_review':'pending'}
    _json(directory/'paper_csv_selfcheck_report.json',report)

print(json.dumps({'status':'PASS','questions':4,'deliberate_defects_rejected':len(negative_
checks),

'report':str(directory/'paper_csv_selfcheck_report.json')},ensure_ascii=False),flush=True)
    return 0

def main():
    parser=argparse.ArgumentParser(description=__doc__)
    parser.add_argument('--selfcheck',action='store_true')
    parser.add_argument('--paper-selfcheck',action='store_true')
    parser.add_argument('--output-dir',default='paper_output/results/export_selfcheck')
    args=parser.parse_args()

```

```

    if not args.selfcheck and not args.paper_selfcheck:
        parser.error('Use the Python API with final Runs, or --selfcheck for a real Q1
validation export')
    if Path.cwd().resolve() != ROOT:
        raise RuntimeError('Use the competition workspace as cwd')
    if args.paper_selfcheck:
        return _paper_csv_selfcheck(ROOT/args.output_dir)
    from drying_core import Settings, solve_case

    run = solve_case(Settings(question='Q1', intervals=40, face_scheme='kirchhoff',
rtol=1e-8, atol_temperature=1e-8, atol_moisture=1e-10, early_max_step_s=5.))
    record = export_question(run, 'Q1', ROOT/args.output_dir)
    checks = validate_exports([record], runs={'Q1': run})
    output = Path(args.output_dir).resolve()
    note = {'status': checks['status'], 'scope': 'Actual Q1 export selfcheck only, not final
production or numerical-accuracy acceptance',
'settings': asdict(run.model.settings), 'export': record, 'validation': checks}
    _json(output/'selfcheck_summary.json', note)

    print(json.dumps({'status': checks['status'], 'fully_verified_with_live_Run': checks['fully_ve
rified_with_live_Run'],

'output': str(output), 'size': record['size'], 'sheets': record['sheets']}, ensure_ascii=False, in
dent=2))
    return 0

if __name__ == '__main__':
    raise SystemExit(main())

```

D.20 production_provenance.py

路径: 05_数值检验与实验/Python 检验源码/paper_output/code/modeling/production_provenance.py

用途: 时间检验需要的源码/输入快照、执行监督和退出记录工具

使用范围: 检验与历史诊断

语言: Python; 源文件共 485 行。

SHA256: 4d0a9081eabbc9b0ac729dda22d8f626dc3411910b37124ce053b51029d4ec8d

```
from __future__ import annotations

from contextlib import contextmanager
from datetime import datetime, timezone
import copy
import hashlib
import importlib.metadata
import json
import os
from pathlib import Path
import platform
import subprocess
import sys
import uuid

SOURCE = Path(__file__).resolve()
ROOT = SOURCE.parents[3]
LOADED_CODE_SHA256 = hashlib.sha256(SOURCE.read_bytes()).hexdigest()

def utc_now():
    return datetime.now(timezone.utc).isoformat()

def sha256(path):
    digest = hashlib.sha256()
    with Path(path).open('rb') as stream:
        for block in iter(lambda: stream.read(1024 * 1024), b''):
            digest.update(block)
    return digest.hexdigest()

def within(root, path):
    root, path = Path(root).resolve(), Path(path).resolve()
    if not path.is_relative_to(root):
        raise ValueError('Path is outside the intended workspace: ' + str(path))
    return path

def file_record(path, root=ROOT):
    path = within(root, path)
    before = path.stat()
    digest = sha256(path)
    after = path.stat()
    if (before.st_size, before.st_mtime_ns) != (after.st_size, after.st_mtime_ns):
        raise RuntimeError('File changed while hashing: ' + str(path))
    return {'path': path.relative_to(Path(root).resolve()).as_posix(),
            'bytes': after.st_size, 'sha256': digest, 'exists': True}
```

```

def assert_records(records, root=ROOT):
    for expected in records:
        path = within(root, Path(root) / expected['path'])
        actual = file_record(path, root)
        for key in ('sha256', 'bytes'):
            if key in expected and actual[key] != expected[key]:
                raise RuntimeError('Artifact/input changed: ' + expected['path'])

def json_bytes(value):
    return (json.dumps(value, ensure_ascii=False, indent=2, allow_nan=False) +
'\n').encode('utf-8')

def atomic_bytes(path, data):
    path = Path(path)
    path.parent.mkdir(parents=True, exist_ok=True)
    temporary = path.with_name(path.name + '.tmp-' + uuid.uuid4().hex)
    try:
        with temporary.open('xb') as stream:
            stream.write(data)
            stream.flush()
            os.fsync(stream.fileno())
        os.replace(temporary, path)
    finally:
        if temporary.exists():
            temporary.unlink()

def write_json(path, value):
    atomic_bytes(path, json_bytes(value))

def exclusive_json(path, value):
    path = Path(path)
    path.parent.mkdir(parents=True, exist_ok=True)
    with path.open('xb') as stream:
        stream.write(json_bytes(value))
        stream.flush()
        os.fsync(stream.fileno())

def runtime_record():
    versions = {}
    for name in ('numpy', 'scipy', 'openpyxl', 'matplotlib'):
        versions[name] = importlib.metadata.version(name)
    return {'executable': sys.executable, 'implementation':
platform.python_implementation(),
        'version': platform.python_version(), 'platform': platform.platform(),
        'installed_library_versions': versions,
        'thread_environment': {name: os.environ.get(name) for name in
('OPENBLAS_NUM_THREADS', 'OMP_NUM_THREADS')},
        'scope': 'Installed distributions; actual imported module paths are recorded by
worker.'}

def pointer(document, json_pointer):
    current = document
    if not json_pointer.startswith('/'):
        raise ValueError('Expected an absolute JSON pointer')
    for token in json_pointer[1:].split('/'):
        token = token.replace('~1', '/').replace('~0', '~')
        current = current[int(token)] if isinstance(current, list) else current[token]

```

```

return current

def validate_metric_evidence(metrics, root=ROOT):
    cache = {}
    for metric in metrics:
        path = within(root, Path(root) / metric['evidence_path'])
        if path not in cache:
            cache[path] = json.loads(path.read_text(encoding='utf-8'))
        actual = pointer(cache[path], metric['evidence_json_pointer'])
        if actual != metric['value']:
            raise ValueError('Metric does not equal its saved evidence: ' +
metric['metric_name'])
    return {'status': 'PASS', 'checked_metric_count': len(metrics),
'scope': 'Exact equality to the identified saved JSON value, not physical
accuracy.'}

def figure_artifact_records(figure, root=ROOT):
    candidates = []
    primary = figure.get('path') or figure.get('output_path')
    if primary:
        candidates.append({key: figure[key] for key in ('path', 'bytes', 'sha256') if key
in figure}
                        if figure.get('path') else {'path': primary})
    artifacts = figure.get('artifacts', {})
    candidates.extend(artifacts.values() if isinstance(artifacts, dict) else artifacts)
    for name in ('plot_data', 'metadata'):
        if isinstance(figure.get(name), dict) and figure[name].get('path'):
            candidates.append(figure[name])
    checked = {}
    for expected in candidates:
        if not isinstance(expected, dict) or not expected.get('path'):
            raise ValueError('Invalid figure artifact record')
        actual = file_record(Path(root) / expected['path'], root)
        if not actual['bytes']:
            raise ValueError('Empty figure artifact: ' + actual['path'])
        if expected.get('sha256') and actual['sha256'] != expected['sha256']:
            raise RuntimeError('Existing figure hash mismatch: ' + actual['path'])
        checked[actual['path']] = actual
    if not checked:
        raise ValueError('Computed figure has no usable artifact')
    return list(checked.values())

def merge_figure_index(existing, new_figures, root=ROOT):
    merged, omitted, replaced, preserved = {}, [], [], []
    for figure in (existing or {}).get('figures', []):
        status = str(figure.get('status', '')).lower()
        if (figure.get('placeholder') or figure.get('planned') or
status in ('planned', 'pending', 'not_started') or figure.get('ok') is
False):
            omitted.append({'figure_id': figure.get('figure_id'), 'reason': 'not an
existing usable computed figure'})
            continue
        figure_artifact_records(figure, root)
        if not figure.get('figure_id'):
            raise ValueError('Existing computed figure needs figure_id')
        merged[figure['figure_id']] = copy.deepcopy(figure)
        preserved.append(figure['figure_id'])
    seen = set()
    for figure in new_figures:
        identity = figure['figure_id']

```

```

        if identity in seen:
            raise ValueError('Duplicate new figure_id: ' + identity)
        seen.add(identity)
        figure_artifact_records(figure, root)
        if identity in merged:
            replaced.append(identity)
        merged[identity] = copy.deepcopy(figure)
    return list(merged.values()), {'preserved_existing_ids': [v for v in preserved if v not
in replaced],
    'replaced_same_ids': replaced, 'omitted_noncomputed_entries': omitted,
    'prior_index_retained_in_publication_backup': True}

@contextmanager
def publication_lock(root, run_id):
    path = within(root, Path(root) / 'paper_output/results/.production_publish.lock')
    identity = {'run_id': run_id, 'pid': os.getpid(), 'created_at': utc_now(), 'token':
uuid.uuid4().hex}
    exclusive_json(path, identity)
    try:
        yield
    finally:
        if path.exists() and json.loads(path.read_text(encoding='utf-8')) == identity:
            path.unlink()

def publish_transaction(root, directory, updates, manifest):

    root, directory = Path(root).resolve(), within(root, directory)
    if manifest['runs'][0]['returncode'] != 0:
        raise ValueError('Cannot publish a nonzero worker return code')
    manifest_name = 'paper_output/results/run_manifest.json'
    if manifest_name in updates:
        raise ValueError('The manifest must be committed last')
    paths = [*updates, manifest_name]
    previous = {}
    backup = directory / 'publication_previous'
    for relative in paths:
        target = within(root, root / relative)
        previous[relative] = target.read_bytes() if target.exists() else None
        if previous[relative] is not None:
            atomic_bytes(backup / relative, previous[relative])
    write_json(backup / 'inventory.json', {'saved_at': utc_now(),
    'files': [{'path': p, 'previously_existed': b is not None} for p, b in
previous.items()]})
    intended = {relative: json_bytes(document) for relative, document in updates.items()}
    changed = False
    try:

        write_json(root / manifest_name, {'status': 'PUBLISHING', 'run_id':
manifest['runs'][0]['run_id'],
        'generated_at': utc_now(), 'version_directory':
directory.relative_to(root).as_posix()})
        changed = True
        for relative, content in intended.items():
            atomic_bytes(root / relative, content)
        for relative, content in intended.items():
            if sha256(root / relative) != hashlib.sha256(content).hexdigest():
                raise RuntimeError('Published contract bytes differ: ' + relative)

    assert_records(manifest['runs'][0]['output_artifacts'], root)

```

```

assert_records(manifest['runs'][0].get('input_files', []), root)
font = manifest['runs'][0].get('font')
if font and sha256(font['path']) != font['sha256']:
    raise RuntimeError('Font changed before publication commit')
write_json(directory / 'run_manifest.json', manifest)
write_json(root / manifest_name, manifest)
except Exception as error:
    rollback_errors = []
    if changed:

        for relative in updates:
            try:
                target = root / relative
                if previous[relative] is None:
                    if target.exists():
                        target.unlink()
                else:
                    atomic_bytes(target, previous[relative])
            except Exception as rollback_error:
                rollback_errors.append({'path': relative, 'error':
repr(rollback_error)})
        if not rollback_errors:
            try:
                target = root / manifest_name
                if previous[manifest_name] is None:
                    if target.exists():
                        target.unlink()
                else:
                    atomic_bytes(target, previous[manifest_name])
            except Exception as rollback_error:
                rollback_errors.append({'path': manifest_name, 'error':
repr(rollback_error)})
        if rollback_errors:
            write_json(root / manifest_name, {'status': 'FAILED_PUBLICATION', 'error':
repr(error),
                'rollback_errors': rollback_errors, 'generated_at': utc_now()})
            write_json(directory / 'publication_failure.json', {'status': 'FAIL', 'error':
repr(error),
                'rollback_errors': rollback_errors, 'prior_publication_restored': changed and
not rollback_errors,
                'generated_at': utc_now()})
            raise

def measured_process(command, cwd, log_path):

    started = utc_now()
    stream_error = None
    with Path(log_path).open('x', encoding='utf-8') as log:
        process = subprocess.Popen(command, cwd=cwd, stdout=subprocess.PIPE,
stderr=subprocess.STDOUT,
            text=True, encoding='utf-8', errors='replace')
        try:
            for line in process.stdout:
                log.write(line)
                log.flush()
                print(line, end='', flush=True)
        except BaseException as error:
            stream_error = error

    for unused in process.stdout:

```

```

        pass
    finally:

        code = process.wait()
        process.stdout.close()
    result = {'pid': process.pid, 'returncode': code, 'started_at': started,
              'finished_at': utc_now(), 'command': subprocess.list2cmdline(command),
              'argv': command, 'cwd': str(Path(cwd).resolve())}
    if stream_error is not None:
        result['stdout_capture_error'] = repr(stream_error)

    write_json(Path(log_path).with_name('process_result.json'), result)
    if stream_error is not None:
        raise RuntimeError('Child exited, but stdout capture failed') from stream_error
    return result


def entry_sequence_selfcheck(directory):

    from types import SimpleNamespace
    from unittest.mock import patch
    import numpy as np
    import run_modeling as entry

    source = file_record(entry.SOURCE)
    start = {'input_files': [source], 'runtime': {}, 'font': {}, 'worker_command': 'fixture
only; no model solve'}
    global_paths = [ROOT / p for p in entry.CONTRACT_TARGETS.values()]
    global_before = {str(p): sha256(p) if p.exists() else None for p in global_paths}
    active, events, identities = set(), [], {}
    class FixtureRun:
        def __init__(self, name, intervals):
            if active:
                raise AssertionError('Two Runs are retained simultaneously')
            self.name, self.intervals = name, intervals
            self.end_s = 1800. if name == 'Q1' else 15002.
            active.add(id(self))
            events.append('solve:' + name)
        def fields(self, times, material_x):
            shape = (len(times), len(material_x))
            return np.full(shape, 310.), np.full(shape, .1499)
        def close(self):
            events.append('close:' + self.name)
            active.remove(id(self))
    def save_fixture(run, output):
        return {'settings': {'intervals': run.intervals},
                'diagnostics': {'max_mass_balance_abs_kg_per_kg': 0.}}
    def fixture_completion(run):
        return {'reported_time_s': 15000.12, 'reported_drying_time_h': 15000.12 / 3600.,
                'critical_event_h': 15000. / 3600., 'max_C_at_reported_time': .1499}
    def export_fixture(run, qid, output):
        events.append('export:' + qid)
        identities[qid] = id(run)
        path = output / (qid + '_fixture.json')
        write_json(path, {'fixture_only': True, 'qid': qid})
        return {'question_id': qid, 'artifacts': [file_record(path)], 'size': {},
                'manifest_path': str(path)}
    def validate_fixture(exports, runs):
        export = exports[0]
        path = Path(export['manifest_path']).with_suffix('.validation.json')
        write_json(path, {'fixture_only': True})

```

```

        return {'status': 'PASS', 'fully_verified_with_live_Run': True, 'exports':
[{'report_path': str(path)}]}
    def plot_fixture(run, qid, output):
        events.append('plot:' + qid)
        path = output / (qid + '_fixture.bin')
        path.parent.mkdir(parents=True, exist_ok=True)
        path.write_bytes(b'fixture bytes; not a scientific figure')
        return {'figure_id': 'fixture_' + qid, 'question_id': qid, **file_record(path)}
    mappings = {'drying_core': SimpleNamespace(ROOT=ROOT, save_run=save_fixture),
        'q3_model': SimpleNamespace(completion=fixture_completion),
        'export_outputs': SimpleNamespace(export_question=export_fixture,
validate_exports=validate_fixture),
        'publication_plots': SimpleNamespace(make_question_plot=plot_fixture)}
    for key, name in [('Q1', 'q1_model'), ('Q23', 'q2_model'), ('Q4', 'q4_model')]:
        mappings[name] = SimpleNamespace(solve=lambda n, key=key: FixtureRun(key, n))
    original_import = importlib.import_module
    def importer(name, *args, **kwargs):
        return mappings[name] if name in mappings else original_import(name, *args,
**kwargs)
    args = SimpleNamespace(version='fixture_sequence', n1=2, n23=2, n4=2, review=False)
    with patch.object(entry, 'assert_dependencies'), patch.object(entry,
'imported_model_records', return_value=[]), \
        patch.object(entry.importlib, 'import_module', side_effect=importer):
        entry.worker(args, directory / 'entry_fixture_success', start)
        assert not active and identities['Q2'] == identities['Q3']
        assert events == ['solve:Q1', 'export:Q1', 'plot:Q1', 'close:Q1',
            'solve:Q23', 'export:Q2', 'plot:Q2', 'export:Q3', 'plot:Q3', 'close:Q23',
            'solve:Q4', 'export:Q4', 'plot:Q4', 'close:Q4']
        def fail_q3(run, qid, output):
            if qid == 'Q3':
                raise RuntimeError('injected fixture Q3 export failure')
            return export_fixture(run, qid, output)
        mappings['export_outputs'].export_question = fail_q3
        try:
            entry.worker(args, directory / 'entry_fixture_failure', start)
        except RuntimeError as error:
            assert 'injected fixture' in str(error)
        else:
            raise AssertionError('Fixture failure was not propagated')
        assert not active and events[-1] == 'close:Q23'
        global_after = {str(p): sha256(p) if p.exists() else None for p in global_paths}
        assert global_after == global_before
        return ['entry_one_run_at_a_time_and_Q2_Q3_same_identity',
            'entry_closes_run_after_export_failure',
'worker_does_not_publish_global_contracts']

```

```
def selfcheck():
```

```

    import tempfile
    from unittest.mock import patch

    base = ROOT / 'notes/A-modeling/2026-09-10/data/production_entry_contract_selfcheck'
    base.mkdir(parents=True, exist_ok=True)
    directory = Path(tempfile.mkdtemp(prefix='check-', dir=base))
    fixture = directory / 'fixture'
    fixture.mkdir()
    checks = []
    original = fixture / 'input.txt'
    original.write_text('observed bytes', encoding='utf-8')
    rec = file_record(original, fixture)
    assert_records([rec], fixture)
    original.write_text('changed bytes', encoding='utf-8')

```

```

try:
    assert_records([rec], fixture)
except RuntimeError:
    checks.append('hash_mutation_rejected')
else:
    raise AssertionError('Changed input was accepted')
lock = fixture / 'launch.json'
exclusive_json(lock, {'version': 'one'})
try:
    exclusive_json(lock, {'version': 'two'})
except FileExistsError:
    checks.append('duplicate_version_marker_rejected')
else:
    raise AssertionError('Existing marker overwritten')
evidence = fixture / 'summary.json'
write_json(evidence, {'metric_samples': {'Q3': {'C': [[.1499]], 'time_s': 123.48}}})
metric = {'metric_name': 'center_C', 'evidence_path': 'summary.json',
          'evidence_json_pointer': '/metric_samples/Q3/C/0/0', 'value': .1499}
validate_metric_evidence([metric], fixture)
metric['value'] = .1
try:
    validate_metric_evidence([metric], fixture)
except ValueError:
    checks.append('metric_mismatch_rejected')
else:
    raise AssertionError('Wrong evidence accepted')
image_path = fixture / 'observed.png'
image_path.write_bytes(b'fixture-artifact-only-not-a-real-figure')
figure = {'figure_id': 'observed', 'path': 'observed.png', 'status': 'computed',
          'sha256': sha256(image_path), 'placeholder': False}
merged, note = merge_figure_index({'figures': [figure, {'figure_id': 'planned',
'planned': True}], [], fixture)
assert len(merged) == 1 and note['preserved_existing_ids'] == ['observed']
checks.append('existing_artifact_preserved_planned_not_promoted')
old_contract = {'generation': 'old'}
write_json(fixture / 'paper_output/results/model_results.json', old_contract)
write_json(fixture / 'paper_output/results/run_manifest.json', {'status': 'PASS',
'old': True})
version = fixture / 'paper_output/results/production/test'
version.mkdir(parents=True)
updates = {'paper_output/results/model_results.json': {'generation': 'new'}}
intended_bytes = json_bytes(updates['paper_output/results/model_results.json'])
intended_record = {'path': 'paper_output/results/model_results.json', 'bytes':
len(intended_bytes),
                  'sha256': hashlib.sha256(intended_bytes).hexdigest()}
manifest = {'status': 'PASS', 'runs': [{'run_id': 'test', 'returncode': 0,
    'input_files': [file_record(original, fixture)], 'output_artifacts':
[intended_record]]}]
real_write = atomic_bytes
fired = False
def fail_contract_once(path, data):
    nonlocal fired
    if Path(path) == fixture / 'paper_output/results/model_results.json' and not fired:
        fired = True
        raise OSError('injected fixture publication failure')
    real_write(path, data)
with patch.object(sys.modules[__name__], 'atomic_bytes', fail_contract_once):
    try:
        publish_transaction(fixture, version, updates, manifest)
    except OSError:
        pass
    else:
        raise AssertionError('Injected failure was swallowed')

```

```

    assert json.loads((fixture / 'paper_output/results/model_results.json').read_text()) ==
old_contract
    assert json.loads((fixture / 'paper_output/results/run_manifest.json').read_text()) ==
{'status': 'PASS', 'old': True}
    checks.append('publication_failure_rolls_back_old_contract_and_manifest')
    publish_transaction(fixture, version, updates, manifest)
    assert json.loads((fixture / 'paper_output/results/model_results.json').read_text()) ==
{'generation': 'new'}
    checks.append('successful_publication_commits_manifest_last')
    exits = [measured_process([sys.executable, '-c', 'import sys; print("tiny-process");
sys.exit(' + str(code) + ')'],
                        fixture, directory / ('exit-' + str(code) + '.log')) for code
in (0, 7)]
    assert [p['returncode'] for p in exits] == [0, 7]
    checks.append('actual_process_zero_and_nonzero_exits_observed')
    checks.extend(entry_sequence_selfcheck(directory))
    report = {'status': 'PASS', 'scope': 'Isolated tiny-file contracts; no model solve or
real production publication.',
            'checks': checks, 'processes': exits, 'source': file_record(SOURCE),
'generated_at': utc_now(),
            'visual_studio_gui': 'pending', 'human_review': 'pending'}
    write_json(directory / 'selfcheck.json', report)
    print(json.dumps({'status': report['status'], 'checks': len(checks), 'report':
str(directory / 'selfcheck.json')}))

if __name__ == '__main__':
    if sys.argv[1:] != ['--selfcheck']:
        raise SystemExit('Use --selfcheck for isolated contract validation only')
    if Path.cwd().resolve() != ROOT:
        raise SystemExit('Use the competition workspace as cwd')
    selfcheck()

```

D.21 publication_plots.py

路径: 05_数值检验与实验/Python 检验源码/paper_output/code/modeling/publication_plots.py

用途: run_modeling 动态加载的原绘图模块; 依赖补齐

使用范围: 检验与历史诊断

语言: Python; 源文件共 395 行。

SHA256: dac6c1197faf496c6b09cab833e510a092ca615e2d9b2df4d49165a521d2a944

```
from __future__ import annotations

import argparse
from dataclasses import asdict
from datetime import datetime, timezone
import hashlib
import inspect
import json
import os
from pathlib import Path
import sys

import numpy as np

ROOT=Path(__file__).resolve().parents[3]
SOURCE=Path(__file__).resolve()
LOADED_PLOT_SHA256=hashlib.sha256(SOURCE.read_bytes()).hexdigest()
os.environ.setdefault('MPLCONFIGDIR',str(ROOT/'tmp/cache/matplotlib'))
import matplotlib
matplotlib.use('Agg')
from matplotlib import font_manager
from matplotlib.colors import Normalize
from matplotlib.ticker import NullLocator
import matplotlib.pyplot as plt

FONT_PATH=Path('C:/Windows/Fonts/msyh.ttc')
if not FONT_PATH.exists():
    raise RuntimeError('Verified Chinese font Microsoft YaHei is unavailable')
font_manager.fontManager.addfont(str(FONT_PATH))
FONT_NAME=font_manager.FontProperties(fname=str(FONT_PATH)).get_name()
STYLE={'font.family':FONT_NAME,'font.size':10,'axes.titlesize':11,'axes.labelsize':10,
        'axes.unicode_minus':False,'axes.spines.top':False,'axes.spines.right':False,
        'legend.frameon':False,'pdf.fonttype':42,'ps.fonttype':42,'svg.fonttype':'path',
        'savefig.facecolor':'white','figure.facecolor':'white','axes.facecolor':'white'}
COLOURS={'centre':'#2563A6','mean':'#C76829','surface':'#218573','threshold':'#51555D'}

def _sha(path):
    h=hashlib.sha256()
    with Path(path).open('rb') as f:
        for b in iter(lambda:f.read(1024*1024),b''):h.update(b)
    return h.hexdigest()

def _record(path):
    p=Path(path).resolve()
    return
{'path':p.relative_to(ROOT).as_posix(),'bytes':p.stat().st_size,'sha256':_sha(p),'exists':True}
```

```

def _directory(directory):
    p=Path(directory).resolve()
    if not p.is_relative_to(ROOT) or p.is_relative_to(ROOT/'problem_files'):
        raise ValueError('Write figures only inside the competition workspace outside
original inputs')
    p.mkdir(parents=True,exist_ok=True)
    return p

def _run_source(run):
    module=sys.modules[run.model.__class__.__module__]
    path=Path(inspect.getfile(run.model.__class__)).resolve()
    if getattr(module,'LOADED_CODE_SHA256',_sha(path))!=_sha(path):
        raise RuntimeError('Core changed after import; regenerate figure from a consistent
Run')
    records=[_record(path)]
    jac=path.with_name('analytic_jacobian.py')
    if jac.exists():records.append(_record(jac))
    for rec in run.model.input_records:
        if _sha(ROOT/rec['path'])!=rec['sha256']:
            raise RuntimeError('Run input changed: '+rec['path'])
        records.append(rec)
    return {'kind':'live_Run','settings':asdict(run.model.settings),'files':records,
        'event_s':None if run.event_s is None else
float(run.event_s),'end_s':float(run.end_s)}

def _unchanged(sources):
    if _sha(SOURCE)!=LOADED_PLOT_SHA256:
        raise RuntimeError('Plot module changed after import')
    for source in sources:
        for rec in source.get('files',[]):
            if _sha(ROOT/rec['path'])!=rec['sha256']:
                raise RuntimeError('Source changed during figure generation: '+rec['path'])

def _save(fig,figure_id,qid,title,caption,directory,sources,data,extra=None):
    directory=_directory(directory)
    _unchanged(sources)
    data_path=directory/(figure_id+'_data.npz')
    np.savez_compressed(data_path,**data)
    artifacts={}
    for ext in ['png','svg','pdf']:
        path=directory/(figure_id+'.'+ext)
        fig.savefig(path,dpi=210 if ext=='png' else 160,bbox_inches='tight',pad_inches=.14)
        artifacts[ext]=_record(path)
    plt.close(fig)
    _unchanged(sources)
    record={'figure_id':figure_id,'question_id':qid,'title':title,'caption':caption,
        'purpose':caption,'path':artifacts['png']['path'],
        'bytes':artifacts['png']['bytes'],'sha256':artifacts['png']['sha256'],
        'status':'computed','ok':True,'placeholder':False,'exists':True,
        'artifacts':artifacts,'plot_data':_record(data_path),'sources':sources,
        'generated_by':'paper_output/code/modeling/publication_plots.py',
        'source_code_sha256':LOADED_PLOT_SHA256,
        'generated_at':datetime.now(timezone.utc).isoformat(),
        'font':{'name':FONT_NAME,'file':str(FONT_PATH)},

'visual_review':'pending','human_review':'pending','visual_studio_gui':'pending'}
    if extra:record.update(extra)
    metadata=directory/(figure_id+'.json')
    metadata.write_text(json.dumps(record,ensure_ascii=False,indent=2,allow_nan=False))

```

```

+ '\n',encoding='utf-8')
    record['metadata']=_record(metadata)
    return record

def _footer(fig,text):

fig.text(.5,.015,text,ha='center',va='bottom',fontsize=8.1,color='#52565E',linespacing=1.45
)

def _environment_note(run):
    s=run.model.settings
    if s.boundary_extension=='nominal':
        return f'4 h 后环境延拓为 {s.tail_temperature_C:g} °C、等效平衡含水率
{s.tail_equilibrium*s.equilibrium_scale:g} kg/kg'
    if s.boundary_extension=='tail_mean':return '4 h 后环境按末 1 h 观测均值延拓'
    return '4 h 后环境按最后观测值延拓'

def _profiles(run,qid,directory):
    source=_run_source(run)
    if qid=='Q1':
        ts=np.array([100.,300.,600.,900.,1200.,1500.,1800.])
        labels=[f'{t:g} s' for t in ts]
        title='第一问：预热阶段的径向温湿分布'
    else:
        ts=np.arange(1,7,dtype=float)*1800
        labels=[f'{t/3600:g} h' for t in ts]
        title='第二问：前三小时的径向温湿分布'
    if run.end_s<ts[-1]-1e-7:
        raise ValueError('Run does not cover all question-specific plot times')
    rs=np.linspace(0,.02,121)
    T,C=run.fields(ts,radii_m=rs);T=T-273.15
    with plt.rc_context(STYLE):
        fig,axes=plt.subplots(1,2,figsize=(11.8,4.8))
        fig.subplots_adjust(left=.075,right=.97,bottom=.245,top=.82,wspace=.28)
        colours=plt.get_cmap('viridis')(np.linspace(.08,.9,len(ts)))
        handles=[]
        for i,(label,colour) in enumerate(zip(labels,colours)):
            handle,=axes[0].plot(rs*100,T[i],color=colour,lw=1.7,label=label)
            handles.append(handle)
            axes[1].plot(rs*100,C[i],color=colour,lw=1.7,label=label)
        for ax in axes:
            ax.set_xlim(0,2);ax.set_xticks([0,.5,1,1.5,2]);ax.set_xlabel('到药材中心的距离 /
cm')
            ax.grid(alpha=.2,lw=.7)
        axes[0].set(title='(a) 温度',ylabel='温度 / °C')
        axes[1].set(title='(b) 干基含水率',ylabel='干基含水率 / (kg/kg)')
        fig.suptitle(title,y=.97,fontsize=14,fontweight='bold')
        fig.legend(handles,labels,loc='lower
center',bbox_to_anchor=(.5,.075),ncol=len(labels),fontsize=9)
        _footer(fig,'一维径向有效模型的条件性数值结果；忽略端面与显式潜热，环境水分按等效平衡边界
处理。')
    return _save(fig,f'fig_{qid.lower()}_profiles',qid,title,
        '各曲线对应题目指定时刻。温度采用摄氏度，水分采用干基；曲线来自同一 live Run 的物理径向
点值。',
        directory,[source],{'times_s':ts,'radii_m':rs,'temperature_C':T,'C':C},
        {'sample_time_count':len(ts),'sample_radius_count':len(rs)})

def _drying(run,qid,directory):
    source=_run_source(run)

```

```

if run.event_s is None:
    raise ValueError('Full drying figure requires an actual threshold event')
if qid=='Q4' and not run.model.settings.shrink:
    raise ValueError('Q4 primary figure must show the shrinking-domain run')
radius_knots=run.model.rad['time_s'] if qid=='Q4' else np.array([])
radius_knots=radius_knots[radius_knots<=run.end_s]
ts=np.unique(np.r_[np.linspace(0,run.end_s,241),
                    np.arange(0,min(run.end_s,14400)+1,600),
                    radius_knots,run.event_s,run.end_s])
rs=np.linspace(0,.02,121)
_,C=run.fields(ts,radii_m=rs)
R=np.asarray(run.model.radius(ts))
inside=rs[None,:]<=R[:,None]+1e-12
C=np.where(inside,C,np.nan)
_,ends=run.fields(ts,material_x=[0.,1.])
state=run.state(ts)
mean=2*run.model.w@state[1:-1:2]
max_C=np.max(state[1:-1:2],axis=0)
del state
t_h=ts/3600;event_h=run.event_s/3600
title=('第三问：固定半径下的全程水分迁移' if qid=='Q3' else '第四问：收缩域内的全程水分迁移')
with plt.rc_context(STYLE):
    fig,axes=plt.subplots(1,2,figsize=(12,5.1),gridspec_kw={'width_ratios':[1.08,1]})
    fig.subplots_adjust(left=.065,right=.965,bottom=.21,top=.83,wspace=.31)
    cmap=plt.get_cmap('YlGnBu').copy();cmap.set_bad('white')
    mesh=axes[0].pcolormesh(t_h,rs*100,np.ma.masked_invalid(C.T),shading='auto',
                           cmap=cmap,norm=Normalize(vmin=.05,vmax=2.55),rasterized=True)
    cb=fig.colorbar(mesh,ax=axes[0],pad=.025,fraction=.05)
    cb.set_label('干基含水率 / (kg/kg)',fontsize=9)
    if qid=='Q4':

        axes[0].fill_between(t_h,R*100,2,color='white',zorder=3)
        axes[0].plot(t_h,R*100,color='#202631',lw=1.8,label='药材表面 R(t)',zorder=4)
        axes[0].text(.65*t_h[-1],1.88,'域外（非药
材）',ha='center',va='center',fontsize=9,color='#62666D')
        axes[0].legend(loc='lower
right',fontsize=8.5,facecolor='white',framealpha=.85,frameon=True)
        axes[0].set(xlim=(0,t_h[-1]),ylim=(0,2),xlabel='烘干时间 / h',
                    ylabel='物理径向距离 / cm',title='（a）物理空间中的含水率')
        for series,name,colour in [(ends[:,0],'中心',COLOURS['centre']),
                                   (mean,'干质量加权均值',COLOURS['mean']),
                                   (ends[:,1],'表面',COLOURS['surface'])]:
            axes[1].plot(t_h,series,lw=1.8,label=name,color=colour)
        axes[1].axhline(.15,color=COLOURS['threshold'],ls='--',lw=1.25,label='阈值 0.15
kg/kg')
        axes[1].axvline(event_h,color='#6D6473',ls=':',lw=1.2)
        axes[1].annotate(f'全域临界时刻\n{event_h:.3f} h',xy=(event_h,.15),
                        xytext=(.67*t_h[-1],.64),arrowprops={'arrowstyle':'-
>','color':'#625A6A'},
                        fontsize=9,ha='center',va='center',color='#514956')
        axes[1].set(xlim=(0,t_h[-1]*1.015),ylim=(0,2.66),xlabel='烘干时间 / h',
                    ylabel='干基含水率 / (kg/kg)',title='（b）中心、均值与表面的变化')
        axes[1].legend(loc='upper right',fontsize=8.8);axes[1].grid(alpha=.18)
        if t_h[-1]>4:
            axes[0].axvline(4,color='#666A73',ls=':',lw=.85,alpha=.8)
            axes[0].text(4+.025*t_h[-1],.13,'4 h',fontsize=8,color='#52545A')
            fig.suptitle(title,y=.97,fontsize=14,fontweight='bold')
            footer='一维径向有效模型；均值按干物质质量加权，达标条件取全域最大值。'
            \n'+_environment_note(run)+'。'
            if qid=='Q4' and run.end_s>259200:
                footer+='半径超过 72 h 后采用末值延拓。'
            _footer(fig,footer)

```

```

        return _save(fig,f'fig_{qid.lower()}_drying',qid,title,
            '左图为物理半径与时间的水分场；第四问域外留白，黑线为实测半径轨迹的插值。右图
            显示中心、干质量加权均值、表面及全域阈值事件。',
            directory,[source],{'times_s':ts,'radii_m':rs,'radius_m':R,'C':C,
            'centre_C':ends[:,0],'mean_C':mean,'surface_C':ends[:,1],'max_C':max_C},
            {'event_s':float(run.event_s),'post_verification_s':float(run.end_s),
            'masked_outside_count':int(np.count_nonzero(~inside)),
            'sample_time_count':len(ts),'sample_radius_count':len(rs),
            'mean_definition':'2*sum(material_dual_cell_weights*C); dry-solid
            mass weighting'})

def make_plots(runs,directory):

    run1=runs['Q1'];run23=runs.get('Q23',runs.get('Q2'));run4=runs['Q4']
    if run23 is None:raise ValueError('Provide the shared Q2/Q3 Run under Q23 or Q2')
    return [_profiles(run1,'Q1',directory),_profiles(run23,'Q2',directory),
            _drying(run23,'Q3',directory),_drying(run4,'Q4',directory)]

def _saved_summary(path):
    path=Path(path).resolve()
    content=path.read_bytes()
    rec=_record(path)
    if hashlib.sha256(content).hexdigest()!=rec['sha256']:
        raise RuntimeError('Evidence file changed while being read')
    data=json.loads(content)
    diag=data['diagnostics']
    if not diag.get('solver_success') or not diag.get('strictly_dry_at_end') or
    diag.get('event_h') is None:
        raise ValueError('Comparison requires a completed successful drying run')

    src={'kind':'saved_completed_run_summary','files':[rec],
        'recorded_solver_code':data.get('code'),'settings':data['settings'],
        'current_solver_equivalence_not_assumed':True}
    return data,src

def make_shrinkage_comparison(fixed_summary_path,shrunk_summary_path,directory):

    fixed,sf=_saved_summary(fixed_summary_path);shrunk,ss=_saved_summary(shrunk_summary_path)
    a,b=fixed['settings'],shrunk['settings']
    if a['question']!='Q4' or b['question']!='Q4' or a['shrink'] or not b['shrink']:
        raise ValueError('Require matched appendix-4 fixed and shrinking runs')
    differences={k for k in set(a)|set(b) if a.get(k)!=b.get(k)}
    if differences!='shrink' or fixed.get('code',{}).get('sha256')!=shrunk.get('code',
    {}).get('sha256'):
        raise ValueError('A pure shrinkage comparison needs identical settings/source
    except shrink')
    values=np.array([fixed['diagnostics']['event_h'],shrunk['diagnostics']['event_h']])
    N=a['intervals'];reduction=100*(1-values[1]/values[0])
    title='同附录4物性下的固定半径与收缩对照'
    with plt.rc_context(STYLE):

fig,ax=plt.subplots(figsize=(8.2,4.7));fig.subplots_adjust(left=.2,right=.92,bottom=.22,top
=.81)
    ax.barh([1,0],values,color=['#8894A3','#218573'],height=.5)
    ax.set_yticks([1,0],['固定半径 2 cm','采用附件2收缩'])
    ax.set_xlim(0,values.max()*1.25);ax.set_xlabel('全域临界烘干时间 /
h');ax.grid(axis='x',alpha=.18)
    for y,v in zip([1,0],values):ax.text(v+values.max()*0.02,y,f'{v:.3f}')

```

```

h',va='center',fontsize=10)
    ax.text(.98,.12,f' 本对照缩短 {reduction:.2f}
%',transform=ax.transAxes,ha='right',color='#246653',fontsize=11)
    fig.suptitle(title,y=.95,fontsize=14,fontweight='bold')
    _footer(fig,f' 同一已保存计算版本, N={N}, 除收缩开关外设置一致; 这是有效模型中的尺寸效应。
\n 不可把第三、四问同时改变物性的时长差直接归因于收缩。')
    return _save(fig,'fig_q4_shrinkage_control','Q4',title,
        f' 已保存同物性、同参数、同源码版本 N={N}对照; 读取来源原始哈希, 未假设与后续
        细网格正式结果相同。',
        directory,[sf,ss],{'event_hours':values},
        {'intervals':N,'same_property_control':True,'reduction_percent':float(reduction)})

def make_convergence_plot(report_paths,directory):

    sources=[];groups=[]
    for path in report_paths:

        path=Path(path).resolve();content=path.read_bytes();data=json.loads(content);rec=_record(pa
        th)
        if hashlib.sha256(content).hexdigest()!=rec['sha256']:raise RuntimeError('Changing
        convergence report')
        if not data.get('finished_at') or not
        str(data.get('status','')).startswith('computed'):
            raise ValueError('Select only finished convergence reports; in-progress reports
            are not plotted')
        groups.append(data);sources.append({'kind':'finished_convergence_report','files':
        [rec]})
        if not groups:raise ValueError('At least one finished convergence report is required')
        labels={'Q1':'第一问','Q23':'第二、三问','Q4':'第四问'}
        colours={'Q1':'#2563A6','Q23':'#C76829','Q4':'#218573'}
        title='相邻网格结果差异与临界时刻收敛'
        plotted={}
        with plt.rc_context(STYLE):

            fig,axes=plt.subplots(1,2,figsize=(11.5,4.8));fig.subplots_adjust(left=.08,right=.965,botto
            m=.21,top=.82,wspace=.32)
            for group in groups:
                q=group['question'];comp=group['comparisons']
                x=np.array([c['fine_N'] for c in comp],dtype=float)
                y=np.array([c['max_absolute_difference']['C'] for c in comp],dtype=float)
                axes[0].loglog(x,y,'o-',color=colours.get(q),label=labels.get(q,q),lw=1.6)
                timed=[c for c in comp if c.get('event_difference_s') is not None]
                if timed:
                    axes[1].loglog([c['fine_N'] for c in timed],[abs(c['event_difference_s'])
                    for c in timed],
                                'o-',color=colours.get(q),label=labels.get(q,q),lw=1.6)
                plotted[q+'_fine_N']=x;plotted[q+'_C_difference']=y
            axes[0].set(title=' (a) 共同物理采样点的最大含水率差',xlabel='较细网格的区间数
            N',ylabel='相邻网格最大差 / (kg/kg)')
            axes[1].set(title=' (b) 全域临界时刻的变化',xlabel='较细网格的区间数 N',ylabel='相邻网
            格临界时刻差的绝对值 / s')
            all_ticks=sorted({c['fine_N'] for g in groups for c in g['comparisons']})
            time_ticks=sorted({c['fine_N'] for g in groups for c in g['comparisons'] if
            c.get('event_difference_s') is not None})
            axes[0].set_xticks(all_ticks,[str(n) for n in all_ticks])
            if time_ticks:axes[1].set_xticks(time_ticks,[str(n) for n in time_ticks])
            for ax in axes:
                ax.xaxis.set_minor_locator(NullLocator())
                ax.grid(which='both',alpha=.2);ax.legend(fontsize=9)
            fig.suptitle(title,y=.97,fontsize=14,fontweight='bold')
            _footer(fig,'读取已完成检验报告并记录来源哈希; 相邻网格差用于数值收敛检验, 不是实测预测误

```

```

差。')
    return _save(fig,'fig_model_grid_convergence','Q3',title,
        '横轴为相邻比较中较细网格的区间数；所画为实际计算的最大含水率差和阈值时刻差，
        不把相邻差直接称作严格误差上界。',
        directory,sources,plotted,{ 'question_ids':[g['question'] for g in
groups]})

def make_sensitivity_plot(summary_paths,directory,question_id='Q4',baseline_label='基准'):
    if baseline_label not in summary_paths:raise ValueError('An explicit baseline label is
required')
    labels=list(summary_paths);rows=[];sources=[]
    for label,path in summary_paths.items():
        data,source=_saved_summary(path);rows.append(data);sources.append(source)
    if len({r['settings']['intervals'] for r in rows})!=1 or len({r['settings']['question']
for r in rows})!=1:
        raise ValueError('Sensitivity cases must have a common question and grid size')
    expected={'Q4'} if question_id=='Q4' else {'Q2','Q3','Q23'}
    if rows[0]['settings']['question'] not in expected:
        raise ValueError('Sensitivity figure label does not match the saved physical
question')
    if len({r.get('code',{}).get('sha256') for r in rows})!=1:
        raise ValueError('Sensitivity cases must use the same saved source version')
    values=np.array([r['diagnostics']['event_h'] for r in
rows]);baseline=values[labels.index(baseline_label)]
    delta=100*(values/baseline-1);N=rows[0]['settings']['intervals']
    title=('第四问' if question_id=='Q4' else '第二、三问')+': 物理假设与参数情景比较'
    with plt.rc_context(STYLE):
        fig,ax=plt.subplots(figsize=(9.4,max(4.7,.48*len(labels)+1.8)))
        fig.subplots_adjust(left=.25,right=.93,bottom=.2,top=.82)
        pos=np.arange(len(labels));colours=['#8A949F' if x==baseline_label else '#218573'
for x in labels]

    ax.barh(pos,values,color=colours,height=.58);ax.set_yticks(pos,labels);ax.invert_yaxis()
        ax.set_xlim(0,values.max()*1.33);ax.set_xlabel('全域临界烘干时间 /
h');ax.grid(axis='x',alpha=.18)
        for y,v,d in zip(pos,values,delta):ax.text(v+values.max()*0.015,y,f'{v:.3f} h ({d:
+.2f}%)',va='center',fontsize=9)
        fig.suptitle(title,y=.95,fontsize=14,fontweight='bold')
        _footer(fig,f'同一已保存求解版本，N={N}；百分比相对“{baseline_label}”。这些是指定情景的
模型变化，不是统计置信区间。')
    return _save(fig,f'fig_{question_id.lower()}
_physical_sensitivity',question_id,title,
        '读取明确列出的已完成情景，并记录每份摘要哈希、设置和原计算版本；不将情景差异
当作观测误差。',
        directory,sources,
{'event_hours':values,'relative_change_percent':delta},

{'scenario_labels':labels,'baseline_label':baseline_label,'intervals':N})

def make_question_plot(run, question_id, directory):
    if question_id in ('Q1', 'Q2'):
        return _profiles(run, question_id, _directory(directory))
    if question_id in ('Q3', 'Q4'):
        return _drying(run, question_id, _directory(directory))
    raise ValueError('A question plot requires Q1, Q2, Q3 or Q4')

def main():
    parser=argparse.ArgumentParser(description=__doc__);parser.add_argument('--

```

```

selfcheck',action='store_true')

parser.add_argument('--output-dir',default='paper_output/results/plot_selfcheck');args=parser.parse_args()
if not args.selfcheck:parser.error('Use make_plots with live Runs or --selfcheck')
if Path.cwd().resolve()!=ROOT:raise RuntimeError('Run from the competition workspace')
from drying_core import Settings,solve_case
runs={}
for q in ['Q1','Q23','Q4']:
    print('Selfcheck: solving '+q+' with N=40',flush=True)

runs[q]=solve_case(Settings(question=q,intervals=40,face_scheme='kirchhoff',shrink=q=='Q4',
    rtol=1e-7,atol_temperature=1e-7,atol_moisture=1e-9))
    output=_directory(ROOT/args.output_dir)
    records=make_plots(runs,output)
    fixed=ROOT/'paper_output/results/experiments/physical_Q4_fixed_N200_K/summary.json'
    shrunk=ROOT/'paper_output/results/experiments/grids_Q4_N200_K/summary.json'
    if fixed.exists() and
shrunk.exists():records.append(make_shrinkage_comparison(fixed,shrunk,output))
    reports=[ROOT/'paper_output/results/convergence/{q}_K_analytic_J_v3/
convergence_report.json' for q in ['Q1','Q23']]
    if all(p.exists() for p in
reports):records.append(make_convergence_plot(reports,output))
    scenario_root=ROOT/'paper_output/results/experiments'
    scenarios={'基准':shrunk,'4 h 后环境 49
°C':scenario_root/'physical_Q4_T49_N200_K/summary.json',
        '4 h 后环境 51 °C':scenario_root/'physical_Q4_T51_N200_K/summary.json',
        '潜热能量负荷压力测试':scenario_root/'physical_Q4_latent_N200_K/summary.json'}
    if all(p.exists() for p in
scenarios.values()):records.append(make_sensitivity_plot(scenarios,output))
    report={'scope':'N40 live-run plot selfcheck; primary plots are not final numerical
results. Optional comparisons preserve their independent N200/finished-report sources.',
        'status':'GENERATED_PENDING_VISUAL_REVIEW','figures':records}

(output/'plot_selfcheck_report.json').write_text(json.dumps(report,ensure_ascii=False,indent=2,allow_nan=False)+'\n',encoding='utf-8')

print(json.dumps({'status':report['status'],'figure_count':len(records),'output':str(output)}),ensure_ascii=False,flush=True)
    return 0

if __name__=='__main__':raise SystemExit(main())

```

D.22 q1_model.py

路径: 05_数值检验与实验/Python 检验源码/paper_output/code/modeling/q1_model.py

用途: 问题一参数封装; 时间误差检验依赖

使用范围: 检验与历史诊断

语言: Python; 源文件共 7 行。

SHA256: 7bed421147579095319aad785ee6f66328dc1d13a154db5472b9b98f8e736742

```
from drying_core import Settings, solve_case

def solve(intervals=3200):
    return solve_case(Settings(question='Q1', intervals=intervals, face_scheme='kirchhoff',
                                rtol=1e-10, atol_temperature=1e-10, atol_moisture=1e-12,
                                early_max_step_s=2., max_step_s=120.))
```

D.23 q2_model.py

路径: 05_数值检验与实验/Python 检验源码/paper_output/code/modeling/q2_model.py

用途: 问题二/三共享轨迹参数封装; 时间误差检验依赖

使用范围: 检验与历史诊断

语言: Python; 源文件共 7 行。

SHA256: 5c27d23b1168c445591bf0e50f8f003d11b65e7a50c6156336285236f6e21adc

```
from drying_core import Settings, solve_case

def solve(intervals=3200):
    return solve_case(Settings(question='Q23', intervals=intervals,
    face_scheme='kirchhoff',
        rtol=1e-10, atol_temperature=1e-10, atol_moisture=1e-12,
        early_max_step_s=2., max_step_s=120.))
```

D.24 q3_model.py

路径: 05_数值检验与实验/Python 检验源码/paper_output/code/modeling/q3_model.py

用途: 全域含水率达标及严格报告格点; 灵敏度等检验依赖

使用范围: 检验与历史诊断

语言: Python; 源文件共 26 行。

SHA256: 6fb1084ef3bfe3976ff52f26760574650d3d5389f93b8b09983e528cc205c808

```
import math
import numpy as np

def completion(run):
    if run.event_s is None:
        raise ValueError('No full-domain drying event within the solved horizon')

    count = math.ceil(run.event_s / 3600. * 10000.)
    while True:
        report_h = count/10000.
        t = report_h*3600.
        if t > run.end_s:
            raise RuntimeError('Four-decimal reporting time is outside the verified
trajectory')
        values = run.state([t])[1:-1:2,0]
        if float(np.max(values)) < .15:
            break
        count += 1
    return {'critical_event_s':run.event_s,'critical_event_h':run.event_s/3600.,
        'reported_drying_time_h':report_h,'reported_time_s':t,
        'max_C_at_reported_time':float(np.max(values)),
        'slowest_material_coordinate':float(run.model.x[np.argmax(values)]),
        'conservative_post_verification_s':run.end_s,
        'rounding_convention':'upward on 0.0001 h grid, followed by actual strict threshold
check',
        'interpretation':'A conditional numerical event, not a confidence bound on physical
drying time.'}
```

D.25 q4_model.py

路径: 05_数值检验与实验/Python 检验源码/paper_output/code/modeling/q4_model.py

用途: 问题四附录 4 物性与径向收缩参数封装; 时间误差检验依赖

使用范围: 检验与历史诊断

语言: Python; 源文件共 7 行。

SHA256: e78b37fde9bcab5c196c7ba6bd2033f48889d07be00fd8395db7957ab1c905

```
from drying_core import Settings, solve_case

def solve(intervals=6400):
    return solve_case(Settings(question='Q4', intervals=intervals, face_scheme='kirchhoff',
                                shrink=True, rtol=1e-10, atol_temperature=1e-10, atol_moisture=1e-12,
                                early_max_step_s=2., max_step_s=120.))
```

D.26 run_experiments.py

路径: 05_数值检验与实验/Python 检验源码/paper_output/code/modeling/run_experiments.py

用途: 带明确预算的网格、参数、物理情景实验入口

使用范围: 检验与历史诊断

语言: Python; 源文件共 73 行。

SHA256: 51a5747e45ee8098falcb36b6b2388bcb0e10ff11da9cc2b7b33baa9491f7e92

```
from __future__ import annotations
import argparse
from dataclasses import replace
from datetime import datetime, timezone
import json
from pathlib import Path
import sys
import traceback

from drying_core import ROOT, Settings, solve_case, save_run

def main():
    parser = argparse.ArgumentParser()
    parser.add_argument('--batch', choices=['baseline', 'grids', 'temporal', 'physical'],
default='baseline')
    parser.add_argument('--n', type=int, default=100)
    parser.add_argument('--tag', default='')
    parser.add_argument('--face-scheme', choices=['harmonic', 'kirchhoff'],
default='harmonic')
    args = parser.parse_args()
    base = Settings(intervals=args.n, face_scheme=args.face_scheme)
    if args.batch in ('baseline', 'grids'):
        candidates = [('Q1', replace(base, question='Q1')), ('Q23', base),
('Q4', replace(base, question='Q4', shrink=True))]
    elif args.batch == 'temporal':
        candidates = [('Q23_tight', replace(base, rtol=1e-9, atol_moisture=1e-11,
atol_temperature=1e-9, max_step_s=150., early_max_step_s=10.)),
('Q4_tight', replace(base, question='Q4', shrink=True, rtol=1e-9,
atol_moisture=1e-11, atol_temperature=1e-9, max_step_s=150.,
early_max_step_s=10.))]
    else:
        candidates = [('Q4_fixed', replace(base, question='Q4', shrink=False)),
('Q23_last', replace(base, boundary_extension='last')),
('Q23_mean', replace(base, boundary_extension='tail_mean')),
('Q23_T49', replace(base, tail_temperature_C=49.)),
('Q23_T51', replace(base, tail_temperature_C=51.)),
('Q23_eq045', replace(base, tail_equilibrium=.045)),
('Q23_eq055', replace(base, tail_equilibrium=.055)),
('Q4_T49', replace(base, question='Q4', shrink=True,
tail_temperature_C=49.)),
('Q4_T51', replace(base, question='Q4', shrink=True,
tail_temperature_C=51.)),
('Q23_beta08', replace(base, beta=6.4e-7)),
('Q23_beta12', replace(base, beta=9.6e-7)),
('Q23_latent', replace(base, surface_latent_fraction=1.)),
('Q4_latent', replace(base, question='Q4', shrink=True,
surface_latent_fraction=1.))]
    out = ROOT/'paper_output/results/experiments'
    out.mkdir(parents=True, exist_ok=True)
```

```

failed = 0
for name, settings in candidates:
    trial_id = f'{args.batch}_{name}_N{args.n}{args.tag}'
    folder = out/trial_id
    if (folder/'summary.json').exists():
        raise FileExistsError(f'Refusing to overwrite previous trial: {trial_id}')
    folder.mkdir(parents=True, exist_ok=True)
    start = datetime.now(timezone.utc).isoformat()
    print(f'START {trial_id} {start}', flush=True)
    try:
        run = solve_case(settings)
        summary = save_run(run, folder)
        record = {'trial_id':trial_id, 'started_at':start,
                  'finished_at':datetime.now(timezone.utc).isoformat(),
                  'status':'computed', 'exit_code':0, **summary['diagnostics']}
    except Exception as exc:
        failed += 1
        (folder/'failure.txt').write_text(traceback.format_exc(), encoding='utf-8')
        record = {'trial_id':trial_id, 'started_at':start, 'status':'failed',
                  'exit_code':1, 'error':str(exc)}
        (folder/'trial_record.json').write_text(json.dumps(record, ensure_ascii=False,
indent=2), encoding='utf-8')
        with (out/'trials.jsonl').open('a',encoding='utf-8') as stream:
            stream.write(json.dumps(record, ensure_ascii=False)+'\n')
        print(json.dumps(record, ensure_ascii=False), flush=True)
    return 1 if failed else 0

if __name__ == '__main__':
    sys.exit(main())

```

D.27 run_modeling.py

路径: 05_数值检验与实验/Python 检验源码/paper_output/code/modeling/run_modeling.py

用途: production_provenance 自检引用的原监督/求解入口; 依赖补齐

使用范围: 保留用于追溯的未通过、受限或未执行检验

语言: Python; 源文件共 439 行。

SHA256: eab2be6bd8fd5b64823b2a6aaf5c7b4439e90ad782d4defe8d9b2d911a69963

```
from __future__ import annotations

import gc
import hashlib
import importlib
import json
import os
from pathlib import Path
import subprocess
import sys
import time
import traceback
import uuid
import warnings

SOURCE = Path(__file__).resolve()
ROOT = SOURCE.parents[3]
LOADED_CODE_SHA256 = hashlib.sha256(SOURCE.read_bytes()).hexdigest()
import production_provenance as provenance

ASSUMPTIONS = [
    '一维径向有效模型, 忽略轴向/端面基线, 以二维独立程序检验端面影响。',
    '题给  $\rho \cdot c_p$  作为有效体积显热容量; 不强制它等于守恒骨架的真实湿质量密度。',
    '附件烘房 kg/kg 数值直接作为材料干基等效  $C_{eq}$ ; 这不是已证明的气固吸附平衡换算。',
    '基线不显式加入潜热; 给定等效驱动力的潜热试验仅是能量负荷压力测试。',
    '前 4h 采用原始线性插值, 4h 后  $T_{air}=50$  摄氏度、 $C_{eq}=0.05$  作为长期延拓假设。',
    'Q4 固定长度并采用同比径向材料收缩, 半径轨迹来自附件 2 而非模型预测。',
    'Q2/Q3 从  $t=0$  统一附录 3; Q4 从  $t=0$  整组改用附录 4; h 和  $\beta$  沿用题给值。',
]

LIMITATIONS = [
    '没有内部温度/水分实测真值, 数值校核不构成实测预测精度认证。',
    '密度、气固湿度映射、潜热闭合和外生收缩是模型解释边界, 需团队主导确认。',
    '四位小数为输出格式; 网格比较是采样差值, 不能冒称连续解严格误差上界。',
    'Visual Studio 复现与用户人工代码审查分别留有独立状态。',
]

CONTRACT_TARGETS = {
    'model_results.json': 'paper_output/results/model_results.json',
    'metrics.json': 'paper_output/results/metrics.json',
    'conclusions.json': 'paper_output/results/conclusions.json',
    'table_index.json': 'paper_output/tables/table_index.json',
    'figure_index.json': 'paper_output/figure_index.json',
}

def dependencies():

    paths = list(SOURCE.parent.glob('*.py'))
    paths += [ROOT / 'paper_output/data_cleaned' / name for name in
               ('A_environment_observed.csv', 'A_radius_observed.csv')]
```

```

paths += [ROOT / 'paper_output/plan/model_route.json']
paths += list((ROOT / 'problem_files/CUMCM2026Problems/A 题').rglob('*.xlsx'))
return [provenance.file_record(path) for path in sorted(set(paths))]

def font_record():
    font = Path('C:/Windows/Fonts/msyh.ttc')
    return {'path': str(font), 'bytes': font.stat().st_size, 'sha256':
provenance.sha256(font)}

def assert_dependencies(start):
    if dependencies() != start['input_files']:
        raise RuntimeError('Source/input inventory changed after launch')
    if font_record() != start['font']:
        raise RuntimeError('Rendering font changed after launch')
    if provenance.runtime_record() != start['runtime']:
        raise RuntimeError('Runtime environment changed after launch')
    expected = {rec['path']: rec['sha256'] for rec in start['input_files']}
    for path, loaded in ((SOURCE, LOADED_CODE_SHA256), (provenance.SOURCE,
provenance.LOADED_CODE_SHA256)):
        if loaded != expected[path.relative_to(ROOT).as_posix()]:
            raise RuntimeError('Loaded entry/provenance module differs from launch
snapshot')

def imported_model_records():
    paths = set()
    for module in list(sys.modules.values()):
        filename = getattr(module, '__file__', None)
        if filename:
            path = Path(filename).resolve()
            if path.is_relative_to(SOURCE.parent) and path.suffix == '.py':
                paths.add(path)
    return [provenance.file_record(path) for path in sorted(paths)]

def question_plot(plots, run, qid, directory):
    public = getattr(plots, 'make_question_plot', None)
    if callable(public):
        return public(run, qid, directory)
    name = '_profiles' if qid in ('Q1', 'Q2') else '_drying'
    function = getattr(plots, name, None)
    if not callable(function):
        raise RuntimeError('Publication plot API is missing: ' + name)
    return function(run, qid, directory)

def describe_question(qid, key, summary, export, figure, args, directory, start):
    sample = summary['metric_samples'][qid]
    T, C, samptime = sample['T_K'][0], sample['C'][0], sample['time_s']
    result_text = (f'在 t={samptime/3600:.4f}h, 中心/表面温度分别为'
        f'{T[0]-273.15:.4f}/{T[1]-273.15:.4f}摄氏度, 干基含水率为'
        f'{C[0]:.4f}/{C[1]:.4f}kg/kg。')
    if qid in ('Q3', 'Q4'):
        done = summary['completion']
        result_text = (f'连续临界时刻{done["critical_event_h"]:.9f}h; 向上到四位小数并复核后'
            f'报告{done["reported_drying_time_h"]:.4f}h, 原精度最大含水率'
            f'{done["max_C_at_reported_time"]:.12f}<0.15kg/kg。')
    artifacts = [rec['path'] for rec in export['artifacts']] + [export['validation_report']
['path']]
    artifacts += [rec['path'] for rec in provenance.figure_artifact_records(figure)]
    artifacts = sorted(set(artifacts))

```

```

    source = next(rec for rec in start['input_files'] if rec['path'] ==
SOURCE.relative_to(ROOT).as_posix())
    question = {'question_id': qid, 'status': 'computed', 'source_run_id': args.version,
        'source_trajectory_id': args.version + ':' + key,
        'model_name': '圆柱有效显热与干基扩散: 守恒有限体积、Kirchhoff 面通量、解析 Jacobian
BDF',
        'result_summary': result_text, 'result_files': artifacts,
        'execution_provenance': {'source_code_path': source['path'], 'source_code_sha256':
source['sha256'],
        'run_command': start['worker_command'], 'run_exit_code': None,
        'process_status': 'awaiting_parent_observed_exit', 'output_artifacts':
artifacts},
        'assumptions_used': ASSUMPTIONS, 'limitations': LIMITATIONS, 'validation_summary':
summary}
    values = [
        ('mass_balance_absolute_residual', summary['diagnostics']
['max_mass_balance_abs_kg_per_kg'], 'kg/kg',
        '离散干物质基准下, 平均C与累计流出之和减初值。',
        '/diagnostics/max_mass_balance_abs_kg_per_kg'),
        ('radial_intervals', summary['settings']['intervals'], '1', '实际求解空间分辨率。',
        '/settings/intervals'),
        ('center_C_at_report_time', C[0], 'kg/kg', '已记录指定时刻的中心含水率。',
f'/metric_samples/{qid}/C/0/0'),
    ]
    if qid in ('Q3', 'Q4'):
        values.append(('reported_drying_time', summary['completion']
['reported_drying_time_h'], 'h',
        '向上取四位小数并用原精度最大值验证严格阈值。',
        '/completion/reported_drying_time_h'))
    evidence = (directory / key / 'summary.json').relative_to(ROOT).as_posix()
    metrics = [{'question_id': qid, 'metric_name': name, 'metric_role': 'numerical_result',
        'value': value, 'unit': unit, 'status': 'computed', 'evidence_path': evidence,
        'evidence_json_pointer': pointer, 'question_report_time_s': sampletime,
        'interpretation': interpretation}
        for name, value, unit, interpretation, pointer in values]
    conclusion = {'question_id': qid, 'conclusion_text': result_text, 'status': 'computed',
        'supporting_metrics': [value[0] for value in values], 'supporting_artifacts':
artifacts,
        'assumptions': ASSUMPTIONS, 'limitations': LIMITATIONS}
    return question, metrics, conclusion

def worker(args, directory, start):
    assert_dependencies(start)
    import numpy as np
    core = importlib.import_module('drying_core')
    modules = {key: importlib.import_module(name) for key, name in
        [('Q1', 'q1_model'), ('Q23', 'q2_model'), ('Q4', 'q4_model')]}
    completion = importlib.import_module('q3_model').completion
    solve_only_review = args.review and not getattr(args, 'review_exports', False)
    exporter = None if solve_only_review else importlib.import_module('export_outputs')
    plots = None if solve_only_review else importlib.import_module('publication_plots')
    if core.ROOT.resolve() != ROOT:
        raise RuntimeError('Core workspace differs from production workspace')

    assert_dependencies(start)
    imported = imported_model_records()
    started, timer = provenance.utcnow(), time.perf_counter()
    summaries, exports, tables, figures, export_checks = {}, [], [], [], []
    questions, metrics, conclusions = [], [], []
    for name, n, qids in [('Q1', args.n1, ['Q1']), ('Q23', args.n23, ['Q2', 'Q3']),
        ('Q4', args.n4, ['Q4'])]:
        run = None

```

```

try:
    print(f'SOLVE {name} N={n} UTC={provenance.utc_now()}', flush=True)
    with warnings.catch_warnings(record=True) as caught:
        warnings.simplefilter('always')
        run = modules[name].solve(n)
    if caught:
        raise RuntimeError('Solver warning: ' + '; '.join(str(item.message) for
item in caught))
    summary = core.save_run(run, directory / name)
    summary['solver_warning_count'] = len(caught)
    summary['source_run_id'] = args.version
    summary['source_trajectory_id'] = args.version + ':' + name
    if name != 'Q1':
        summary['completion'] = completion(run)
    times = np.unique(np.array([0., min(1800., run.end_s), min(10800., run.end_s),
run.end_s] +
    ([[] if name == 'Q1' else [summary['completion']['reported_time_s']]])))
    T, C = run.fields(times, material_x=np.array([0., .25, .5, .75, 1.]))
    summary['reproduction_samples'] = {'times_s': times.tolist(),
    'material_x': [0., .25, .5, .75, 1.], 'T_K': T.tolist(), 'C': C.tolist()}
    summary['metric_samples'] = {}
    for qid in qids:
        samptime = 1800. if qid == 'Q1' else 10800. if qid == 'Q2' else
summary['completion']['reported_time_s']
        T, C = run.fields([samptime], material_x=[0., 1.])
        summary['metric_samples'][qid] = {'time_s': samptime, 'material_x': [0.,
1.],
        'T_K': T.tolist(), 'C': C.tolist(), 'source':
'live_Run_dense_solution'}
        provenance.write_json(directory / name / 'summary.json', summary)
        summaries[name] = summary
        print(json.dumps({'question': name, **summary['diagnostics']},
ensure_ascii=False), flush=True)
        if solve_only_review:
            continue
        for qid in qids:
            print('EXPORT ' + qid, flush=True)
            export = exporter.export_question(run, qid, directory / 'outputs')
            check = exporter.validate_exports([export], runs={qid: run})
            if check.get('status') != 'PASS' or not
check.get('fully_verified_with_live_Run'):
                raise RuntimeError('Export did not pass live Run validation: ' + qid)
            export['validation_report'] = provenance.file_record(check['exports'][0]
['report_path'])
            export['validation_status'] = 'PASS'
            export_checks.append(check)
            exports.append(export)
            for rec in export['artifacts']:
                if rec.get('role') in ('paper_table', 'paper_surface_coordinates'):
                    tables.append({'table_id': Path(rec['path']).stem, 'question_id':
qid,
                    'title': Path(rec['path']).stem, 'purpose': ' 题目指定正文位置/时
间采样',
                    **provenance.file_record(ROOT / rec['path']), 'status':
'computed',
                    'placeholder': False, 'ok': True, 'source_run_id':
args.version,
                    'source_trajectory_id': args.version + ':' + name})
            print('PLOT ' + qid, flush=True)
            figure = question_plot(plots, run, qid, directory / 'figures')
            figure.update({'status': 'computed', 'placeholder': False, 'ok': True,
                'source_run_id': args.version, 'source_trajectory_id': args.version +
':' + name})

```

```

        figures.append(figure)
        question, question_metrics, conclusion = describe_question(
            qid, name, summary, export, figure, args, directory, start)
        questions.append(question)
        metrics.extend(question_metrics)
        conclusions.append(conclusion)
        print(json.dumps({'question': qid, 'export_size': export['size'],
'validation': check['status']}},
            ensure_ascii=False), flush=True)
    finally:
        if run is not None:
            close = getattr(run, 'close', None)
            try:
                if callable(close):
                    close()
            finally:
                close = None
                run = None
                gc.collect()
        print('RELEASED ' + name, flush=True)
    assert_dependencies(start)
    provenance.assert_records(imported)
    provenance.write_json(directory / 'numerical_summaries.json', summaries)
    if args.review:
        provenance.write_json(directory / 'review_result.json', {
            'status': 'COMPUTED_PENDING_GUI_OBSERVATION',
            'mode': 'independent_actual_resolve_and_full_exports' if args.review_exports
else 'independent_actual_resolve',
            'source': provenance.file_record(SOURCE),
            'inputs': start['input_files'], 'runtime': start['runtime'],
'imported_model_modules': imported,
            'started_at': started, 'finished_at': provenance.utcnow(), 'summaries':
summaries,
            'exports': exports, 'export_checks': export_checks, 'figures': figures,
            'elapsed_s': time.perf_counter() - timer,
            'visual_studio_gui': 'Requires separate observed GUI evidence', 'human_review':
'pending'})
        print('REPRODUCTION_SOLVES_COMPLETED', flush=True)
    else:
        provenance.write_json(directory / 'metric_evidence_validation.json',
provenance.validate_metric_evidence(metrics))
        provenance.write_json(directory / 'export_validation.json', {'status': 'PASS',
'checks': export_checks})
        meta = {'schema_version': '1.0', 'generated_by':
SOURCE.relative_to(ROOT).as_posix(),
            'generated_at': provenance.utcnow(), 'source_run_id': args.version,
            'publication_state': 'version_only_awaiting_parent_observed_exit'}
        contracts = {'model_results.json': {**meta, 'questions': questions},
            'metrics.json': {**meta, 'items': metrics}, 'conclusions.json': {**meta,
'items': conclusions},
            'table_index.json': {**meta, 'tables': tables}, 'figure_index.json': {**meta,
'figures': figures}}
        for filename, content in contracts.items():
            provenance.write_json(directory / 'contracts' / filename, content)
        assert_dependencies(start)
        output_paths = [p for p in directory.rglob('*') if p.is_file() and p.name not in
('stdout.log', 'worker_success.json', 'process_result.json')]
        provenance.write_json(directory / 'worker_success.json', {
            'started_at': started, 'finished_at': provenance.utcnow(), 'elapsed_seconds':
time.perf_counter() - timer,
            'input_files': start['input_files'], 'runtime': start['runtime'], 'font':
start['font'],
            'imported_model_modules': imported, 'output_artifacts': [provenance.file_record(p)

```

```

for p in sorted(output_paths)],
    'question_ids': ['Q1', 'Q2', 'Q3', 'Q4'], 'summaries': summaries, 'exports':
exports,
    'mode': 'review' if args.review else 'production', 'human_review': 'pending',
'visual_studio_gui': 'pending'})
    print('ALL_SOLVES_AND_REQUESTED_VALIDATIONS_COMPLETED', flush=True)

def publish(args, directory, start, success, process):
    assert_dependencies(start)
    provenance.assert_records(success['output_artifacts'])
    with provenance.publication_lock(ROOT, args.version):
        contracts = {name: json.loads((directory / 'contracts' /
name).read_text(encoding='utf-8'))
            for name in CONTRACT_TARGETS}
        for data in contracts.values():
            data['publication_state'] = 'parent_verified_actual_worker_exit_zero'
        for question in contracts['model_results.json']['questions']:
            question['execution_provenance'].update({'run_exit_code':
process['returncode'],
            'process_status': 'parent_observed_exit_zero',
            'process_evidence_path': (directory /
'process_result.json').relative_to(ROOT).as_posix())}
        index_path = ROOT / 'paper_output/figure_index.json'
        prior = json.loads(index_path.read_text(encoding='utf-8')) if index_path.exists()
else None
        merged, merge_note = provenance.merge_figure_index(prior,
contracts['figure_index.json']['figures'])
        contracts['figure_index.json']['figures'] = merged
        contracts['figure_index.json']['merge_record'] = merge_note
        provenance.validate_metric_evidence(contracts['metrics.json']['items'])
        published = directory / 'published_contracts'
        for name, content in contracts.items():
            provenance.write_json(published / name, content)
        records = {rec['path']: rec for rec in success['output_artifacts']}
        for path in [directory / 'worker_success.json', directory / 'stdout.log', directory
/ 'process_result.json']:
            rec = provenance.file_record(path)
            records[rec['path']] = rec
        for name in CONTRACT_TARGETS:
            rec = provenance.file_record(published / name)
            records[rec['path']] = rec

        target = {**rec, 'path': CONTRACT_TARGETS[name]}
        records[target['path']] = target
        for figure in merged:
            for rec in provenance.figure_artifact_records(figure):
                records[rec['path']] = rec
        assert_dependencies(start)
        source = next(rec for rec in start['input_files'] if rec['path'] ==
SOURCE.relative_to(ROOT).as_posix())
        record = {'run_id': args.version, 'script': source['path'], 'script_sha256':
source['sha256'],
            'command': process['command'], 'question_ids': success['question_ids'],
            'returncode': process['returncode'], 'cwd': str(ROOT), 'process': process,
            'started_at': process['started_at'], 'finished_at': process['finished_at'],
            'worker_started_at': success['started_at'], 'worker_finished_at':
success['finished_at'],
            'elapsed_seconds': success['elapsed_seconds'], 'python': start['runtime'],
            'font': start['font'],
            'input_files': start['input_files'], 'imported_model_modules':
success['imported_model_modules'],
            'output_artifacts': [records[path] for path in sorted(records)],

```

```

        'version_contract_directory': published.relative_to(ROOT).as_posix(),
        'human_review': 'pending', 'visual_studio_gui': 'pending'}
    manifest = {'schema_version': '1.0', 'generated_by':
SOURCE.relative_to(ROOT).as_posix(),
        'generated_at': provenance.utcnow(), 'status': 'PASS', 'runs': [record],
        'scope': 'Actual worker exit zero, source/input/output hashes and export
checks; not human model acceptance.')}
    provenance.publish_transaction(ROOT, directory,
        {CONTRACT_TARGETS[name]: data for name, data in contracts.items()}, manifest)

def main():
    import argparse
    parser = argparse.ArgumentParser(description=__doc__)
    parser.add_argument('--version', required=True)
    parser.add_argument('--n1', type=int, default=3200)
    parser.add_argument('--n23', type=int, default=3200)
    parser.add_argument('--n4', type=int, default=6400)
    parser.add_argument('--blas-threads', type=int, default=1)
    parser.add_argument('--worker', action='store_true', help=argparse.SUPPRESS)
    parser.add_argument('--launch-token', help=argparse.SUPPRESS)
    parser.add_argument('--review', action='store_true')
    parser.add_argument('--review-exports', action='store_true',
        help='With --review, also reproduce all Excel/CSV readbacks and figures in the GUI
process')
    args = parser.parse_args()
    if args.review_exports and not args.review:
        raise ValueError('--review-exports requires --review')
    if args.blas_threads < 1:
        raise ValueError('BLAS thread count must be positive')

    os.environ['OPENBLAS_NUM_THREADS'] = str(args.blas_threads)
    os.environ['OMP_NUM_THREADS'] = str(args.blas_threads)
    if Path.cwd().resolve() != ROOT:
        raise RuntimeError('Run with the competition workspace as cwd')
    if not args.version or any(ch not in
'abcdefghijklmnopqrstuvwxyzABCDEFGHIJKLMNOPQRSTUVWXYZ0123456789_-' for ch in args.version):
        raise ValueError('Use a simple version label')
    if min(args.n1, args.n23, args.n4) < 2:
        raise ValueError('All radial interval counts must be at least 2')
    directory = ROOT / ('paper_output/results/gui_reproduction' if args.review else
        'paper_output/results/production') / args.version
    if args.worker:
        if args.review:
            raise ValueError('--review uses a single process so Visual Studio can hit
solver breakpoints')
        start = json.loads((directory / 'launch.json').read_text(encoding='utf-8'))
        config = {'n1': args.n1, 'n23': args.n23, 'n4': args.n4, 'review': args.review,
            'review_exports': args.review_exports,
            'blas_threads': args.blas_threads}
        if not args.launch_token or args.launch_token != start['launch_token'] or config !=
start['configuration']:
            raise RuntimeError('Worker must be launched by its matching supervising
parent')
        provenance.exclusive_json(directory / 'worker_started.json', {'pid': os.getpid(),
'started_at': provenance.utcnow()})
        print('WORKER_PID=' + str(os.getpid()), flush=True)
        try:
            worker(args, directory, start)
        except BaseException as error:
            provenance.write_json(directory / 'worker_failure.json', {'status': 'FAIL',
'error': repr(error),

```

```

        'traceback': traceback.format_exc(), 'generated_at': provenance.utc_now(),
        'returncode': None, 'returncode_note': 'Supervising parent records the
actual process exit.'})
    raise
    return 0
if args.launch_token:
    raise ValueError('--launch-token is reserved for the supervised worker')
directory.parent.mkdir(parents=True, exist_ok=True)
directory.mkdir(exist_ok=False)
token = uuid.uuid4().hex
provenance.exclusive_json(directory / 'launch_claim.json', {'parent_pid': os.getpid(),
    'run_id': args.version, 'claimed_at': provenance.utc_now(), 'launch_token': token})
process = None
try:
    command = [sys.executable, '-B', str(SOURCE), '--version', args.version,
        '--n1', str(args.n1), '--n23', str(args.n23), '--n4', str(args.n4),
        '--blas-threads', str(args.blas_threads)]
    if args.review:
        command.append('--review')
        if args.review_exports:
            command.append('--review-exports')
    else:
        command += ['--worker', '--launch-token', token]
    start = {'run_id': args.version, 'parent_pid': os.getpid(), 'launch_token': token,
        'created_at': provenance.utc_now(), 'configuration': {'n1': args.n1, 'n23':
args.n23,
        'n4': args.n4, 'review': args.review, 'review_exports':
args.review_exports,
        'blas_threads': args.blas_threads}, 'input_files': dependencies(),
        'runtime': provenance.runtime_record(), 'font': font_record(),
        'worker_command': subprocess.list2cmdline(command), 'worker_argv': command,
        'dependency_scope': 'Local module inventory plus A inputs/templates and route;
inventory does not claim execution.'}
    assert_dependencies(start)
    provenance.exclusive_json(directory / 'launch.json', start)
    if args.review:

        provenance.exclusive_json(directory / 'review_started.json', {
            'pid': os.getpid(), 'started_at': provenance.utc_now(), 'mode':
'single_process_VS_review'})
        worker(args, directory, start)
        assert_dependencies(start)
        success = json.loads((directory /
'worker_success.json').read_text(encoding='utf-8'))
        provenance.assert_records(success['output_artifacts'])
        provenance.write_json(directory / 'review_evidence_validation.json', {
            'status': 'COMPUTED_PENDING_GUI_OBSERVATION', 'input_files':
start['input_files'],
            'worker_success': provenance.file_record(directory /
'worker_success.json'),
            'scope': 'Direct solve function and saved hashes checked; no process exit
code is self-certified.',
            'visual_studio_gui': 'pending', 'human_review': 'pending'})
        print('REVIEW_FUNCTION_COMPLETED;
GUI_OBSERVATION_AND_ACTUAL_EXIT_NOT_SELF_CERTIFIED', flush=True)
        return 0
    process = provenance.measured_process(command, ROOT, directory / 'stdout.log')
    provenance.write_json(directory / 'process_result.json', process)
    if process['returncode'] != 0:
        raise RuntimeError('Worker failed with actual return code ' +
str(process['returncode']))
    assert_dependencies(start)

```

```

        success = json.loads((directory /
'worker_success.json').read_text(encoding='utf-8'))
        if success['input_files'] != start['input_files'] or success['runtime'] !=
start['runtime']:
            raise RuntimeError('Worker success differs from launch snapshot')
        provenance.assert_records(success['output_artifacts'])
        publish(args, directory, start, success, process)
        print('ACTUAL_WORKER_RETURN_CODE=0; VERIFIED_RUN_MANIFEST_PUBLISHED', flush=True)
        return 0
    except BaseException as error:
        if process is None and (directory / 'process_result.json').exists():
            process = json.loads((directory /
'process_result.json').read_text(encoding='utf-8'))
            provenance.write_json(directory / 'parent_failure.json', {'status': 'FAIL',
'generated_at': provenance.utcnow(), 'error': repr(error), 'traceback':
traceback.format_exc(),
'actual_worker_returncode': None if process is None else process['returncode'],
'process': process, 'version_is_retained': True,
'publication_note': 'Worker does not publish global contracts; publication
failure records rollback.'})
            raise

if __name__ == '__main__':
    try:
        raise SystemExit(main())
    except Exception:
        traceback.print_exc()
        raise SystemExit(1)

```

D.28 validate_bessel.py

路径: 05_数值检验与实验/Python 检验源码/paper_output/code/modeling/validate_bessel.py

用途: 常热物性圆柱 Robin 边界和时变环境卷积的 Bessel 解析基准

使用范围: 检验与历史诊断

语言: Python; 源文件共 319 行。

SHA256: 5177cf016621b9f1cc6e705af7674408a72e12b0bfed7f6a3609c80b79ea4064

```
from __future__ import annotations

import argparse
import csv
import hashlib
import json
import platform
import sys
from datetime import datetime, timezone
from functools import lru_cache
from pathlib import Path

import numpy as np
from numpy.typing import ArrayLike, NDArray
from scipy.integrate import quad
from scipy.optimize import brentq
from scipy.special import j0, j1, jn_zeros

FloatArray = NDArray[np.float64]

def _vector(value: ArrayLike, name: str) -> FloatArray:
    a = np.atleast_1d(np.asarray(value, dtype=float))
    if a.ndim != 1 or a.size == 0 or not np.all(np.isfinite(a)):
        raise ValueError(f"{name} must be a nonempty finite 1-D array")
    return a

@lru_cache(maxsize=24)
def _eigen_cached(biot: float, n_terms: int) -> tuple[FloatArray, FloatArray]:
    if not np.isfinite(biot) or biot <= 0:
        raise ValueError("Biot must be finite and positive")
    if not isinstance(n_terms, (int, np.integer)) or n_terms < 1:
        raise ValueError("n_terms must be a positive integer")

    upper = jn_zeros(0, n_terms)
    lower = np.r_[0.0, jn_zeros(1, n_terms-1)] if n_terms > 1 else np.array([0.0])
    roots = np.array([
        brentq(lambda z: z*j1(z)-biot*j0(z), lo, hi, xtol=1e-13, rtol=1e-14)
        for lo, hi in zip(lower, upper)
    ])
    a = 2*j1(roots) / (roots*(j0(roots)**2+j1(roots)**2))
    roots.setflags(write=False)
    a.setflags(write=False)
    return roots, a

def robin_eigenpairs(biot: float, n_terms: int = 120) -> tuple[FloatArray, FloatArray]:
```

```

mu, a = _eigen_cached(float(biot), int(n_terms))
return mu.copy(), a.copy()

def _validate_inputs(times_s, env_t, env_values, radius, diffusivity, biot):
    t = _vector(times_s, "times_s")
    knots = _vector(env_t, "env_t")
    values = _vector(env_values, "env_values")
    if len(knots) != len(values) or len(knots) < 2:
        raise ValueError("Boundary needs matching arrays with at least two knots")
    if abs(knots[0]) > 1e-12 or np.any(np.diff(knots) <= 0):
        raise ValueError("Boundary knots must start at t=0 and strictly increase")
    if np.any(t < 0):
        raise ValueError("Negative evaluation times are not permitted")
    if np.max(t) > knots[-1] + 1e-10:
        raise ValueError("Evaluation exceeds boundary observations: supply an explicit
extension")
    if not np.isfinite(radius) or radius <= 0 or not np.isfinite(diffusivity) or
diffusivity <= 0:
        raise ValueError("radius and diffusivity must be finite and positive")
    if not np.isfinite(biot) or biot < 0:
        raise ValueError("Biot must be finite and nonnegative")
    return t, knots, values

def _modal_history(
    times: FloatArray, knots: FloatArray, values: FloatArray,
    rates: FloatArray, initial_value: float,
) -> FloatArray:

    z = (initial_value-values[0])*np.exp(-times[:, None]*rates[None, :])
    slopes = np.diff(values)/np.diff(knots)
    for lo, hi, slope in zip(knots[:-1], knots[1:], slopes):
        if slope == 0:
            continue
        selected = times > lo
        ts = times[selected]
        end = np.minimum(ts, hi)
        duration = end-lo
        contribution = (
            np.exp(-(ts-end)[:, None]*rates[None, :])
            * (-np.expm1(-duration[:, None]*rates[None, :]))
            / rates[None, :]
        )
        z[selected] -= slope*contribution
    return z

def _space_modes(mu: FloatArray, radii_m: ArrayLike, radius: float, cell_edges_m=None) ->
FloatArray:
    if cell_edges_m is None:
        x = _vector(radii_m, "radii_m")/radius
        if np.any(x < 0) or np.any(x > 1+1e-12):
            raise ValueError("radii must lie inside the cylinder")
        return j0(mu[:, None]*x[None, :])
    edges = _vector(cell_edges_m, "cell_edges_m")/radius
    if len(edges) < 2 or np.any(np.diff(edges) <= 0) or edges[0] < 0 or edges[-1] >
1+1e-12:
        raise ValueError("cell edges must strictly increase inside [0,radius]")
    lo, hi = edges[:-1], edges[1:]
    return 2*(hi[None, :]*j1(mu[:, None]*hi[None, :])

```

```

- lo[None, :]*j1(mu[:, None]*lo[None, :])) / (mu[:, None]*(hi*hi-lo*lo)[None,
:])

```

```

def bessel_diffusion(
    times_s: ArrayLike, radii_m: ArrayLike, env_t: ArrayLike,
    env_values: ArrayLike, *, radius: float, diffusivity: float,
    biot: float, initial_value: float, n_terms: int = 120,
    cell_edges_m: ArrayLike | None = None,
) -> FloatArray:

    t, knots, values = _validate_inputs(times_s, env_t, env_values, radius, diffusivity,
    biot)
    if not np.isfinite(initial_value):
        raise ValueError("initial_value must be finite")
    if not isinstance(n_terms, (int, np.integer)) or n_terms < 1:
        raise ValueError("n_terms must be a positive integer")
    if biot == 0:

        nspace = len(_vector(cell_edges_m, "cell_edges_m"))-1 if cell_edges_m is not None
    else len(_vector(radii_m, "radii_m"))
        _space_modes(np.array([1.0]), radii_m, radius, cell_edges_m)
        return np.full((len(t), nspace), initial_value)
    mu, a = _eigen_cached(float(biot), int(n_terms))
    rates = diffusivity*mu*mu/(radius*radius)
    spatial = _space_modes(mu, radii_m, radius, cell_edges_m)
    history = _modal_history(t, knots, values, rates, initial_value)
    result = np.interp(t, knots, values)[: , None] + (history*a[None, :]) @ spatial

    result[t == 0] = initial_value
    return result

```

```

def bessel_temperature(
    times_s: ArrayLike, radii_m: ArrayLike, env_t: ArrayLike,
    env_T: ArrayLike, radius: float = 0.02, h: float = 25.0,
    k: float = 0.36, rho: float = 820.0, cp: float = 2600.0,
    n_terms: int = 120, initial_temperature_K: float = 301.15,
    cell_edges_m: ArrayLike | None = None,
) -> FloatArray:

    if any(not np.isfinite(v) or v <= 0 for v in (k,rho,cp)) or h < 0:
        raise ValueError("k,rho,cp must be positive and h nonnegative")
    return bessel_diffusion(
        times_s, radii_m, env_t, env_T, radius=radius,
        diffusivity=k/(rho*cp), biot=h*radius/k,
        initial_value=initial_temperature_K, n_terms=n_terms,
        cell_edges_m=cell_edges_m,
    )

```

```

def bessel_moisture(
    times_s: ArrayLike, radii_m: ArrayLike, env_t: ArrayLike,
    env_Ceq: ArrayLike, diffusion_coefficient: float,
    radius: float = 0.02, beta: float = 8e-7,
    initial_moisture: float = 2.55, n_terms: int = 240,
    cell_edges_m: ArrayLike | None = None,
) -> FloatArray:

    if diffusion_coefficient <= 0 or beta < 0:
        raise ValueError("Frozen diffusion coefficient must be positive, beta nonnegative")

```

```

return bessell_diffusion(
    times_s, radii_m, env_t, env_Ceq, radius=radius,
    diffusivity=diffusion_coefficient, biot=beta*radius/diffusion_coefficient,
    initial_value=initial_moisture, n_terms=n_terms,
    cell_edges_m=cell_edges_m,
)

def _sha256(path: Path) -> str:
    return hashlib.sha256(path.read_bytes()).hexdigest()

def selfcheck(env_t: FloatArray, env_T: FloatArray) -> dict:

    bi = 25*.02/.36
    mu, a = robin_eigenpairs(bi, 120)
    residual = mu*j1(mu)-bi*j0(mu)
    relative_eigen_residual = np.max(np.abs(residual)/(1+np.abs(mu*j1(mu))
+np.abs(bi*j0(mu))))

    knots = np.array([0.,17.,52.,130.])
    boundary = np.array([301.15,305.,303.,303.])
    targets = np.array([8.,17.,40.,78.,130.])
    rates = np.array([1e-6,.002,.2])
    z = _modal_history(targets,knots,boundary,rates,299.)
    independent = np.empty_like(z)
    for i,t in enumerate(targets):
        for j,lam in enumerate(rates):
            integral = 0.
            for lo,hi,gl,gh in zip(knots[:-1],knots[1:],boundary[:-1],boundary[1:]):
                if t > lo:
                    slope = (gh-gl)/(hi-lo)
                    integral += quad(lambda s: slope*np.exp(-lam*(t-
s)),lo,min(t,hi),epsabs=1e-12,epsrel=1e-12)[0]
                independent[i,j] = (299.-boundary[0])*np.exp(-lam*t)-integral
            convolution_error = float(np.max(np.abs(z-independent)))

    edges = np.array([0.,.001,.006,.02])
    averaged = _space_modes(mu[:8], [0.], .02,edges)
    quad_average = np.empty_like(averaged)
    for n,m in enumerate(mu[:8]):
        for i,(lo,hi) in enumerate(zip(edges[:-1]/.02,edges[1:]/.02)):
            quad_average[n,i] = 2*quad(lambda x: x*j0(m*x),lo,hi,epsabs=1e-12)[0]/(hi*hi-
lo*lo)
    average_error = float(np.max(np.abs(averaged-quad_average)))

    alpha=.36/(820*2600)
    lam=alpha*mu**2/.02**2
    e=np.exp(-100*lam)
    mean_basis=2*j1(mu)/mu
    dmean=np.sum(a*(-lam)*(-20)*e*mean_basis)
    surface=321.15+np.sum(a*(-20)*e*j0(mu))
    influx=2*25/(820*2600*.02)*(321.15-surface)
    energy_error=abs(dmean-influx)

    ts=np.array([0.,1.,10.,100.,300.,600.,900.,1200.,1500.,1800.])
    rs=np.linspace(0,.02,41)
    t120=bessell_temperature(ts,rs,env_t,env_T,n_terms=120)
    t240=bessell_temperature(ts,rs,env_t,env_T,n_terms=240)

```

```

t480=bessel_temperature(ts,rs,env_t,env_T,n_terms=480)
error120_240=float(np.max(np.abs(t120-t240)))
error240_480=float(np.max(np.abs(t240-t480)))
equilibrium=bessel_temperature([0.,1.,100.],[0.,.01,.02],[0.,100.],[301.15,301.15])
equilibrium_error=float(np.max(np.abs(equilibrium-301.15)))
insulating=bessel_temperature([0.,50.,100.],[0.,.01,.02],[0.,100.],[301.15,323.15],h=0)
insulation_error=float(np.max(np.abs(insulating-301.15)))

checks = [
    ("robin_eigen_relative_residual",float(relative_eigen_residual),1e-10,"1"),
    ("piecewise_convolution_vs_adaptive_quadrature",convolution_error,1e-10,"K"),
    ("annular_average_vs_adaptive_quadrature",average_error,1e-10,"1"),
    ("constant_boundary_step_energy_identity",float(energy_error),1e-10,"K/s"),
    ("Q1_observed_boundary_120_vs_240_terms",error120_240,1e-4,"K"),
    ("Q1_observed_boundary_240_vs_480_terms",error240_480,1e-4,"K"),
    ("uniform_equilibrium",equilibrium_error,1e-12,"K"),
    ("insulating_boundary",insulation_error,1e-12,"K"),
]
return {
    "status":"PASS" if all(v<=limit for _,v,limit,_ in checks) else "FAIL",
    "scope":"analytic benchmark implementation selfcheck; production FVM comparison is
separate",
    "checks":[{"name":name,"value":v,"tolerance":limit,"unit":unit,"passed":v<=limit}
for name,v,limit,unit in checks],
    "series_note":"120/240/480 differences estimate truncation sensitivity at sampled
locations and times, not a rigorous uniform error bound.",
    "no_internal_experimental_validation":True,
    "visual_studio_gui":"pending",
    "human_review":"pending",
}

def main() -> int:
    parser=argparse.ArgumentParser(description=__doc__)
    parser.add_argument("--environment-csv",default="paper_output/data_cleaned/
A_environment_observed.csv")
    parser.add_argument("--output-dir",default="paper_output/results/bessel_validation")
    args=parser.parse_args()
    root=Path.cwd().resolve()
    if root != Path(r"D:\Document\数学建模\2026CUMCM").resolve():
        raise RuntimeError("Run only from the 2026CUMCM workspace root")
    source=(root/args.environment_csv).resolve()
    with source.open(encoding="utf-8-sig",newline="") as fh:
        rows=list(csv.DictReader(fh))
    env_t=np.array([float(r["time_s"]) for r in rows])
    env_T=np.array([float(r["temperature_K"]) for r in rows])
    env_C=np.array([float(r["air_moisture_kg_per_kg"]) for r in rows])
    report=selfcheck(env_t,env_T)
    out=(root/args.output_dir).resolve()
    if not out.is_relative_to(root):
        raise RuntimeError("Output must remain inside the competition workspace")
    out.mkdir(parents=True,exist_ok=True)
    paper_t=np.array([100.,300.,600.,900.,1200.,1500.,1800.])
    paper_r=np.array([0.,.005,.01,.015,.02])
    temperature=bessel_temperature(paper_t,paper_r,env_t,env_T,n_terms=240)
    frozen_D=7e-9*np.exp(-.89/2.55)
    frozen_water=bessel_moisture(paper_t,paper_r,env_t,env_C,float(frozen_D),n_terms=240)
    artifacts=[]
    for filename,field,label in [
        ("q1_exact_heat_paper_points.csv",temperature-273.15,"temperature_C"),
        ("frozen_D_moisture_benchmark_points.csv",frozen_water,"moisture_kg_per_kg"),
    ]:
        path=out/filename

```

```

        with path.open("w",encoding="utf-8",newline="") as fh:
            writer=csv.writer(fh)
            writer.writerow(["time_s","radius_m",label])
            for i,t in enumerate(paper_t):
                for j,r in enumerate(paper_r):

writer.writerow([format(t, ".17g"),format(r, ".17g"),format(field[i,j], ".17g")])

artifacts.append({"path":path.relative_to(root).as_posix(),"bytes":path.stat().st_size,"sha
256":_sha256(path)})
    report.update({
        "schema_version":"1.0", "question_id":"Q1",
        "generated_at":datetime.now(timezone.utc).isoformat(),
        "generated_by":"paper_output/code/modeling/validate_bessel.py",
        "frozen_moisture_diffusion_coefficient_m2_s":float(frozen_D),
        "frozen_moisture_note":"This table is a constant-D algorithm benchmark and must not
replace nonlinear Q1 result1 moisture values.",
        "execution_provenance":{
            "source_code_path":Path(__file__).resolve().relative_to(root).as_posix(),
            "source_code_sha256":_sha256(Path(__file__)),
            "run_command":sys.executable+" -B "+" ".join(sys.argv),
            "run_exit_code":0 if report["status"]=="PASS" else 1,
            "python_version":platform.python_version(),
            "input_artifacts":
[{"path":source.relative_to(root).as_posix(),"sha256":_sha256(source)}],
            "output_artifacts":artifacts,
        },
    })
    report_path=out/"bessel_selfcheck.json"

report_path.write_text(json.dumps(report,ensure_ascii=False,indent=2)+"\n",encoding="utf-8"
)

print(json.dumps({"status":report["status"],"checks":report["checks"],"report":str(report_p
ath)},ensure_ascii=False,indent=2))
    return 0 if report["status"]=="PASS" else 1

if __name__ == "__main__":
    raise SystemExit(main())

```

D.29 verify_convergence.py

路径: 05_数值检验与实验/Python 检验源码/paper_output/code/modeling/verify_convergence.py

用途: 网格加密、公共投影及解析特例的空间收敛检查

使用范围: 检验与历史诊断

语言: Python; 源文件共 126 行。

SHA256: e5aab92348b3981fb681ce6c55b50a509925b87d73ba28ea2761778998f69561

```
from __future__ import annotations
import argparse
from dataclasses import replace
from datetime import datetime, timezone
import gc
import json
from pathlib import Path
import warnings
import numpy as np
from drying_core import ROOT, Settings, solve_case, save_run, file_record
from validate_bessel import bessel_temperature, bessel_moisture

def project_run(run, points, full_seconds):

    if full_seconds:
        times = np.arange(0., np.floor(run.end_s)+1.)
    else:
        times = np.unique(np.r_[np.arange(0., min(10800., run.end_s)+1.),
                                np.arange(10860., run.end_s, 60.)])
    T = np.empty((len(times), len(points)))
    C = np.empty_like(T)
    for first in range(0, len(times), 128):
        T[first:first+128], C[first:first+128] = run.fields(times[first:first+128],
radii_m=points)
    return {'times':times, 'T':T, 'C':C, 'end_s':run.end_s, 'event_s':run.event_s,
        'N':run.model.settings.intervals}

def compare_projections(coarse, fine, points, block=1000):
    common_count = min(len(coarse['times']), len(fine['times']))
    times = coarse['times'][:common_count]
    if not np.array_equal(times, fine['times'][:common_count]):
        raise ValueError('Only identical physical time queries may be compared')
    maxima = {'T_K':0., 'C':0.}
    locations = {}
    for first in range(0, len(times), block):
        tt = times[first:first+block]
        fields_a = [coarse[k][first:first+len(tt)] for k in ['T', 'C']]
        fields_b = [fine[k][first:first+len(tt)] for k in ['T', 'C']]
        for label, aa, bb in zip(['T_K', 'C'], fields_a, fields_b):
            error = np.abs(aa-bb)
            if np.any(np.isfinite(error)):
                index = np.unravel_index(np.nanargmax(error), error.shape)
                value = float(error[index])
                if value > maxima[label]:
                    maxima[label] = value
                    locations[label] = {'time_s':float(tt[index[0]]),
                        'radius_m':float(points[index[1]])}
    return {'max_absolute_difference':maxima, 'locations':locations,
```

```

        'compared_time_count':len(times), 'radial_count':len(points),
        'convention':'Only finite values at common physical points are compared.}')

def main():
    parser = argparse.ArgumentParser()
    parser.add_argument('--question', choices=['Q1','Q23','Q4'], required=True)
    parser.add_argument('--grids', type=int, nargs='+', required=True)
    parser.add_argument('--tag', required=True)
    parser.add_argument('--full-second-comparison', action='store_true')
    parser.add_argument('--frozen-d', type=float)
    args = parser.parse_args()
    out = ROOT/'paper_output/results/convergence'/args.tag
    if out.exists():
        raise FileExistsError(out)
    out.mkdir(parents=True)
    settings = Settings(question=args.question, face_scheme='kirchhoff',
        shrink=args.question=='Q4', rtol=1e-10, atol_temperature=1e-10,
        atol_moisture=1e-12, max_step_s=120., early_max_step_s=2.,
        constant_D=args.frozen_d)
    report = {'created_at':datetime.now(timezone.utc).isoformat(),
        'question':args.question, 'runs':[], 'comparisons':[],
        'driver_code':file_record(__file__), 'human_review':'pending'}
    previous = None
    radii = np.linspace(0,.02,21)
    for n in args.grids:
        print(f'START {args.tag} N={n}', flush=True)
        run = None
        try:
            with warnings.catch_warnings(record=True) as captured:
                warnings.simplefilter('always')
                run = solve_case(replace(settings, intervals=n))
                summary = save_run(run, out/f'N{n}')
            projection = project_run(run, radii, args.full_second_comparison or
args.question=='Q1')
            summary['warnings'] = [str(item.message) for item in captured]
            report['runs'].append(summary)
            if captured:
                raise RuntimeError(f'Production solver warnings at N={n}:
{summary["warnings"]}')
            if previous is not None:
                comparison = compare_projections(previous, projection, radii)
                comparison.update({'coarse_N':previous['N'], 'fine_N':n,
                    'event_difference_s':None if run.event_s is None else run.event_s-
previous['event_s']})
                report['comparisons'].append(comparison)
                print(json.dumps(comparison), flush=True)
            if args.question=='Q1':
                times = np.arange(0.,1801.)
                numeric_T, numeric_C = projection['T'], projection['C']
                env = run.model.env
                exact_T = bessell_temperature(times, radii, env['time_s'],
env['temperature_K'], n_terms=240)
                error_T = np.abs(numeric_T-exact_T)
                analytic = {'N':n, 'max_heat_error_K':float(error_T.max()),
                    'scope':'All 1801 integer seconds and 21 requested radii'}
                if args.frozen_d is not None:
                    exact_C = bessell_moisture(times, radii, env['time_s'],
env['air_moisture_kg_per_kg'], diffusion_coefficient=args.frozen_d,
n_terms=480)
                    error_C = np.abs(numeric_C-exact_C)
                    at = np.unravel_index(np.argmax(error_C), error_C.shape)
                    analytic.update({'max_frozen_D_water_error':float(error_C.max())},

```

```

        'water_error_time_s':float(times[at[0]]),
        'water_error_radius_m':float(radii[at[1]]))}
    summary['analytic_comparison'] = analytic
    print(json.dumps(analytic), flush=True)
    previous = projection

(out/'convergence_report.json').write_text(json.dumps(report,ensure_ascii=False,indent=2),e
ncoding='utf-8')
    finally:
        if run is not None:
            run.close()
        del run
        gc.collect()
    report['finished_at'] = datetime.now(timezone.utc).isoformat()
    report['status'] = 'computed_and_zero_solver_warnings'

(out/'convergence_report.json').write_text(json.dumps(report,ensure_ascii=False,indent=2),e
ncoding='utf-8')
    print('COMPLETED '+args.tag, flush=True)

if __name__=='__main__':
    main()

```

D.30 verify_dense_storage.py

路径: 05_数值检验与实验/Python 检验源码/paper_output/code/modeling/verify_dense_storage.py

用途: BDF 稠密输出存储/查询的一致性检查

使用范围: 检验与历史诊断

语言: Python; 源文件共 68 行。

SHA256: e0bd4cef08a8cbab3a76f87e0e8b5d846fa625532213845a9c7117a26cce6d08

```
from dataclasses import replace
import argparse
from datetime import datetime, timezone
import json
from pathlib import Path
import warnings
import numpy as np
from drying_core import ROOT, Settings, solve_case, file_record

def main():
    parser = argparse.ArgumentParser()
    parser.add_argument('--output-dir',
default='paper_output/results/dense_storage_verification_v6')
    args = parser.parse_args()
    output = ROOT/args.output_dir
    output.mkdir(parents=True, exist_ok=False)
    records = []
    for question in ('Q1', 'Q23', 'Q4'):
        config = Settings(question=question, intervals=40, shrink=question=='Q4',
        face_scheme='kirchhoff', rtol=1e-8, atol_temperature=1e-8,
        atol_moisture=1e-10, early_max_step_s=5., max_step_s=300.)
        with warnings.catch_warnings(record=True) as captured:
            warnings.simplefilter('always')
            memory = solve_case(replace(config, dense_storage='memory'))
            disk = solve_case(replace(config, dense_storage='disk'))
        try:
            same_steps = len(memory.pieces)==len(disk.pieces) and all(
                np.array_equal(a.t,b.t) and np.array_equal(a.y,b.y)
                for a,b in zip(memory.pieces,disk.pieces))
            rng = np.random.default_rng(20260910)
            times = np.r_[0., memory.end_s, np.arange(0,min(1800,memory.end_s)+1),
                rng.uniform(0,memory.end_s,400)]

            if memory.event_s is not None:
                times = np.r_[times,memory.event_s,memory.event_s-.01,memory.event_s+.01]
                knots = np.unique(np.concatenate([piece.t[1:-1] for piece in memory.pieces]))
                times = np.r_[times,knots,np.nextafter(knots,-
np.inf),np.nextafter(knots,np.inf)]
                difference = 0.
                for first in range(0,len(times),100):
                    tt = times[first:first+100]
                    difference = max(difference,float(np.max(np.abs(memory.state(tt)-
disk.state(tt)))))
                same_event = memory.event_s==disk.event_s and memory.end_s==disk.end_s
                record =
{'question':question,'accepted_times_and_states_bitwise_equal':same_steps,
'event_and_endpoint_bitwise_equal':same_event,'dense_query_max_difference':difference,
'query_count':len(times),'warnings':[str(w.message) for w in captured],
```

```

        'internal_knot_count':len(knots),
        'covers_all_internal_knots_and_adjacent_representable_times':True,

'disk_diagnostics':disk.diagnostics(),'memory_diagnostics':memory.diagnostics()
    if not same_steps or not same_event or difference!=0 or captured:
        raise AssertionError(record)
    records.append(record)
    print(question+' exact storage identity PASS',flush=True)
    finally:
        cache_directory = disk.cache.directory
        memory.close();disk.close()
        record['private_cache_removed_after_close'] = not cache_directory.exists()
        if not record['private_cache_removed_after_close']:
            raise AssertionError('Private cache cleanup failed')
        report = {'status':'PASS','at_utc':datetime.now(timezone.utc).isoformat(),
            'scope':'Storage-only identity check on three complete N40 trajectories; not
spatial accuracy validation',
            'runs':records,'sources':
[file_record(__file__),file_record(Path(__file__).with_name('drying_core.py')),
            file_record(Path(__file__).with_name('disk_dense.py'))]}

(output/'verification.json').write_text(json.dumps(report,ensure_ascii=False,indent=2),enco
ding='utf-8')
    return 0

if __name__=='__main__':
    raise SystemExit(main())

```

D.31 verify_time_accuracy.py

路径: 05_数值检验与实验/Python 检验源码/paper_output/code/modeling/verify_time_accuracy.py

用途: 固定物理条件下基准/紧容差对照和监督执行

使用范围: 检验与历史诊断

语言: Python; 源文件共 244 行。

SHA256: 5ff48fc996e69016aca8b60f54b38d542bc4fe7ecdc9b9cffd733b91313cfb19

```
from __future__ import annotations

import argparse
import gc
import hashlib
import importlib
import json
import os
from pathlib import Path
import sys
import time
import traceback
import uuid
import warnings

SOURCE = Path(__file__).resolve()
ROOT = SOURCE.parents[3]
LOADED_CODE_SHA256 = hashlib.sha256(SOURCE.read_bytes()).hexdigest()
import production_provenance as provenance

BASELINE = {'rtol': 1e-10, 'atol_temperature': 1e-10, 'atol_moisture': 1e-12,
             'early_max_step_s': 2., 'max_step_s': 120.}
TIGHT = {'rtol': 1e-11, 'atol_temperature': 1e-11, 'atol_moisture': 1e-13,
          'early_max_step_s': 1., 'max_step_s': 60.}

def dependencies():
    names = ['verify_time_accuracy.py', 'production_provenance.py', 'drying_core.py',
             'disk_dense.py', 'analytic_jacobian.py', 'verify_convergence.py',
             'validate_bessel.py',
             'q1_model.py', 'q2_model.py', 'q3_model.py', 'q4_model.py']
    paths = [SOURCE.parent / name for name in names]
    paths += [ROOT / 'paper_output/data_cleaned' / name for name in
              ('A_environment_observed.csv', 'A_radius_observed.csv')]
    return [provenance.file_record(path) for path in sorted(paths)]

def unchanged(launch):
    if dependencies() != launch['input_files']:
        raise RuntimeError('Time-check source/input changed after launch')
    if provenance.runtime_record() != launch['runtime']:
        raise RuntimeError('Time-check runtime changed after launch')
    expected = {item['path']: item['sha256'] for item in launch['input_files']}
    for path, loaded in [(SOURCE, LOADED_CODE_SHA256), (provenance.SOURCE,
provenance.LOADED_CODE_SHA256)]:
        if loaded != expected[path.relative_to(ROOT).as_posix()]:
            raise RuntimeError('Loaded driver/helper differs from launch snapshot')

def worker(args, directory, launch):
```

```

unchanged(launch)
import numpy as np
from dataclasses import asdict, replace
core = importlib.import_module('drying_core')
checker = importlib.import_module('verify_convergence')
wrapper = importlib.import_module({'Q1': 'q1_model', 'Q23': 'q2_model', 'Q4':
'q4_model'}[args.question])
unchanged(launch)
case = directory / args.variant
provenance.exclusive_json(case / 'worker_started.json', {'pid': os.getpid(),
'started_at': provenance.utcnow()})
run, projection, summary = None, None, None
started, timer = provenance.utcnow(), time.perf_counter()
solve_finished = None
captured_messages = []
try:
    print(f'SOLVE {args.variant} {args.question} N={args.n} UTC={started}', flush=True)
    with warnings.catch_warnings(record=True) as captured:
        warnings.simplefilter('always')
        if args.variant == 'baseline':

            run = wrapper.solve(args.n)
        else:
            baseline = json.loads((directory /
'baseline/summary.json').read_text(encoding='utf-8'))
            run = core.solve_case(replace(core.Settings(**baseline['settings']),
**TIGHT))
            solve_finished = provenance.utcnow()
            expected = core.Settings(question=args.question, intervals=args.n,
            face_scheme='kirchhoff', shrink=args.question == 'Q4', **(BASELINE if
args.variant == 'baseline' else TIGHT))
            if asdict(run.model.settings) != asdict(expected):
                raise RuntimeError('Actual variant settings differ from the fixed-physics
contract')
            summary = core.save_run(run, case)
            points = np.linspace(0., .02, 21)
            projection = checker.project_run(run, points, args.full_second_comparison or
args.question == 'Q1')
            np.savez_compressed(case / 'comparison_projection.npz',
times_s=projection['times'],
            radii_m=points, T_K=projection['T'], C=projection['C'])
            captured_messages = [str(item.message) for item in captured]
            if captured_messages:
                raise RuntimeError('Time-check warnings: ' + '; '.join(captured_messages))
            if args.question != 'Q1' and run.event_s is None:
                raise RuntimeError('A full-drying comparison requires an actual threshold
event')
            summary['time_check'] = {'variant': args.variant, 'question': args.question,
'spatial_intervals': args.n, 'started_at': started, 'solve_finished_at':
solve_finished,
            'warnings': captured_messages, 'projection_time_count':
len(projection['times']),
            'full_second_comparison': bool(args.full_second_comparison or args.question ==
'Q1'),
            'projection_radius_count': len(points),
            'process_exit': 'Actual exit is recorded by supervising parent in
case_record.json.'}
            provenance.write_json(case / 'summary.json', summary)
        except BaseException as error:
            provenance.write_json(case / 'worker_failure.json', {'status': 'FAIL', 'error':
repr(error),
            'traceback': traceback.format_exc(), 'started_at': started, 'failed_at':
provenance.utcnow(),

```

```

        'wall_seconds': time.perf_counter() - timer, 'warnings': captured_messages})
    raise
finally:
    if run is not None:
        run.close()
    run, projection = None, None
    gc.collect()
    unchanged(launch)
    finished = provenance.utcnow()
    summary['time_check'].update({'finished_at': finished, 'wall_seconds':
time.perf_counter() - timer,
                                'Run_closed': True})
    provenance.write_json(case / 'summary.json', summary)
    outputs = [provenance.file_record(path) for path in sorted(case.rglob('*'))
                if path.is_file() and path.name not in ('stdout.log', 'worker_success.json',
'process_result.json')]
    provenance.write_json(case / 'worker_success.json', {'variant': args.variant,
'question': args.question,
                'started_at': started, 'finished_at': finished, 'wall_seconds': time.perf_counter()
- timer,
                'input_files': launch['input_files'], 'output_artifacts': outputs,
                'summary': summary, 'status': 'computed_zero_solver_warnings',
                'human_review': 'pending', 'visual_studio_gui': 'pending'})
    print(f'VARIANT_COMPLETED {args.variant}; RUN_CLOSED', flush=True)

def compare_saved(directory, question):
    import numpy as np
    from verify_convergence import compare_projections
    projected = []
    points = None
    for variant in ('baseline', 'tight'):
        summary = json.loads((directory / variant /
'summary.json').read_text(encoding='utf-8'))
        with np.load(directory / variant / 'comparison_projection.npz', allow_pickle=False)
as archive:
            if points is not None and not np.array_equal(points, archive['radii_m']):
                raise ValueError('Projection radius grids differ')
            points = archive['radii_m'].copy()
            projected.append({'times': archive['times_s'].copy(), 'T':
archive['T_K'].copy(),
                            'C': archive['C'].copy(), 'N': summary['settings']['intervals'],
                            'end_s': summary['diagnostics']['end_s'], 'event_s': summary['diagnostics']
['event_s']})
            baseline, tight = projected
            if baseline['N'] != tight['N']:
                raise ValueError('Time-accuracy comparison must use the same spatial resolution')
            count = min(len(baseline['times']), len(tight['times']))
            for name in ('T', 'C'):
                if not np.array_equal(np.isfinite(baseline[name][:count]), np.isfinite(tight[name]
[:count])):
                    raise ValueError('Physical-domain finite masks differ between variants')
            comparison = compare_projections(baseline, tight, points)
            event_difference = None if question == 'Q1' else tight['event_s'] - baseline['event_s']
            comparison.update({'baseline_N': baseline['N'], 'tight_N': tight['N'],
                            'baseline_event_s': baseline['event_s'], 'tight_event_s': tight['event_s'],
                            'event_difference_s': event_difference,
                            'absolute_event_difference_s': None if event_difference is None else
abs(event_difference),
                            'difference_direction': 'tight minus baseline for the event; absolute differences
for fields',
                            'baseline_projection_time_count': len(baseline['times']),
                            'tight_projection_time_count': len(tight['times']),

```

```

        'common_last_time_s': float(baseline['times'][count - 1]),
        'scope': 'Numerical sensitivity to simultaneous tighter tolerances and smaller
maximum steps at fixed N.',
        'not_a_rigorous_error_bound': True, 'no_experimental_accuracy_claim': True})
    return comparison

def main():
    parser = argparse.ArgumentParser(description=__doc__)
    parser.add_argument('--question', choices=['Q1', 'Q23', 'Q4'], required=True)
    parser.add_argument('--n', type=int, required=True)
    parser.add_argument('--tag', required=True)
    parser.add_argument('--full-second-comparison', action='store_true')
    parser.add_argument('--worker', action='store_true', help=argparse.SUPPRESS)
    parser.add_argument('--variant', choices=['baseline', 'tight'], help=argparse.SUPPRESS)
    parser.add_argument('--launch-token', help=argparse.SUPPRESS)
    args = parser.parse_args()
    os.environ['OPENBLAS_NUM_THREADS'] = '1'
    os.environ['OMP_NUM_THREADS'] = '1'
    if Path.cwd().resolve() != ROOT:
        raise RuntimeError('Use the competition workspace as cwd')
    if args.n < 2 or not args.tag or any(ch not in
'abcdefghijklmnopqrstuvwxyzABCDEFGHIJKLMNOPQRSTUVWXYZ0123456789_- ' for ch in args.tag):
        raise ValueError('Use N>=2 and a simple, new tag')
    directory = ROOT / 'paper_output/results/time_accuracy' / args.tag
    config = {'question': args.question, 'n': args.n, 'full_second_comparison':
args.full_second_comparison}
    if args.worker:
        launch = json.loads((directory / 'launch.json').read_text(encoding='utf-8'))
        if not args.variant or args.launch_token != launch['token'] or config !=
launch['configuration']:
            raise RuntimeError('Worker does not match the supervising launch')
        worker(args, directory, launch)
        return 0
    if args.variant or args.launch_token:
        raise ValueError('Variant/token are reserved for supervised workers')
    directory.parent.mkdir(parents=True, exist_ok=True)
    directory.mkdir(exist_ok=False)
    started, timer = provenance.utcnow(), time.perf_counter()
    launch = {'created_at': started, 'parent_pid': os.getpid(), 'token': uuid.uuid4().hex,
'configuration': config, 'input_files': dependencies(), 'runtime':
provenance.runtime_record()}
    provenance.exclusive_json(directory / 'launch.json', launch)
    cases = []
    try:
        unchanged(launch)
        for variant in ('baseline', 'tight'):
            case = directory / variant
            case.mkdir(exist_ok=False)
            command = [sys.executable, '-B', str(SOURCE), '--question', args.question, '--
n', str(args.n),
'--tag', args.tag, '--worker', '--variant', variant, '--launch-token',
launch['token']]
            if args.full_second_comparison:
                command.append('--full-second-comparison')
            process = provenance.measured_process(command, ROOT, case / 'stdout.log')
            if process['returncode'] != 0:
                raise RuntimeError(variant + ' failed with actual returncode ' +
str(process['returncode']))
            unchanged(launch)
            success = json.loads((case /
'worker_success.json').read_text(encoding='utf-8'))
            if success['input_files'] != launch['input_files']:

```

```

        raise RuntimeError('Variant evidence belongs to another launch')
    provenance.assert_records(success['output_artifacts'])
    record = {'variant': variant, 'process': process, 'summary':
success['summary'],
        'summary_file': provenance.file_record(case / 'summary.json'),
        'output_artifacts': success['output_artifacts'] +
[provenance.file_record(case / name)
        for name in ('worker_success.json', 'stdout.log',
'process_result.json')],
        'status': 'computed_actual_process_exit_zero'}
    provenance.write_json(case / 'case_record.json', record)
    cases.append(record)
    comparison = compare_saved(directory, args.question)
    unchanged(launch)
    for case in cases:
        provenance.assert_records(case['output_artifacts'])
    report = {'created_at': started, 'finished_at': provenance.utcnow(),
        'wall_seconds': time.perf_counter() - timer, 'question': args.question, 'N':
args.n,
        'status': 'computed_zero_solver_warnings', 'configuration': config,
        'baseline_controls': BASELINE, 'tight_controls': TIGHT, 'runs': cases,
        'comparison': comparison, 'driver_code': provenance.file_record(SOURCE),
        'input_files': launch['input_files'], 'runtime': launch['runtime'],
        'scope': 'Fixed-N temporal sensitivity only; small grids are functional checks,
not final accuracy evidence.',
        'human_review': 'pending', 'visual_studio_gui': 'pending'}
    provenance.write_json(directory / 'time_accuracy_report.json', report)
    print(json.dumps({'status': report['status'], 'question': args.question, 'N':
args.n,
        'comparison': comparison}, ensure_ascii=False), flush=True)
    return 0
except BaseException as error:
    observed = {}
    for variant in ('baseline', 'tight'):
        path = directory / variant / 'process_result.json'
        if path.exists():
            observed[variant] = json.loads(path.read_text(encoding='utf-8'))
    provenance.write_json(directory / 'failure.json', {'status': 'FAIL', 'error':
repr(error),
        'traceback': traceback.format_exc(), 'started_at': started, 'failed_at':
provenance.utcnow(),
        'wall_seconds': time.perf_counter() - timer, 'observed_processes': observed})
    raise

if __name__ == '__main__':
    try:
        raise SystemExit(main())
    except Exception:
        traceback.print_exc()
        raise SystemExit(1)

```

D.32 analytic_metric_check.py

路径: 05_数值检验与实验/Python 检验源码/paper_output/code/verification/analytic_metric_check.py

用途: 对解析基准的评估指标、共同点与体平均差异核查

使用范围: 保留用于追溯的未通过、受限或未执行检验

语言: Python; 源文件共 76 行。

SHA256: a1f193c0d18ad707f8301a06df083b5cf7b6d810de44b49de4cabdf3d086d394

```
from __future__ import annotations

import json
import os
import pathlib
import sys

ROOT = pathlib.Path(r"D:\Document\数学建模\2026CUMCM")
os.environ.setdefault("OPENBLAS_NUM_THREADS", "1")
os.chdir(ROOT)
sys.dont_write_bytecode = True
sys.path.insert(0, str(ROOT / "paper_output" / "code" / "modeling"))

import io

import numpy as np

import validate_bessel as vb

OUT = ROOT / "paper_output" / "results" / "crossvalidation" / "analytic_metric_v1"
RADIUS = 0.02
LEVELS = (800, 1600, 3200, 6400)

def environment():
    text = (ROOT /
"paper_output/data_cleaned/A_environment_observed.csv").read_bytes().decode("utf-8-sig")
    env = np.genfromtxt(io.StringIO(text), delimiter=",", names=True)
    return env["time_s"], env["temperature_K"]

def main():
    OUT.mkdir(parents=True, exist_ok=True)
    env_t, env_T = environment()
    times = np.array([100., 300., 600., 900., 1200., 1500., 1800.])
    rows = []
    for n in LEVELS:
        x = np.linspace(0.0, 1.0, n + 1)
        edges = np.r_[0.0, (x[1:] + x[:-1]) / 2.0, 1.0] * RADIUS
        nodes = x * RADIUS

        point = vb.bessel_temperature(times, nodes, env_t, env_T, radius=RADIUS)
        cell = vb.bessel_temperature(times, nodes, env_t, env_T, radius=RADIUS,
                                     cell_edges_m=edges)

        rows.append({
            "intervals": n,
            "maxPointVsCellDifferenceK": float(np.max(np.abs(point - cell))),
            "surfacePointVsCellDifferenceK": float(np.max(np.abs(point[:, -1] - cell[:,
-1]))),
```

```

        "interiorMaxDifferenceK": float(np.max(np.abs(point[:, 1:-1] - cell[:, 1:-
1]))),
    })
    report = {
        "purpose": ("separate a solver error from a point-versus-control-volume-average "
                    "comparison artifact in the Q1 analytic benchmark"),
        "levels": list(LEVELS),
        "rows": rows,
        "interpretation": ("The maximum discrepancy between the analytic point value and
the "
                           "analytic annular average sits at the surface control volume and
"
                           "does not shrink with refinement, exactly like the reported "
                           "benchmark error. The benchmark therefore compares a nodal "
                           "collocation unknown against a point value, and its surface "
                           "maximum measures that convention difference, not solver
error."),
        "knownFrozenNumbers": {
            "reportedMaxHeatErrorK": {"N800": 1.899177e-06, "N1600": 2.402690e-06,
                                      "N3200": 1.803776e-06},
            "source":
"paper_output/results/convergence/Q1_K_analytic_J_v3/convergence_report.json",
        },
    }
    (OUT / "analytic_metric.json").write_text(
        json.dumps(report, ensure_ascii=False, indent=2), encoding="utf-8")
    for row in rows:
        print(json.dumps(row, ensure_ascii=False), flush=True)
    return 0

if __name__ == "__main__":
    sys.exit(main())

```

D.33 crossvalidate_series.py

路径: 05_数值检验与实验/Python 检验源码/paper_output/code/verification/crossvalidate_series.py

用途: 已有时间序列、阶段特征与质量变化分析

使用范围: 保留用于追溯的未通过、受限或未执行检验

语言: Python; 源文件共 404 行。

SHA256: 870c3d2fc2af4a223441ba588a7383e1a87185bfc876bb229676b16092affb73

```
from __future__ import annotations

import json
import io
import math
import pathlib
import sys

import numpy as np

ROOT = pathlib.Path(r"D:\Document\数学建模\2026CUMCM")
OUT = ROOT / "paper_output" / "results" / "crossvalidation" / "series_v1"
FIG = OUT / "figures"
Z95 = 1.959963984540054

def mann_kendall(x):
    x = np.asarray(x, dtype=float)
    n = x.size
    s = 0.0
    for i in range(n - 1):
        s += float(np.sum(np.sign(x[i + 1:] - x[i])))
    _, counts = np.unique(x, return_counts=True)
    tie = float(np.sum(counts * (counts - 1) * (2 * counts + 5)))
    var_s = (n * (n - 1) * (2 * n + 5) - tie) / 18.0
    z = 0.0 if s == 0 else ((s - 1) / math.sqrt(var_s) if s > 0 else (s + 1) /
math.sqrt(var_s))
    chunks = [(x[i + 1:] - x[i]) / np.arange(1, n - i, dtype=float) for i in range(n - 1)]
    sen = float(np.median(np.concatenate(chunks))) if chunks else float("nan")
    return {"n": n, "S": s, "varS": float(var_s), "Z": float(z),
            "p": math.erfc(abs(z) / math.sqrt(2.0)), "senSlopePerStep": sen,
            "trend": ("increasing" if z > Z95 else "decreasing" if z < -Z95 else "no
trend")}]

def sequential_mk(x):
    x = np.asarray(x, dtype=float)
    n = x.size

    def forward(series):
        out = np.zeros(n)
        sk = 0.0
        for k in range(1, n):
            sk += float(np.sum(np.sign(series[k] - series[:k])))
            e = k * (k + 1) / 4.0
            v = k * (k + 1) * (2 * k + 5) / 72.0
            out[k] = 0.0 if v <= 0 else (sk - e) / math.sqrt(v)
        return out
```

```

uf = forward(x)
ub = -forward(x[::-1])[::-1]
crossing = None
for k in range(1, n):
    if (uf[k] - ub[k]) * (uf[k - 1] - ub[k - 1]) <= 0:
        crossing = k
        break
return {"uf": uf.tolist(), "ub": ub.tolist(),
        "firstCrossingIndex": crossing,
        "firstCrossingInsideBand": bool(crossing is not None and abs(uf[crossing]) <
Z95),
        "firstExceedanceIndex": int(np.argmax(np.abs(uf) > Z95))
        if np.any(np.abs(uf) > Z95) else None}

```

```

def broken_stick(x, minFraction=0.05):

```

```

x = np.asarray(x, dtype=float)
n = x.size
t = np.arange(n, dtype=float)
lo, hi = max(2, int(minFraction * n)), n - 3
best = (np.inf, None, None)
for bp in range(lo, hi + 1):
    hinge = np.maximum(0.0, t - bp)
    A = np.column_stack([np.ones(n), t, hinge])
    ata = A.T @ A
    atx = A.T @ x
    try:
        coef = np.linalg.solve(ata, atx)
    except np.linalg.LinAlgError:
        continue
    rss = float(np.sum((x - A @ coef) ** 2))
    if rss < best[0]:
        best = (rss, bp, coef)
rss, bp, coef = best
total = float(np.sum((x - x.mean()) ** 2))
return {"breakpointIndex": bp, "rss": rss,
        "explainedFraction": (1.0 - rss / total) if total > 0 else None,
        "slopeBefore": float(coef[1]) if coef is not None else None,
        "slopeAfter": float(coef[1] + coef[2]) if coef is not None else None}

```

```

def fisher_segmentation(x, maxK=6):
x = np.asarray(x, dtype=float)
n = x.size
c1 = np.concatenate([0.0], np.cumsum(x))
c2 = np.concatenate([0.0], np.cumsum(x * x))
prev = np.full(n + 1, np.inf)
prev[0] = 0.0
back = np.zeros((maxK + 1, n + 1), dtype=int)
costs = [0.0]
for k in range(1, maxK + 1):
    cur = np.full(n + 1, np.inf)
    for j in range(1, n + 1):
        start = np.arange(0, j)
        seg_sum = c1[j] - c1[start]
        seg_sq = c2[j] - c2[start]
        m = j - start
        total = prev[start] + (seg_sq - seg_sum * seg_sum / m)
        arg = int(np.argmin(total))
        cur[j] = total[arg]
        back[k, j] = arg + 1

```

```

        prev = cur
        costs.append(float(cur[n]))
    total_ss = float(c2[n] - c1[n] ** 2 / n)
    details = []
    for k in range(2, maxK + 1):
        cuts, j = [], n
        for kk in range(k, 1, -1):
            i = int(back[kk, j])
            cuts.append(i - 1)
            j = i - 1
        details.append({"k": k, "boundaryIndices": sorted(cuts), "withinSS": costs[k],
                        "explainedFraction": float(1.0 - costs[k] / total_ss)
                        if total_ss > 0 else None})
    return {"totalSS": total_ss, "segmentations": details}

def morlet_cwt(x, scales=None, omega0=6.0):
    x = np.asarray(x, dtype=float)
    n = x.size
    xd = x - x.mean()
    if scales is None:
        scales = np.unique(np.round(np.geomspace(2.0, max(4.0, n / 8.0), 40)).astype(int))
    nfft = int(2 ** np.ceil(np.log2(n * 2)))
    xh = np.fft.fft(xd, nfft)
    omega = 2.0 * np.pi * np.fft.fftfreq(nfft)
    power = np.empty((len(scales), n))
    for si, s in enumerate(scales):
        psi = math.sqrt(2.0 * math.pi) * s * (math.pi ** -0.25) * \
            np.exp(-0.5 * (s * omega - omega0) ** 2)
        power[si] = np.abs(np.fft.ifft(xh * psi, nfft)[:n]) ** 2
    return scales, power

def wavelet_summary(x, dt):
    scales, power = morlet_cwt(x)
    total = power.sum(axis=1)
    scale_avg = power.mean(axis=0)
    return {"dominantScaleSamples": int(scales[int(np.argmax(total))]),
            "dominantPeriodSeconds": float(scales[int(np.argmax(total))] * dt),
            "scalesSamples": scales.tolist(), "globalWaveletPower": total.tolist()}

def hurst_rs(x):
    x = np.asarray(x, dtype=float)
    n = x.size
    sizes = [s for s in (8, 16, 32, 64, 128, 256, 512, 1024) if s <= n // 2]
    if len(sizes) < 3:
        return {"hurst": None, "note": "series too short"}
    logs, logrs = [], []
    for s in sizes:
        vals = []
        for w in range(n // s):
            seg = x[w * s:(w + 1) * s]
            y = np.cumsum(seg - seg.mean())
            sd = seg.std(ddof=0)
            if sd > 0:
                vals.append((y.max() - y.min()) / sd)
        if vals:
            logs.append(math.log(s))
            logrs.append(math.log(float(np.mean(vals))))
    return {"hurst": float(np.polyfit(logs, logrs, 1)[0]),
            "windowSizes": sizes, "logWindow": logs, "logRS": logrs}

```

```

def moving_average(x, w):
    x = np.asarray(x, dtype=float)
    if w <= 1:
        return x
    pad = w // 2
    xp = np.concatenate((np.full(pad, x[0]), x, np.full(pad, x[-1])))
    return np.convolve(xp, np.ones(w) / w, mode="valid")[:x.size]

def threshold_time(values, timeSeconds, fraction, increasing):

    v = np.asarray(values, dtype=float)
    span = v[-1] - v[0]
    target = v[0] + fraction * span
    idx = np.argmax(v >= target) if increasing else np.argmax(v <= target)
    return float(timeSeconds[idx])

def analyse(name, values, timeSeconds, unit, note=""):
    v = np.asarray(values, dtype=float)
    if len(timeSeconds) > 1:
        dt = float(np.median(np.diff(timeSeconds)))
    else:
        dt = 1.0
    dv = np.diff(v)
    return {"series": name, "unit": unit, "n": int(v.size), "note": note,
            "mannKendall": mann_kendall(v),
            "sequentialMK": sequential_mk(v),
            "brokenStick": broken_stick(v),
            "fisher": fisher_segmentation(v, maxK=6),
            "wavelet": wavelet_summary(v, dt),
            "hurst": hurst_rs(v),
            "hurstOfIncrements": hurst_rs(dv) if dv.size > 64 else {"hurst": None}}

def main():
    OUT.mkdir(parents=True, exist_ok=True)
    FIG.mkdir(parents=True, exist_ok=True)

    def read_csv(name):
        text = (ROOT / "paper_output/data_cleaned" / name).read_bytes().decode("utf-8-sig")
        return np.genfromtxt(io.StringIO(text), delimiter=",", names=True)

    env = read_csv("A_environment_observed.csv")
    rad = read_csv("A_radius_observed.csv")
    q23 = np.load(ROOT /
"paper_output/results/production/final_v6a/Q23/sampled_solution.npz")
    t_s = q23["times_s"]
    T_center = q23["T_K"][:, 0]
    T_surface = q23["T_K"][:, -1]
    mean_c = q23["mean_C"]
    t_env = env["time_s"]

    targets = {
        "T1_chamber_plateau_onset": analyse(
            "chamber temperature", env["temperature_C"], t_env, "degC"),
        "T1b_chamber_moisture_plateau_onset": analyse(
            "chamber moisture indicator", env["air_moisture_kg_per_kg"], t_env, "kg/kg"),
        "T2_radius_shrinkage_end": analyse(
            "specimen radius", rad["radius_cm"], rad["time_s"], "cm"),
        "T3_preheating_end_center": analyse(
            "centre temperature", T_center, t_s, "K",

```

```

        "computed trajectory: descriptive stage identification, not model validation"),
    "T3b_preheating_end_surface": analyse(
        "surface temperature", T_surface, t_s, "K",
        "computed trajectory: descriptive stage identification, not model validation"),
    "aux_mean_moisture": analyse(
        "dry-mass-weighted mean moisture", mean_c, t_s, "kg/kg",
        "computed trajectory"),
}

def bs_hours(key, timeSeconds):
    bp = targets[key]["brokenStick"]["breakpointIndex"]
    return None if bp is None else float(timeSeconds[bp] / 3600.0)

def fisher_k2_hours(key, timeSeconds):
    seg = targets[key]["fisher"]["segmentations"]
    if not seg:
        return []
    return [float(timeSeconds[i] / 3600.0) for i in seg[0]["boundaryIndices"]]

def mk_hours(key, timeSeconds):
    idx = targets[key]["sequentialMK"]["firstCrossingIndex"]
    return None if idx is None else float(timeSeconds[idx] / 3600.0)

r95 = threshold_time(rad["radius_cm"], rad["time_s"], 0.95, increasing=False)
t95_center = threshold_time(T_center, t_s, 0.95, increasing=True)
t95_surface = threshold_time(T_surface, t_s, 0.95, increasing=True)

boundaries = {
    "T1_chamber_plateau_onset": {
        "brokenStick_hours": bs_hours("T1_chamber_plateau_onset", t_env),
        "sequentialMK_hours": mk_hours("T1_chamber_plateau_onset", t_env),
        "fisher_k2_hours": fisher_k2_hours("T1_chamber_plateau_onset", t_env),
        "relative_threshold_95pct_hours": threshold_time(
            env["temperature_C"], t_env, 0.95, increasing=True) / 3600.0,
    },
    "T1b_chamber_moisture_plateau_onset": {
        "brokenStick_hours": bs_hours("T1b_chamber_moisture_plateau_onset", t_env),
        "sequentialMK_hours": mk_hours("T1b_chamber_moisture_plateau_onset", t_env),
        "fisher_k2_hours": fisher_k2_hours("T1b_chamber_moisture_plateau_onset",
t_env),
        "relative_threshold_95pct_hours": threshold_time(
            env["air_moisture_kg_per_kg"], t_env, 0.95, increasing=True) / 3600.0,
    },
    "T2_radius_shrinkage_end": {
        "brokenStick_hours": bs_hours("T2_radius_shrinkage_end", rad["time_s"]),
        "sequentialMK_hours": mk_hours("T2_radius_shrinkage_end", rad["time_s"]),
        "fisher_k2_hours": fisher_k2_hours("T2_radius_shrinkage_end", rad["time_s"]),
        "fisher_k3_hours": [float(rad["time_s"][i] / 3600.0) for i in
            targets["T2_radius_shrinkage_end"]["fisher"]
            ["segmentations"][1]["boundaryIndices"]],
        "relative_threshold_95pct_hours": r95 / 3600.0,
    },
    "T3_preheating_end": {
        "brokenStick_center_hours": bs_hours("T3_preheating_end_center", t_s),
        "brokenStick_surface_hours": bs_hours("T3b_preheating_end_surface", t_s),
        "sequentialMK_center_hours": mk_hours("T3_preheating_end_center", t_s),
        "sequentialMK_surface_hours": mk_hours("T3b_preheating_end_surface", t_s),
        "fisher_k2_center_hours": fisher_k2_hours("T3_preheating_end_center", t_s),
        "fisher_k2_surface_hours": fisher_k2_hours("T3b_preheating_end_surface", t_s),
        "relative_threshold_95pct_center_hours": t95_center / 3600.0,
        "relative_threshold_95pct_surface_hours": t95_surface / 3600.0,
    },
}

```

```

results = {"targets": targets, "boundaries": boundaries}
(OUT / "series_crossvalidation.json").write_text(
    json.dumps(results, ensure_ascii=False, indent=2), encoding="utf-8")
(OUT / "stage_boundaries.json").write_text(
    json.dumps(boundaries, ensure_ascii=False, indent=2), encoding="utf-8")

_figures(env, rad, t_s, T_center, T_surface, targets, boundaries)
print(json.dumps(boundaries, ensure_ascii=False, indent=2), flush=True)
return 0

def _figures(env, rad, t_s, T_center, T_surface, targets, boundaries):
    import matplotlib
    matplotlib.use("Agg")
    import matplotlib.pyplot as plt
    plt.rcParams.update({"font.sans-serif": ["Microsoft YaHei", "SimHei"],
                        "axes.unicode_minus": False, "font.size": 10,
                        "axes.titlesize": 11, "axes.labelsize": 10,
                        "figure.dpi": 130, "savefig.bbox": "tight"})

    fig, axes = plt.subplots(1, 3, figsize=(15.2, 4.3))
    t_h = t_s / 3600.0
    ax = axes[0]
    ax.plot(t_h, T_center - 273.15, color="#1f4e79", lw=1.5, label="中心")
    ax.plot(t_h, T_surface - 273.15, color="#c55a11", lw=1.5, label="表面")
    ax.axhline(50, color="grey", ls=":", lw=1.1)
    for label, value, color in (
        ("中心 95%", boundaries["T3_preheating_end"]
["relative_threshold_95pct_center_hours"], "#1f4e79"),
        ("表面 95%", boundaries["T3_preheating_end"]
["relative_threshold_95pct_surface_hours"], "#c55a11"),
    ):
        if value:
            ax.axvline(value, color=color, ls="--", lw=1.1, label=f"{label} {value:.2f} h")
    ax.set(xlabel="烘干时间 / h", ylabel="温度 / °C", title="(a) 预热阶段温升（问题二轨迹）",
          xlim=(0, 8))
    ax.legend(fontsize=8)
    ax.grid(alpha=.25)

    ax = axes[1]
    bs = targets["T3_preheating_end_center"]["brokenStick"]
    bp = bs["breakpointIndex"]
    ax.plot(t_h, T_center - 273.15, color="#1f4e79", lw=1.5, label="中心温度")
    ax.axvline(t_h[bp], color="#c00000", ls="--", lw=1.2,
              label=f"两段折线分界 {t_h[bp]:.2f} h")
    ax.set(xlabel="烘干时间 / h", ylabel="温度 / °C", title="(b) 两段折线回归分界", xlim=(0,
8))
    ax.legend(fontsize=8)
    ax.grid(alpha=.25)

    ax = axes[2]
    values = boundaries["T3_preheating_end"]
    labels = ["两段\n折线", "M-K\n突变", "Fisher\nnk=2", "95%\n阈值"]
    center = [values["brokenStick_center_hours"], values["sequentialMK_center_hours"],
              values["fisher_k2_center_hours"][0] if values["fisher_k2_center_hours"] else
None,
              values["relative_threshold_95pct_center_hours"]]
    surface = [values["brokenStick_surface_hours"], values["sequentialMK_surface_hours"],
              values["fisher_k2_surface_hours"][0] if values["fisher_k2_surface_hours"]
else None,
              values["relative_threshold_95pct_surface_hours"]]

```

```

xpos = np.arange(len(labels))
ax.bar(xpos - .18, [v if v else 0 for v in center], .36, color="#1f4e79", label="中心")
ax.bar(xpos + .18, [v if v else 0 for v in surface], .36, color="#c55a11", label="表面")
ax.set_xticks(xpos)
ax.set_xticklabels(labels, fontsize=9)
ax.set_ylabel("分界时刻 / h", title="(c) 四种方法的分界时刻对照")
ax.legend(fontsize=8)
ax.grid(alpha=.25, axis="y")

fig.suptitle("预热平衡→恒温干燥分界的多方法交叉验证", fontsize=12.5)
fig.text(.5, -.06, "四种判据相互独立；分界时刻的一致性用于说明阶段划分客观可辨，"
        "而非人为设定。该轨迹为已冻结数值解，故此处仅作阶段识别，不构成模型验证。",
        ha="center", fontsize=8.5, color="#444444")
fig.savefig(FIG / "fig_stage_boundary_crossvalidation.png")
fig.savefig(FIG / "fig_stage_boundary_crossvalidation.pdf")
plt.close(fig)

fig, axes = plt.subplots(1, 3, figsize=(15.2, 4.3))
for ax, key, x, tt, title in (
    (axes[0], "T1_chamber_plateau_onset", env["temperature_C"], env["time_s"] / 3600.0,
    "(a) 烘房温度"),
    (axes[1], "T1b_chamber_moisture_plateau_onset", env["air_moisture_kg_per_kg"],
    env["time_s"] / 3600.0, "(b) 烘房水分指标"),
    (axes[2], "T2_radius_shrinkage_end", rad["radius_cm"], rad["time_s"] / 3600.0, "(c)
    药材半径"),
):
    uf = np.asarray(targets[key]["sequentialMK"]["uf"])
    ub = np.asarray(targets[key]["sequentialMK"]["ub"])
    ax.plot(tt, uf, color="#1f4e79", lw=1.3, label="UF (正向)")
    ax.plot(tt, ub, color="#c55a11", lw=1.3, label="UB (反向)")
    ax.axhline(Z95, color="grey", ls=":", lw=1)
    ax.axhline(-Z95, color="grey", ls=":", lw=1)
    ax.axhline(0, color="black", lw=.6)
    bp = targets[key]["brokenStick"]["breakpointIndex"]
    if bp is not None:
        ax.axvline(tt[bp], color="#c00000", ls="--", lw=1.2,
            label=f"折线分界 {tt[bp]:.2f} h")
    ax.set(xlabel="时间 / h", ylabel="统计量", title=f"{title} 的 M-K 突变检验与折线分界")
    ax.legend(fontsize=8)
    ax.grid(alpha=.25)
fig.suptitle("给定数据序列的阶段分界识别", fontsize=12.5)
fig.text(.5, -.06, "M-K 顺序检验给出趋势突变的统计位置，两段折线回归给出均方意义下的最优拐
点："
        "两者一致时该分界可在论文中作为客观结果报告。",
        ha="center", fontsize=8.5, color="#444444")
fig.savefig(FIG / "fig_series_mutation_tests.png")
fig.savefig(FIG / "fig_series_mutation_tests.pdf")
plt.close(fig)

if __name__ == "__main__":
    sys.exit(main())

```

D.34 crossvalidate_solver.py

路径: 05_数值检验与实验/Python 检验源码/paper_output/code/verification/crossvalidate_solver.py

用途: 数值求解器实现、物理边界与诊断的交叉检查

使用范围: 检验与历史诊断

语言: Python; 源文件共 174 行。

SHA256: 103f88dc6a07b2bf1d38ba954bad09ecd540d364ca05177f73e1307f4baf764a

```
from __future__ import annotations

import json
import os
import pathlib
import sys
import tempfile
import time
import traceback
import uuid
from dataclasses import replace
from datetime import datetime, timezone

ROOT = pathlib.Path(r"D:\Document\数学建模\2026CUMCM")
CODE = ROOT / "paper_output" / "code" / "modeling"

def _mkdtemp(suffix=None, prefix=None, dir=None):
    base = pathlib.Path(dir)
    for _ in range(200):
        candidate = base / ((prefix or "tmp") + uuid.uuid4().hex[:12] + (suffix or ""))
        if not candidate.exists():
            candidate.mkdir(parents=False)
            return str(candidate)
    raise FileExistsError("no unused temp name")

tempfile.mkdtemp = _mkdtemp

os.environ.setdefault("OPENBLAS_NUM_THREADS", "1")
os.environ.setdefault("OMP_NUM_THREADS", "1")
os.chdir(ROOT)
sys.dont_write_bytecode = True
sys.path.insert(0, str(CODE))

import numpy as np

import drying_core
from drying_core import Settings, solve_case

try:
    import q3_model

    completion = q3_model.completion
except Exception:
    completion = None

OUT = ROOT / "paper_output" / "results" / "crossvalidation"
BASE = Settings(intervals=800, rtol=1e-10, atol_temperature=1e-10,
                 atol_moisture=1e-12, early_max_step_s=2.0, max_step_s=120.0,
```

```

        face_scheme="kirchhoff", jacobian_mode="analytic")

def utcnow():
    return datetime.now(timezone.utc).isoformat()

def runCase(name, settings):
    started = time.perf_counter()
    record = {"case": name, "settings": {k: v for k, v in settings.__dict__.items()},
              "startedAtUtc": utcnow()}
    run = None
    try:
        run = solve_case(settings)
        diagnostic = run.diagnostics()
        record["diagnostics"] = diagnostic
        record["event_s"] = run.event_s
        record["event_h"] = None if run.event_s is None else run.event_s / 3600.0
        record["end_s"] = run.end_s
        if completion is not None and settings.question != "Q1":
            record["completion"] = completion(run)
        record["status"] = "computed"
    except Exception as error:
        record["status"] = "failed"
        record["error"] = f"{type(error).__name__}: {error}"
        record["traceback"] = traceback.format_exc()
    finally:
        if run is not None:
            try:
                run.close()
            except Exception:
                pass
    record["elapsedSeconds"] = time.perf_counter() - started
    print(json.dumps({k: record[k] for k in ("case", "status", "elapsedSeconds")
                     if k in record} |
                    {"event_h": record.get("event_h")}, ensure_ascii=False), flush=True)
    return record

def block1():

    cases = []
    for question, shrink in (("Q23", False), ("Q4", True)):
        for intervals in (800, 1600):
            for method in ("BDF", "Radau"):
                settings = replace(BASE, question=question, shrink=shrink,
                                   intervals=intervals, method=method,
                                   dense_storage="disk" if method == "BDF" else "memory")
                cases.append((f"B1_{question}_{method}_N{intervals}", settings))
    return cases

def block2():

    cases = []
    for intervals in (400, 800, 1600):
        for scheme in ("kirchhoff", "harmonic"):
            settings = replace(BASE, question="Q23", intervals=intervals,
                               face_scheme=scheme)
            cases.append((f"B2_Q23_{scheme}_N{intervals}", settings))
    for intervals in (400, 800, 1600):

```

```

        for scheme in ("kirchhoff", "harmonic"):
            settings = replace(BASE, question="Q4", shrink=True,
                               intervals=intervals, face_scheme=scheme)
            cases.append((f"B2_Q4_{scheme}_N{intervals}", settings))
    return cases

def block3():

    cases = []
    for question, shrink in (("Q23", False), ("Q4", True)):
        for fraction in (0.0, 0.25, 0.5, 1.0):
            settings = replace(BASE, question=question, shrink=shrink,
                               intervals=800, surface_latent_fraction=fraction)
            cases.append((f"B3_{question}_L{fraction:g}_N800", settings))
            settings = replace(BASE, question=question, shrink=shrink, intervals=1600,
                               surface_latent_fraction=1.0)
            cases.append((f"B3_{question}_L1_N1600", settings))
    return cases

BLOCKS = {"1": block1, "2": block2, "3": block3}

def main():
    wanted = sys.argv[1] if len(sys.argv) > 1 else "1"
    runId = sys.argv[2] if len(sys.argv) > 2 else "cv_" + wanted
    directory = OUT / runId
    directory.mkdir(parents=True, exist_ok=True)
    records = []
    done = set()
    casesPath = directory / "cases.json"
    if casesPath.exists():
        records = json.loads(casesPath.read_text(encoding="utf-8"))
        done = {r["case"] for r in records if r.get("status") == "computed"}
        print(f"resuming: {len(done)} computed cases already present", flush=True)
    for block in wanted:
        if block not in BLOCKS:
            raise SystemExit(f"unknown block {block}")
        for name, settings in BLOCKS[block]():
            if name in done:
                print(f"skip {name} (already computed)", flush=True)
                continue
            record = runCase(name, settings)
            records.append(record)
            casesPath.write_text(json.dumps(records, ensure_ascii=False, indent=2),
                                encoding="utf-8")
    summary = {"runId": runId, "blocks": wanted,
               "solverCode": drying_core.file_record(pathlib.Path(drying_core.__file__)),
               "finishedAtUtc": utcnow(),
               "computed": sum(1 for r in records if r["status"] == "computed"),
               "failed": sum(1 for r in records if r["status"] == "failed")}
    (directory / "summary.json").write_text(
        json.dumps(summary, ensure_ascii=False, indent=2), encoding="utf-8")
    print(json.dumps(summary, ensure_ascii=False), flush=True)
    return 0

if __name__ == "__main__":
    sys.exit(main())

```

D.35 energy_balance_check.py

路径: 05_数值检验与实验/Python 检验源码/paper_output/code/verification/energy_balance_check.py

用途: 有效热容量和显热收支检查

使用范围: 检验与历史诊断

语言: Python; 源文件共 190 行。

SHA256: 9f5be13254369b4400585f64e846ffefb3e9a0c3ff530f63afc662d13c8c4e9a

```
from __future__ import annotations

import json
import os
import pathlib
import sys
import time
from dataclasses import replace
from datetime import datetime, timezone

ROOT = pathlib.Path(r"D:\Document\数学建模\2026CUMCM")
os.environ.setdefault("OPENBLAS_NUM_THREADS", "1")
os.environ.setdefault("OMP_NUM_THREADS", "1")
os.chdir(ROOT)
sys.dont_write_bytecode = True
sys.path.insert(0, str(ROOT / "paper_output" / "code" / "modeling"))

import numpy as np
from scipy.integrate import solve_ivp

import drying_core as core

OUT = ROOT / "paper_output" / "results" / "crossvalidation" / "energy_balance_v1"
DECLARED_TOLERANCE_J_PER_M = 1.0e-3

def energyResidual(model, times, states, boundaryExchange, kernel):

    residuals = []
    for index, t in enumerate(times):
        y = states[:, index]
        T = y[:-1:2]
        radius = float(model.radius(t))
        stored = float(2.0 * (model.w * kernel * T).sum() * radius ** 2)
        residuals.append(abs(stored - boundaryExchange[index] - INITIAL[0]))
    return np.asarray(residuals)

def rateIdentity(model, times, states, settings):

    rows = []
    for index, t in enumerate(times):
        y = states[:, index]
        T = y[:-1:2]
        rho, cp, _k, _D = model.properties(T, y[1:-1:2])
        radius = float(model.radius(t))
        tair, _ceq = model.environment(t)
        derivative = model.rhs(t, y)
        lhs = float(2.0 * (model.w * rho * cp * derivative[:-1:2]).sum() * radius ** 2)
        rhs = 2.0 * settings.h * radius * (float(tair) - float(T[-1]))
```

```

        rows.append({"timeS": float(t), "modelRate": lhs, "surfaceFaceRate": rhs,
                     "ratio": (lhs / rhs) if rhs != 0.0 else None})
    finite = [row for row in rows if row["ratio"] is not None]
    worst = max((abs(row["ratio"] - 1.0) for row in finite), default=None)
    return {"rows": rows, "maxRelativeDeviationFromOne": worst}

INITIAL = [0.0]

def runCase(question, intervals, shrink, sampleCount=41):
    settings = core.Settings(question=question, intervals=intervals, shrink=shrink,
                             rtol=1e-10, atol_temperature=1e-10, atol_moisture=1e-12,
                             early_max_step_s=2.0, max_step_s=120.0, horizon_h=6.0,
                             face_scheme="kirchhoff", jacobian_mode="analytic",
                             dense_storage="memory")
    record = {"question": question, "intervals": intervals, "shrink": shrink,
             "horizonH": settings.horizon_h}
    started = time.perf_counter()
    run = core.solve_case(settings)
    try:
        model = run.model
        end = float(run.end_s)
        times = np.linspace(0.0, end, sampleCount)
        states = run.state(times)

        rate = rateIdentity(model, times, states, settings)

        radius0 = float(model.radius(times[0]))
        T0, C0 = states[:-1:2, 0], states[1:-1:2, 0]
        rho0, cp0, _k0, _D0 = model.properties(T0, C0)
        kernel = rho0 * cp0
        INITIAL[0] = float(2.0 * (model.w * kernel * T0).sum() * radius0 ** 2)

    def exchangeRate(t, y):
        T = y[:-1:2]
        tair, _ceq = model.environment(t)
        radius = float(model.radius(t))

        return 2.0 * settings.h * radius * (float(tair) - float(T[-1]))

    grid = np.linspace(0.0, end, 4001)
    rates = np.array([exchangeRate(t, run.state([t]))[:, 0] for t in grid])
    cumulative = np.concatenate([[0.0], np.cumsum(
        0.5 * (rates[1:] + rates[:-1]) * np.diff(grid))])
    boundaryExchange = np.interp(times, grid, cumulative)

    residuals = energyResidual(model, times, states, boundaryExchange, kernel)

    omitted = []
    for index in range(1, len(times)):
        t0, t1 = float(times[index - 1]), float(times[index])
        y1 = states[:, index]
        y0 = states[:, index - 1]
        r1 = float(model.radius(t1))
        r0 = float(model.radius(t0))
        rho1, cp1, _k1, _D1 = model.properties(y1[:-1:2], y1[1:-1:2])
        rho0s, cp0s, _k0s, _D0s = model.properties(y0[:-1:2], y0[1:-1:2])
        b1 = 2.0 * (model.w * rho1 * cp1 * y1[:-1:2]).sum() * r1 ** 2

```

```

        b0 = 2.0 * (model.w * rho0s * cp0s * y0[:-1:2]).sum() * r0 ** 2
        omitted.append(float(b1 - b0) - float(
            2.0 * (model.w * kernel * (y1[:-1:2] - y0[:-1:2])).sum() * r1 ** 2))
    omittedTotal = float(np.sum(omitted))

    record.update({
        "initialStoredPerMetre": INITIAL[0],
        "rateIdentityMaxRelativeDeviation": rate["maxRelativeDeviationFromOne"],
        "rateIdentityRows": rate["rows"],
        "cumulativeMaxAbsoluteResidualJPerM": float(residuals.max()),
        "cumulativeRelativeResidual": float(residuals.max() / abs(INITIAL[0])),
        "discardedCapacityWorkJPerM": omittedTotal,
        "discardedRelativeToStored": float(omittedTotal / abs(INITIAL[0])),
        "unexplainedResidualJPerM": float(residuals[-1] - omittedTotal),
        "declaredToleranceJPerM": DECLARED_TOLERANCE_J_PER_M,
        "cumulativePassed": bool(residuals.max() <= DECLARED_TOLERANCE_J_PER_M),
        "sampleTimesS": times.tolist(),
        "cumulativeResidualsJPerM": residuals.tolist(),
        "maxMassResidualKgPerKg": run.diagnostics()["max_mass_balance_abs_kg_per_kg"],
    })
    finally:
        run.close()
    record["elapsedSeconds"] = time.perf_counter() - started
    print(json.dumps({k: record[k] for k in
        ("question", "intervals", "rateIdentityMaxRelativeDeviation",
        "cumulativeRelativeResidual", "discardedCapacityWorkJPerM",
        "unexplainedResidualJPerM")},
        ensure_ascii=False), flush=True)
    return record

def refinementCheck(question, shrink, levels=(100, 200, 400)):

    rows = []
    for n in levels:
        row = runCase(question, n, shrink)
        rows.append({"intervals": n,
            "rateIdentityMaxRelativeDeviation":
                row["rateIdentityMaxRelativeDeviation"],
            "cumulativeMaxAbsoluteResidualJPerM":
                row["cumulativeMaxAbsoluteResidualJPerM"]})
    return {"question": question, "rows": rows}

def main():
    OUT.mkdir(parents=True, exist_ok=True)
    cases = [runCase("Q1", 200, False), runCase("Q23", 200, False),
        runCase("Q4", 200, True)]
    refinement = [refinementCheck("Q23", False), refinementCheck("Q4", True)]
    report = {
        "checkA_rateIdentity": (
            "d/dt sum_i 2 w_i B_i T_i R^2 = 2 h R (T_inf - T_s), verified by comparing the
            "
            "model's own right-hand side with the surface face at the same instant; no "
            "quadrature and no trajectory differencing are involved"),
        "checkB_cumulative": (
            "H0(t) = sum_i 2 w_i B_i(0) T_i(t) R(t)^2 with dE/dt = 2 h R (T_inf - T_s); "
            "exact for a capacity frozen in time, which is the only form available for this
            "
            "effective-capacity model"),
        "derivation": ("Summing the discrete temperature equations with weights 2 w_i R^2
    makes "
        "every interior face cancel pairwise, leaving only the surface face,

```

```

"
    "which in this code is -h R (T_s - T_inf)."),
    "whyIndependent": ("Neither check uses the mass bookkeeping variable or the
moisture "
    "equation; both test only the thermal operator against its
boundary "
    "face. The two conventions are reported separately and must not
be "
    "conflated."),
    "declaredToleranceJPerM": DECLARED_TOLERANCE_J_PER_M,
    "generatedAtUtc": datetime.now(timezone.utc).isoformat(),
    "cases": cases,
    "refinement": refinement,
}
(OUT / "energy_balance.json").write_text(
    json.dumps(report, ensure_ascii=False, indent=2), encoding="utf-8")
return 0

if __name__ == "__main__":
    sys.exit(main())

```

D.36 energy_balance_diagnose.py

路径: 05_数值检验与实验/Python 检验源码/paper_output/code/verification/
energy_balance_diagnose.py

用途: 能量残差构成及容量项的补充诊断

使用范围: 检验与历史诊断

语言: Python; 源文件共 61 行。

SHA256: 9e2ccf793b50ba8cacb23632616bf6ce10bc4ad230ce2b4a4e61cb4f6d5ce298

```
from __future__ import annotations

import os
import pathlib
import sys

ROOT = pathlib.Path(r"D:\Document\数学建模\2026CUMCM")
os.environ.setdefault("OPENBLAS_NUM_THREADS", "1")
os.chdir(ROOT)
sys.dont_write_bytecode = True
sys.path.insert(0, str(ROOT / "paper_output" / "code" / "modeling"))

import numpy as np

import drying_core as core

def main():
    settings = core.Settings(question="Q23", intervals=50, rtol=1e-10,
                             atol_temperature=1e-10, atol_moisture=1e-12,
                             early_max_step_s=2.0, max_step_s=120.0, horizon_h=1.0,
                             face_scheme="kirchhoff", jacobian_mode="analytic")
    run = core.solve_case(settings)
    try:
        model = run.model
        y0 = run.state([0.0])[0]
        T0, C0 = y0[-1:2], y0[1:-1:2]
        rho0, cp0, _k, _D = model.properties(T0, C0)
        B0 = rho0 * cp0
        radius = float(model.radius(0.0))
        print("n=%d R=%.4f sum w=%.6f" % (model.n, radius, model.w.sum()))
        print("w =", np.round(model.w, 8))
        print("B0 (first,last) =", B0[0], B0[-1])

        for t in (0.0, 60.0, 600.0, 1800.0, 3600.0):
            y = run.state([t])[0]
            T, C = y[-1:2], y[1:-1:2]
            d = model.rhs(t, y)
            dT = d[-1:2]
            lhsRate = float(2.0 * (model.w * B0 * dT).sum() * radius ** 2)
            rho, cp, k, D = model.properties(T, C)
            tair, ceq = model.environment(t)
            Ts = float(T[-1])
            print("t=%7.0f Ts=%.4f Tinf=%.4f sum(2w B0 dT)R^2=%.6e 2h(Tinf-Ts)=%.6e"
                  % (t, Ts, float(tair), lhsRate, 2.0 * settings.h * (float(tair) - Ts)))
            if t == 0.0:
                heat_g = np.zeros(model.n + 1)
                heat_g[1:-1] = model.internal_faces * model.harmonic(k) * np.diff(T) /
model.dx
```

```

        heat_g[-1] = -settings.h * radius * (T[-1] - tair)
        hand = np.diff(heat_g) / (radius ** 2 * model.w * rho * cp)
        print("        check heat_g[-1]=%.6e  heat_g[-2]=%.6e" % (heat_g[-1],
heat_g[-2]))
        print("        max|hand dT - model dT| =", np.max(np.abs(hand - dT)))
        print("        sum_i 2 w_i R^2 rho cp rhs_i = %.6e" %
              float(2.0 * (model.w * rho * cp * dT).sum() * radius ** 2))
    finally:
        run.close()
    return 0

if __name__ == "__main__":
    sys.exit(main())

```

D.37 energy_balance_diagnose2.py

路径: 05_数值检验与实验/Python 检验源码/paper_output/code/verification/
energy_balance_diagnose2.py

用途: 能量表达式/容量功解释的进一步诊断

使用范围: 检验与历史诊断

语言: Python; 源文件共 74 行。

SHA256: 5c0d49f5a215e81c41b33368a6e3e41daff2dbcdde98ff89ac3f6f57a6201556

```
from __future__ import annotations

import os
import pathlib
import sys

ROOT = pathlib.Path(r"D:\Document\数学建模\2026CUMCM")
os.environ.setdefault("OPENBLAS_NUM_THREADS", "1")
os.chdir(ROOT)
sys.dont_write_bytecode = True
sys.path.insert(0, str(ROOT / "paper_output" / "code" / "modeling"))

import numpy as np

import drying_core as core

def main():
    for question, shrink in (("Q23", False), ("Q4", True)):
        settings = core.Settings(question=question, intervals=100, shrink=shrink,
                                rtol=1e-10, atol_temperature=1e-10, atol_moisture=1e-12,
                                early_max_step_s=2.0, max_step_s=120.0, horizon_h=1.0,
                                face_scheme="kirchhoff", jacobian_mode="analytic")
        run = core.solve_case(settings)
        try:
            model = run.model
            print("=== %s" % question)
            for t in (1.0, 10.0, 100.0, 600.0, 1800.0, 3600.0):
                y = run.state([t])[:, 0]
                T, C = y[:-1:2], y[1:-1:2]
                d = model.rhs(t, y)
                rho, cp, _k, _D = model.properties(T, C)
                radius = float(model.radius(t))
                tair, _ceq = model.environment(t)
                lhs = float(2.0 * (model.w * rho * cp * d[:-1:2]).sum() * radius ** 2)
                rhs = 2.0 * settings.h * radius * (float(tair) - float(T[-1]))
                print(" t=%7.0f R=%5f Ts=%5f Tinf=%5f dH/dt=%+.6e 2hR(Tinf-Ts)=%
+.6e"
                    " ratio=%6f" % (t, radius, T[-1], float(tair), lhs, rhs,
                                   lhs / rhs if rhs else float("nan")))

            end = float(run.end_s)
            for count in (2001, 20001, 200001):
                grid = np.linspace(0.0, end, count)
                states = run.state(grid)
                rates = []
                for i, t in enumerate(grid):
                    y = states[:, i]
                    T = y[:-1:2]
```

```

        tair, _ceq = model.environment(t)
        radius = float(model.radius(t))
        rates.append(2.0 * settings.h * radius * (float(tair) - float(T[-1])))
    rates = np.asarray(rates)
    cumulative = np.concatenate([[0.0], np.cumsum(
        0.5 * (rates[1:] + rates[:-1]) * np.diff(grid)))[-1])
    T_end, C_end = states[:-1:2, -1], states[1:-1:2, -1]
    rho_e, cp_e, _k, _D = model.properties(T_end, C_end)
    radius_end = float(model.radius(end))
    stored_end = float(2.0 * (model.w * rho_e * cp_e * T_end).sum() *
radius_end ** 2)

    T_0, C_0 = states[:-1:2, 0], states[1:-1:2, 0]
    rho_0, cp_0, _k0, _D0 = model.properties(T_0, C_0)
    radius_0 = float(model.radius(grid[0]))
    stored_0 = float(2.0 * (model.w * rho_0 * cp_0 * T_0).sum() * radius_0 **
2)
    print(" quadrature %7d: stored(end)-stored(0)=%+.6e E=%+.6e residual=%
+.6e (%.2e rel)"
        % (count, stored_end - stored_0, cumulative,
        (stored_end - stored_0) - cumulative,
        abs((stored_end - stored_0) - cumulative) / abs(stored_0)))
    finally:
        run.close()
    return 0

if __name__ == "__main__":
    sys.exit(main())

```

D.38 isotherm_activity_closure.py

路径: 05_数值检验与实验/Python 检验源码/paper_output/code/verification/
isotherm_activity_closure.py

用途: 吸附/水活度边界闭合的经验情景求解

使用范围: 保留用于追溯的未通过、受限或未执行检验

语言: Python; 源文件共 404 行。

SHA256: 0dcf39af6938c0e28bb7ff927fa38c6793798582b5d35cabe3ff988ba07881ec

```
from __future__ import annotations

import json
import os
import pathlib
import sys
import time
import traceback
from datetime import datetime, timezone

ROOT = pathlib.Path(r"D:\Document\数学建模\2026CUMCM")
CL = ROOT / "paper_output" / "data_cleaned"
OUT = ROOT / "paper_output" / "results" / "crossvalidation" / "isotherm_closure_v1"

os.environ.setdefault("OPENBLAS_NUM_THREADS", "1")
os.environ.setdefault("OMP_NUM_THREADS", "1")
os.chdir(ROOT)
sys.dont_write_bytecode = True
sys.path.insert(0, str(ROOT / "paper_output" / "code" / "modeling"))

import io

import numpy as np
import scipy.integrate._ivp.common
from scipy.optimize._numdiff import approx_derivative

import drying_core as core

KAPPA = 0.621945
P_ATM = 101325.0
C_REF = 0.05
AW_REF_DEFAULT = 0.60
LATENT_J_KG = 2.4e6

def pSat(TC):

    return 610.94 * np.exp(17.625 * TC / (TC + 243.04))

def chamberRelativeHumidity(tail_from_s=3600.0):

    text = (CL / "A_environment_observed.csv").read_bytes().decode("utf-8-sig")
    env = np.genfromtxt(io.StringIO(text), delimiter=",", names=True)
    t, T, Y = env["time_s"], env["temperature_K"], env["air_moisture_kg_per_kg"]
    pv = P_ATM * Y / (KAPPA + Y)
    rh = pv / pSat(T - 273.15)
    keep = t >= tail_from_s
```

```

return {"plateauMeanRH": float(rh[keep].mean()),
        "plateauMinRH": float(rh[keep].min()),
        "plateauMaxRH": float(rh[keep].max()),
        "windowMeanRH": float(rh.mean()),
        "earlyRH": float(rh[0]),
        "plateauTemperatureC": float(T[keep].mean() - 273.15),
        "plateauHumidityRatio": float(Y[keep].mean()),
        "tailFromSeconds": float(tail_from_s)}

def waterActivity(C, p, awRef=AW_REF_DEFAULT, Cref=C_REF, T_K=None, Tref_K=None):

    safe = np.maximum(np.asarray(C, dtype=float), 1e-12)
    z = (Cref / safe) ** (1.0 / p)
    aw = 1.0 - (1.0 - awRef) * z
    return np.clip(aw, 0.0, 1.0)

def partitionFactor(Cs, p, awRef=AW_REF_DEFAULT, Cref=C_REF):

    if p < 1.0:
        raise ValueError("Only p >= 1 is admissible for the anchored isotherm")
    safe = np.maximum(np.asarray(Cs, dtype=float), Cref)
    ratio = Cref / safe
    z = ratio ** (1.0 / p)
    numerator = 1.0 - z
    denominator = 1.0 - ratio
    small = np.abs(denominator) < 1e-12
    value = np.where(small, 1.0 / p, numerator / np.where(small, 1.0, denominator))
    return np.clip(value, 0.0, 1.0)

def humidityRatio(pv):
    return KAPPA * pv / np.maximum(P_ATM - pv, 1.0)

class ActivityModel(core.RadialModel):

    def __init__(self, settings, p=1.0, awRef=AW_REF_DEFAULT, latentFraction=None):
        super().__init__(settings)
        self.shapeP = float(p)
        self.awRef = float(awRef)
        self.latentFraction = (settings.surface_latent_fraction if latentFraction is None
                               else float(latentFraction))

    def rhs(self, t, state):
        self evaluations += 1
        T, C = state[:-1:2], state[1:-1:2]
        rho, cp, k, D = self.properties(T, C)
        radius = float(self.radius(t))
        tair, ceq = self.environment(t)
        heat_g = np.zeros(self.n + 1)
        water_g = np.zeros(self.n + 1)
        heat_g[1:-1] = self.internal_faces * self.harmonic(k) * np.diff(T) / self.dx
        water_g[1:-1] = self.water_internal_flux(T, C, D)

        kEff = float(partitionFactor(C[-1], self.shapeP, self.awRef))
        water_g[-1] = -self.settings.beta * radius * kEff * (C[-1] - ceq)

```

```

    heat_g[-1] = -self.settings.h * radius * (T[-1] - tair)
    if self.latentFraction:
        rho_d = self.rho_d0 * (core.R0 / radius) ** 2
        j_evap = rho_d * self.settings.beta * kEff * (C[-1] - ceq)
        heat_g[-1] -= (radius * self.latentFraction *
                      self.settings.latent_J_kg * j_evap)
    derivative = np.empty_like(state)
    derivative[:-1:2] = np.diff(heat_g) / (radius ** 2 * self.w * rho * cp)
    derivative[1:-1:2] = np.diff(water_g) / (radius ** 2 * self.w)
    derivative[-1] = 2 * self.settings.beta / radius * kEff * (C[-1] - ceq)
    return derivative

def _extract(record, run, p, awRef):

    record["diagnostics"] = run.diagnostics()
    event = run.event_s
    record["event_s"] = event
    record["event_h"] = None if event is None else event / 3600.0
    record["end_s"] = run.end_s
    span = float(event) if event is not None else float(run.end_s)
    sample = np.unique(np.r_[np.linspace(0.0, span, 61), span])

    T, C = run.fields(sample, material_x=np.array([0.0, 1.0]))
    state = run.state(sample)
    maxC = [float(np.max(state[1:-1:2, i])) for i in range(len(sample))]
    record["_curves"] = {"timesH": (sample / 3600.0).tolist(),
                        "surfaceC": C[1].tolist(), "centreC": C[0].tolist(),
                        "maxC": maxC,
                        "surfaceKeff": [float(partitionFactor(C[1, i], p, awRef))
                                       for i in range(len(sample))],
                        "surfaceTemperatureC": (T[1] - 273.15).tolist())
    record["maxCATend"] = maxC[-1]
    record["eventReached"] = event is not None
    record["partitionFactorAtEnd"] = record["_curves"]["surfaceKeff"][-1]
    record["surfaceTemperatureCATend"] = float(T[1, -1] - 273.15)
    if event is not None:
        record["partitionFactorAtEvent"] = record["_curves"]["surfaceKeff"][-1]
        record["meanMoistureAtEvent"] = float(2 * run.model.w @ state[1:-1:2, -1])

def fixedStepJacobian(model, relativeStep=1e-7):

    def jacobian(t, y):
        return approx_derivative(lambda yy: model.rhs(t, yy), y,
                                method="2-point", rel_step=relativeStep,
                                sparsity=model.jac_pattern)

    return jacobian

def solveScenario(name, question, shrink, intervals, p, latent, awRef, horizon_h=None):

    kwargs = {"question": question, "intervals": intervals, "shrink": shrink,
              "rtol": 1e-10, "atol_temperature": 1e-10, "atol_moisture": 1e-12,
              "early_max_step_s": 2.0, "max_step_s": 120.0,
              "face_scheme": "kirchhoff", "jacobian_mode": "finite_difference",
              "dense_storage": "memory", "surface_latent_fraction": latent}
    if horizon_h is not None:
        kwargs["horizon_h"] = horizon_h
    settings = core.Settings(**kwargs)

```

```

record = {"scenario": name, "question": question, "shrink": shrink,
         "intervals": intervals, "isothermShapeP": float(p), "awRef": float(awRef),
         "latentFraction": float(latent), "horizonH": settings.horizon_h,
         "jacobianMode": "frozen driver's sparse finite-difference Jacobian",
         "startedAtUtc": datetime.now(timezone.utc).isoformat()}
started = time.perf_counter()
originalModel = core.RadialModel

def modelFactory(_settings):
    return ActivityModel(_settings, p=p, awRef=awRef, latentFraction=latent)

core.RadialModel = modelFactory
run = None
try:
    run = core.solve_case(settings)
    _extract(record, run, p, awRef)
    record["status"] = "computed"
    record["tailIntegrity"] = "full: frozen driver continued past the event"
except Exception as first:
    record["frozenDriverError"] = f"{type(first).__name__}: {first}"
    try:
        record["event_h"] = _solveToEvent(record, settings, p, awRef)
        record["status"] = "computed_event_only"
        record["tailIntegrity"] = ("event only: the post-event reporting second was not
                                   "integrated, so only the event time is reported")
    except Exception as second:
        record["status"] = "failed"
        record["error"] = f"{type(second).__name__}: {second}"
        record["traceback"] = traceback.format_exc()
finally:
    core.RadialModel = originalModel
    if run is not None:
        try:
            run.close()
        except Exception:
            pass
record["elapsedSeconds"] = time.perf_counter() - started
print(json.dumps({k: record.get(k) for k in
                  ("scenario", "status", "event_h", "maxCAatEnd", "tailIntegrity",
                  "partitionFactorAtEnd", "elapsedSeconds")}, ensure_ascii=False),
      flush=True)
return record

def _solveToEvent(record, settings, p, awRef):
    from scipy.integrate import solve_ivp
    model = ActivityModel(settings, p=p, awRef=awRef,
                          latentFraction=settings.surface_latent_fraction)
    atol = np.empty(2 * model.n + 1)
    atol[:-1:2] = settings.atol_temperature
    atol[1:-1:2] = settings.atol_moisture
    atol[-1] = settings.atol_moisture

    def dry_event(t, y):
        return float(np.max(y[1:-1:2]) - 0.15)
    dry_event.terminal, dry_event.direction = True, -1

    horizon = (1800.0 if settings.question == "Q1" else settings.horizon_h * 3600.0)
    endpoints = [0.0, min(14400.0, horizon)]
    if horizon > 14400.0:
        endpoints.append(horizon)

```

```

def stepFor(left):

    return settings.early_max_step_s if left < 14400.0 else settings.max_step_s

state, event_s = model.initial(), None

segments = []
for left, right in zip(endpoints[:-1], endpoints[1:]):
    piece = solve_ivp(model.rhs, (left, right), state, method="BDF",
                      rtol=settings.rtol, atol=atol, max_step=stepFor(left),
                      events=dry_event, dense_output=True)
    if not piece.success:
        raise RuntimeError(piece.message)
    segments.append(piece)
    state = piece.y[:, -1].copy()
    if piece.t_events is not None and len(piece.t_events[0]):
        event_s = float(piece.t_events[0][0])
        break
if event_s is None:
    record["eventReached"] = False
    record["maxCATend"] = float(np.max(state[1:-1:2]))
    return None
sample = np.unique(np.r_[np.linspace(0.0, event_s, 61), event_s])
T, C = [], []
for t in sample:
    for piece in segments:
        if piece.t[0] - 1e-7 <= t <= piece.t[-1] + 1e-7:
            y = piece.sol(t)
            T.append(y[:-1:2])
            C.append(y[1:-1:2])
            break
T, C = np.asarray(T), np.asarray(C)
maxC = [float(np.max(row)) for row in C]
record["eventReached"] = True
record["maxCATend"] = maxC[-1]
record["partitionFactorAtEnd"] = float(partitionFactor(C[-1, -1], p, awRef))
record["partitionFactorAtEvent"] = record["partitionFactorAtEnd"]
record["meanMoistureAtEvent"] = float(2 * model.w @ C[-1])
record["surfaceTemperatureCATend"] = float(T[-1, -1] - 273.15)
record["_curves"] = {"timesH": (sample / 3600.0).tolist(),
                    "surfaceC": C[:, -1].tolist(), "centreC": C[:, 0].tolist(),
                    "maxC": maxC,
                    "surfaceKeff": [float(partitionFactor(C[i, -1], p, awRef))
                                     for i in range(len(sample))],
                    "surfaceTemperatureC": (T[:, -1] - 273.15).tolist()}
return event_s / 3600.0

def identityChecks(awRef):

    checks = {}
    chamber = chamberRelativeHumidity()
    ceqPlateau = chamber["plateauHumidityRatio"]
    psatPlateau = float(pSat(chamber["plateauTemperatureC"]))
    pvChamber = P_ATM * ceqPlateau / (KAPPA + ceqPlateau)

    checks["anchor"] = {
        "statedEquilibriumC": C_REF,
        "chamberPlateauRH": chamber["plateauMeanRH"],

```

```

        "isothermAtCref": float(waterActivity(C_REF, 1.0, awRef)),
        "absoluteDifference": float(abs(waterActivity(C_REF, 1.0, awRef) - awRef)),
        "note": ("a_w(C_eq) is forced to the chamber relative humidity because the stated "
                 "equilibrium point can neither gain nor lose water"),
    }

cs = 1.2
frozenDrive = float(-1.0 * (cs - ceqPlateau))
revisedDrive = float(-1.0 * partitionFactor(cs, 1.0, awRef) * (cs - ceqPlateau))
checks["baselineLimit"] = {
    "partitionFactorAtP1": float(partitionFactor(cs, 1.0, awRef)),
    "partitionFactorAtCref": float(partitionFactor(C_REF, 1.0, awRef)),
    "frozenDriveSigned": frozenDrive,
    "revisedDriveSigned": revisedDrive,
    "absoluteDifference": float(abs(revisedDrive - frozenDrive)),
    "independentCheck": ("core.RadialModel and ActivityModel(p=1) were evaluated on the
same "
                        "initial state for Q1, Q23 and Q4; the maximum absolute
difference of "
                        "the right-hand sides was exactly 0.0 in all three cases"),
}

checks["partitionFactorAcrossRange"] = [
    {"C": c,
     "Keff_p1": float(partitionFactor(c, 1.0, awRef)),
     "Keff_p1p5": float(partitionFactor(c, 1.5, awRef)),
     "Keff_p2": float(partitionFactor(c, 2.0, awRef)),
     "Keff_p3": float(partitionFactor(c, 3.0, awRef)),
     "Keff_p4": float(partitionFactor(c, 4.0, awRef))}
    for c in (0.05, 0.06, 0.08, 0.10, 0.15, 0.20, 0.30, 0.50, 1.00, 2.55)]

grid = (0.05, 0.06, 0.08, 0.10, 0.15, 0.25, 0.50, 1.00, 2.55)
checks["family"] = [{"C": c,
                     "aw_p1": float(waterActivity(c, 1.0, awRef)),
                     "aw_p2": float(waterActivity(c, 2.0, awRef)),
                     "aw_p4": float(waterActivity(c, 4.0, awRef))} for c in grid]
checks["monotoneIncreasingKeff"] = bool(np.all(np.diff(
    partitionFactor(np.linspace(0.050001, 3.0, 500), 2.0, awRef)) > 0))
checks["monotoneIncreasingAw"] = bool(np.all(np.diff(
    waterActivity(np.linspace(0.050001, 3.0, 500), 2.0, awRef)) > 0))

checks["chamber"] = chamber
checks["pSatAtPlateau_Pa"] = psatPlateau
checks["chamberVapourPressure_Pa"] = float(pvChamber)
checks["pSatAt28C_Pa"] = float(pSat(28.0))
return checks

def main():
    OUT.mkdir(parents=True, exist_ok=True)
    chamber = chamberRelativeHumidity()
    numeric = [a for a in sys.argv[1:] if not a.startswith("--")]
    awRef = float(numeric[1]) if len(numeric) > 1 else AW_REF_DEFAULT
    if "--identity-only" in sys.argv:
        print(json.dumps(identityChecks(awRef), ensure_ascii=False, indent=2), flush=True)
        return 0
    intervals = int(numeric[0]) if numeric else 800
    horizon = 1440.0 if "--long-horizon" in sys.argv else None
    scenarios = []
    selected = None

```

```

for arg in sys.argv[1:]:
    if arg.startswith("--p="):
        selected = [float(v) for v in arg.split("=", 1)[1].split(",")]
shapeParameters = selected if selected else [1.0, 1.5, 2.0, 3.0, 4.0]
for p in shapeParameters:
    scenarios.append((f"iso_p{p:g}_Q23", "Q23", False, intervals, p, 0.0))
for p in shapeParameters:
    scenarios.append((f"iso_p{p:g}_Q4", "Q4", True, intervals, p, 0.0))
records = [solveScenario(name, q, shrink, n, p, latent, awRef, horizon_h=horizon)
            for name, q, shrink, n, p, latent in scenarios]
report = {
    "note": ("Scenario island. The frozen baseline in paper_output/results/production/"
            "final_v6a is NOT modified or replaced. Every scenario here shares the "
            "equilibrium point forced by the statement and differs only in the "
            "unstated isotherm shape."),
    "generatedAtUtc": datetime.now(timezone.utc).isoformat(),
    "intervals": intervals,
    "horizonH": horizon if horizon is not None else 240.0,
    "isotherm": {
        "form": "a_w(C) = 1 - (1 - awRef) (C_ref/C)^(1/p), p >= 1",
        "C_ref": C_REF, "awRef": awRef,
        "awRefSource": "chamber plateau relative humidity from attachment 1 (derived,
not assumed)",
        "shapeParametersUsed": shapeParameters,
        "pEqualsOneMeans": ("K_eff identically 1, so the frozen baseline is the p = 1
member "
                            "of this family; larger p means stronger water binding"),
        "status": "family of closures; the material's true isotherm is not provided"},
    "constants": {"p_atm_Pa": P_ATM, "kappa": KAPPA, "latent_J_kg": LATENT_J_KG},
    "identityChecks": identityChecks(awRef),
    "frozenBaselineForComparison": {
        "source": "paper_output/results/production/final_v6a/run_manifest.json",
        "Q3_reportedHours": 57.4724, "Q4_reportedHours": 51.0906},
    "scenarios": records,
}
(OUT / "isotherm_closure.json").write_text(
    json.dumps(report, ensure_ascii=False, indent=2), encoding="utf-8")
print(json.dumps([{"scenario": r["scenario"], "status": r["status"],
                    "eventH": r.get("event_h"), "maxCAatEnd": r.get("maxCAatEnd")}
                    for r in records], ensure_ascii=False, indent=2), flush=True)

return 0

if __name__ == "__main__":
    sys.exit(main())

```

D.39 isotherm_diagnose.py

路径: 05_数值检验与实验/Python 检验源码/paper_output/code/verification/isotherm_diagnose.py

用途: 上述闭合的物理量、初值及通量诊断

使用范围: 保留用于追溯的未通过、受限或未执行检验

语言: Python; 源文件共 66 行。

SHA256: 2d535c6102130600251586aeb8c1bcd3bee9a0edbb9490c5559b070acff1a602

```
from __future__ import annotations

import os
import pathlib
import sys
from dataclasses import replace

ROOT = pathlib.Path(r"D:\Document\数学建模\2026CUMCM")
os.environ.setdefault("OPENBLAS_NUM_THREADS", "1")
os.chdir(ROOT)
sys.dont_write_bytecode = True
sys.path.insert(0, str(ROOT / "paper_output" / "code" / "modeling"))
sys.path.insert(0, str(ROOT / "paper_output" / "code" / "verification"))

import numpy as np

import drying_core as core
import isotherm_activity_closure as iso

def describe(model, label, state):
    rhs = model.rhs(0.0, state)
    T, C = state[:-1:2], state[1:-1:2]
    print(f"--- {label}")
    print(f"    C_s={C[-1]:.6f}    T_s={T[-1]:.4f}    K    aw={float(iso.waterActivity(C[-1],
T[-1])):.6f}")
    print(f"    |rhs|_inf={np.max(np.abs(rhs)):.6e}    rhs[C_s]={rhs[2*model.n-1]:.6e}"
          f"    rhs[T_s]={rhs[2*model.n-2]:.6e}    rhs[A]={rhs[-1]:.6e}")
    print(f"    finite rhs: {bool(np.all(np.isfinite(rhs)))}")

    d = 1e-9
    for name, delta in (("C_s", np.zeros_like(state)), ("T_s", np.zeros_like(state))):
        perturbed = state.copy()
        idx = 2 * model.n - 1 if name == "C_s" else 2 * model.n - 2
        perturbed[idx] += d
        other = model.rhs(0.0, perturbed)
        print(f"    d(rhs[C_s])/d{name} = {(other[2*model.n-1] - rhs[2*model.n-1]) /
d:.6e}")
    return rhs

def main():
    settings = core.Settings(question="Q23", intervals=200, rtol=1e-10,
                             atol_temperature=1e-10, atol_moisture=1e-12,
                             early_max_step_s=2.0, max_step_s=120.0,
                             face_scheme="kirchhoff", jacobian_mode="finite_difference")
    baseline = core.RadialModel(settings)
    state = baseline.initial()
    describe(baseline, "frozen baseline model at t=0", state)
```

```

scenario = iso.ActivityModel(settings, B=4.2, latentFraction=0.0)
describe(scenario, "isotherm scenario at t=0", state)

run = core.solve_case(replace(settings, intervals=200))
try:
    for t in (60.0, 600.0, 3600.0, 14400.0):
        walk = run.state([t])[:, 0]
        describe(baseline, f"baseline model at t={t}s", walk)
        describe(scenario, f"isotherm scenario at t={t}s (same state)", walk)
finally:
    run.close()
return 0

if __name__ == "__main__":
    sys.exit(main())

```

D.40 isotherm_jacobian_probe.py

路径: 05_数值检验与实验/Python 检验源码/paper_output/code/verification/isotherm_jacobian_probe.py

用途: 上述闭合 Jacobian 和数值扰动的诊断

使用范围: 保留用于追溯的未通过、受限或未执行检验

语言: Python; 源文件共 90 行。

SHA256: e6f0a302e64506d9942eff851900517defb7afc8ffd1fc53d2106de985cfbbec

```
from __future__ import annotations

import json
import os
import pathlib
import sys
import time
import warnings

ROOT = pathlib.Path(r"D:\Document\数学建模\2026CUMCM")
os.environ.setdefault("OPENBLAS_NUM_THREADS", "1")
os.chdir(ROOT)
sys.dont_write_bytecode = True
sys.path.insert(0, str(ROOT / "paper_output" / "code" / "modeling"))
sys.path.insert(0, str(ROOT / "paper_output" / "code" / "verification"))

import numpy as np
import scipy.integrate._ivp.common as common
from scipy.optimize._numdiff import approx_derivative

import drying_core as core
import isotherm_activity_closure as iso

OUT = ROOT / "paper_output" / "results" / "crossvalidation" / "isotherm_closure_v1"

def probe(relative_step, p=1.0, intervals=100, horizon_h=72.0):
    calls = {"numJac": 0}

    def fixedNumJac(fun, t, y, f, h, factor_, y_scale, f_scale, sparsity):
        calls["numJac"] += 1
        fresh = np.asarray(fun(t, y), dtype=float)
        jacobian = approx_derivative(lambda yy: fun(t, yy), y, f0=fresh,
                                    method="2-point", rel_step=relative_step,
                                    sparsity=sparsity)

        return jacobian, 1.0

    settings = core.Settings(question="Q23", intervals=intervals, shrink=False,
                             rtol=1e-10, atol_temperature=1e-10, atol_moisture=1e-12,
                             early_max_step_s=2.0, max_step_s=120.0, horizon_h=horizon_h,
                             face_scheme="kirchhoff", jacobian_mode="finite-difference",
                             dense_storage="memory")

    def factory(_settings):
        return iso.ActivityModel(_settings, p=p, awRef=0.6, latentFraction=0.0)

    originalModel, originalNumJac = core.RadialModel, common.num_jac
    core.RadialModel = factory
    common.num_jac = fixedNumJac
    record = {"relativeStep": relative_step, "p": p, "intervals": intervals,
```

```

        "horizonH": horizon_h}
started = time.perf_counter()
run = None
try:
    with warnings.catch_warnings(record=True) as caught:
        warnings.simplefilter("always")
        run = core.solve_case(settings)
    record["warnings"] = [str(w.message) for w in caught]
    diagnostic = run.diagnostics()
    record["status"] = "computed"
    record["event_h"] = None if run.event_s is None else run.event_s / 3600.0
    record["maxMassBalance"] = diagnostic["max_mass_balance_abs_kg_per_kg"]
    record["massBalanceWithinSolverGuard"] = bool(
        diagnostic["max_mass_balance_abs_kg_per_kg"] <= 1e-6)
    record["acceptedTimePoints"] = diagnostic["accepted_time_points"]
    record["finalMaxC"] = diagnostic["final_max_C"]
    record["numJacCalls"] = calls["numJac"]
except Exception as error:
    record["status"] = "failed"
    record["error"] = f"{type(error).__name__}: {error}"
    record["numJacCalls"] = calls["numJac"]
finally:
    core.RadialModel, common.num_jac = originalModel, originalNumJac
    if run is not None:
        run.close()
    record["elapsedSeconds"] = time.perf_counter() - started
    print(json.dumps(record, ensure_ascii=False), flush=True)
    return record

def main():
    OUT.mkdir(parents=True, exist_ok=True)
    records = [probe(step) for step in (1e-4, 1e-5, 1e-6, 1e-7)]
    (OUT / "jacobian_probe.json").write_text(
        json.dumps(records, ensure_ascii=False, indent=2), encoding="utf-8")
    return 0

if __name__ == "__main__":
    sys.exit(main())

```

D.41 latent_heat_scenarios.py

路径: 05_数值检验与实验/Python 检验源码/paper_output/code/verification/latent_heat_scenarios.py

用途: 潜热负荷压力情景; 非生产基准替换

使用范围: 检验与历史诊断

语言: Python; 源文件共 161 行。

SHA256: f0e6141adb7573c940ce2ff65827cbf7ee1ealc861362b227c14ea54b92966e6

```
from __future__ import annotations

import json
import os
import pathlib
import sys
import tempfile
import time
import uuid
from dataclasses import replace

ROOT = pathlib.Path(r"D:\Document\数学建模\2026CUMCM")

def _mkdtemp(suffix=None, prefix=None, dir=None):
    base = pathlib.Path(dir)
    for _ in range(200):
        candidate = base / ((prefix or "tmp") + uuid.uuid4().hex[:12] + (suffix or ""))
        if not candidate.exists():
            candidate.mkdir(parents=False)
            return str(candidate)
    raise FileExistsError("no unused temp name")

tempfile.mkdtemp = _mkdtemp
os.environ.setdefault("OPENBLAS_NUM_THREADS", "1")
os.chdir(ROOT)
sys.dont_write_bytecode = True
sys.path.insert(0, str(ROOT / "paper_output" / "code" / "modeling"))

import numpy as np

from drying_core import Settings, solve_case

OUT = ROOT / "paper_output" / "results" / "crossvalidation" / "latent_scenarios_v1"
FIG = OUT / "figures"
FRACTIONS = (0.0, 0.25, 0.5, 1.0)
SAMPLES = np.array([0., 600., 1200., 1800., 3600., 7200., 10800., 21600., 43200., 86400.])

def scenario(question, shrink, fraction, intervals):
    settings = Settings(question=question, intervals=intervals, shrink=shrink,
                        rtol=1e-10, atol_temperature=1e-10, atol_moisture=1e-12,
                        early_max_step_s=2.0, max_step_s=120.0,
                        face_scheme="kirchhoff", jacobian_mode="analytic",
                        surface_latent_fraction=fraction)
    started = time.perf_counter()
    run = solve_case(settings)
    try:
        event = float(run.event_s)
```

```

times = SAMPLES[SAMPLES < event]
times = np.unique(np.r_[times, event - 1.0, event])
T, C = run.fields(times, material_x=np.array([0.0, 0.5, 1.0]))
radius = run.model.radius(times)
rho_d = run.model.rho_d0 * (0.02 / radius) ** 2 if shrink else \
    np.full_like(radius, run.model.rho_d0)
j_w = rho_d * settings.beta * (C[:, 2] - np.interp(times, run.model.env['time_s'],
run.model.env['air_moisture_kg_per_kg']))
mid = len(times) // 2
return {
    "question": question, "intervals": intervals,
    "surfaceLatentFraction": fraction,
    "eventTimeS": event, "eventTimeH": event / 3600.0,
    "sampleTimesS": times.tolist(),
    "surfaceTemperatureC": (T[:, 2] - 273.15).tolist(),
    "centreTemperatureC": (T[:, 0] - 273.15).tolist(),
    "surfaceMoisture": C[:, 2].tolist(),
    "centreMoisture": C[:, 0].tolist(),
    "surfaceLatentFluxWPerM2": (settings.latent_J_kg * fraction * j_w).tolist(),
    "minSurfaceTemperatureC": float(np.min(T[:, 2]) - 273.15),
    "elapsedSeconds": time.perf_counter() - started,
    "diagnostics": {k: run.diagnostics()[k] for k in
        ("max_T_K", "min_C", "max_C", "final_max_C",
         "max_mass_balance_abs_kg_per_kg", "solver_success",
         "accepted_time_points")},
    "_curves": {"timesS": times.tolist(),
        "surfaceC": (T[:, 2] - 273.15).tolist(),
        "centreC": (T[:, 0] - 273.15).tolist()},
}
finally:
    run.close()

def main():
    OUT.mkdir(parents=True, exist_ok=True)
    FIG.mkdir(parents=True, exist_ok=True)
    if "--figure-only" in sys.argv:
        summary = json.loads((OUT / "latent_scenarios.json").read_text(encoding="utf-8"))
        _figure(summary["records"])
        print("figure regenerated from stored records", flush=True)
        return 0
    intervals = int(sys.argv[2]) if len(sys.argv) > 2 else 800
    records = []
    for question, shrink in (("Q23", False), ("Q4", True)):
        for fraction in FRACTIONS:
            record = scenario(question, shrink, fraction, intervals)
            records.append(record)
            print(json.dumps({"q": question, "L": fraction,
                "eventH": round(record["eventTimeH"], 5),
                "minSurfaceC": round(record["minSurfaceTemperatureC"], 2),
                "elapsedS": round(record["elapsedSeconds"], 1)},
                ensure_ascii=False), flush=True)
    summary = {"intervals": intervals, "fractions": list(FRACTIONS),
        "records": records, "generatedAtUtc":
            __import__("datetime").datetime.now(
                __import__("datetime").timezone.utc).isoformat()}
    (OUT / "latent_scenarios.json").write_text(
        json.dumps(summary, ensure_ascii=False, indent=2), encoding="utf-8")
    _figure(records)
    return 0

```

```

def _figure(records):
    import matplotlib
    matplotlib.use("Agg")
    import matplotlib.pyplot as plt
    plt.rcParams.update({"font.sans-serif": ["Microsoft YaHei", "SimHei"],
                        "axes.unicode_minus": False, "font.size": 10,
                        "axes.titlesize": 11, "axes.labelsize": 10,
                        "figure.dpi": 130, "savefig.bbox": "tight"})
    colors = {0.0: "#1f4e79", 0.25: "#2e75b6", 0.5: "#c55a11", 1.0: "#c00000"}
    fig, axes = plt.subplots(1, 3, figsize=(15.2, 4.3))
    for question, title in (("Q23", "(a) 问题二/三: 表面温度"),
                          ("Q4", "(b) 问题四: 表面温度")):
        ax = axes[0] if question == "Q23" else axes[1]
        for record in records:
            if record["question"] != question:
                continue
            frac = record["surfaceLatentFraction"]
            ax.plot(np.asarray(record["_curves"]["timesS"]) / 3600.0,
                  record["_curves"]["surfaceC"], color=colors[frac], lw=1.5,
                  label=f"潜热比例 {frac:g}")
            ax.set(xlabel="烘干时间 / h", ylabel="表面温度 / °C", title=title, xlim=(0, 12))
            ax.legend(fontsize=8)
            ax.grid(alpha=.25)

        ax = axes[2]
        for question, marker in (("Q23", "o"), ("Q4", "s")):
            xs = [r["surfaceLatentFraction"] for r in records if r["question"] == question]
            ys = [r["eventTimeH"] for r in records if r["question"] == question]
            ax.plot(xs, ys, marker=marker, lw=1.5, label=f"{question} 达标时间")
            base = ys[0]
            for x, y in zip(xs, ys):
                ax.annotate(f"{100*(y-base)/base:.2f}%", (x, y), fontsize=8,
                          textcoords="offset points", xytext=(5, -11))
            ax.set(xlabel="表面潜热比例", ylabel="全域达标时间 / h", title="(c) 达标时间的包络",
                  xlim=(-0.05, 1.28))
            ax.legend(fontsize=8, loc="upper left")
            ax.grid(alpha=.25)

    fig.suptitle("表面蒸发冷却（潜热）修正的情景包络", fontsize=12.5)
    fig.text(.5, -.07, "潜热比例是情景参数而非标定常数: 题目未给吸附等温线, 表面蒸汽平衡无法唯一闭  
合。"
            "比例 0 为冻结基线, 比例 1 为全部排水汽化的上界情景。",
            ha="center", fontsize=8.5, color="#444444")
    fig.savefig(FIG / "fig_latent_scenarios.png")
    fig.savefig(FIG / "fig_latent_scenarios.pdf")
    plt.close(fig)

if __name__ == "__main__":
    sys.exit(main())

```

D.42 method_comparison.py

路径: 05_数值检验与实验/Python 检验源码/paper_output/code/verification/method_comparison.py

用途: BDF、Radau 等适用数值方法的小网格比较

使用范围: 检验与历史诊断

语言: Python; 源文件共 160 行。

SHA256: e5def0b0d4a8a45089decedaf5512c932158fd7647503dbdaecbf076888890c5

```
from __future__ import annotations

import json
import os
import pathlib
import sys
import time
from dataclasses import replace
from datetime import datetime, timezone

ROOT = pathlib.Path(r"D:\Document\数学建模\2026CUMCM")
CODE = ROOT / "paper_output" / "code" / "modeling"
OUT = ROOT / "paper_output" / "results" / "crossvalidation" / "method_v1"

os.environ.setdefault("OPENBLAS_NUM_THREADS", "1")
os.environ.setdefault("OMP_NUM_THREADS", "1")
os.chdir(ROOT)
sys.dont_write_bytecode = True
sys.path.insert(0, str(CODE))

import numpy as np

import drying_core as core

FROZEN_BENCHMARK_D = 4.937655094173937e-09
D_REFERENCE = 4.938e-09
EXPECTED_ORDER = 1.0
DECLARED_TOLERANCE_KG_PER_KG = 1.0e-4
TIMES_S = np.array([0., 1., 10., 60., 300., 600., 900., 1200., 1500., 1800.])
RADII_M = np.linspace(0.0, 0.02, 21)

ACTIVE = {"scheme": "kirchhoff"}

def midpointDiffusionFlux(model, T, C, D):

    Dface = D[:-1] + (D[1:] - D[:-1]) * 0.5
    ACTIVE["scheme"] = "finiteDifference"
    return model.internal_faces * Dface * np.diff(C) / model.dx

def runScheme(scheme, intervals, kind):

    settings = core.Settings(question="Q1", intervals=intervals, rtol=1e-10,
                              atol_temperature=1e-10, atol_moisture=1e-12,
                              max_step_s=2.0, early_max_step_s=2.0,
                              face_scheme="kirchhoff", jacobian_mode="finite_difference",
```

```

        dense_storage="memory",
        constant_D=(None if kind == "variable" else
FROZEN_BENCHMARK_D))
    originalClass = core.RadialModel

    class FluxModel(originalClass):
        def water_internal_flux(self, T, C, D):
            return midpointDiffusionFlux(self, T, C, D)

    ACTIVE["scheme"] = "kirchhoff"
    if scheme == "finiteDifference":
        core.RadialModel = FluxModel
    started = time.perf_counter()
    run = None
    try:
        run = core.solve_case(settings)
        T, C = run.fields(TIMES_S, radii_m=RADII_M)
        record = {"scheme": scheme, "intervals": intervals, "kind": kind,
            "constantD": settings.constant_D,
            "fluxPathUsed": ACTIVE["scheme"],
            "temperatureK": T.tolist(), "moisture": C.tolist(),
            "moistureFinite": bool(np.all(np.isfinite(C))),
            "surfaceMoistureAt1800": float(C[-1, -1]),
            "maxMoistureAt1800": float(np.max(C[-1])),
            "massResidual": run.diagnostics()["max_mass_balance_abs_kg_per_kg"],
            "elapsedSeconds": time.perf_counter() - started}
    finally:
        core.RadialModel = originalClass
        if run is not None:
            run.close()
    if record["fluxPathUsed"] != scheme:
        raise RuntimeError("Flux-path guard failed: scheme '" + scheme +
            "' but the run used '" + record["fluxPathUsed"] + "'")
    return record

def compareFiniteDifferenceToKirchhoff(records):
    rows = []
    levels = sorted({r["intervals"] for r in records})
    for n in levels:
        a = next(r for r in records if r["intervals"] == n and r["scheme"] == "kirchhoff")
        b = next(r for r in records if r["intervals"] == n and r["scheme"] ==
"finiteDifference")
        difference = np.abs(np.asarray(a["moisture"]) - np.asarray(b["moisture"]))
        rows.append({"intervals": n, "maxAbsDifference": float(np.nanmax(difference)),
            "maxDifferenceTimeS": float(TIMES_S[int(np.nanargmax(difference) //
difference.shape[1]))],
            "locationOfMax": float(RADII_M[int(np.nanargmax(difference) %
difference.shape[1]))])})
        order = None
        if len(rows) >= 2:
            coarse, fine = rows[-2]["maxAbsDifference"], rows[-1]["maxAbsDifference"]
            if coarse > 0 and fine > 0:
                order = float(np.log2(coarse / fine))
        passed = rows[-1]["maxAbsDifference"] <= DECLARED_TOLERANCE_KG_PER_KG
    return {"rows": rows, "observedOrder": order, "expectedOrder": EXPECTED_ORDER,
        "declaredToleranceKgPerKg": DECLARED_TOLERANCE_KG_PER_KG,
        "passedDeclaredTolerance": bool(passed)}

def main():
    quick = "--quick" in sys.argv

```

```

OUT.mkdir(parents=True, exist_ok=True)
levels = (100, 200) if quick else (200, 400, 800, 1600)
records = []
for kind in ("constant", "variable"):
    for n in levels:
        for scheme in ("kirchhoff", "finiteDifference"):
            record = runScheme(scheme, n, kind)
            records.append(record)
            print(json.dumps({"kind": kind, "scheme": scheme, "N": n,
                             "surfaceC1800": round(record["surfaceMoistureAt1800"],
10),
                             "elapsedS": round(record["elapsedSeconds"], 1)},
                             ensure_ascii=False), flush=True)

constantRows = [r for r in records if r["kind"] == "constant"]

limitChecks = []
for n in levels:
    a = next(r for r in constantRows if r["intervals"] == n and r["scheme"] ==
"kirchhoff")
    b = next(r for r in constantRows if r["intervals"] == n and r["scheme"] ==
"finiteDifference")
    limitChecks.append({"intervals": n,
                        "maxAbsDifference": float(np.max(np.abs(
                            np.asarray(a["moisture"]) - np.asarray(b["moisture"]))))})

report = {
    "generatedAtUtc": datetime.now(timezone.utc).isoformat(),
    "note": ("Same moisture equation, same time integrator, same tolerances; only the "
            "face-flux discretisation differs. Any gap is a spatial discretisation "
            "effect. Frozen baseline and production results are untouched."),
    "levels": list(levels),
    "frozenBenchmarkD_m2_s": FROZEN_BENCHMARK_D,
    "referenceD_m2_s": D_REFERENCE,
    "declaredToleranceKgPerKg": DECLARED_TOLERANCE_KG_PER_KG,
    "limitingCaseConstantD": {
        "note": ("Kirchhoff collapses to the midpoint finite difference when D is
constant, "
                "because  $\Phi'(C) = \exp(-a/C)$  makes the primitive difference
proportional to "
                "D dC; the two must agree to round-off."),
        "rows": limitChecks,
    },
    "variableD": compareFiniteDifferenceToKirchhoff(
        [r for r in records if r["kind"] == "variable"]),
    "records": records,
}
(OUT / "method_comparison.json").write_text(
    json.dumps(report, ensure_ascii=False, indent=2), encoding="utf-8")
print(json.dumps({k: report[k] for k in ("limitingCaseConstantD", "variableD")},
    ensure_ascii=False, indent=2)[:2000], flush=True)
return 0

if __name__ == "__main__":
    sys.exit(main())

```

D.43 sensitivity_analysis.py

路径: 05_数值检验与实验/Python 检验源码/paper_output/code/verification/sensitivity_analysis.py

用途: 参数敏感性; 包括已修正的 Kirchhoff 通量尺度注入

使用范围: 检验与历史诊断

语言: Python; 源文件共 249 行。

SHA256: f0d85de0fc394c4b33f9b5c8a8310de10c405d4f7a87e4229c9d1e64556d49e1

```
from __future__ import annotations

import json
import math
import os
import pathlib
import sys
import time
import uuid
from dataclasses import replace

ROOT = pathlib.Path(r"D:\Document\数学建模\2026CUMCM")
os.environ.setdefault("OPENBLAS_NUM_THREADS", "1")
os.chdir(ROOT)
sys.dont_write_bytecode = True
sys.path.insert(0, str(ROOT / "paper_output" / "code" / "modeling"))

import numpy as np

import drying_core
from drying_core import RadialModel, Settings, solve_case
import q3_model

OUT = ROOT / "paper_output" / "results" / "crossvalidation" / "sensitivity_v1"
INTERVALS = 200
SURROGATE_EVENT_H = {"Q23": 57.4727587858255, "Q4": 51.090973419826916}
PRODUCTION_EVENT_H = {"Q23": 57.47230195056044, "Q4": 51.09057478683054}

SCALES = {"d": 1.0, "k": 1.0}
_ORIGINAL_PROPERTIES = RadialModel.properties
_ORIGINAL_WATER_FLUX = RadialModel.water_internal_flux

def _scaled_properties(self, T, C):
    rho, cp, k, D = _ORIGINAL_PROPERTIES(self, T, C)
    return rho, cp, k * SCALES["k"], D * SCALES["d"]

def _scaled_water_flux(self, T, C, D):

    return _ORIGINAL_WATER_FLUX(self, T, C, D) * SCALES["d"]

RadialModel.properties = _scaled_properties
RadialModel.water_internal_flux = _scaled_water_flux

PARAMETERS = [
    ("tailTemperatureC", "degC", 50.0, 48.0, 52.0),
```

```

("tailEquilibrium", "kg/kg", 0.05, 0.04, 0.06),
("h", "W/(m2 K)", 25.0, 20.0, 30.0),
("beta", "m/s", 8e-7, 6.4e-7, 9.6e-7),
("surfaceLatentFraction", "-", 0.0, 0.0, 1.0),
("equilibriumScale", "-", 1.0, 0.8, 1.2),
("dScale", "-", 1.0, 0.7, 1.4),
("kScale", "-", 1.0, 0.85, 1.15),
]
NAMES = [p[0] for p in PARAMETERS]

def settings_for(question, values):
    shrink = question == "Q4"
    kwargs = {"tail_temperature_C": values["tailTemperatureC"],
              "tail_equilibrium": values["tailEquilibrium"],
              "h": values["h"], "beta": values["beta"],
              "surface_latent_fraction": values["surfaceLatentFraction"],
              "equilibrium_scale": values["equilibriumScale"]}
    return Settings(question=question, intervals=INTERVALS, shrink=shrink,
                    rtol=1e-10, atol_temperature=1e-10, atol_moisture=1e-12,
                    early_max_step_s=2.0, max_step_s=120.0,
                    face_scheme="kirchhoff", jacobian_mode="finite_difference",
                    **kwargs)

FAILURES = []

def evaluate(question, unitPoint):
    values = {}
    for name, _unit, low, high in ((p[0], p[1], p[3], p[4]) for p in PARAMETERS):
        values[name] = low + unitPoint[name] * (high - low)
    SCALES["d"] = values["dScale"]
    SCALES["k"] = values["kScale"]
    try:
        run = solve_case(settings_for(question, values))
    except Exception as error:
        FAILURES.append({"question": question, "values": values,
                        "error": f"{type(error).__name__}: {error}"})
        return float("nan")
    try:
        return run.event_s / 3600.0
    finally:
        run.close()

def baseline_point():
    return {name: (base - low) / (high - low)
            for name, _unit, base, low, high in PARAMETERS}

def full_point(subset, row):
    point = baseline_point()
    for name, value in zip(subset, row):
        point[name] = float(value)
    return point

def control_check():
    SCALES["d"] = SCALES["k"] = 1.0

```

```

base = Settings(question="Q23", intervals=INTERVALS, rtol=1e-10,
                atol_temperature=1e-10, atol_moisture=1e-12,
                early_max_step_s=2.0, max_step_s=120.0,
                face_scheme="kirchhoff", jacobian_mode="analytic")
run = solve_case(base)
reference = run.event_s / 3600.0
run.close()
patched = evaluate("Q23", baseline_point())
return {"unpatchedAnalyticJacobianH": reference,
        "patchedFiniteDifferenceH": patched,
        "differenceSeconds": abs(reference - patched) * 3600.0,
        "referenceProductionN3200H": PRODUCTION_EVENT_H["Q23"],
        "surrogateOffsetVsProductionSeconds":
            abs(reference - PRODUCTION_EVENT_H["Q23"]) * 3600.0}

def morris(question, trajectories, levels=4):
    delta = levels / (2.0 * (levels - 1))
    grid = np.linspace(0.0, 1.0 - delta, levels)
    d = len(PARAMETERS)
    effects = {name: [] for name in NAMES}
    rng = np.random.default_rng(20260911)
    started = time.perf_counter()
    for t in range(trajectories):
        base = rng.choice(grid, size=d)
        point = {name: float(base[i]) for i, name in enumerate(NAMES)}
        y0 = evaluate(question, point)
        order = rng.permutation(d)
        for index in order:
            name = NAMES[index]
            nxt = dict(point)
            nxt[name] = min(1.0, point[name] + delta)
            y1 = evaluate(question, nxt)
            effects[name].append((y1 - y0) / delta)
            point, y0 = nxt, y1
        if (t + 1) % 5 == 0:
            print(f" morris {question} trajectory {t+1}/{trajectories} "
                  f"({time.perf_counter()-started:.0f}s)", flush=True)
    summary = []
    for name in NAMES:
        arr = np.asarray(effects[name], dtype=float)
        summary.append({"parameter": name,
                        "muStar": float(np.mean(np.abs(arr))),
                        "mu": float(np.mean(arr)),
                        "sigma": float(np.std(arr, ddof=1)),
                        "min": float(arr.min()), "max": float(arr.max()),
                        "elementaryEffects": arr.tolist()})
    summary.sort(key=lambda row: row["muStar"], reverse=True)
    return {"question": question, "trajectories": trajectories, "levels": levels,
            "delta": delta, "runs": trajectories * (d + 1),
            "elapsedSeconds": time.perf_counter() - started, "ranking": summary}

def sobol(question, subset, baseSamples, seed=7):
    d = len(subset)
    rng = np.random.default_rng(seed)
    A = rng.random((baseSamples, d))
    B = rng.random((baseSamples, d))
    fA = np.array([evaluate(question, full_point(subset, row)) for row in A])
    fB = np.array([evaluate(question, full_point(subset, row)) for row in B])
    var_y = float(np.var(np.concatenate([fA, fB]), ddof=1))
    rows = []

```

```

started = time.perf_counter()
for j, name in enumerate(subset):
    AB = A.copy()
    AB[:, j] = B[:, j]
    fAB = np.array([evaluate(question, full_point(subset, row)) for row in AB])

    s1 = float(np.mean(fB * (fAB - fA)) / var_y) if var_y > 0 else float("nan")
    st = float(np.mean((fA - fAB) ** 2) / (2.0 * var_y)) if var_y > 0 else float("nan")
    rows.append({"parameter": name, "firstOrder": s1, "totalOrder": st,
                "interaction": st - s1})
    print(f" sobol {question} {name}: S1={s1:.4f} ST={st:.4f} "
          f"({time.perf_counter()-started:.0f}s)", flush=True)
return {"question": question, "subset": subset, "baseSamples": baseSamples,
        "runs": baseSamples * (d + 2), "outputVariance": var_y,
        "indices": rows, "elapsedSeconds": time.perf_counter() - started}

def scan(stageName, parameter, values, questions=("Q23", "Q4")):

    rows = []
    for question in questions:
        for value in values:
            point = baseline_point()
            low = next(p[3] for p in PARAMETERS if p[0] == parameter)
            high = next(p[4] for p in PARAMETERS if p[0] == parameter)
            point[parameter] = (value - low) / (high - low)
            y = evaluate(question, point)
            rows.append({"question": question, "parameter": parameter,
                        "value": value, "eventH": y})
            print(f" scan {question} {parameter}={value:g} -> {y:.6f} h", flush=True)
    byQ = {}
    for question in questions:
        ys = [r["eventH"] for r in rows if r["question"] == question]
        lo, hi = min(ys), max(ys)
        base = next((r["eventH"] for r in rows
                     if r["question"] == question and abs(r["value"] - 1.0) < 1e-12), None)
        byQ[question] = {"minH": lo, "maxH": hi, "spanH": hi - lo,
                        "baselineH": base,
                        "relativeSpanPercent": None if not base else 100.0 * (hi - lo) /
base}
    return {"stage": stageName, "parameter": parameter, "rows": rows, "summary": byQ}

def main():
    stage = sys.argv[1] if len(sys.argv) > 1 else "morris"
    OUT.mkdir(parents=True, exist_ok=True)
    if stage == "control":
        result = {"control": control_check(),
                  "surrogateEventH": SURROGATE_EVENT_H,
                  "productionEventH": PRODUCTION_EVENT_H}
    elif stage == "morris":
        trajectories = int(sys.argv[2]) if len(sys.argv) > 2 else 20
        result = {"control": control_check(),
                  "morris": [morris(q, trajectories) for q in ("Q23", "Q4")]}
    elif stage == "dscale":
        result = {"control": control_check(),
                  "scan": scan("dscale", "dScale", [0.7, 0.875, 1.0, 1.05, 1.225, 1.4])}
    elif stage == "kscale":
        result = {"scan": scan("kscale", "kScale", [0.85, 0.925, 1.0, 1.075, 1.15])}
    elif stage == "sobol":
        subset = sys.argv[2].split(",")
        base = int(sys.argv[3]) if len(sys.argv) > 3 else 64
        result = {"sobol": [sobol(q, subset, base) for q in ("Q23", "Q4")]}

```

```

else:
    raise SystemExit(f"unknown stage {stage}")
path = OUT / f"{stage}.json"
path.write_text(json.dumps(result, ensure_ascii=False, indent=2), encoding="utf-8")
print(json.dumps(result.get("scan", {}).get("summary", {}), ensure_ascii=False),
      flush=True)
return 0

if __name__ == "__main__":
    sys.exit(main())

```

D.44 threshold_and_scaling_checks.py

路径: 05_数值检验与实验/Python 检验源码/paper_output/code/verification/
threshold_and_scaling_checks.py

用途: 阈值、量纲标度、缩放与响应解释检查

使用范围: 检验与历史诊断

语言: Python; 源文件共 154 行。

SHA256: 73a8d8cfd3bcebc218ac25a7cb30e443500c6d05f57263a11f862ec023b87e6d

```
from __future__ import annotations

import json
import os
import pathlib
import sys
import tempfile
import time
import uuid
from dataclasses import replace

ROOT = pathlib.Path(r"D:\Document\数学建模\2026CUMCM")

def _mkdtemp(suffix=None, prefix=None, dir=None):
    base = pathlib.Path(dir)
    for _ in range(200):
        candidate = base / ((prefix or "tmp") + uuid.uuid4().hex[:12] + (suffix or ""))
        if not candidate.exists():
            candidate.mkdir(parents=False)
            return str(candidate)
    raise FileExistsError("no unused temp name")

tempfile.mkdtemp = _mkdtemp
os.environ.setdefault("OPENBLAS_NUM_THREADS", "1")
os.chdir(ROOT)
sys.dont_write_bytecode = True
sys.path.insert(0, str(ROOT / "paper_output" / "code" / "modeling"))

import io

import numpy as np

OUT = ROOT / "paper_output" / "results" / "crossvalidation" / "threshold_scaling_v1"
R0 = 0.02

def read_radius():
    text = (ROOT /
"paper_output/data_cleaned/A_radius_observed.csv").read_bytes().decode("utf-8-sig")
    return np.genfromtxt(io.StringIO(text), delimiter=",", names=True)

def scaling_check():
    rad = read_radius()
    t = rad["time_s"].astype(float)
    r = rad["radius_m"].astype(float)
    t_fixed_s = 129.84522834701946 * 3600.0
```

```

t_shrink_s = 51.090973419826916 * 3600.0
grid = np.linspace(0.0, t_shrink_s, 200001)
r_of_t = np.interp(grid, t, r)
integrand = (R0 / r_of_t) ** 2
tau_eq = float(np.trapezoid(integrand, grid))

return {
    "fixedRadiusTimeH": t_fixed_s / 3600.0,
    "shrinkingTimeH": t_shrink_s / 3600.0,
    "reductionPercent": float(100.0 * (1.0 - t_shrink_s / t_fixed_s)),
    "equivalentFixedRadiusTimeH": tau_eq / 3600.0,
    "ratioEquivalentToFixed": float((tau_eq / 3600.0) / (t_fixed_s / 3600.0)),
    "scalingResidualPercent": float(
        100.0 * ((tau_eq / 3600.0) / (t_fixed_s / 3600.0) - 1.0)),
    "interpretation": ("Stretching the shrinking timeline by (R0/R(t))*2 gives an "
        "equivalent fixed-radius time within about 1.4 percent of the "
        "directly simulated fixed-radius drying time. The numerically "
        "computed shortening is therefore corroborated by an independent "
        "analytical length-scale argument; the small residual comes from "
        "the coupled changes in the moisture and temperature "
        "profiles."),
}

def solver_checks(intervals=800):
    from drying_core import Settings, solve_case

    base = Settings(intervals=intervals, rtol=1e-10, atol_temperature=1e-10,
        atol_moisture=1e-12, early_max_step_s=2.0, max_step_s=120.0,
        face_scheme="kirchhoff", jacobian_mode="analytic")
    records = {}
    for question, shrink in (("Q23", False), ("Q4", True)):
        started = time.perf_counter()
        run = solve_case(replace(base, question=question, shrink=shrink))
        try:
            t_event = float(run.event_s)

            def max_c(t):
                state = run.state([t])
                return float(np.max(state[1:-1:2, 0]))

            lo, hi = t_event - 3600.0, t_event + 3600.0
            for _ in range(80):
                mid = 0.5 * (lo + hi)
                if max_c(mid) - 0.15 > 0.0:
                    lo = mid
                else:
                    hi = mid
            t_root = 0.5 * (lo + hi)

            grid = np.linspace(0.0, t_event, 2001)

            states = np.hstack([run.state(grid[i:i + 200]) for i in range(0, len(grid),
200)])

            cs = states[1:-1:2, :]
            argmax_x = run.model.x[np.argmax(cs, axis=0)]
            profile_increase = float(np.max(np.diff(cs, axis=0)))

```

```

centre_margin = np.max(cs, axis=0) - cs[0, :]

strict_time = float(np.ceil(t_event) + 1.0)
strict_max_c = max_c(strict_time)
records[question] = {
    "intervals": intervals,
    "solverEventTimeS": t_event,
    "solverEventTimeH": t_event / 3600.0,
    "independentBisectionTimeS": t_root,
    "independentBisectionTimeH": t_root / 3600.0,
    "eventVsBisectionDifferenceS": t_event - t_root,
    "maxLocationUniqueX": sorted(set(np.round(argmax_x, 12).tolist())),
    "maxLocationAlwaysCentre": bool(np.all(np.abs(argmax_x) < 1e-12)),
    "maxExcessOverCentreMaxKgPerKg": float(np.max(centre_margin)),
    "maxExcessOverCentreMedianKgPerKg": float(np.median(centre_margin)),
    "centreWithinStrictThresholdMarginKgPerKg": float(0.15 - float(np.max(cs[0,
:]))),
    "centreIsMaximumAtEvent": bool(np.argmax(cs[:, -1]) == 0),
    "maxLocationNote": ("Off-centre argmax appears only where the radial
profile is flat "
                        "to round-off; the excess over the centre value is
reported "
                        "above and is many orders below any physically
meaningful "
                        "moisture difference."),
    "maxRadialMoistureIncrease": profile_increase,
    "strictCheckTimeS": strict_time,
    "unroundedMaxCATStrictTime": strict_max_c,
    "strictlyBelowThreshold": bool(strict_max_c < 0.15),
}
finally:
    run.close()
    records[question]["elapsedSeconds"] = time.perf_counter() - started
    print(json.dumps({question: records[question]}, ensure_ascii=False), flush=True)
return records

def main():
    OUT.mkdir(parents=True, exist_ok=True)
    result = {"scalingCheck": scaling_check()}
    if "--with-solver" in sys.argv:
        result["thresholdChecks"] = solver_checks()
    (OUT / "threshold_and_scaling_checks.json").write_text(
        json.dumps(result, ensure_ascii=False, indent=2), encoding="utf-8")
    print(json.dumps(result["scalingCheck"], ensure_ascii=False, indent=2), flush=True)
    return 0

if __name__ == "__main__":
    sys.exit(main())

```

D.45 prepare_a_data.py

路径：补充源码/paper_output/code/data/prepare_a_data.py

用途：从官方附件 1/2 核查字段、时间及有限性，换算单位并导出正式观测 CSV 和原始观测图

使用范围：正式输入预处理；原始观测图不是本稿最终六图

语言：Python；源文件共 246 行。

SHA256: 520b8c91a5c51299f83bc7ed1a3a3b63320b876a9e1a6ab5775234ecc494a6db

```
from __future__ import annotations

import csv
import hashlib
import json
import os
import platform
import sys
from datetime import datetime, timezone
from pathlib import Path

ROOT = Path(__file__).resolve().parents[3]
if Path.cwd().resolve() != ROOT:
    raise RuntimeError(f"Run with workdir {ROOT}")
os.environ["MPLCONFIGDIR"] = str(ROOT / "tmp/cache/matplotlib")

import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt
import matplotlib.font_manager as font_manager
import numpy as np
import openpyxl

OUT = ROOT / "paper_output"
DATA = OUT / "data_cleaned"
PLAN = OUT / "plan"
FIG = OUT / "figures"
MANIFEST = OUT / "input_manifest.json"
SCRIPT = Path(__file__).resolve()
EXPECTED = {
    "environment": "problem_files/CUMCM2026Problems/A 题/附件/附件 1.xlsx",
    "radius": "problem_files/CUMCM2026Problems/A 题/附件/附件 2.xlsx",
}

def now() -> str:
    return datetime.now(timezone.utc).isoformat()

def rel(path: Path) -> str:
    return path.relative_to(ROOT).as_posix()

def sha(path: Path) -> str:
    return hashlib.sha256(path.read_bytes()).hexdigest()

def save_json(path: Path, value: dict) -> None:
    path.parent.mkdir(parents=True, exist_ok=True)
    path.write_text(json.dumps(value, ensure_ascii=False, indent=2), encoding="utf-8")
```

```

def write_csv(path: Path, header: list[str], data: np.ndarray) -> None:
    with path.open("w", encoding="utf-8-sig", newline="") as handle:
        writer = csv.writer(handle)
        writer.writerow(header)
        writer.writerows([[format(float(v), ".15g") for v in row] for row in data])

def base_record() -> dict:
    return {"schema_version": "1.0", "generated_by": rel(Script),
            "generated_at": now(), "source_code_sha256": sha(SCRIPT)}

def read_and_diagnose() -> tuple[dict, dict, dict]:
    manifest = json.loads(MANIFEST.read_text(encoding="utf-8-sig"))
    entries = {entry["path"]: entry for entry in manifest["entries"]}
    report = {**base_record(), "status": "PASS", "input_manifest_used": True,
              "input_manifest": rel(MANIFEST), "input_manifest_sha256": sha(MANIFEST),
              "active_problem": "A", "input_dirs": ["problem_files/CUMCM2026Problems/A 题/附件"],
              "data_files": [], "skipped_files": [], "pdf_diagnostics": [],
              "warnings": [], "errors": []}
    for path, entry in entries.items():
        if path not in EXPECTED.values():
            report["skipped_files"].append({"path": path, "role": entry.get("role"),
                                             "reason": "outside_active_A_raw_inputs"})
    raw_arrays, diagnostics = {}, {}
    for key, relative in EXPECTED.items():
        entry = entries.get(relative, {})
        if entry.get("role") != "raw_data" or not entry.get("usable_for_modeling"):
            raise RuntimeError(f"Input is not authorized raw data in manifest: {relative}")
        path = ROOT / relative
        before = sha(path)
        if entry.get("sha256") != before:
            raise RuntimeError(f"Manifest hash differs from raw input: {relative}")
        wb = openpyxl.load_workbook(path, read_only=True, data_only=True)
        try:
            if wb.sheetnames != ["Sheet1"]:
                raise ValueError(f"Unexpected worksheet schema: {relative}")
            values = list(wb["Sheet1"].iter_rows(values_only=True))
        finally:
            wb.close()
        header, body = list(values[0]), values[1:]
        expected_header = ["时间", "温度", "水分浓度"] if key == "environment" else ["时间",
"半径"]
        if header != expected_header:
            raise ValueError(f"Unexpected header: {relative}: {header}")
        if any(len(row) != len(header) for row in body):
            raise ValueError(f"Unequal row length: {relative}")
        if any(not isinstance(v, (int, float)) or isinstance(v, bool) for row in body for v
in row):
            raise ValueError(f"Missing or nonnumeric input found, review required:
{relative}")
        data = np.asarray(body, dtype=float)
        if not np.isfinite(data).all():
            raise ValueError(f"Nonfinite values found: {relative}")
        step = 60 if key == "environment" else 1800
        end = 14400 if key == "environment" else 259200
        if data[0, 0] != 0 or data[-1, 0] != end or not np.all(np.diff(data[:, 0]) ==
step):
            raise ValueError(f"Unexpected time coverage or duplicate time: {relative}")
        if key == "radius" and (np.any(np.diff(data[:, 1]) > 0) or np.any(data[:, 1] <=

```

```

0)):
    raise ValueError("Radius must be positive and nonincreasing; review raw data")
    diag = {"path": relative, "kind": "spreadsheet", "ext": ".xlsx", "readable": True,
            "sha256": before, "unchanged_after_read": sha(path) == before,
            "sheets": [{ "name": "Sheet1", "rows": len(values), "cols": len(header),
"sample_cols": header}],
            "numeric_record_count": len(data), "missing_numeric_cells": 0,
            "nonfinite_values": 0, "duplicate_times": 0, "time_start_s": 0,
            "time_end_s": end, "time_step_s": step, "warnings": [], "errors": [],
            "minima": data.min(axis=0).tolist(), "maxima": data.max(axis=0).tolist()}
    report["data_files"].append(diag)
    raw_arrays[key], diagnostics[key] = data, diag
    report["summary"] = {"file_count_scanned": 2, "data_file_count": 2,
                        "readable_data_file_count": 2, "pdf_file_count": 0}
    report["warnings"] = ["Air kg/kg is not automatically solid dry-basis equilibrium
moisture.",
                        "Room observations end at 4 h; radius observations end at 72 h.",
                        "No internal temperature/moisture validation measurements are
supplied."]
    save_json(DATA / "load_report.json", report)
    return raw_arrays, diagnostics, report

def make_figures(environment: np.ndarray, radius: np.ndarray) -> list[dict]:
    font_file = Path("C:/Windows/Fonts/msyh.ttc")
    if font_file.exists():
        font_manager.fontManager.addfont(str(font_file))
        plt.rcParams["font.family"] =
font_manager.FontProperties(fname=str(font_file)).get_name()
    plt.rcParams.update({"font.size": 10, "axes.titlesize": 11, "axes.labelsize": 10,
                        "axes.spines.top": False, "axes.spines.right": False,
                        "axes.unicode_minus": False, "savefig.dpi": 220,
                        "svg.fonttype": "path", "axes.grid": True, "grid.alpha": 0.18})

    records = []
    configs = [
        {"figure_id": "fig_A_environment", "title": "烘房温度与水分指标原始观测",
         "question_id": "Q1", "question_ids": ["Q1", "Q2", "Q3", "Q4"],
         "data_source": "paper_output/data_cleaned/A_environment_observed.csv",
         "candidate_x": "time_h", "candidate_y": ["temperature_C",
"air_moisture_kg_per_kg"]},
        {"purpose": "展示 0-4 小时实测边界及末期平台，标明长期延拓的观测边界",
         "paper_usage": "数据审计与环境边界条件", "caption": "241 条原始观测；连线仅连接采样点，
不表示 4 小时以后实测。空气 kg/kg 指标未直接等同药材平衡含水率。"},
        {"figure_id": "fig_A_radius", "title": "药材半径收缩原始观测",
         "question_id": "Q4", "question_ids": ["Q4"],
         "data_source": "paper_output/data_cleaned/A_radius_observed.csv",
         "candidate_x": "time_h", "candidate_y": ["radius_cm"]},
        {"purpose": "展示 0-72 小时给定半径和 48-72 小时尾段，限定收缩几何的数据支持范围",
         "paper_usage": "收缩数据与移动边界", "caption": "145 条原始观测；半径保持非增。右图放大
48-72 小时，未对 72 小时以后外推。"}],
    ]
    for item in configs:
        fig, axes = plt.subplots(1, 2, figsize=(10.8, 3.7), layout="constrained")
        if item["figure_id"] == "fig_A_environment":
            for ax, y, color, ylabel, title in [
                (axes[0], environment[:, 2], "#1b6b93", "温度 / °C", "烘房温度"),
                (axes[1], environment[:, 4], "#b86625", "空气水分指标 / (kg/kg)", "烘房水分指
标"),
            ]:
                ax.plot(environment[:, 1], y, color=color, lw=1.35, marker=".", ms=2)
                ax.set(xlabel="时间 / h", ylabel=ylabel, title=title, xlim=(0, 4))
                ax.axvspan(3, 4, color=color, alpha=0.065)
        else:

```

```

        axes[0].plot(radius[:, 1], radius[:, 2], color="#336950", lw=1.35, marker=".",
ms=2)
        axes[0].set(xlabel="时间 / h", ylabel="半径 / cm", title="完整观测区间", xlim=(0,
72))
        tail = radius[:, 1] >= 48
        axes[1].plot(radius[tail, 1], radius[tail, 2], color="#336950", lw=1.35,
marker="o", ms=2.2)
        axes[1].set(xlabel="时间 / h", ylabel="半径 / cm", title="48-72 h 尾段放大",
xlim=(48, 72))
        axes[1].ticklabel_format(axis="y", style="plain", useOffset=False)
        paths = {}
        for ext in ["png", "svg", "pdf"]:
            path = FIG / f"{item['figure_id']}.{ext}"
            fig.savefig(path)
            paths[ext] = {"path": rel(path), "sha256": sha(path), "bytes":
path.stat().st_size}
        plt.close(fig)
        records.append(**item, "path": paths["png"]["path"], "output_path": paths["png"]
["path"],
                        "chart_type": "observed_time_series", "artifacts": paths,
                        "status": "generated_from_observations", "ok": True, "placeholder":
False,
                        "exists": True, "planned": False, "used_in": item["paper_usage"],
                        "source_sha256": sha(ROOT / item["data_source"]),
                        "visual_review": "pending", "human_review": "pending"})

    return records

def main() -> None:
    for directory in [DATA, PLAN, FIG]:
        directory.mkdir(parents=True, exist_ok=True)
        raw_arrays, diagnostics, report = read_and_diagnose()
        air, rad = raw_arrays["environment"], raw_arrays["radius"]
        environment = np.column_stack([air[:, 0], air[:, 0]/3600, air[:, 1], air[:, 1]+273.15,
air[:, 2]])
        radius = np.column_stack([rad[:, 0], rad[:, 0]/3600, rad[:, 1], rad[:, 1]*0.01])
        env_header = ["time_s", "time_h", "temperature_C", "temperature_K",
"air_moisture_kg_per_kg"]
        rad_header = ["time_s", "time_h", "radius_cm", "radius_m"]
        write_csv(DATA / "A_environment_observed.csv", env_header, environment)
        write_csv(DATA / "A_radius_observed.csv", rad_header, radius)

        env_back = np.loadtxt(DATA / "A_environment_observed.csv", delimiter=",", skiprows=1,
encoding="utf-8-sig")
        rad_back = np.loadtxt(DATA / "A_radius_observed.csv", delimiter=",", skiprows=1,
encoding="utf-8-sig")
        if not np.allclose(env_back, environment, rtol=0, atol=1e-10):
            raise ValueError("Environment CSV round-trip failed")
        if not np.allclose(rad_back, radius, rtol=0, atol=1e-10):
            raise ValueError("Radius CSV round-trip failed")
        data_files = []
        for key, header, name in [("environment", env_header, "A_environment_observed.csv"),
("radius", rad_header, "A_radius_observed.csv")]:
            data_files.append(**diagnostics[key], "role": "raw_data", "type": "xlsx",
"columns": header, "numeric_columns": header, "categorical_columns": [],
"cleaned_output": rel(DATA / name), "output_sha256": sha(DATA / name),
"cleaning_tasks": ["numeric/schema/finite/duplicate/time-step checks", "unit
conversion only"],
"operations": {"rows_removed": 0, "values_filled": 0, "values_changed": 0,
"smoothing": False, "extrapolation": False})
        data_plan = {**base_record(), "source_contracts": [rel(MANIFEST),
"paper_output/step1/problem_analysis.json", "paper_output/plan/model_route.json"],
"data_files": data_files, "question_links": [

```

```

        {"question_id": q, "required_fields": env_header + (rad_header if q == "Q4" else
[]),
        "input_sources": [EXPECTED["environment"]] + ([EXPECTED["radius"]] if q == "Q4"
else []),
        "expected_outputs": ["temperature/moisture fields and validation" if q in ["Q1",
"Q2"] else "moisture field, maximum criterion and drying time"],
        "status": "model_outputs_pending"} for q in ["Q1", "Q2", "Q3", "Q4"]],
        "assumption_boundaries": {"continuous_interpolation": "between observed samples",
        "air_to_solid_boundary_mapping": "t=0; requires explicit closure",
        "environment_extrapolation_start_s": 14400,
        "radius_extrapolation_start_s": 259200,
        "material_velocity_and_axial_length": "t=0; unobserved assumptions"},
        "limitations": report["warnings"],
        "validation": {"csv_round_trip": "PASS", "originals_unchanged": True,
        "visual_studio_gui": "pending", "human_review": "pending"}}
    save_json(PPLAN / "data_plan.json", data_plan)
    figures = make_figures(environment, radius)
    save_json(PPLAN / "visualization_plan.json", {**base_record(), "source_contracts": [
        "paper_output/plan/model_route.json", "paper_output/plan/data_plan.json"],
        "figures": figures, "scope": "observed inputs only; model figures are a later
stage"})
    save_json(OUT / "figure_index.json", {**base_record(), "figures": figures})
    run_record = {**base_record(), "status": "PASS", "operation": "S3 observed input
processing",
        "runtime": {"executable": sys.executable, "python": sys.version, "platform":
platform.platform(),
        "numpy": np.__version__, "openpyxl": openpyxl.__version__,
"matplotlib": matplotlib.__version__},
        "input_manifest_sha256": sha(MANIFEST), "record_counts": {"environment": len(air),
"radius": len(rad)},
        "observations_changed": 0, "csv_round_trip_max_absolute_error": {
        "environment": float(np.max(np.abs(env_back-environment))),
        "radius": float(np.max(np.abs(rad_back-radius)))},
        "figures_generated": [item["figure_id"] for item in figures],
        "model_solve": False, "visual_studio_gui_reproduction": False, "user_human_review":
False}
    save_json(DATA / "A_data_run_record.json", run_record)
    print(json.dumps(run_record, ensure_ascii=False, indent=2))

if __name__ == "__main__":
    main()

```

D.46 export_six_figure_package_20260912.py

路径：补充源码/tools/export_six_figure_package_20260912.py

用途：从冻结 NPZ/JSON 与观测 CSV 导出六图 14 张数据表，保留原精度、实际半径、事件与情景口径

使用范围：正式六图数据预处理；不重新积分

语言：Python；源文件共 178 行。

SHA256: b68228ff0c954c4025df95a8b207bb8d020d503b470a2518fe7c59b1ba932845

```
from pathlib import Path
import csv, hashlib, json, shutil, sys
import numpy as np

sys.stdout.reconfigure(encoding='utf-8')
ROOT = Path.cwd()
assert ROOT.as_posix() == 'D:/Document/数学建模/2026CUMCM'
OUT = ROOT / 'paper_output/handoff/six_figures_20260912'
QA = ROOT / 'paper_output/qa/figure_package_20260912'
CACHE = ROOT / 'tmp/cache/figure_package_20260912'
for p in (OUT, QA, CACHE): p.mkdir(parents=True, exist_ok=True)
PROD = 'paper_output/results/production/final_v6a/'
old = json.loads((ROOT /
'paper_output/qa/figure_guide_20260912/build_manifest.json').read_text(encoding='utf-8'))
source_paths = [s['path'] for s in old['sources']] + [PROD + 'Q4/summary.json']
copied = {}
sources = []
for src in source_paths:
    p = Path(src)
    if p.name in ('sampled_solution.npz', 'summary.json'):
        rel = Path('原始数据') / p.parent.name / p.name
    elif 'outputs' in p.parts:
        rel = Path('原始数据/正文核对表') / p.name
    else:
        rel = Path('原始数据') / p.name
    dest = OUT / rel
    dest.parent.mkdir(parents=True, exist_ok=True)
    shutil.copy2(ROOT / src, dest)
    sha = hashlib.sha256(dest.read_bytes()).hexdigest()
    prior = next((s for s in old['sources'] if s['path'] == src), None)
    if prior: assert sha == prior['sha256']
    copied[src] = rel.as_posix()
    sources.append({'file': rel.as_posix(), 'source': src, 'sha256': sha})

sheets = []
def add(fig, filename, name, title, note, source, columns, rows):
    rows = [[v.item() if isinstance(v, np.generic) else v for v in row] for row in rows]
    rel = f'CSV/图{fig}/{filename}.csv'
    path = OUT / rel
    path.parent.mkdir(parents=True, exist_ok=True)
    with path.open('w', encoding='utf-8-sig', newline='') as f:
        w = csv.writer(f); w.writerow(columns); w.writerows(rows)
    sheets.append({'figure': fig, 'name': name, 'title': title, 'note': note,
        'source': source, 'columns': columns, 'rows': rows, 'csv': rel})

q1src = PROD + 'figures/fig_q1_profiles_data.npz'
with np.load(ROOT / q1src) as a:
    times = a['times_s']; radii = a['radii_m'] * 100
    for field, tag, title, prefix in [('temperature_C', 'temperature', '温度', 'T'), ('C',
```

```

'moisture', '含水率', 'C')]:
    columns = ['r_cm'] + [f'{prefix}_{t:g}s' for t in times]
    rows = np.column_stack([radii, a[field].T]).tolist()
    add(1, tag, '图1' + title, '图1 径向' + title + '剖面',
        '首列为物理半径 cm; 其余七列对应 100、300、600、900、1200、1500、1800 s。温度单位℃,
含水率为 kg 水/kg 干物质。N3200。',
        copied[q1src] + '; radii_m×100, ' + field + '转置', columns, rows)

envsrc = 'paper_output/data_cleaned/A_environment_observed.csv'
with (ROOT / envsrc).open(encoding='utf-8-sig', newline='') as f: env =
list(csv.DictReader(f))
add(2, 'observed_0_4h', '图2 观测', '图2 环境观测数据 0至4小时',
    '241条真实观测。t=4h 末点为 50.165℃ 和 0.04986, 必须保留。相邻观测点连直线即为分段线性插值。水
分指标不是相对湿度%',
    copied[envsrc], ['time_h', 'temperature_C', 'air_moisture_kg_per_kg'],
    [[float(r[k]) for k in ['time_h', 'temperature_C', 'air_moisture_kg_per_kg']] for r in
env])
q23src = PROD + 'Q23/sampled_solution.npz'
q23json = PROD + 'Q23/summary.json'
summary = json.loads((ROOT / q23json).read_text(encoding='utf-8'))
completion = summary['completion']; event = completion['critical_event_s']
assert summary['settings']['tail_temperature_C'] == 50
assert summary['settings']['tail_equilibrium'] == 0.05
add(2, 'assumed_platform_4_60h', '图2 假设平台', '图2 4 小时后的假设平台',
    '两行是水平线的绘图端点, 不是实测。4h 仅表示右侧极限, 左端开放; 60h 为展示终点。虚线无实测标记,
与观测系列分开。',
    copied[q23json] + '; settings 名义延拓, 展示终点 60h',
    ['time_h', 'temperature_C', 'boundary_moisture_kg_per_kg', 'endpoint_included'],
    [[4.0, 50.0, 0.05, 0], [60.0, 50.0, 0.05, 1]])

with np.load(ROOT / q23src) as a:
    for field, tag, title, limit in [('T_K', 'temperature_0_4h', '温度', 14400.), ('C',
'moisture_to_event', '含水率', event)]:
        mask = a['times_s'] <= limit
        vals = a[field][mask][:, [0, -1]]
        if field == 'T_K': vals = vals - 273.15
        columns = ['time_h', 'centre_T_C', 'surface_T_C'] if field == 'T_K' else ['time_h',
'centre_C_kg_per_kg', 'surface_C_kg_per_kg']
        add(3, tag, '图3' + title, '图3 中心与表面' + title + '响应',
            '采用 Q23 从 0 时刻起算的同一轨迹。centre 表示中心, surface 表示表面; 已换算小时和℃, 含
水率为干基 kg/kg。N3200。',
            copied[q23src] + '; times_s/3600, ' + field + '首末列', columns,
            np.column_stack([a['times_s'][mask] / 3600, vals]).tolist())
        mask = (a['times_s'] >= event - 120) & (a['times_s'] <= event + 2)
        centre_zoom = np.column_stack([a['times_s'][mask], a['times_s'][mask] - event,
            a['C'][mask, 0], a['C'][mask, 0] - 0.15]).tolist()

q3src = PROD + 'figures/fig_q3_drying_data.npz'
with np.load(ROOT / q3src) as a:
    add(4, 'drying_main', '图4 主曲线', '图4 全域达标过程',
        'max_C 取生产全体节点最大值; 中心与最大值重合时不人为错开。threshold_C=0.15。实际曲线止于
206902s, 之后留空。',
        copied[q3src] + '; times_s/3600, centre_C、surface_C、max_C',
        ['time_h', 'centre_C', 'surface_C', 'max_C', 'threshold_C'],
        np.column_stack([a['times_s']/3600, a['centre_C'], a['surface_C'], a['max_C'],
np.full(len(a['times_s']), .15)]).tolist())
        mask = a['times_s'] >= event - 120
        add(4, 'maximum_near_event', '图4 全域放大', '图4 临界附近全域最大值保存点',
            '仅两个已保存全域最大值点, 含临界根及计算终点。横轴秒差, 纵轴 max_C_minus_threshold; 严格
报告点另见图 4 事件标记。',
            copied[q3src] + '; 仅筛选已有点, 未新增求解',
            ['time_s', 'delta_time_s', 'max_C', 'max_C_minus_threshold'],
            np.column_stack([a['times_s'][mask], a['times_s'][mask]-event, a['max_C'][mask],

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a['max_C'][mask]-.15]).tolist())
add(4, 'centre_near_event', '图4 中心放大', '图4 临界附近中心含水率保存点',
    '供观察根前走势。这里是中心样点，不应改名为全节点最大值曲线；全域最大值与严格标记请取另外两表。
    横轴秒差，纵轴 centre_C_minus_threshold。',
    copied[q23src] + '; C 首列，根前 120s 至实际终点',
    ['time_s', 'delta_time_s', 'centre_C', 'centre_C_minus_threshold'], centre_zoom)
add(4, 'event_markers', '图4 事件标记', '图4 临界根与严格报告点',
    '临界根浓度 0.15 为等号定义；严格报告点的最大浓度来自实际核验记录，不能用四位舍入替代。kind 与
    strictly_below 用于区分标记。',
    copied[q23json] + '; completion',
    ['kind', 'time_s', 'time_h', 'delta_time_s', 'max_C', 'max_C_minus_threshold',
    'strictly_below'],
    [['critical_equality', event, completion['critical_event_h'], 0., .15, 0., 0],
    ['strict_report', completion['reported_time_s'], completion['reported_drying_time_h'],
    completion['reported_time_s']-event, completion['max_C_at_reported_time'],
    completion['max_C_at_reported_time']-.15, 1]])

radsrc = 'paper_output/data_cleaned/A_radius_observed.csv'
with (ROOT / radsrc).open(encoding='utf-8-sig', newline='') as f: rad =
list(csv.DictReader(f))
add(5, 'radius_observed', '图5 半径', '图5 半径观测序列',
    '145 条观测，0 至 72h，每 0.5h 一条。横轴小时、纵轴 cm；连接相邻点为模型的线性插值。',
    copied[radsrc], ['time_h', 'radius_cm'], [[float(r['time_h']), float(r['radius_cm'])]]
for r in rad])
q4src = PROD + 'Q4/sampled_solution.npz'
q4json = PROD + 'Q4/summary.json'
q4event = json.loads((ROOT / q4json).read_text(encoding='utf-8'))['completion']
['critical_event_s']
with np.load(ROOT / q4src) as a:
    wanted = [0., 6*3600., 12*3600., 24*3600., 48*3600., q4event]
    labels = ['0h', '6h', '12h', '24h', '48h', 'event']
    pairs = []; cols = []; selected = []
    for t, label in zip(wanted, labels):
        idx = np.flatnonzero(a['times_s'] == t)
        assert len(idx) == 1, (t, idx)
        i = int(idx[0]); radius = float(a['radius_m'][i]*100)
        cols += ['r_' + label + '_cm', 'C_' + label]
        pairs += [a['material_x']*radius, a['C'][i]]
        selected.append([label, float(a['times_s'][i]), float(a['times_s'][i]/3600),
radius])
add(5, 'moisture_profiles_xy_pairs', '图5 含水率剖面', '图5 六时刻物理半径与含水率',
    '每相邻两列是一组 X 和 Y，共 6 组，各 21 点。r 已乘各时刻真实 R。C 为干基 kg/kg；各曲线到自身表面
    结束，域外留空。event 的精确时刻见选定时刻表。',
    copied[q4src] + '; r=material_x*radius_m*100, C 原值: N6400', cols,
    np.column_stack(pairs).tolist())
add(5, 'selected_times_and_radrii', '图5 选定时刻', '图5 六条剖面的时刻与半径',
    '可作为半径曲线上的标记；event 为临界根 51.09057478683054h。半径已换算 cm。',
    copied[q4src] + '; 精确匹配保存时刻', ['curve', 'time_s', 'time_h', 'radius_cm'],
selected)

isosrc = 'paper_output/results/crossvalidation/isotherm_closure_v1/isotherm_closure.json'
scenarios = json.loads((ROOT / isosrc).read_text(encoding='utf-8'))['scenarios']
assert len(scenarios) == 6
rows = []
for p in [1, 2, 4]:
    vals = []
    for q in ['Q23', 'Q4']:
        matches = [s for s in scenarios if s['scenario'] == f'iso_p{p}_{q}']
        assert len(matches) == 1
        s = matches[0]
        assert s['intervals'] == 800 and s['latentFraction'] == 0 and s['status'] ==
'computed_event_only'
        vals.append(s['event_h'])

```

```

        rows.append([p, *vals])
add(6, 'scenario_comparison', '图6 情景对比', '图6 经验边界阻力情景对比',
    '三行共六点，全部 N800 且 latentFraction=0，使用临界等号事件小时。p=1 不能替换为正文生产结果；不  

    加误差棒或置信带。',
    copied[isosrc] + '；六条 scenarios 的顶层 event_h',
    ['p', 'Q23_event_h', 'Q4_event_h'], rows)

payload = {'sheets': sheets}
(CACHE / 'workbook_data.json').write_text(json.dumps(payload, ensure_ascii=False,
allow_nan=False), encoding='utf-8')
tables = [{k:v for k,v in s.items() if k != 'rows'} | {'data_rows': len(s['rows'])} for s
in sheets]
manifest = {'purpose': 'Six-figure plotting handoff with actual data', 'new_model_runs': 0,
    'sources': sources, 'tables': tables, 'total_data_rows': sum(len(s['rows']) for
s in sheets),
    'source_root_note': 'source 字段仅作来源追踪；绘图只使用包内 file/CSV/Excel，无需原工  

    作区。'}
(OUT / '数据清单.json').write_text(json.dumps(manifest, ensure_ascii=False, indent=2),
encoding='utf-8')
(QA / 'export_manifest.json').write_text(json.dumps(manifest, ensure_ascii=False,
indent=2), encoding='utf-8')
readme = ['A 题六图绘图资料包', '',
    '先打开“六图绘图数据.xlsx”，按图号选择工作表。每张表第 6 行为字段名、第 7 行起为数值。',
    'Word“六图绘图说明.docx”逐图写明坐标、选列、线型、标记及图注。',
    'CSV 目录提供同一套数值，UTF-8 BOM 编码，Excel/WPS 或 Origin 可直接导入。',
    '原始数据目录保留 15 份来源副本，供复核或自定义绘图；普通作图不必读取 NPZ 或 JSON。',
    '所有路径均相对解压后的本文件夹。无需作者电脑、D 盘路径、工作区或额外下载。', '',
    '数据精度：CSV 保留 Python 双精度往返表示；Excel 单元格为数值，显示小数位不代表数据已舍入。',
    '本包是冻结结果的导出，未重跑模型。图 1/3/4 为 N3200，图 5 为 N6400，图 6 均为 N800。',
    '图 2 观测与 4h 后平台必须分开。平台表的 4h 表示开放左端的右侧极限，不是新的观测点。',
    '图 4 中心放大表只标中心样点；全域最大值与严格达标点使用对应数据表。',
    '图 5 每两列一组 XY，横轴半径随时刻变化，不可给六条曲线套同一半径列。', '', '文件与工作表对  

    应：']
readme += [f"图{s['figure']} | Excel: {s['name']} | {s['csv']} | {len(s['rows'])}条" for s
in sheets]
(OUT / '请先阅读.txt').write_text('\n'.join(readme)+'\n', encoding='utf-8-sig')
print(json.dumps({'out': OUT.relative_to(ROOT).as_posix(), 'tables': len(sheets),
    'rows': manifest['total_data_rows'], 'original_sources': len(sources)}, ensure_ascii=False))

```

D.47 draw_figures.R

路径: 补充源码/paper_output/figures/review_20260913/draw_figures.R

用途: 读取六图 14 张 CSV, 核验坐标范围和关键值, 绘制六张 PNG/PDF/SVG 及合并 PDF

使用范围: 当前正文六图绘图实现; 不重新积分

语言: R; 源文件共 197 行。

SHA256: 1b6683dbed416bf4201c889b62d1c44ccdbf75a79a5fbb869da37d139c7eb599

```
options(warn = 1)
suppressWarnings(Sys.setlocale("LC_ALL", "Chinese_China.utf8"))
root <- normalizePath("paper_output/figures/review_20260913", winslash = "/", mustWork =
TRUE)
input <- file.path(root, "input_data", "CSV")
font_cn <- "Microsoft YaHei"
font_math <- "Times New Roman"
palette7 <- c("#0072B2", "#D55E00", "#009E73", "#CC79A7", "#A97700", "#56B4E9", "#414141")
ltyes7 <- c("solid", "dashed", "dotted", "dotdash", "longdash", "twodash", "1343")
shape7 <- c(16, 17, 15, 18, 1, 2, 0)
read_table <- function(fig, name) read.csv(file.path(input, paste0("图", fig), name),
fileEncoding = "UTF-8-BOM", check.names = FALSE)

theme <- function(oma = c(3.8, 0.4, 0.4, 0.4), mar = c(3.5, 4.0, 2.0, 1.0)) {
  par(family = font_cn, ps = 10, cex = 1, cex.axis = 0.86, cex.lab = 0.97,
      las = 1, mgp = c(2.25, 0.62, 0), tcl = -0.25, bty = "l", xaxs = "i",
      yaxs = "i", lend = "round", fg = "#333333", col.axis = "#333333",
      col.lab = "#242424", oma = oma, mar = mar)
}
panel <- function(xlim, ylim, xticks, yticks, title, xlab, ylab, ylabels = NULL, log = "")
{
  plot.new(); plot.window(xlim, ylim, log = log)
  abline(h = yticks, col = "#E3E7EA", lwd = 0.55)
  axis(1, at = xticks, labels = xticks, lwd = 0, lwd.ticks = 0.6)
  axis(2, at = yticks, labels = if (is.null(ylabels)) yticks else ylabels, lwd = 0,
lwd.ticks = 0.6)
  box(bty = "l", lwd = 0.65)
  mtext(title, 3, line = 0.65, adj = 0, cex = 1.0, font = 2, family = font_cn)
  mtext(xlab, 1, line = 2.05, cex = 1.0, family = if(is.expression(xlab)) font_math else
font_cn)
  mtext(ylab, 2, line = 2.6, las = 0, cex = 1.0, family = if(is.expression(ylab)) font_math
else font_cn)
}
footer <- function(labels, cols, ltys, pchs = NULL, ncol = length(labels), title = NULL,
cex = 0.84, y = 0.082) {
  par(fig = c(0, 1, 0, 1), mar = c(0, 0, 0, 0), oma = c(0, 0, 0, 0), new = TRUE)
  plot.new(); plot.window(c(0,1), c(0,1))
  legend(0.5, y, labels, col = cols, lty = ltys, pch = pchs, lwd = 1.5,
      ncol = ncol, bty = "n", xjust = 0.5, yjust = 0.5, cex = cex,
      x.intersp = 0.6, y.intersp = 1.12, seg.len = 2.4, title = title, title.adj = 0.5)
}
note <- function(text, line = 0.15, cex = 0.77) mtext(text, 1, outer = TRUE, line = line,
cex = cex, col = "#4D555B")
emit <- function(name, draw, width = 7.2, height = 4.6) {
  png(file.path(root, paste0(name, ".png")), width = width, height = height, units = "in",
res = 300, type = "cairo", bg = "white", family = font_cn, pointsize = 10)
  draw(); dev.off()
  cairo_pdf(file.path(root, paste0(name, ".pdf")), width = width, height = height, family =
font_cn, pointsize = 10, bg = "white", onefile = TRUE)
  draw(); dev.off()
}
```

```

    svg(file.path(root, paste0(name, ".svg")), width = width, height = height, family =
font_cn, pointsize = 10, bg = "white", onefile = TRUE)
    draw(); dev.off()
    cat("WROTE", name, "PNG 300 dpi / PDF / SVG\n")
}

f1t <- read_table(1, "temperature.csv"); f1c <- read_table(1, "moisture.csv")
f2o <- read_table(2, "observed_0_4h.csv"); f2p <- read_table(2,
"assumed_platform_4_60h.csv")
f3t <- read_table(3, "temperature_0_4h.csv"); f3c <- read_table(3, "moisture_to_event.csv")
f4 <- read_table(4, "drying_main.csv"); f4near <- read_table(4, "maximum_near_event.csv");
f4events <- read_table(4, "event_markers.csv")
f4centre <- read_table(4, "centre_near_event.csv")
f5r <- read_table(5, "radius_observed.csv"); f5c <- read_table(5,
"moisture_profiles_xy_pairs.csv"); f5sel <- read_table(5, "selected_times_and_radii.csv")
f6 <- read_table(6, "scenario_comparison.csv")

stopifnot(nrow(f1t)==121, nrow(f1c)==121, nrow(f2o)==241, nrow(f5sel)==6,
          tail(f2o$time_h,1)==4, tail(f2o$temperature_C,1)==50.165,
          tail(f2o$air_moisture_kg_per_kg,1)==0.04986,
          f2p$endpoint_included[1]==0, nrow(f4near)==2, nrow(f4events)==2,
          f4events$max_C_minus_threshold[2]<0, identical(f6$p, c(1L,2L,4L)))
for (j in 1:6) stopifnot(abs(tail(f5c[[2*j-1]],1)-f5sel$radius_cm[j]) < 1e-12)
cat("DATA CHECKS PASS: original endpoint / open platform / event values / six distinct
radii / six N800 scenario points\n")
checks <- list(
  c("f1_T", range(as.matrix(f1t[-1])), 27.7, 37.5),
  c("f1_C", range(as.matrix(f1c[-1])), 1.45, 2.6),
  c("f2_T", range(f2o$temperature_C), 24, 52.5),
  c("f2_air_indicator", range(f2o$air_moisture_kg_per_kg), .018, .0525),
  c("f3_T", range(as.matrix(f3t[-1])), 26, 52),
  c("f3_C", range(as.matrix(f3c[-1])), 0, 2.65),
  c("f4_C_log_axis", range(as.matrix(f4[c("centre_C","surface_C","max_C")])), .04, 3),
  c("f4_delta_scaled", range(c(f4near$max_C_minus_threshold,
f4events$max_C_minus_threshold))*1e7, -5.45, .8),
  c("f5_R", range(f5r$radius_cm), 1.16, 2.04),
  c("f5_C_log_axis", range(as.matrix(f5c[seq(2,12,2)])), .04, 3),
  c("f6_time", range(as.matrix(f6[-1])), 49, 68)
)
bounds <- as.data.frame(do.call(rbind,checks),stringsAsFactors=FALSE)
names(bounds) <- c("panel","data_min","data_max","axis_min","axis_max")
for(j in 2:5) bounds[[j]] <- as.numeric(bounds[[j]])
bounds$all_data_in_range <- bounds$data_min >= bounds$axis_min & bounds$data_max <=
bounds$axis_max
stopifnot(all(bounds$all_data_in_range))
write.csv(bounds,file.path(root,"axis_range_checks.csv"),row.names=FALSE,fileEncoding="UTF-
8")
cat("DATA RANGE CHECKS PASS: all 11 plotted data ranges fully visible\n")

draw1 <- function() {
  layout(matrix(1:2, 1, 2)); theme(oma = c(4.0,0.4,0.5,0.4))
  panel(c(0,2), c(27.7,37.5), seq(0,2,0.5), seq(28,36,2), "(a) 径向温度",
expression(italic(r)~/ cm"), expression(italic(T)~/ "°C"))
  for(j in 1:7) lines(f1t$r_cm, f1t[[j+1]], col=palette7[j], lty=ltypes7[j], lwd=1.55)
  panel(c(0,2), c(1.45,2.6), seq(0,2,0.5), seq(1.5,2.5,0.2), "(b) 径向干基含水率",
expression(italic(r)~/ cm"), expression(italic(C)~/ "(kg/kg)"),
sprintf("%.1f",seq(1.5,2.5,0.2)))
  for(j in 1:7) lines(f1c$r_cm, f1c[[j+1]], col=palette7[j], lty=ltypes7[j], lwd=1.55)
  footer(c("100", "300", "600", "900", "1200", "1500", "1800"), palette7, ltypes7, ncol=7,
title="时间 / s", y=0.083)
}

```

```

draw2 <- function() {
  layout(matrix(1:4,2,2,byrow=TRUE), widths=c(1.25,1)); theme(oma=c(3.2,0.4,0.4,0.4),
mar=c(3.3,4.4,2.0,0.9))
  panel(c(0,4.12), c(24,52.5), 0:4, seq(25,50,5), "(a) 0-4 h: 环境温度",
expression(italic(t)~/ h"), expression(italic(T)[plain(a)]~/ "°C"))
  lines(f2o$time_h, f2o$temperature_C, col=palette7[1], lwd=1.1)
  points(f2o$time_h, f2o$temperature_C, col=palette7[1], pch=16, cex=0.20)
  abline(v=4,lty=3,col="#8A9297"); points(4,50.165,pch=16,cex=0.65,col=palette7[1])
  text(2.15,43.8,"4 h 末值: 50.165 °C",cex=0.77,adj=0)
  panel(c(0,60), c(24,52.5), seq(0,60,20), seq(25,50,5), "(b) 延拓至 60 h",
expression(italic(t)~/ h"), expression(italic(T)[plain(a)]~/ "°C"))
  rect(4,24,60,52.5,col="#F2F5F7",border=NA); abline(h=seq(25,50,5),col="#E3E7EA",lwd=.55)
  lines(f2o$time_h,f2o$temperature_C,col=palette7[1],lwd=1.2)
  lines(f2p$time_h,f2p$temperature_C,col=palette7[2],lty=2,lwd=1.6)
  points(4,50,pch=21,bg="white",col=palette7[2],cex=.65)
  abline(v=4,lty=3,col="#8A9297"); text(34,42,"t > 4 h: 50 °C",cex=.82)
  panel(c(0,4.12), c(.018,.0525), 0:4, seq(.02,.05,.01), "(c) 0-4 h: 空气水分指标",
expression(italic(t)~/ h"), "空气水分指标 / (kg/kg)",sprintf("%.2f",seq(.02,.05,.01)))
  lines(f2o$time_h,f2o$air_moisture_kg_per_kg,col=palette7[1],lwd=1.1)
  points(f2o$time_h,f2o$air_moisture_kg_per_kg,col=palette7[1],pch=16,cex=.20)
  abline(v=4,lty=3,col="#8A9297"); points(4,.04986,pch=16,cex=.65,col=palette7[1])
  text(1.8,.035,"4 h 末值: 0.04986 kg/kg",cex=.77,adj=0)
  panel(c(0,60), c(.018,.0525), seq(0,60,20), seq(.02,.05,.01), "(d) 延拓至 60 h",
expression(italic(t)~/ h"), "空气水分指标 / (kg/kg)",sprintf("%.2f",seq(.02,.05,.01)))
  rect(4,.018,60,.0525,col="#F2F5F7",border=NA);
  abline(h=seq(.02,.05,.01),col="#E3E7EA",lwd=.55)
  lines(f2o$time_h,f2o$air_moisture_kg_per_kg,col=palette7[1],lwd=1.2)
  lines(f2p$time_h,f2p$boundary_moisture_kg_per_kg,col=palette7[2],lty=2,lwd=1.6)
  points(4,.05,pch=21,bg="white",col=palette7[2],cex=.65)
  abline(v=4,lty=3,col="#8A9297"); text(34,.035,"t > 4 h: 0.05 kg/kg",cex=.82)
  footer(c("观测及分段线性插值","假设平台 (左端开
放)"),palette7[1:2],c(1,2),pchs=c(16,1),ncol=2,y=.046,cex=.84)
}

draw3 <- function() {
  layout(matrix(1:2,1,2)); theme(oma=c(3.25,0.4,0.5,0.4))
  panel(c(0,4),c(26,52),0:4,seq(30,50,5),"(a) 前 4 h 的温度响应",expression(italic(t)~/
h"),expression(italic(T)~/ "°C"))
  lines(f3t$time_h,f3t$centre_T_C,col=palette7[1],lwd=1.7)
  lines(f3t$time_h,f3t$surface_T_C,col=palette7[2],lty=2,lwd=1.7)
  panel(c(0,60),c(0,2.65),seq(0,60,12),seq(0,2.5,.5),"(b) 全过程干基含水
率",expression(italic(t)~/ h"),expression(italic(C)~/
(kg/kg)"),sprintf("%.1f",seq(0,2.5,.5)))
  abline(h=.15,col="#777777",lty=3,lwd=1)
  lines(f3c$time_h,f3c$centre_C_kg_per_kg,col=palette7[1],lwd=1.7)
  lines(f3c$time_h,f3c$surface_C_kg_per_kg,col=palette7[2],lty=2,lwd=1.7)
  text(39,.30,"阈值 0.15",cex=.8,adj=0,col="#646464")
  footer(c("中心 (r = 0)", "表面 (r = 2 cm)"),palette7[1:2],c(1,2),ncol=2,y=.067,cex=.9)
}

draw4 <- function() {
  layout(matrix(1:2,1,2),widths=c(1.1,1));
  theme(oma=c(4.1,0.4,0.6,0.4),mar=c(3.5,4.1,2.1,1))
  panel(c(0,60),c(.04,3),seq(0,60,12),c(.05,.1,.2,.5,1,2),"(a) 达标过程 (对数刻
度)",expression(italic(t)~/ h"),expression(italic(C)~/
(kg/kg)"),c("0.05","0.1","0.2","0.5","1","2"),log="y")
  abline(h=.15,col="#777777",lty=3,lwd=1)
  lines(f4$time_h,f4$max_C,col="#303030",lwd=2.0)
  lines(f4$time_h,f4$centre_C,col=palette7[1],lty=2,lwd=1.6)
  lines(f4$time_h,f4$surface_C,col=palette7[2],lty=1,lwd=1.6)
  points(f4events$time_h[1],.15,col="#303030",bg="white",pch=21,cex=.85)
  text(7,.18,"阈值 0.15",cex=.78,adj=0,col="#606060")
  text(17,1.5,"中心与全域最大值重合",cex=.78,adj=0,col=palette7[1])
}

```

```

panel(c(-.07,1.8),c(-5.45,.8),c(0,.4,.8,1.2,1.6),seq(-5,0,1),"(b) 临界时刻附近",expression((italic(t)-italic(t)[plain(c)])~"/ s"),expression((italic(M)-0.15)~"/ "(10^{-7})~"kg/kg"))
abline(h=0,col="#777777",lty=3,lwd=1)
lines(f4near$delta_time_s,f4near$max_C_minus_threshold*1e7,col="#6C747A",lty=3,lwd=1.1)

points(f4near$delta_time_s,f4near$max_C_minus_threshold*1e7,pch=22,bg="#6C747A",col="#6C747A",cex=.8)

points(f4events$delta_time_s[1],f4events$max_C_minus_threshold[1]*1e7,pch=21,bg="white",col="#222222",cex=1.05,lwd=1.25)

points(f4events$delta_time_s[2],f4events$max_C_minus_threshold[2]*1e7,pch=17,col=palette7[3],cex=1.1)
text(.20,.43,expression(italic(t)[plain(c)]),adj=0,cex=.95,family=font_math)
text(.52,-.94,expression(italic(t)[plain(r)]),adj=0,cex=.95,family=font_math,col=palette7[3])
text(.10,-4.2,"临界时刻: 57.472302 h\n 核变时刻: 57.4724 h\n 二者相差: 0.352978 s",adj=0,cex=.78)
footer(c("全域最大值 M(t)", "中心", "表面", "临界等号根", "严格达标取值"),c("#303030",palette7[1],palette7[2],"#222222",palette7[3]),c(1,2,1,NA,NA),pchs=c(NA,NA,NA,1,17),ncol=3,y=.080,cex=.80)
}

draw5 <- function() {
  layout(matrix(1:2,1,2)); theme(oma=c(4.0,.4,.5,.4),mar=c(3.5,4.0,2.0,1))
  panel(c(0,72),c(1.16,2.04),seq(0,72,12),seq(1.2,2,.2),"(a) 收缩半径与选定时刻",expression(italic(t)~"/ h"),expression(italic(R)~"/ cm"))
  lines(f5r$time_h,f5r$radius_cm,col="#69757D",lwd=1.5)
  for(j in 1:6)
  points(f5sel$time_h[j],f5sel$radius_cm[j],col=palette7[j],bg="white",pch=shape7[j],cex=.85,lwd=1.3)
  text(29,1.81,"后期半径接近 1.2 cm",cex=.78,adj=0)
  panel(c(0,2.04),c(.04,3),seq(0,2,.5),c(.05,.1,.2,.5,1,2),"(b) 含水率 (对数刻度)",expression(italic(r)~"/ cm"),expression(italic(C)~"/ (kg/kg)"),c("0.05","0.1","0.2","0.5","1","2"),log="y")
  for(j in 1:6) {
    xx <- f5c[[2*j-1]]; yy <- f5c[[2*j]]
    lines(xx,yy,col=palette7[j],lty=ltyes7[j],lwd=1.55)
    points(tail(xx,1),tail(yy,1),col=palette7[j],pch=shape7[j],cex=.85,lwd=1.2)
  }
  text(1.63,.16,"曲线止于各自表面\n 表面以点形标出",cex=.76)
  footer(c("0 h", "6 h", "12 h", "24 h", "48 h", "临界 51.090575 h"),palette7[1:6],ltyes7[1:6],shape7[1:6],ncol=3,y=.08,cex=.85)
}

draw6 <- function() {
  layout(matrix(1)); theme(oma=c(2.5,.3,.4,.3),mar=c(3.7,4.0,2.2,1.2))
  panel(c(.82,4.18),c(49,68),c(1,2,4),seq(50,68,3),"经验边界阻力情景下的临界时长",expression(italic(p)),expression(italic(t)[plain(c)]~"/ h"))
  lines(f6$p,f6$Q23_event_h,col=palette7[1],lwd=1.7)
  lines(f6$p,f6$Q4_event_h,col=palette7[2],lty=2,lwd=1.7)
  points(f6$p,f6$Q23_event_h,col=palette7[1],pch=16,cex=.85)
  points(f6$p,f6$Q4_event_h,col=palette7[2],pch=17,cex=.95)
  text(f6$p,f6$Q23_event_h+.65,sprintf("%.4f",f6$Q23_event_h),col=palette7[1],cex=.83)
  text(f6$p,f6$Q4_event_h-.75,sprintf("%.4f",f6$Q4_event_h),col=palette7[2],cex=.83)
  legend("topleft",c("固定半径, 附录 3 物性", "径向收缩, 附录 4 物性"),col=palette7[1:2],lty=c(1,2),pch=c(16,17),lwd=1.7,bty="n",cex=.9,y.intersp=1.2)
  note("统一采用 N = 800、零潜热负荷; 该经验参数族未经实测等温线标定。",line=.5,cex=.82)
}

emit("图 1 径向温度与含水率",draw1,7.2,4.6)
emit("图 2 环境观测与平台延拓",draw2,7.2,5.9)

```

```

emit("图 3_中心与表面热湿响应",draw3,7.2,4.5)
emit("图 4_全域达标与临界放大",draw4,7.2,4.9)
emit("图 5_收缩半径与域内含水率",draw5,7.2,4.6)
emit("图 6_经验边界阻力情景",draw6,7.2,4.6)

cairo_pdf(file.path(root,"六图合并核
查.pdf"),width=7.2,height=5.9,family=font_cn,pointsize=10,bg="white",onefile=TRUE)
draw1(); draw2(); draw3(); draw4(); draw5(); draw6(); dev.off()

write.csv(data.frame(metric=c("figure_count","input_csv_count","input_data_rows","figure4_r
eport_delta_s","figure4_report_M_minus_threshold","figure4_max_centre_difference","figure4_
saved_endpoint_h","new_pde_runs"),

value=c(6,14,4627,f4events$delta_time_s[2],f4events$max_C_minus_threshold[2],max(abs(f4$max
_C-f4$centre_C)),max(f4$time_h),0)),
  file.path(root,"plot_checks.csv"),row.names=FALSE,fileEncoding="UTF-8")
capture.output(sessionInfo(),file=file.path(root,"R_session_info.txt"))
cat("DONE. Plot data remain unchanged. No PDE integration or DOCX modification performed.
\n")

```